

Detects pixelated numbers

First I **generated my | data using code from AI** that I prompt, it was tasked to blur images of numbers 0-9 in blocks of 32X32.

The code was changed a bit by me to fit, and it's added in in the project files (**'num_gen.py'**). This code outputs the data seen also in the project files (**'0_9_blur'**) (default name for data output is 'my_dataset')

The goal here is to recognise not only numbers from images (too easy), but to classify blurred numbers. This has real world use as it can be used to "see" passwords that were blurred in a video (for example on youtube).

The most popular blur is gaussian blur, but in theory any blur type can be used to train this model, or even another model to classify the blur type and then pass it on to a specialized model.

```
In [1]: import ssl
ssl._create_default_https_context = ssl._create_unverified_context
```

```
In [89]: from IPython.display import Image
```

```
In [2]: import numpy as np
import matplotlib.pyplot as plt
import keras
from keras import layers
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import random
import tensorflow_datasets as tfds
```

```
In [3]: from keras.utils import np_utils
```

```
In [4]: import gc
```

```
In [5]: import tensorflow as tf
```

```
In [6]: def augment(image, label):
    image = tf.image.random_flip_left_right(image)
    image = tf.image.random_brightness(image, max_delta=0.1)
    return image, label
```

```
In [7]: def training_progress(history):
    plt.figure(figsize=(12, 5))

    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Accuracy')
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
    plt.title('CNN Model Accuracy')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()
```

```

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('CNN Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()

plt.tight_layout()
plt.show()

return

```

Load and see the data

```

In [8]: train_dir = '0_9_blur/train'
        test_dir = '0_9_blur/test'

```

```

In [9]: IMG_SIZE = (32, 32)

```

```

In [10]: train_ds = tf.keras.utils.image_dataset_from_directory(
            train_dir,
            labels='inferred',
            label_mode='int',
            image_size=IMG_SIZE,
            batch_size=32,
            shuffle=False
        )

```

Found 4000 files belonging to 10 classes.

```

In [11]: test_ds = tf.keras.utils.image_dataset_from_directory(
            test_dir,
            labels='inferred',
            label_mode='int',
            image_size=IMG_SIZE,
            batch_size=32,
            shuffle=False
        )

```

Found 1000 files belonging to 10 classes.

```

In [12]: print("Converting datasets to NumPy arrays...")
        x_train_cnn = np.concatenate([x for x, y in train_ds], axis=0).astype('uint8')
        y_train_cnn = np.concatenate([y for x, y in train_ds], axis=0).reshape(-1, 1)

        x_test_cnn = np.concatenate([x for x, y in test_ds], axis=0).astype('uint8')
        y_test_cnn = np.concatenate([y for x, y in test_ds], axis=0).reshape(-1, 1)

```

Converting datasets to NumPy arrays...

```

In [13]: print("Train size:", x_train_cnn.shape)
        print("Test size:", x_test_cnn.shape)
        print("Image shape:", x_train_cnn[0].shape)
        print("Number of classes:", len(np.unique(y_train_cnn)))

```

Train size: (4000, 32, 32, 3)
Test size: (1000, 32, 32, 3)
Image shape: (32, 32, 3)
Number of classes: 10

Show data samples:

```
In [14]: x_train_cnn_size = x_train_cnn.shape[0]
x_test_cnn_size = x_test_cnn.shape[0]
rand_x_train_size = random.randint(1, x_train_cnn_size)
rand_x_test_size = random.randint(1, x_test_cnn_size)

plt.subplot(1,3,1)
plt.imshow(x_train_cnn[rand_x_train_size], cmap='gray')
plt.title(f'Random {y_train_cnn[rand_x_train_size]} from train data (No.{rand_x_')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(x_test_cnn[rand_x_test_size], cmap='gray')
plt.title(f'Random {y_test_cnn[rand_x_test_size]} from train data (No.{rand_x_tr')
plt.axis('off')

plt.show()
```

Random [9] from train data (No.3649)

Random [3] from train data (No.3649)



Get the data and model ready and start training

Add a channel dimension and normalize

```
In [15]: num_classes = 10
x_train_cnn = x_train_cnn.reshape(x_train_cnn.shape[0], 32, 32, 3).astype('float32')
x_test_cnn = x_test_cnn.reshape(x_test_cnn.shape[0], 32, 32, 3).astype('float32')
print("CNN Train size:", x_train_cnn.shape)
print("CNN Test size:", x_test_cnn.shape)
```

CNN Train size: (4000, 32, 32, 3)
CNN Test size: (1000, 32, 32, 3)

OneHotEncoding

```
In [16]: y_train_cnn_oh = keras.utils.np_utils.to_categorical(y_train_cnn, num_classes)
y_test_cnn_oh = keras.utils.np_utils.to_categorical(y_test_cnn, num_classes)
print("CNN Train labels shape:", y_train_cnn_oh.shape)
print("CNN Test labels shape:", y_test_cnn_oh.shape)
```

CNN Train labels shape: (4000, 10)
CNN Test labels shape: (1000, 10)

```
In [17]: indices = np.arange(x_train_cnn.shape[0])
```

```

np.random.shuffle(indices)

x_train_cnn = x_train_cnn[indices]
y_train_cnn_oh = y_train_cnn_oh[indices]

```

Build a CNN model (convolution, pooling, dense layers)

```

In [80]: def single_model_trainer(epochs_,batch_size_,validation_split_):
    model_cnn = keras.Sequential()
    model_cnn.add(keras.Input(shape=(32, 32, 3)))
    model_cnn.add(layers.Rescaling(1./255))

    model_cnn.add(layers.Conv2D(32, (3, 3), padding="same", activation='relu',
                                kernel_regularizer=keras.regularizers.l2(0.002))
    model_cnn.add(layers.BatchNormalization())
    model_cnn.add(layers.MaxPooling2D((2, 2)))

    model_cnn.add(layers.Conv2D(64, (3, 3), padding="same", activation='relu',
                                kernel_regularizer=keras.regularizers.l2(0.002))
    model_cnn.add(layers.BatchNormalization())
    model_cnn.add(layers.MaxPooling2D((2, 2)))

    model_cnn.add(layers.Flatten())
    model_cnn.add(layers.Dense(128, activation='relu',
                                kernel_regularizer=keras.regularizers.l2(0.002)))
    model_cnn.add(layers.BatchNormalization())
    model_cnn.add(layers.Dropout(0.4))
    model_cnn.add(layers.Dense(10, activation='softmax'))

    opt = tf.keras.optimizers.Adam(learning_rate=0.0003)
    model_cnn.compile(optimizer=opt, loss='categorical_crossentropy', metrics=['

    lr_reducer = keras.callbacks.ReduceLROnPlateau(
        monitor='val_loss',
        factor=0.5,
        patience=4,
        verbose=1
    )

    early_stop = keras.callbacks.EarlyStopping(
        monitor='val_accuracy',
        patience=12,
        restore_best_weights=True
    )
    history = model_cnn.fit(x_train_cnn, y_train_cnn_oh, epochs=epochs_,batch_si

    return model_cnn, history

```

Copy for training multiple loop:

```

In [66]: def model_trainer(epochs_,batch_size_,validation_split_):
    model_cnn = keras.Sequential()
    model_cnn.add(keras.Input(shape=(32, 32, 3)))
    model_cnn.add(layers.Rescaling(1./255))

```

```

model_cnn.add(layers.Conv2D(32, (3, 3), padding="same", activation='relu',
                             kernel_regularizer=keras.regularizers.l2(0.002))
model_cnn.add(layers.BatchNormalization())
model_cnn.add(layers.MaxPooling2D((2, 2)))

model_cnn.add(layers.Conv2D(64, (3, 3), padding="same", activation='relu',
                             kernel_regularizer=keras.regularizers.l2(0.002))
model_cnn.add(layers.BatchNormalization())
model_cnn.add(layers.MaxPooling2D((2, 2)))

model_cnn.add(layers.Flatten())
model_cnn.add(layers.Dense(128, activation='relu',
                             kernel_regularizer=keras.regularizers.l2(0.002)))
model_cnn.add(layers.BatchNormalization())
model_cnn.add(layers.Dropout(0.4))
model_cnn.add(layers.Dense(10, activation='softmax'))

opt = tf.keras.optimizers.Adam(learning_rate=0.0003)
model_cnn.compile(optimizer=opt, loss='categorical_crossentropy', metrics=['

lr_reducer = keras.callbacks.ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.5,
    patience=4,
    verbose=1
)

early_stop = keras.callbacks.EarlyStopping(
    monitor='val_accuracy',
    patience=12,
    restore_best_weights=True
)
history = model_cnn.fit(x_train_cnn, y_train_cnn_oh, epochs=epochs_, batch_si

del model_cnn

return history

```

```

In [67]: validation_split_arr = np.linspace(0.2, 0.4, 5)
batch_size_arr = np.linspace(8, 64, 8)
curr_model = [100, batch_size_arr, validation_split_arr]

```

```

In [68]: history = []
my_history = []

```

```

In [69]: counter_for_train = 0

```

```

In [64]: history_cnn = model_trainer(100, 64, 0.35)

```

Epoch 1/100
41/41 [=====] - 2s 23ms/step - loss: 3.2062 - accuracy: 0.1500 - val_loss: 3.1517 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 2/100
41/41 [=====] - 0s 10ms/step - loss: 2.7260 - accuracy: 0.2388 - val_loss: 3.7573 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 3/100
41/41 [=====] - 0s 10ms/step - loss: 2.4447 - accuracy: 0.3127 - val_loss: 4.6187 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 4/100
41/41 [=====] - 0s 10ms/step - loss: 2.0989 - accuracy: 0.4212 - val_loss: 5.3667 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 5/100
41/41 [=====] - ETA: 0s - loss: 1.7675 - accuracy: 0.5435
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
41/41 [=====] - 0s 11ms/step - loss: 1.7675 - accuracy: 0.5435 - val_loss: 5.6912 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 6/100
41/41 [=====] - 0s 11ms/step - loss: 1.4491 - accuracy: 0.6715 - val_loss: 6.1290 - val_accuracy: 0.1129 - lr: 1.5000e-04
Epoch 7/100
41/41 [=====] - 0s 11ms/step - loss: 1.2918 - accuracy: 0.7319 - val_loss: 6.9603 - val_accuracy: 0.1029 - lr: 1.5000e-04
Epoch 8/100
41/41 [=====] - 0s 11ms/step - loss: 1.1565 - accuracy: 0.7823 - val_loss: 8.2984 - val_accuracy: 0.1029 - lr: 1.5000e-04
Epoch 9/100
37/41 [=====>...] - ETA: 0s - loss: 1.0821 - accuracy: 0.7952
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
41/41 [=====] - 1s 14ms/step - loss: 1.0782 - accuracy: 0.7958 - val_loss: 9.4258 - val_accuracy: 0.1107 - lr: 1.5000e-04
Epoch 10/100
41/41 [=====] - 1s 13ms/step - loss: 0.9627 - accuracy: 0.8538 - val_loss: 10.8822 - val_accuracy: 0.1386 - lr: 7.5000e-05
Epoch 11/100
41/41 [=====] - 1s 14ms/step - loss: 0.9159 - accuracy: 0.8635 - val_loss: 10.8979 - val_accuracy: 0.1307 - lr: 7.5000e-05
Epoch 12/100
41/41 [=====] - 1s 13ms/step - loss: 0.8626 - accuracy: 0.8881 - val_loss: 10.9510 - val_accuracy: 0.1114 - lr: 7.5000e-05
Epoch 13/100
39/41 [=====>...] - ETA: 0s - loss: 0.8410 - accuracy: 0.8910
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
41/41 [=====] - 1s 13ms/step - loss: 0.8418 - accuracy: 0.8912 - val_loss: 10.4454 - val_accuracy: 0.1029 - lr: 7.5000e-05
Epoch 14/100
41/41 [=====] - 1s 16ms/step - loss: 0.7919 - accuracy: 0.9173 - val_loss: 10.6493 - val_accuracy: 0.1029 - lr: 3.7500e-05
Epoch 15/100
41/41 [=====] - 1s 15ms/step - loss: 0.7676 - accuracy: 0.9265 - val_loss: 7.9718 - val_accuracy: 0.1757 - lr: 3.7500e-05
Epoch 16/100
41/41 [=====] - 1s 13ms/step - loss: 0.7477 - accuracy: 0.9392 - val_loss: 5.8539 - val_accuracy: 0.1957 - lr: 3.7500e-05
Epoch 17/100
41/41 [=====] - ETA: 0s - loss: 0.7279 - accuracy: 0.9392

Epoch 17: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
41/41 [=====] - 1s 15ms/step - loss: 0.7279 - accuracy: 0.9392 - val_loss: 5.5051 - val_accuracy: 0.1064 - lr: 3.7500e-05
Epoch 18/100
41/41 [=====] - 1s 14ms/step - loss: 0.7141 - accuracy: 0.9492 - val_loss: 6.1030 - val_accuracy: 0.1271 - lr: 1.8750e-05
Epoch 19/100
41/41 [=====] - 1s 14ms/step - loss: 0.7184 - accuracy: 0.9450 - val_loss: 3.7354 - val_accuracy: 0.2071 - lr: 1.8750e-05
Epoch 20/100
41/41 [=====] - 1s 13ms/step - loss: 0.7041 - accuracy: 0.9519 - val_loss: 2.0850 - val_accuracy: 0.4364 - lr: 1.8750e-05
Epoch 21/100
41/41 [=====] - 1s 12ms/step - loss: 0.6887 - accuracy: 0.9519 - val_loss: 2.3486 - val_accuracy: 0.3693 - lr: 1.8750e-05
Epoch 22/100
41/41 [=====] - 1s 15ms/step - loss: 0.6895 - accuracy: 0.9473 - val_loss: 2.1858 - val_accuracy: 0.4336 - lr: 1.8750e-05
Epoch 23/100
41/41 [=====] - 1s 15ms/step - loss: 0.6728 - accuracy: 0.9612 - val_loss: 1.6272 - val_accuracy: 0.5493 - lr: 1.8750e-05
Epoch 24/100
41/41 [=====] - 1s 15ms/step - loss: 0.6692 - accuracy: 0.9596 - val_loss: 1.5188 - val_accuracy: 0.6493 - lr: 1.8750e-05
Epoch 25/100
41/41 [=====] - 1s 15ms/step - loss: 0.6654 - accuracy: 0.9562 - val_loss: 1.5574 - val_accuracy: 0.5929 - lr: 1.8750e-05
Epoch 26/100
41/41 [=====] - 1s 14ms/step - loss: 0.6568 - accuracy: 0.9635 - val_loss: 1.2705 - val_accuracy: 0.6829 - lr: 1.8750e-05
Epoch 27/100
41/41 [=====] - 1s 13ms/step - loss: 0.6505 - accuracy: 0.9627 - val_loss: 1.0534 - val_accuracy: 0.7943 - lr: 1.8750e-05
Epoch 28/100
41/41 [=====] - 1s 14ms/step - loss: 0.6410 - accuracy: 0.9665 - val_loss: 0.9854 - val_accuracy: 0.8086 - lr: 1.8750e-05
Epoch 29/100
41/41 [=====] - 1s 13ms/step - loss: 0.6383 - accuracy: 0.9692 - val_loss: 1.3117 - val_accuracy: 0.6893 - lr: 1.8750e-05
Epoch 30/100
41/41 [=====] - 1s 14ms/step - loss: 0.6407 - accuracy: 0.9650 - val_loss: 0.9887 - val_accuracy: 0.7993 - lr: 1.8750e-05
Epoch 31/100
41/41 [=====] - 1s 13ms/step - loss: 0.6344 - accuracy: 0.9677 - val_loss: 0.9929 - val_accuracy: 0.8057 - lr: 1.8750e-05
Epoch 32/100
41/41 [=====] - 1s 13ms/step - loss: 0.6205 - accuracy: 0.9700 - val_loss: 0.9666 - val_accuracy: 0.8221 - lr: 1.8750e-05
Epoch 33/100
41/41 [=====] - 1s 13ms/step - loss: 0.6192 - accuracy: 0.9708 - val_loss: 1.0281 - val_accuracy: 0.7857 - lr: 1.8750e-05
Epoch 34/100
41/41 [=====] - 0s 12ms/step - loss: 0.6082 - accuracy: 0.9735 - val_loss: 0.8841 - val_accuracy: 0.8521 - lr: 1.8750e-05
Epoch 35/100
41/41 [=====] - 0s 12ms/step - loss: 0.5965 - accuracy: 0.9788 - val_loss: 0.9742 - val_accuracy: 0.8157 - lr: 1.8750e-05
Epoch 36/100
41/41 [=====] - 1s 12ms/step - loss: 0.5981 - accuracy: 0.9781 - val_loss: 0.9351 - val_accuracy: 0.8321 - lr: 1.8750e-05

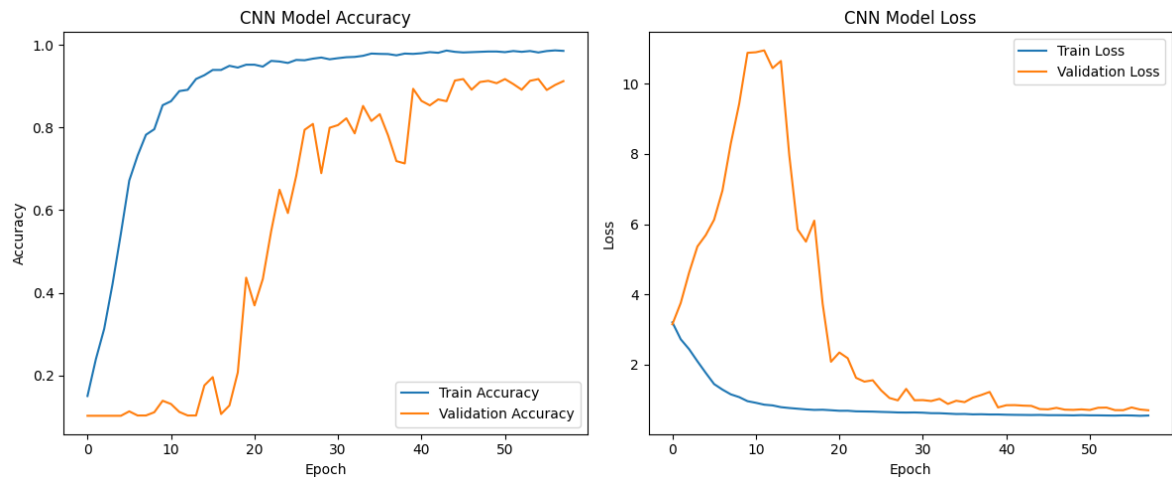
Epoch 37/100
41/41 [=====] - 1s 13ms/step - loss: 0.5868 - accuracy: 0.9777 - val_loss: 1.0640 - val_accuracy: 0.7807 - lr: 1.8750e-05
Epoch 38/100
37/41 [=====>...] - ETA: 0s - loss: 0.5905 - accuracy: 0.9738
Epoch 38: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
41/41 [=====] - 0s 12ms/step - loss: 0.5897 - accuracy: 0.9746 - val_loss: 1.1383 - val_accuracy: 0.7186 - lr: 1.8750e-05
Epoch 39/100
41/41 [=====] - 1s 13ms/step - loss: 0.5830 - accuracy: 0.9788 - val_loss: 1.2252 - val_accuracy: 0.7129 - lr: 9.3750e-06
Epoch 40/100
41/41 [=====] - 1s 13ms/step - loss: 0.5831 - accuracy: 0.9781 - val_loss: 0.7837 - val_accuracy: 0.8936 - lr: 9.3750e-06
Epoch 41/100
41/41 [=====] - 1s 13ms/step - loss: 0.5766 - accuracy: 0.9796 - val_loss: 0.8471 - val_accuracy: 0.8643 - lr: 9.3750e-06
Epoch 42/100
41/41 [=====] - 1s 15ms/step - loss: 0.5727 - accuracy: 0.9823 - val_loss: 0.8496 - val_accuracy: 0.8536 - lr: 9.3750e-06
Epoch 43/100
41/41 [=====] - 1s 17ms/step - loss: 0.5705 - accuracy: 0.9808 - val_loss: 0.8355 - val_accuracy: 0.8679 - lr: 9.3750e-06
Epoch 44/100
38/41 [=====>...] - ETA: 0s - loss: 0.5674 - accuracy: 0.9864
Epoch 44: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
41/41 [=====] - 1s 16ms/step - loss: 0.5677 - accuracy: 0.9862 - val_loss: 0.8280 - val_accuracy: 0.8636 - lr: 9.3750e-06
Epoch 45/100
41/41 [=====] - 1s 17ms/step - loss: 0.5707 - accuracy: 0.9831 - val_loss: 0.7377 - val_accuracy: 0.9136 - lr: 4.6875e-06
Epoch 46/100
41/41 [=====] - 1s 14ms/step - loss: 0.5635 - accuracy: 0.9815 - val_loss: 0.7297 - val_accuracy: 0.9171 - lr: 4.6875e-06
Epoch 47/100
41/41 [=====] - 1s 14ms/step - loss: 0.5634 - accuracy: 0.9823 - val_loss: 0.7718 - val_accuracy: 0.8914 - lr: 4.6875e-06
Epoch 48/100
41/41 [=====] - 1s 14ms/step - loss: 0.5621 - accuracy: 0.9831 - val_loss: 0.7209 - val_accuracy: 0.9100 - lr: 4.6875e-06
Epoch 49/100
41/41 [=====] - 1s 14ms/step - loss: 0.5579 - accuracy: 0.9838 - val_loss: 0.7150 - val_accuracy: 0.9129 - lr: 4.6875e-06
Epoch 50/100
41/41 [=====] - 1s 15ms/step - loss: 0.5646 - accuracy: 0.9838 - val_loss: 0.7288 - val_accuracy: 0.9071 - lr: 4.6875e-06
Epoch 51/100
41/41 [=====] - 1s 17ms/step - loss: 0.5591 - accuracy: 0.9823 - val_loss: 0.7124 - val_accuracy: 0.9171 - lr: 4.6875e-06
Epoch 52/100
41/41 [=====] - 1s 15ms/step - loss: 0.5584 - accuracy: 0.9850 - val_loss: 0.7750 - val_accuracy: 0.9050 - lr: 4.6875e-06
Epoch 53/100
41/41 [=====] - 1s 17ms/step - loss: 0.5546 - accuracy: 0.9831 - val_loss: 0.7796 - val_accuracy: 0.8914 - lr: 4.6875e-06
Epoch 54/100
41/41 [=====] - 1s 16ms/step - loss: 0.5519 - accuracy: 0.9850 - val_loss: 0.7078 - val_accuracy: 0.9129 - lr: 4.6875e-06


```

Epoch 55/100
41/41 [=====] - 1s 16ms/step - loss: 0.5581 - accuracy:
0.9815 - val_loss: 0.7059 - val_accuracy: 0.9171 - lr: 4.6875e-06
Epoch 56/100
41/41 [=====] - 1s 15ms/step - loss: 0.5548 - accuracy:
0.9850 - val_loss: 0.7834 - val_accuracy: 0.8907 - lr: 4.6875e-06
Epoch 57/100
41/41 [=====] - 1s 14ms/step - loss: 0.5461 - accuracy:
0.9865 - val_loss: 0.7256 - val_accuracy: 0.9029 - lr: 4.6875e-06
Epoch 58/100
41/41 [=====] - 1s 15ms/step - loss: 0.5537 - accuracy:
0.9854 - val_loss: 0.7031 - val_accuracy: 0.9121 - lr: 4.6875e-06

```

In [65]: `training_progress(history_cnn)`



After I found a model that seems promising, I test the hyperparameters in a loop to find the optimal one:

```

In [70]: for batch_array_number in range(0, len(batch_size_arr)):
          for validation_array_number in range(0, len(validation_split_arr)):
              counter_for_train = counter_for_train + 1
              print([curr_model[0], int(curr_model[1][batch_array_number]), curr_model[2]
              print(f' {(100*counter_for_train)/(len(batch_size_arr)*len(validation_spl
              # Save the data to remake if found
              my_history.append([curr_model[0], int(curr_model[1][batch_array_number]),
              # Save the data
              history.append(model_trainer(curr_model[0], int(curr_model[1][batch_array
              gc.collect()

```

[100, 8, 0.2]
2.5%
Epoch 1/100
400/400 [=====] - 6s 9ms/step - loss: 2.8984 - accuracy:
0.1863 - val_loss: 8.1513 - val_accuracy: 0.1112 - lr: 3.0000e-04
Epoch 2/100
400/400 [=====] - 4s 11ms/step - loss: 2.0882 - accuracy
: 0.4056 - val_loss: 24.2131 - val_accuracy: 0.1112 - lr: 3.0000e-04
Epoch 3/100
400/400 [=====] - 4s 10ms/step - loss: 1.5164 - accuracy
: 0.5938 - val_loss: 20.2035 - val_accuracy: 0.1075 - lr: 3.0000e-04
Epoch 4/100
400/400 [=====] - 3s 9ms/step - loss: 1.0148 - accuracy:
0.7912 - val_loss: 13.3179 - val_accuracy: 0.1725 - lr: 3.0000e-04
Epoch 5/100
400/400 [=====] - 3s 7ms/step - loss: 0.7079 - accuracy:
0.9106 - val_loss: 7.9740 - val_accuracy: 0.1900 - lr: 3.0000e-04
Epoch 6/100
400/400 [=====] - 4s 9ms/step - loss: 0.5334 - accuracy:
0.9541 - val_loss: 7.5515 - val_accuracy: 0.2275 - lr: 3.0000e-04
Epoch 7/100
400/400 [=====] - 4s 10ms/step - loss: 0.4548 - accuracy
: 0.9663 - val_loss: 12.2565 - val_accuracy: 0.1838 - lr: 3.0000e-04
Epoch 8/100
400/400 [=====] - 4s 10ms/step - loss: 0.4013 - accuracy
: 0.9744 - val_loss: 10.3321 - val_accuracy: 0.2612 - lr: 3.0000e-04
Epoch 9/100
400/400 [=====] - 3s 8ms/step - loss: 0.3693 - accuracy:
0.9772 - val_loss: 50.0309 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 10/100
400/400 [=====] - 4s 10ms/step - loss: 0.3201 - accuracy
: 0.9844 - val_loss: 3.2556 - val_accuracy: 0.4087 - lr: 3.0000e-04
Epoch 11/100
400/400 [=====] - 4s 10ms/step - loss: 0.2999 - accuracy
: 0.9847 - val_loss: 59.1950 - val_accuracy: 0.2763 - lr: 3.0000e-04
Epoch 12/100
400/400 [=====] - 4s 10ms/step - loss: 0.2734 - accuracy
: 0.9841 - val_loss: 117.8520 - val_accuracy: 0.0975 - lr: 3.0000e-04
Epoch 13/100
400/400 [=====] - 4s 10ms/step - loss: 0.2424 - accuracy
: 0.9906 - val_loss: 8.1097 - val_accuracy: 0.3250 - lr: 3.0000e-04
Epoch 14/100
398/400 [=====>.] - ETA: 0s - loss: 0.2347 - accuracy: 0.9
890
Epoch 14: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
400/400 [=====] - 4s 10ms/step - loss: 0.2343 - accuracy
: 0.9891 - val_loss: 222.0292 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 15/100
400/400 [=====] - 4s 10ms/step - loss: 0.2025 - accuracy
: 0.9916 - val_loss: 10.2757 - val_accuracy: 0.2113 - lr: 1.5000e-04
Epoch 16/100
400/400 [=====] - 4s 10ms/step - loss: 0.1807 - accuracy
: 0.9956 - val_loss: 17.0033 - val_accuracy: 0.1088 - lr: 1.5000e-04
Epoch 17/100
400/400 [=====] - 4s 9ms/step - loss: 0.1647 - accuracy:
0.9962 - val_loss: 3.6808 - val_accuracy: 0.4487 - lr: 1.5000e-04
Epoch 18/100
398/400 [=====>.] - ETA: 0s - loss: 0.1619 - accuracy: 0.9
950
Epoch 18: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.

400/400 [=====] - 4s 10ms/step - loss: 0.1628 - accuracy
: 0.9944 - val_loss: 5.7650 - val_accuracy: 0.5587 - lr: 1.5000e-04
Epoch 19/100
400/400 [=====] - 4s 10ms/step - loss: 0.1462 - accuracy
: 0.9978 - val_loss: 2.7857 - val_accuracy: 0.5838 - lr: 7.5000e-05
Epoch 20/100
400/400 [=====] - 4s 10ms/step - loss: 0.1344 - accuracy
: 0.9987 - val_loss: 0.9532 - val_accuracy: 0.7300 - lr: 7.5000e-05
Epoch 21/100
400/400 [=====] - 4s 10ms/step - loss: 0.1313 - accuracy
: 0.9981 - val_loss: 1.2970 - val_accuracy: 0.6800 - lr: 7.5000e-05
Epoch 22/100
400/400 [=====] - 4s 10ms/step - loss: 0.1315 - accuracy
: 0.9969 - val_loss: 14.8637 - val_accuracy: 0.1000 - lr: 7.5000e-05
Epoch 23/100
400/400 [=====] - 4s 10ms/step - loss: 0.1282 - accuracy
: 0.9953 - val_loss: 17.3166 - val_accuracy: 0.0950 - lr: 7.5000e-05
Epoch 24/100
400/400 [=====] - ETA: 0s - loss: 0.1142 - accuracy: 0.9
978
Epoch 24: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
400/400 [=====] - 4s 10ms/step - loss: 0.1142 - accuracy
: 0.9978 - val_loss: 13.2757 - val_accuracy: 0.1238 - lr: 7.5000e-05
Epoch 25/100
400/400 [=====] - 4s 10ms/step - loss: 0.1104 - accuracy
: 0.9984 - val_loss: 0.1531 - val_accuracy: 0.9800 - lr: 3.7500e-05
Epoch 26/100
400/400 [=====] - 4s 10ms/step - loss: 0.1042 - accuracy
: 0.9991 - val_loss: 15.0379 - val_accuracy: 0.1150 - lr: 3.7500e-05
Epoch 27/100
400/400 [=====] - 4s 10ms/step - loss: 0.1043 - accuracy
: 0.9987 - val_loss: 1.3726 - val_accuracy: 0.6975 - lr: 3.7500e-05
Epoch 28/100
400/400 [=====] - 4s 10ms/step - loss: 0.1013 - accuracy
: 0.9984 - val_loss: 16.3204 - val_accuracy: 0.1037 - lr: 3.7500e-05
Epoch 29/100
398/400 [=====>.] - ETA: 0s - loss: 0.0948 - accuracy: 0.9
987
Epoch 29: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
400/400 [=====] - 4s 10ms/step - loss: 0.0948 - accuracy
: 0.9987 - val_loss: 1.3031 - val_accuracy: 0.6762 - lr: 3.7500e-05
Epoch 30/100
400/400 [=====] - 4s 10ms/step - loss: 0.0926 - accuracy
: 0.9997 - val_loss: 1.4774 - val_accuracy: 0.6650 - lr: 1.8750e-05
Epoch 31/100
400/400 [=====] - 4s 10ms/step - loss: 0.0919 - accuracy
: 0.9991 - val_loss: 0.0955 - val_accuracy: 0.9925 - lr: 1.8750e-05
Epoch 32/100
400/400 [=====] - 4s 10ms/step - loss: 0.0898 - accuracy
: 0.9981 - val_loss: 0.1057 - val_accuracy: 0.9962 - lr: 1.8750e-05
Epoch 33/100
400/400 [=====] - 4s 10ms/step - loss: 0.0885 - accuracy
: 0.9984 - val_loss: 0.1299 - val_accuracy: 0.9825 - lr: 1.8750e-05
Epoch 34/100
400/400 [=====] - 4s 10ms/step - loss: 0.0867 - accuracy
: 0.9991 - val_loss: 0.5173 - val_accuracy: 0.8687 - lr: 1.8750e-05
Epoch 35/100
397/400 [=====>.] - ETA: 0s - loss: 0.0843 - accuracy: 0.9
994
Epoch 35: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.

400/400 [=====] - 4s 10ms/step - loss: 0.0843 - accuracy
: 0.9994 - val_loss: 0.1551 - val_accuracy: 0.9787 - lr: 1.8750e-05
Epoch 36/100
400/400 [=====] - 4s 10ms/step - loss: 0.0810 - accuracy
: 1.0000 - val_loss: 0.0805 - val_accuracy: 0.9987 - lr: 9.3750e-06
Epoch 37/100
400/400 [=====] - 4s 10ms/step - loss: 0.0815 - accuracy
: 0.9991 - val_loss: 0.1148 - val_accuracy: 0.9900 - lr: 9.3750e-06
Epoch 38/100
400/400 [=====] - 4s 9ms/step - loss: 0.0822 - accuracy:
0.9994 - val_loss: 0.0895 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 39/100
400/400 [=====] - 4s 10ms/step - loss: 0.0814 - accuracy
: 0.9997 - val_loss: 0.0775 - val_accuracy: 0.9987 - lr: 9.3750e-06
Epoch 40/100
400/400 [=====] - 4s 10ms/step - loss: 0.0795 - accuracy
: 0.9997 - val_loss: 0.0816 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 41/100
400/400 [=====] - 4s 10ms/step - loss: 0.0774 - accuracy
: 0.9997 - val_loss: 0.0769 - val_accuracy: 0.9987 - lr: 9.3750e-06
Epoch 42/100
400/400 [=====] - 4s 10ms/step - loss: 0.0793 - accuracy
: 0.9984 - val_loss: 0.0941 - val_accuracy: 0.9937 - lr: 9.3750e-06
Epoch 43/100
400/400 [=====] - 4s 10ms/step - loss: 0.0784 - accuracy
: 0.9994 - val_loss: 0.0857 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 44/100
400/400 [=====] - 4s 10ms/step - loss: 0.0750 - accuracy
: 1.0000 - val_loss: 0.0895 - val_accuracy: 0.9937 - lr: 9.3750e-06
Epoch 45/100
400/400 [=====] - 4s 10ms/step - loss: 0.0732 - accuracy
: 1.0000 - val_loss: 0.0731 - val_accuracy: 0.9987 - lr: 9.3750e-06
Epoch 46/100
400/400 [=====] - 4s 10ms/step - loss: 0.0748 - accuracy
: 0.9987 - val_loss: 0.0868 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 47/100
400/400 [=====] - 4s 11ms/step - loss: 0.0745 - accuracy
: 0.9994 - val_loss: 0.0750 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 48/100
400/400 [=====] - 4s 11ms/step - loss: 0.0735 - accuracy
: 0.9991 - val_loss: 0.0862 - val_accuracy: 0.9937 - lr: 9.3750e-06
[100, 8, 0.25]
5.0%
Epoch 1/100
375/375 [=====] - 5s 11ms/step - loss: 2.9939 - accuracy
: 0.1730 - val_loss: 6.7404 - val_accuracy: 0.1160 - lr: 3.0000e-04
Epoch 2/100
375/375 [=====] - 4s 11ms/step - loss: 2.1969 - accuracy
: 0.3403 - val_loss: 6.2769 - val_accuracy: 0.0960 - lr: 3.0000e-04
Epoch 3/100
375/375 [=====] - 4s 11ms/step - loss: 1.5032 - accuracy
: 0.6040 - val_loss: 17.5432 - val_accuracy: 0.1460 - lr: 3.0000e-04
Epoch 4/100
375/375 [=====] - 4s 11ms/step - loss: 0.9925 - accuracy
: 0.8030 - val_loss: 17.7337 - val_accuracy: 0.1030 - lr: 3.0000e-04
Epoch 5/100
375/375 [=====] - 4s 11ms/step - loss: 0.7631 - accuracy
: 0.8873 - val_loss: 3.4745 - val_accuracy: 0.3870 - lr: 3.0000e-04
Epoch 6/100
375/375 [=====] - 4s 11ms/step - loss: 0.5769 - accuracy

: 0.9410 - val_loss: 1.9260 - val_accuracy: 0.5010 - lr: 3.0000e-04
Epoch 7/100
375/375 [=====] - 4s 11ms/step - loss: 0.4874 - accuracy
: 0.9517 - val_loss: 11.5159 - val_accuracy: 0.2530 - lr: 3.0000e-04
Epoch 8/100
375/375 [=====] - 4s 11ms/step - loss: 0.4053 - accuracy
: 0.9760 - val_loss: 33.3976 - val_accuracy: 0.0990 - lr: 3.0000e-04
Epoch 9/100
375/375 [=====] - 4s 11ms/step - loss: 0.4009 - accuracy
: 0.9663 - val_loss: 12.4078 - val_accuracy: 0.3460 - lr: 3.0000e-04
Epoch 10/100
373/375 [=====>.] - ETA: 0s - loss: 0.3478 - accuracy: 0.9
755
Epoch 10: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
375/375 [=====] - 4s 11ms/step - loss: 0.3475 - accuracy
: 0.9757 - val_loss: 17.2225 - val_accuracy: 0.2020 - lr: 3.0000e-04
Epoch 11/100
375/375 [=====] - 4s 11ms/step - loss: 0.2795 - accuracy
: 0.9920 - val_loss: 2.6385 - val_accuracy: 0.5310 - lr: 1.5000e-04
Epoch 12/100
375/375 [=====] - 4s 11ms/step - loss: 0.2541 - accuracy
: 0.9937 - val_loss: 8.1342 - val_accuracy: 0.2760 - lr: 1.5000e-04
Epoch 13/100
375/375 [=====] - 4s 11ms/step - loss: 0.2327 - accuracy
: 0.9953 - val_loss: 15.4015 - val_accuracy: 0.2080 - lr: 1.5000e-04
Epoch 14/100
370/375 [=====>.] - ETA: 0s - loss: 0.2256 - accuracy: 0.9
912
Epoch 14: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
375/375 [=====] - 4s 11ms/step - loss: 0.2252 - accuracy
: 0.9913 - val_loss: 10.9453 - val_accuracy: 0.2050 - lr: 1.5000e-04
Epoch 15/100
375/375 [=====] - 4s 11ms/step - loss: 0.2109 - accuracy
: 0.9950 - val_loss: 1.1606 - val_accuracy: 0.6720 - lr: 7.5000e-05
Epoch 16/100
375/375 [=====] - 4s 10ms/step - loss: 0.1998 - accuracy
: 0.9957 - val_loss: 0.2550 - val_accuracy: 0.9820 - lr: 7.5000e-05
Epoch 17/100
375/375 [=====] - 4s 10ms/step - loss: 0.1922 - accuracy
: 0.9943 - val_loss: 9.5115 - val_accuracy: 0.1560 - lr: 7.5000e-05
Epoch 18/100
375/375 [=====] - 4s 10ms/step - loss: 0.1861 - accuracy
: 0.9967 - val_loss: 8.0104 - val_accuracy: 0.2230 - lr: 7.5000e-05
Epoch 19/100
375/375 [=====] - 4s 11ms/step - loss: 0.1734 - accuracy
: 0.9967 - val_loss: 15.2792 - val_accuracy: 0.1030 - lr: 7.5000e-05
Epoch 20/100
375/375 [=====] - ETA: 0s - loss: 0.1646 - accuracy: 0.9
967
Epoch 20: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
375/375 [=====] - 4s 11ms/step - loss: 0.1646 - accuracy
: 0.9967 - val_loss: 1.6121 - val_accuracy: 0.5630 - lr: 7.5000e-05
Epoch 21/100
375/375 [=====] - 4s 10ms/step - loss: 0.1516 - accuracy
: 0.9977 - val_loss: 1.3607 - val_accuracy: 0.7020 - lr: 3.7500e-05
Epoch 22/100
375/375 [=====] - 4s 10ms/step - loss: 0.1494 - accuracy
: 0.9987 - val_loss: 0.3947 - val_accuracy: 0.9130 - lr: 3.7500e-05
Epoch 23/100
375/375 [=====] - 4s 11ms/step - loss: 0.1442 - accuracy

: 0.9980 - val_loss: 1.7023 - val_accuracy: 0.5930 - lr: 3.7500e-05
Epoch 24/100
372/375 [=====>.] - ETA: 0s - loss: 0.1389 - accuracy: 0.9983
Epoch 24: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
375/375 [=====] - 4s 10ms/step - loss: 0.1389 - accuracy: 0.9983 - val_loss: 0.4709 - val_accuracy: 0.8990 - lr: 3.7500e-05
Epoch 25/100
375/375 [=====] - 4s 10ms/step - loss: 0.1356 - accuracy: 0.9980 - val_loss: 0.1719 - val_accuracy: 0.9870 - lr: 1.8750e-05
Epoch 26/100
375/375 [=====] - 4s 11ms/step - loss: 0.1318 - accuracy: 0.9990 - val_loss: 0.5121 - val_accuracy: 0.8550 - lr: 1.8750e-05
Epoch 27/100
375/375 [=====] - 4s 10ms/step - loss: 0.1268 - accuracy: 0.9993 - val_loss: 0.1217 - val_accuracy: 0.9970 - lr: 1.8750e-05
Epoch 28/100
375/375 [=====] - 4s 11ms/step - loss: 0.1313 - accuracy: 0.9970 - val_loss: 0.2252 - val_accuracy: 0.9660 - lr: 1.8750e-05
Epoch 29/100
375/375 [=====] - 4s 10ms/step - loss: 0.1249 - accuracy: 0.9990 - val_loss: 0.1234 - val_accuracy: 0.9960 - lr: 1.8750e-05
Epoch 30/100
375/375 [=====] - 4s 11ms/step - loss: 0.1192 - accuracy: 0.9990 - val_loss: 0.2318 - val_accuracy: 0.9520 - lr: 1.8750e-05
Epoch 31/100
371/375 [=====>.] - ETA: 0s - loss: 0.1181 - accuracy: 0.9990
Epoch 31: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
375/375 [=====] - 4s 11ms/step - loss: 0.1180 - accuracy: 0.9990 - val_loss: 0.1266 - val_accuracy: 0.9910 - lr: 1.8750e-05
Epoch 32/100
375/375 [=====] - 4s 10ms/step - loss: 0.1123 - accuracy: 0.9993 - val_loss: 0.1150 - val_accuracy: 0.9950 - lr: 9.3750e-06
Epoch 33/100
375/375 [=====] - 4s 11ms/step - loss: 0.1117 - accuracy: 0.9990 - val_loss: 0.1303 - val_accuracy: 0.9880 - lr: 9.3750e-06
Epoch 34/100
375/375 [=====] - 4s 10ms/step - loss: 0.1105 - accuracy: 0.9993 - val_loss: 0.1085 - val_accuracy: 0.9960 - lr: 9.3750e-06
Epoch 35/100
375/375 [=====] - 4s 11ms/step - loss: 0.1083 - accuracy: 0.9997 - val_loss: 0.1093 - val_accuracy: 0.9970 - lr: 9.3750e-06
Epoch 36/100
375/375 [=====] - 4s 11ms/step - loss: 0.1094 - accuracy: 0.9983 - val_loss: 0.1080 - val_accuracy: 0.9990 - lr: 9.3750e-06
Epoch 37/100
375/375 [=====] - 4s 10ms/step - loss: 0.1066 - accuracy: 0.9987 - val_loss: 0.1239 - val_accuracy: 0.9900 - lr: 9.3750e-06
Epoch 38/100
375/375 [=====] - 4s 11ms/step - loss: 0.1038 - accuracy: 0.9990 - val_loss: 0.6918 - val_accuracy: 0.7990 - lr: 9.3750e-06
Epoch 39/100
375/375 [=====] - 4s 10ms/step - loss: 0.1024 - accuracy: 0.9993 - val_loss: 0.1126 - val_accuracy: 0.9890 - lr: 9.3750e-06
Epoch 40/100
375/375 [=====] - 4s 10ms/step - loss: 0.1026 - accuracy: 0.9990 - val_loss: 0.1059 - val_accuracy: 0.9930 - lr: 9.3750e-06
Epoch 41/100
375/375 [=====] - 4s 10ms/step - loss: 0.1013 - accuracy:

: 0.9990 - val_loss: 0.1885 - val_accuracy: 0.9760 - lr: 9.3750e-06
Epoch 42/100
375/375 [=====] - 5s 12ms/step - loss: 0.1012 - accuracy
: 0.9980 - val_loss: 0.2106 - val_accuracy: 0.9610 - lr: 9.3750e-06
Epoch 43/100
375/375 [=====] - 6s 16ms/step - loss: 0.0986 - accuracy
: 0.9997 - val_loss: 0.1001 - val_accuracy: 0.9960 - lr: 9.3750e-06
Epoch 44/100
375/375 [=====] - 4s 12ms/step - loss: 0.0980 - accuracy
: 0.9990 - val_loss: 0.1189 - val_accuracy: 0.9880 - lr: 9.3750e-06
Epoch 45/100
375/375 [=====] - 4s 11ms/step - loss: 0.0965 - accuracy
: 0.9990 - val_loss: 0.1026 - val_accuracy: 0.9960 - lr: 9.3750e-06
Epoch 46/100
375/375 [=====] - 4s 10ms/step - loss: 0.0960 - accuracy
: 0.9990 - val_loss: 0.5110 - val_accuracy: 0.8470 - lr: 9.3750e-06
Epoch 47/100
371/375 [=====>.] - ETA: 0s - loss: 0.0947 - accuracy: 0.9
987
Epoch 47: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
375/375 [=====] - 4s 10ms/step - loss: 0.0946 - accuracy
: 0.9987 - val_loss: 0.1062 - val_accuracy: 0.9940 - lr: 9.3750e-06
Epoch 48/100
375/375 [=====] - 4s 11ms/step - loss: 0.0991 - accuracy
: 0.9973 - val_loss: 0.0949 - val_accuracy: 0.9980 - lr: 4.6875e-06
[100, 8, 0.30000000000000004]
7.5%
Epoch 1/100
350/350 [=====] - 5s 12ms/step - loss: 2.9276 - accuracy
: 0.1571 - val_loss: 5.6054 - val_accuracy: 0.1142 - lr: 3.0000e-04
Epoch 2/100
350/350 [=====] - 4s 12ms/step - loss: 2.2512 - accuracy
: 0.3279 - val_loss: 21.4620 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 3/100
350/350 [=====] - 4s 11ms/step - loss: 1.7165 - accuracy
: 0.5129 - val_loss: 8.1642 - val_accuracy: 0.1517 - lr: 3.0000e-04
Epoch 4/100
350/350 [=====] - 4s 11ms/step - loss: 1.2930 - accuracy
: 0.6725 - val_loss: 13.9365 - val_accuracy: 0.1183 - lr: 3.0000e-04
Epoch 5/100
350/350 [=====] - ETA: 0s - loss: 0.9659 - accuracy: 0.7
982
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
350/350 [=====] - 4s 10ms/step - loss: 0.9659 - accuracy
: 0.7982 - val_loss: 21.1890 - val_accuracy: 0.1017 - lr: 3.0000e-04
Epoch 6/100
350/350 [=====] - 4s 10ms/step - loss: 0.7345 - accuracy
: 0.8829 - val_loss: 2.0830 - val_accuracy: 0.4050 - lr: 1.5000e-04
Epoch 7/100
350/350 [=====] - 4s 11ms/step - loss: 0.6187 - accuracy
: 0.9200 - val_loss: 2.6812 - val_accuracy: 0.3792 - lr: 1.5000e-04
Epoch 8/100
350/350 [=====] - 4s 11ms/step - loss: 0.5297 - accuracy
: 0.9471 - val_loss: 8.3255 - val_accuracy: 0.1600 - lr: 1.5000e-04
Epoch 9/100
350/350 [=====] - 4s 11ms/step - loss: 0.4684 - accuracy
: 0.9589 - val_loss: 34.4748 - val_accuracy: 0.0958 - lr: 1.5000e-04
Epoch 10/100
348/350 [=====>.] - ETA: 0s - loss: 0.4171 - accuracy: 0.9
673

Epoch 10: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
350/350 [=====] - 4s 11ms/step - loss: 0.4168 - accuracy
: 0.9675 - val_loss: 15.1637 - val_accuracy: 0.1008 - lr: 1.5000e-04
Epoch 11/100
350/350 [=====] - 4s 11ms/step - loss: 0.3817 - accuracy
: 0.9761 - val_loss: 14.1505 - val_accuracy: 0.2492 - lr: 7.5000e-05
Epoch 12/100
350/350 [=====] - 4s 11ms/step - loss: 0.3503 - accuracy
: 0.9796 - val_loss: 4.5287 - val_accuracy: 0.2067 - lr: 7.5000e-05
Epoch 13/100
350/350 [=====] - 4s 11ms/step - loss: 0.3191 - accuracy
: 0.9879 - val_loss: 9.7488 - val_accuracy: 0.2417 - lr: 7.5000e-05
Epoch 14/100
347/350 [=====>.] - ETA: 0s - loss: 0.3029 - accuracy: 0.9
892
Epoch 14: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
350/350 [=====] - 4s 11ms/step - loss: 0.3025 - accuracy
: 0.9893 - val_loss: 5.1572 - val_accuracy: 0.3517 - lr: 7.5000e-05
Epoch 15/100
350/350 [=====] - 4s 11ms/step - loss: 0.2881 - accuracy
: 0.9889 - val_loss: 1.3488 - val_accuracy: 0.7008 - lr: 3.7500e-05
Epoch 16/100
350/350 [=====] - 4s 11ms/step - loss: 0.2707 - accuracy
: 0.9932 - val_loss: 4.6050 - val_accuracy: 0.4692 - lr: 3.7500e-05
Epoch 17/100
350/350 [=====] - 4s 11ms/step - loss: 0.2609 - accuracy
: 0.9932 - val_loss: 16.3554 - val_accuracy: 0.1008 - lr: 3.7500e-05
Epoch 18/100
350/350 [=====] - 4s 11ms/step - loss: 0.2556 - accuracy
: 0.9904 - val_loss: 2.5941 - val_accuracy: 0.5683 - lr: 3.7500e-05
Epoch 19/100
349/350 [=====>.] - ETA: 0s - loss: 0.2445 - accuracy: 0.9
928
Epoch 19: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
350/350 [=====] - 4s 11ms/step - loss: 0.2445 - accuracy
: 0.9929 - val_loss: 1.6501 - val_accuracy: 0.5250 - lr: 3.7500e-05
Epoch 20/100
350/350 [=====] - 4s 11ms/step - loss: 0.2362 - accuracy
: 0.9939 - val_loss: 0.3025 - val_accuracy: 0.9667 - lr: 1.8750e-05
Epoch 21/100
350/350 [=====] - 4s 11ms/step - loss: 0.2357 - accuracy
: 0.9925 - val_loss: 3.7844 - val_accuracy: 0.4192 - lr: 1.8750e-05
Epoch 22/100
350/350 [=====] - 4s 11ms/step - loss: 0.2292 - accuracy
: 0.9943 - val_loss: 5.0229 - val_accuracy: 0.3600 - lr: 1.8750e-05
Epoch 23/100
350/350 [=====] - 4s 11ms/step - loss: 0.2163 - accuracy
: 0.9975 - val_loss: 3.2773 - val_accuracy: 0.4292 - lr: 1.8750e-05
Epoch 24/100
348/350 [=====>.] - ETA: 0s - loss: 0.2147 - accuracy: 0.9
957
Epoch 24: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
350/350 [=====] - 4s 11ms/step - loss: 0.2146 - accuracy
: 0.9957 - val_loss: 2.1525 - val_accuracy: 0.4708 - lr: 1.8750e-05
Epoch 25/100
350/350 [=====] - 4s 11ms/step - loss: 0.2088 - accuracy
: 0.9964 - val_loss: 0.2134 - val_accuracy: 0.9842 - lr: 9.3750e-06
Epoch 26/100
350/350 [=====] - 4s 11ms/step - loss: 0.2065 - accuracy
: 0.9957 - val_loss: 1.1057 - val_accuracy: 0.6883 - lr: 9.3750e-06

Epoch 27/100
350/350 [=====] - 4s 11ms/step - loss: 0.2066 - accuracy : 0.9975 - val_loss: 0.2158 - val_accuracy: 0.9850 - lr: 9.3750e-06
Epoch 28/100
350/350 [=====] - 4s 11ms/step - loss: 0.2025 - accuracy : 0.9971 - val_loss: 0.2280 - val_accuracy: 0.9850 - lr: 9.3750e-06
Epoch 29/100
349/350 [=====>.] - ETA: 0s - loss: 0.1983 - accuracy: 0.9975
Epoch 29: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
350/350 [=====] - 4s 11ms/step - loss: 0.1982 - accuracy : 0.9975 - val_loss: 0.2223 - val_accuracy: 0.9858 - lr: 9.3750e-06
Epoch 30/100
350/350 [=====] - 4s 11ms/step - loss: 0.1960 - accuracy : 0.9979 - val_loss: 0.1890 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 31/100
350/350 [=====] - 4s 11ms/step - loss: 0.1943 - accuracy : 0.9968 - val_loss: 0.2067 - val_accuracy: 0.9917 - lr: 4.6875e-06
Epoch 32/100
350/350 [=====] - 4s 11ms/step - loss: 0.1942 - accuracy : 0.9968 - val_loss: 0.2156 - val_accuracy: 0.9842 - lr: 4.6875e-06
Epoch 33/100
350/350 [=====] - 4s 11ms/step - loss: 0.1949 - accuracy : 0.9950 - val_loss: 0.1855 - val_accuracy: 0.9942 - lr: 4.6875e-06
Epoch 34/100
350/350 [=====] - 4s 11ms/step - loss: 0.1907 - accuracy : 0.9971 - val_loss: 0.2164 - val_accuracy: 0.9867 - lr: 4.6875e-06
Epoch 35/100
350/350 [=====] - 4s 11ms/step - loss: 0.1933 - accuracy : 0.9957 - val_loss: 0.2186 - val_accuracy: 0.9800 - lr: 4.6875e-06
Epoch 36/100
350/350 [=====] - 4s 11ms/step - loss: 0.1900 - accuracy : 0.9964 - val_loss: 0.2003 - val_accuracy: 0.9883 - lr: 4.6875e-06
Epoch 37/100
347/350 [=====>.] - ETA: 0s - loss: 0.1897 - accuracy: 0.9964
Epoch 37: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
350/350 [=====] - 4s 11ms/step - loss: 0.1901 - accuracy : 0.9961 - val_loss: 0.3055 - val_accuracy: 0.9550 - lr: 4.6875e-06
Epoch 38/100
350/350 [=====] - 4s 11ms/step - loss: 0.1860 - accuracy : 0.9989 - val_loss: 0.1869 - val_accuracy: 0.9950 - lr: 2.3438e-06
Epoch 39/100
350/350 [=====] - 4s 11ms/step - loss: 0.1853 - accuracy : 0.9975 - val_loss: 0.2011 - val_accuracy: 0.9908 - lr: 2.3438e-06
Epoch 40/100
350/350 [=====] - 4s 11ms/step - loss: 0.1869 - accuracy : 0.9968 - val_loss: 0.1934 - val_accuracy: 0.9892 - lr: 2.3438e-06
Epoch 41/100
348/350 [=====>.] - ETA: 0s - loss: 0.1840 - accuracy: 0.9964
Epoch 41: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
350/350 [=====] - 4s 12ms/step - loss: 0.1839 - accuracy : 0.9964 - val_loss: 0.1905 - val_accuracy: 0.9933 - lr: 2.3438e-06
Epoch 42/100
350/350 [=====] - 4s 12ms/step - loss: 0.1851 - accuracy : 0.9993 - val_loss: 0.1821 - val_accuracy: 0.9933 - lr: 1.1719e-06
[100, 8, 0.35000000000000003]
10.0%
Epoch 1/100

325/325 [=====] - 5s 14ms/step - loss: 2.8705 - accuracy
: 0.1816 - val_loss: 5.5768 - val_accuracy: 0.0928 - lr: 3.0000e-04
Epoch 2/100
325/325 [=====] - 4s 11ms/step - loss: 2.2014 - accuracy
: 0.3255 - val_loss: 8.9546 - val_accuracy: 0.0928 - lr: 3.0000e-04
Epoch 3/100
325/325 [=====] - 4s 13ms/step - loss: 1.5837 - accuracy
: 0.5648 - val_loss: 11.3421 - val_accuracy: 0.0928 - lr: 3.0000e-04
Epoch 4/100
325/325 [=====] - 4s 12ms/step - loss: 1.1325 - accuracy
: 0.7441 - val_loss: 10.3216 - val_accuracy: 0.2391 - lr: 3.0000e-04
Epoch 5/100
322/325 [=====>.] - ETA: 0s - loss: 0.8375 - accuracy: 0.8
482
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
325/325 [=====] - 4s 13ms/step - loss: 0.8363 - accuracy
: 0.8484 - val_loss: 23.4871 - val_accuracy: 0.1949 - lr: 3.0000e-04
Epoch 6/100
325/325 [=====] - 4s 12ms/step - loss: 0.6242 - accuracy
: 0.9327 - val_loss: 1.7350 - val_accuracy: 0.6824 - lr: 1.5000e-04
Epoch 7/100
325/325 [=====] - 4s 12ms/step - loss: 0.5538 - accuracy
: 0.9515 - val_loss: 10.3278 - val_accuracy: 0.1913 - lr: 1.5000e-04
Epoch 8/100
325/325 [=====] - 4s 13ms/step - loss: 0.4882 - accuracy
: 0.9619 - val_loss: 7.1612 - val_accuracy: 0.1271 - lr: 1.5000e-04
Epoch 9/100
325/325 [=====] - 4s 12ms/step - loss: 0.4565 - accuracy
: 0.9688 - val_loss: 19.6445 - val_accuracy: 0.0971 - lr: 1.5000e-04
Epoch 10/100
322/325 [=====>.] - ETA: 0s - loss: 0.4215 - accuracy: 0.9
720
Epoch 10: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
325/325 [=====] - 4s 12ms/step - loss: 0.4211 - accuracy
: 0.9723 - val_loss: 33.9418 - val_accuracy: 0.0971 - lr: 1.5000e-04
Epoch 11/100
325/325 [=====] - 4s 13ms/step - loss: 0.3778 - accuracy
: 0.9800 - val_loss: 9.4196 - val_accuracy: 0.2491 - lr: 7.5000e-05
Epoch 12/100
325/325 [=====] - 4s 14ms/step - loss: 0.3609 - accuracy
: 0.9854 - val_loss: 12.6572 - val_accuracy: 0.0921 - lr: 7.5000e-05
Epoch 13/100
325/325 [=====] - 4s 13ms/step - loss: 0.3282 - accuracy
: 0.9927 - val_loss: 10.0749 - val_accuracy: 0.1485 - lr: 7.5000e-05
Epoch 14/100
321/325 [=====>.] - ETA: 0s - loss: 0.3233 - accuracy: 0.9
871
Epoch 14: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
325/325 [=====] - 4s 14ms/step - loss: 0.3232 - accuracy
: 0.9873 - val_loss: 7.0348 - val_accuracy: 0.2941 - lr: 7.5000e-05
Epoch 15/100
325/325 [=====] - 5s 14ms/step - loss: 0.3104 - accuracy
: 0.9885 - val_loss: 0.5941 - val_accuracy: 0.8751 - lr: 3.7500e-05
Epoch 16/100
325/325 [=====] - 4s 13ms/step - loss: 0.2949 - accuracy
: 0.9900 - val_loss: 2.3255 - val_accuracy: 0.5846 - lr: 3.7500e-05
Epoch 17/100
325/325 [=====] - 4s 13ms/step - loss: 0.2763 - accuracy
: 0.9938 - val_loss: 2.3075 - val_accuracy: 0.4168 - lr: 3.7500e-05
Epoch 18/100

325/325 [=====] - 5s 14ms/step - loss: 0.2682 - accuracy : 0.9931 - val_loss: 5.2842 - val_accuracy: 0.1927 - lr: 3.7500e-05
Epoch 19/100
325/325 [=====] - 4s 14ms/step - loss: 0.2597 - accuracy : 0.9938 - val_loss: 0.3730 - val_accuracy: 0.9515 - lr: 3.7500e-05
Epoch 20/100
325/325 [=====] - 4s 13ms/step - loss: 0.2530 - accuracy : 0.9919 - val_loss: 0.9881 - val_accuracy: 0.7409 - lr: 3.7500e-05
Epoch 21/100
325/325 [=====] - 4s 13ms/step - loss: 0.2368 - accuracy : 0.9958 - val_loss: 1.0945 - val_accuracy: 0.7316 - lr: 3.7500e-05
Epoch 22/100
325/325 [=====] - 5s 14ms/step - loss: 0.2271 - accuracy : 0.9962 - val_loss: 0.5867 - val_accuracy: 0.8551 - lr: 3.7500e-05
Epoch 23/100
324/325 [=====>.] - ETA: 0s - loss: 0.2291 - accuracy: 0.9954
Epoch 23: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
325/325 [=====] - 4s 13ms/step - loss: 0.2292 - accuracy : 0.9954 - val_loss: 0.4094 - val_accuracy: 0.9258 - lr: 3.7500e-05
Epoch 24/100
325/325 [=====] - 4s 13ms/step - loss: 0.2275 - accuracy : 0.9935 - val_loss: 0.2788 - val_accuracy: 0.9729 - lr: 1.8750e-05
Epoch 25/100
325/325 [=====] - 4s 13ms/step - loss: 0.2155 - accuracy : 0.9950 - val_loss: 6.8738 - val_accuracy: 0.1834 - lr: 1.8750e-05
Epoch 26/100
325/325 [=====] - 4s 14ms/step - loss: 0.2119 - accuracy : 0.9962 - val_loss: 0.2869 - val_accuracy: 0.9657 - lr: 1.8750e-05
Epoch 27/100
325/325 [=====] - 4s 13ms/step - loss: 0.2014 - accuracy : 0.9977 - val_loss: 0.2211 - val_accuracy: 0.9843 - lr: 1.8750e-05
Epoch 28/100
325/325 [=====] - 4s 13ms/step - loss: 0.2015 - accuracy : 0.9965 - val_loss: 1.4583 - val_accuracy: 0.6831 - lr: 1.8750e-05
Epoch 29/100
325/325 [=====] - 4s 14ms/step - loss: 0.1911 - accuracy : 0.9969 - val_loss: 3.1767 - val_accuracy: 0.4147 - lr: 1.8750e-05
Epoch 30/100
325/325 [=====] - 4s 13ms/step - loss: 0.1905 - accuracy : 0.9954 - val_loss: 0.6298 - val_accuracy: 0.8301 - lr: 1.8750e-05
Epoch 31/100
325/325 [=====] - 4s 13ms/step - loss: 0.1844 - accuracy : 0.9988 - val_loss: 0.2005 - val_accuracy: 0.9843 - lr: 1.8750e-05
Epoch 32/100
325/325 [=====] - 4s 13ms/step - loss: 0.1816 - accuracy : 0.9954 - val_loss: 0.4571 - val_accuracy: 0.8837 - lr: 1.8750e-05
Epoch 33/100
325/325 [=====] - 4s 14ms/step - loss: 0.1770 - accuracy : 0.9973 - val_loss: 0.2541 - val_accuracy: 0.9679 - lr: 1.8750e-05
Epoch 34/100
325/325 [=====] - 4s 13ms/step - loss: 0.1733 - accuracy : 0.9985 - val_loss: 2.2251 - val_accuracy: 0.5339 - lr: 1.8750e-05
Epoch 35/100
324/325 [=====>.] - ETA: 0s - loss: 0.1707 - accuracy: 0.9973
Epoch 35: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
325/325 [=====] - 4s 13ms/step - loss: 0.1707 - accuracy : 0.9973 - val_loss: 2.3862 - val_accuracy: 0.4725 - lr: 1.8750e-05
Epoch 36/100

325/325 [=====] - 4s 13ms/step - loss: 0.1684 - accuracy : 0.9973 - val_loss: 2.0014 - val_accuracy: 0.6217 - lr: 9.3750e-06
Epoch 37/100
325/325 [=====] - 4s 13ms/step - loss: 0.1640 - accuracy : 0.9977 - val_loss: 0.2453 - val_accuracy: 0.9707 - lr: 9.3750e-06
Epoch 38/100
325/325 [=====] - 4s 13ms/step - loss: 0.1584 - accuracy : 0.9981 - val_loss: 1.1709 - val_accuracy: 0.6681 - lr: 9.3750e-06
Epoch 39/100
325/325 [=====] - 4s 13ms/step - loss: 0.1570 - accuracy : 0.9981 - val_loss: 0.1725 - val_accuracy: 0.9907 - lr: 9.3750e-06
Epoch 40/100
325/325 [=====] - 4s 14ms/step - loss: 0.1584 - accuracy : 0.9985 - val_loss: 0.6377 - val_accuracy: 0.8180 - lr: 9.3750e-06
Epoch 41/100
325/325 [=====] - 4s 13ms/step - loss: 0.1512 - accuracy : 0.9992 - val_loss: 0.1663 - val_accuracy: 0.9886 - lr: 9.3750e-06
Epoch 42/100
325/325 [=====] - 4s 13ms/step - loss: 0.1555 - accuracy : 0.9977 - val_loss: 0.1905 - val_accuracy: 0.9807 - lr: 9.3750e-06
Epoch 43/100
325/325 [=====] - 4s 13ms/step - loss: 0.1562 - accuracy : 0.9965 - val_loss: 3.8278 - val_accuracy: 0.3783 - lr: 9.3750e-06
Epoch 44/100
325/325 [=====] - 4s 13ms/step - loss: 0.1503 - accuracy : 0.9985 - val_loss: 1.0904 - val_accuracy: 0.6545 - lr: 9.3750e-06
Epoch 45/100
324/325 [=====>.] - ETA: 0s - loss: 0.1448 - accuracy: 0.9988
Epoch 45: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
325/325 [=====] - 4s 13ms/step - loss: 0.1448 - accuracy : 0.9988 - val_loss: 2.7736 - val_accuracy: 0.6338 - lr: 9.3750e-06
Epoch 46/100
325/325 [=====] - 4s 13ms/step - loss: 0.1464 - accuracy : 0.9985 - val_loss: 0.3020 - val_accuracy: 0.9415 - lr: 4.6875e-06
Epoch 47/100
325/325 [=====] - 4s 13ms/step - loss: 0.1409 - accuracy : 0.9985 - val_loss: 0.1506 - val_accuracy: 0.9929 - lr: 4.6875e-06
Epoch 48/100
325/325 [=====] - 4s 13ms/step - loss: 0.1430 - accuracy : 0.9977 - val_loss: 0.1596 - val_accuracy: 0.9864 - lr: 4.6875e-06
Epoch 49/100
325/325 [=====] - 4s 13ms/step - loss: 0.1391 - accuracy : 0.9992 - val_loss: 0.1468 - val_accuracy: 0.9907 - lr: 4.6875e-06
Epoch 50/100
325/325 [=====] - 4s 13ms/step - loss: 0.1428 - accuracy : 0.9981 - val_loss: 0.1831 - val_accuracy: 0.9829 - lr: 4.6875e-06
Epoch 51/100
325/325 [=====] - 4s 13ms/step - loss: 0.1360 - accuracy : 0.9992 - val_loss: 0.1685 - val_accuracy: 0.9807 - lr: 4.6875e-06
Epoch 52/100
325/325 [=====] - 4s 13ms/step - loss: 0.1400 - accuracy : 0.9981 - val_loss: 0.8449 - val_accuracy: 0.7509 - lr: 4.6875e-06
Epoch 53/100
325/325 [=====] - 4s 13ms/step - loss: 0.1357 - accuracy : 0.9992 - val_loss: 0.1459 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 54/100
325/325 [=====] - 4s 13ms/step - loss: 0.1348 - accuracy : 0.9985 - val_loss: 0.2758 - val_accuracy: 0.9458 - lr: 4.6875e-06
Epoch 55/100

325/325 [=====] - 4s 13ms/step - loss: 0.1338 - accuracy
: 0.9988 - val_loss: 0.1424 - val_accuracy: 0.9921 - lr: 4.6875e-06
Epoch 56/100
325/325 [=====] - 4s 14ms/step - loss: 0.1319 - accuracy
: 0.9992 - val_loss: 0.1468 - val_accuracy: 0.9893 - lr: 4.6875e-06
Epoch 57/100
325/325 [=====] - 4s 13ms/step - loss: 0.1350 - accuracy
: 0.9981 - val_loss: 0.1699 - val_accuracy: 0.9836 - lr: 4.6875e-06
Epoch 58/100
325/325 [=====] - 4s 13ms/step - loss: 0.1317 - accuracy
: 0.9996 - val_loss: 0.1549 - val_accuracy: 0.9886 - lr: 4.6875e-06
Epoch 59/100
325/325 [=====] - 4s 14ms/step - loss: 0.1267 - accuracy
: 1.0000 - val_loss: 0.1410 - val_accuracy: 0.9907 - lr: 4.6875e-06
[100, 8, 0.4]
12.5%
Epoch 1/100
300/300 [=====] - 5s 14ms/step - loss: 3.0165 - accuracy
: 0.1583 - val_loss: 4.3087 - val_accuracy: 0.0875 - lr: 3.0000e-04
Epoch 2/100
300/300 [=====] - 4s 13ms/step - loss: 2.4001 - accuracy
: 0.2783 - val_loss: 8.7631 - val_accuracy: 0.1000 - lr: 3.0000e-04
Epoch 3/100
300/300 [=====] - 4s 13ms/step - loss: 1.9039 - accuracy
: 0.4429 - val_loss: 140.6400 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 4/100
300/300 [=====] - 4s 13ms/step - loss: 1.3774 - accuracy
: 0.6558 - val_loss: 8.9823 - val_accuracy: 0.0975 - lr: 3.0000e-04
Epoch 5/100
297/300 [=====>.] - ETA: 0s - loss: 1.0040 - accuracy: 0.8
043
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
300/300 [=====] - 4s 13ms/step - loss: 1.0036 - accuracy
: 0.8042 - val_loss: 11.5154 - val_accuracy: 0.1219 - lr: 3.0000e-04
Epoch 6/100
300/300 [=====] - 4s 13ms/step - loss: 0.7671 - accuracy
: 0.8958 - val_loss: 1.1207 - val_accuracy: 0.6775 - lr: 1.5000e-04
Epoch 7/100
300/300 [=====] - 4s 13ms/step - loss: 0.6644 - accuracy
: 0.9296 - val_loss: 1.9436 - val_accuracy: 0.4787 - lr: 1.5000e-04
Epoch 8/100
300/300 [=====] - 4s 13ms/step - loss: 0.5852 - accuracy
: 0.9492 - val_loss: 4.5802 - val_accuracy: 0.3231 - lr: 1.5000e-04
Epoch 9/100
300/300 [=====] - 4s 13ms/step - loss: 0.5220 - accuracy
: 0.9613 - val_loss: 14.3735 - val_accuracy: 0.0988 - lr: 1.5000e-04
Epoch 10/100
298/300 [=====>.] - ETA: 0s - loss: 0.4754 - accuracy: 0.9
719
Epoch 10: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
300/300 [=====] - 4s 13ms/step - loss: 0.4745 - accuracy
: 0.9721 - val_loss: 3.5499 - val_accuracy: 0.3325 - lr: 1.5000e-04
Epoch 11/100
300/300 [=====] - 4s 13ms/step - loss: 0.4264 - accuracy
: 0.9783 - val_loss: 6.5731 - val_accuracy: 0.2025 - lr: 7.5000e-05
Epoch 12/100
300/300 [=====] - 4s 13ms/step - loss: 0.4144 - accuracy
: 0.9792 - val_loss: 3.6590 - val_accuracy: 0.2425 - lr: 7.5000e-05
Epoch 13/100
300/300 [=====] - 4s 13ms/step - loss: 0.3858 - accuracy

: 0.9854 - val_loss: 0.9528 - val_accuracy: 0.7013 - lr: 7.5000e-05
Epoch 14/100
300/300 [=====] - 4s 13ms/step - loss: 0.3675 - accuracy
: 0.9846 - val_loss: 1.3630 - val_accuracy: 0.5319 - lr: 7.5000e-05
Epoch 15/100
300/300 [=====] - 4s 13ms/step - loss: 0.3468 - accuracy
: 0.9875 - val_loss: 1.6635 - val_accuracy: 0.5813 - lr: 7.5000e-05
Epoch 16/100
300/300 [=====] - 4s 13ms/step - loss: 0.3246 - accuracy
: 0.9892 - val_loss: 8.6346 - val_accuracy: 0.1881 - lr: 7.5000e-05
Epoch 17/100
298/300 [=====>.] - ETA: 0s - loss: 0.3098 - accuracy: 0.9
883
Epoch 17: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
300/300 [=====] - 4s 13ms/step - loss: 0.3097 - accuracy
: 0.9883 - val_loss: 1.2738 - val_accuracy: 0.6169 - lr: 7.5000e-05
Epoch 18/100
300/300 [=====] - 4s 13ms/step - loss: 0.2843 - accuracy
: 0.9925 - val_loss: 0.7503 - val_accuracy: 0.8012 - lr: 3.7500e-05
Epoch 19/100
300/300 [=====] - 4s 13ms/step - loss: 0.2824 - accuracy
: 0.9883 - val_loss: 3.1182 - val_accuracy: 0.2688 - lr: 3.7500e-05
Epoch 20/100
300/300 [=====] - 4s 13ms/step - loss: 0.2643 - accuracy
: 0.9937 - val_loss: 2.3423 - val_accuracy: 0.4381 - lr: 3.7500e-05
Epoch 21/100
300/300 [=====] - 4s 14ms/step - loss: 0.2538 - accuracy
: 0.9962 - val_loss: 1.5738 - val_accuracy: 0.7262 - lr: 3.7500e-05
Epoch 22/100
300/300 [=====] - 4s 14ms/step - loss: 0.2481 - accuracy
: 0.9971 - val_loss: 0.3088 - val_accuracy: 0.9650 - lr: 3.7500e-05
Epoch 23/100
300/300 [=====] - 4s 13ms/step - loss: 0.2409 - accuracy
: 0.9950 - val_loss: 2.3385 - val_accuracy: 0.4200 - lr: 3.7500e-05
Epoch 24/100
300/300 [=====] - 4s 14ms/step - loss: 0.2369 - accuracy
: 0.9954 - val_loss: 0.3158 - val_accuracy: 0.9688 - lr: 3.7500e-05
Epoch 25/100
300/300 [=====] - 4s 14ms/step - loss: 0.2319 - accuracy
: 0.9925 - val_loss: 0.8961 - val_accuracy: 0.7194 - lr: 3.7500e-05
Epoch 26/100
300/300 [=====] - 4s 14ms/step - loss: 0.2179 - accuracy
: 0.9971 - val_loss: 0.2968 - val_accuracy: 0.9613 - lr: 3.7500e-05
Epoch 27/100
300/300 [=====] - 4s 14ms/step - loss: 0.2099 - accuracy
: 0.9967 - val_loss: 0.2840 - val_accuracy: 0.9638 - lr: 3.7500e-05
Epoch 28/100
300/300 [=====] - 4s 14ms/step - loss: 0.2024 - accuracy
: 0.9962 - val_loss: 0.5304 - val_accuracy: 0.8725 - lr: 3.7500e-05
Epoch 29/100
300/300 [=====] - 4s 14ms/step - loss: 0.2015 - accuracy
: 0.9950 - val_loss: 2.4531 - val_accuracy: 0.5131 - lr: 3.7500e-05
Epoch 30/100
300/300 [=====] - 4s 14ms/step - loss: 0.1862 - accuracy
: 0.9967 - val_loss: 0.2837 - val_accuracy: 0.9600 - lr: 3.7500e-05
Epoch 31/100
300/300 [=====] - 4s 14ms/step - loss: 0.1861 - accuracy
: 0.9950 - val_loss: 1.4613 - val_accuracy: 0.6181 - lr: 3.7500e-05
Epoch 32/100
300/300 [=====] - 4s 15ms/step - loss: 0.1743 - accuracy

: 0.9971 - val_loss: 6.7957 - val_accuracy: 0.1713 - lr: 3.7500e-05
Epoch 33/100
300/300 [=====] - 4s 14ms/step - loss: 0.1737 - accuracy
: 0.9962 - val_loss: 0.9865 - val_accuracy: 0.7319 - lr: 3.7500e-05
Epoch 34/100
299/300 [=====>.] - ETA: 0s - loss: 0.1653 - accuracy: 0.9
971
Epoch 34: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
300/300 [=====] - 4s 13ms/step - loss: 0.1653 - accuracy
: 0.9971 - val_loss: 0.4104 - val_accuracy: 0.8819 - lr: 3.7500e-05
Epoch 35/100
300/300 [=====] - 4s 14ms/step - loss: 0.1625 - accuracy
: 0.9971 - val_loss: 0.1743 - val_accuracy: 0.9894 - lr: 1.8750e-05
Epoch 36/100
300/300 [=====] - 4s 14ms/step - loss: 0.1646 - accuracy
: 0.9958 - val_loss: 9.6709 - val_accuracy: 0.0994 - lr: 1.8750e-05
Epoch 37/100
300/300 [=====] - 4s 13ms/step - loss: 0.1552 - accuracy
: 0.9975 - val_loss: 0.1888 - val_accuracy: 0.9844 - lr: 1.8750e-05
Epoch 38/100
300/300 [=====] - 4s 13ms/step - loss: 0.1484 - accuracy
: 0.9996 - val_loss: 1.2748 - val_accuracy: 0.6556 - lr: 1.8750e-05
Epoch 39/100
300/300 [=====] - ETA: 0s - loss: 0.1521 - accuracy: 0.9
983
Epoch 39: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
300/300 [=====] - 4s 14ms/step - loss: 0.1521 - accuracy
: 0.9983 - val_loss: 1.8464 - val_accuracy: 0.5350 - lr: 1.8750e-05
Epoch 40/100
300/300 [=====] - 4s 14ms/step - loss: 0.1461 - accuracy
: 0.9983 - val_loss: 0.2089 - val_accuracy: 0.9725 - lr: 9.3750e-06
Epoch 41/100
300/300 [=====] - 4s 14ms/step - loss: 0.1434 - accuracy
: 0.9987 - val_loss: 0.1633 - val_accuracy: 0.9887 - lr: 9.3750e-06
Epoch 42/100
300/300 [=====] - 4s 13ms/step - loss: 0.1425 - accuracy
: 0.9996 - val_loss: 0.1799 - val_accuracy: 0.9850 - lr: 9.3750e-06
Epoch 43/100
300/300 [=====] - 4s 14ms/step - loss: 0.1398 - accuracy
: 0.9983 - val_loss: 0.1502 - val_accuracy: 0.9931 - lr: 9.3750e-06
Epoch 44/100
300/300 [=====] - 4s 14ms/step - loss: 0.1443 - accuracy
: 0.9962 - val_loss: 0.1556 - val_accuracy: 0.9912 - lr: 9.3750e-06
Epoch 45/100
300/300 [=====] - 4s 14ms/step - loss: 0.1453 - accuracy
: 0.9979 - val_loss: 0.1720 - val_accuracy: 0.9831 - lr: 9.3750e-06
Epoch 46/100
300/300 [=====] - 4s 13ms/step - loss: 0.1381 - accuracy
: 0.9992 - val_loss: 0.2010 - val_accuracy: 0.9775 - lr: 9.3750e-06
Epoch 47/100
300/300 [=====] - 4s 12ms/step - loss: 0.1347 - accuracy
: 0.9996 - val_loss: 0.1453 - val_accuracy: 0.9931 - lr: 9.3750e-06
Epoch 48/100
300/300 [=====] - 4s 12ms/step - loss: 0.1338 - accuracy
: 1.0000 - val_loss: 0.2272 - val_accuracy: 0.9594 - lr: 9.3750e-06
Epoch 49/100
300/300 [=====] - 3s 11ms/step - loss: 0.1351 - accuracy
: 0.9992 - val_loss: 0.8904 - val_accuracy: 0.7400 - lr: 9.3750e-06
Epoch 50/100
300/300 [=====] - 3s 11ms/step - loss: 0.1311 - accuracy

: 1.0000 - val_loss: 0.1408 - val_accuracy: 0.9937 - lr: 9.3750e-06
Epoch 51/100
300/300 [=====] - 3s 11ms/step - loss: 0.1317 - accuracy
: 0.9996 - val_loss: 0.1712 - val_accuracy: 0.9856 - lr: 9.3750e-06
Epoch 52/100
300/300 [=====] - 3s 11ms/step - loss: 0.1304 - accuracy
: 0.9996 - val_loss: 0.1441 - val_accuracy: 0.9912 - lr: 9.3750e-06
Epoch 53/100
300/300 [=====] - 3s 11ms/step - loss: 0.1291 - accuracy
: 0.9992 - val_loss: 0.1951 - val_accuracy: 0.9756 - lr: 9.3750e-06
Epoch 54/100
298/300 [=====>.] - ETA: 0s - loss: 0.1334 - accuracy: 0.9
983
Epoch 54: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
300/300 [=====] - 3s 12ms/step - loss: 0.1334 - accuracy
: 0.9983 - val_loss: 0.1525 - val_accuracy: 0.9894 - lr: 9.3750e-06
Epoch 55/100
300/300 [=====] - 3s 12ms/step - loss: 0.1290 - accuracy
: 0.9983 - val_loss: 0.1520 - val_accuracy: 0.9912 - lr: 4.6875e-06
Epoch 56/100
300/300 [=====] - 3s 11ms/step - loss: 0.1246 - accuracy
: 0.9996 - val_loss: 0.1350 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 57/100
300/300 [=====] - 3s 11ms/step - loss: 0.1289 - accuracy
: 0.9975 - val_loss: 0.1358 - val_accuracy: 0.9931 - lr: 4.6875e-06
Epoch 58/100
300/300 [=====] - 3s 11ms/step - loss: 0.1251 - accuracy
: 0.9996 - val_loss: 0.1343 - val_accuracy: 0.9931 - lr: 4.6875e-06
Epoch 59/100
300/300 [=====] - 3s 12ms/step - loss: 0.1240 - accuracy
: 0.9996 - val_loss: 0.1716 - val_accuracy: 0.9825 - lr: 4.6875e-06
Epoch 60/100
300/300 [=====] - 3s 12ms/step - loss: 0.1234 - accuracy
: 0.9992 - val_loss: 0.1362 - val_accuracy: 0.9925 - lr: 4.6875e-06
Epoch 61/100
300/300 [=====] - 3s 11ms/step - loss: 0.1199 - accuracy
: 1.0000 - val_loss: 0.1305 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 62/100
300/300 [=====] - 3s 11ms/step - loss: 0.1235 - accuracy
: 0.9987 - val_loss: 0.1358 - val_accuracy: 0.9931 - lr: 4.6875e-06
Epoch 63/100
300/300 [=====] - 3s 12ms/step - loss: 0.1217 - accuracy
: 1.0000 - val_loss: 0.1364 - val_accuracy: 0.9919 - lr: 4.6875e-06
Epoch 64/100
300/300 [=====] - 3s 12ms/step - loss: 0.1224 - accuracy
: 0.9996 - val_loss: 0.1389 - val_accuracy: 0.9931 - lr: 4.6875e-06
Epoch 65/100
300/300 [=====] - 3s 12ms/step - loss: 0.1256 - accuracy
: 0.9975 - val_loss: 0.1285 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 66/100
300/300 [=====] - 3s 11ms/step - loss: 0.1196 - accuracy
: 0.9996 - val_loss: 0.1508 - val_accuracy: 0.9875 - lr: 4.6875e-06
Epoch 67/100
300/300 [=====] - 3s 11ms/step - loss: 0.1212 - accuracy
: 0.9992 - val_loss: 0.2012 - val_accuracy: 0.9744 - lr: 4.6875e-06
Epoch 68/100
300/300 [=====] - 3s 12ms/step - loss: 0.1185 - accuracy
: 0.9996 - val_loss: 0.1284 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 69/100
298/300 [=====>.] - ETA: 0s - loss: 0.1177 - accuracy: 1.0

000

Epoch 69: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.

300/300 [=====] - 4s 12ms/step - loss: 0.1178 - accuracy : 1.0000 - val_loss: 0.1408 - val_accuracy: 0.9894 - lr: 4.6875e-06

Epoch 70/100

300/300 [=====] - 3s 11ms/step - loss: 0.1202 - accuracy : 1.0000 - val_loss: 0.1334 - val_accuracy: 0.9931 - lr: 2.3438e-06

Epoch 71/100

300/300 [=====] - 4s 12ms/step - loss: 0.1167 - accuracy : 0.9996 - val_loss: 0.1342 - val_accuracy: 0.9906 - lr: 2.3438e-06

Epoch 72/100

300/300 [=====] - 4s 12ms/step - loss: 0.1162 - accuracy : 0.9996 - val_loss: 0.1282 - val_accuracy: 0.9950 - lr: 2.3438e-06

Epoch 73/100

300/300 [=====] - 4s 12ms/step - loss: 0.1175 - accuracy : 1.0000 - val_loss: 0.1248 - val_accuracy: 0.9931 - lr: 2.3438e-06

Epoch 74/100

300/300 [=====] - 4s 12ms/step - loss: 0.1164 - accuracy : 0.9996 - val_loss: 0.1279 - val_accuracy: 0.9931 - lr: 2.3438e-06

Epoch 75/100

300/300 [=====] - 3s 12ms/step - loss: 0.1171 - accuracy : 1.0000 - val_loss: 0.1237 - val_accuracy: 0.9969 - lr: 2.3438e-06

Epoch 76/100

300/300 [=====] - 4s 12ms/step - loss: 0.1164 - accuracy : 0.9996 - val_loss: 0.1267 - val_accuracy: 0.9950 - lr: 2.3438e-06

Epoch 77/100

300/300 [=====] - 4s 12ms/step - loss: 0.1159 - accuracy : 1.0000 - val_loss: 0.1266 - val_accuracy: 0.9944 - lr: 2.3438e-06

Epoch 78/100

300/300 [=====] - 4s 12ms/step - loss: 0.1157 - accuracy : 1.0000 - val_loss: 0.1261 - val_accuracy: 0.9937 - lr: 2.3438e-06

Epoch 79/100

299/300 [=====>.] - ETA: 0s - loss: 0.1163 - accuracy: 0.9992

Epoch 79: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.

300/300 [=====] - 4s 12ms/step - loss: 0.1163 - accuracy : 0.9992 - val_loss: 0.1245 - val_accuracy: 0.9944 - lr: 2.3438e-06

Epoch 80/100

300/300 [=====] - 3s 12ms/step - loss: 0.1144 - accuracy : 1.0000 - val_loss: 0.1249 - val_accuracy: 0.9956 - lr: 1.1719e-06

Epoch 81/100

300/300 [=====] - 4s 12ms/step - loss: 0.1162 - accuracy : 0.9992 - val_loss: 0.1254 - val_accuracy: 0.9956 - lr: 1.1719e-06

Epoch 82/100

300/300 [=====] - 4s 12ms/step - loss: 0.1160 - accuracy : 0.9996 - val_loss: 0.1246 - val_accuracy: 0.9944 - lr: 1.1719e-06

Epoch 83/100

299/300 [=====>.] - ETA: 0s - loss: 0.1141 - accuracy: 1.0000

Epoch 83: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.

300/300 [=====] - 4s 12ms/step - loss: 0.1140 - accuracy : 1.0000 - val_loss: 0.1255 - val_accuracy: 0.9944 - lr: 1.1719e-06

Epoch 84/100

300/300 [=====] - 3s 11ms/step - loss: 0.1159 - accuracy : 0.9987 - val_loss: 0.1260 - val_accuracy: 0.9931 - lr: 5.8594e-07

Epoch 85/100

300/300 [=====] - 3s 11ms/step - loss: 0.1150 - accuracy : 0.9996 - val_loss: 0.1234 - val_accuracy: 0.9956 - lr: 5.8594e-07

Epoch 86/100

300/300 [=====] - 3s 12ms/step - loss: 0.1140 - accuracy

: 1.0000 - val_loss: 0.1236 - val_accuracy: 0.9956 - lr: 5.8594e-07
Epoch 87/100
300/300 [=====] - 4s 12ms/step - loss: 0.1159 - accuracy
: 0.9987 - val_loss: 0.1237 - val_accuracy: 0.9956 - lr: 5.8594e-07
[100, 16, 0.2]
15.0%
Epoch 1/100
200/200 [=====] - 3s 10ms/step - loss: 3.0008 - accuracy
: 0.1816 - val_loss: 7.1514 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 2/100
200/200 [=====] - 2s 10ms/step - loss: 2.2761 - accuracy
: 0.3366 - val_loss: 10.3443 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 3/100
200/200 [=====] - 2s 9ms/step - loss: 1.5964 - accuracy:
0.5944 - val_loss: 12.5352 - val_accuracy: 0.1175 - lr: 3.0000e-04
Epoch 4/100
200/200 [=====] - 2s 11ms/step - loss: 1.0235 - accuracy
: 0.8159 - val_loss: 11.1189 - val_accuracy: 0.1187 - lr: 3.0000e-04
Epoch 5/100
198/200 [=====>.] - ETA: 0s - loss: 0.7110 - accuracy: 0.9
293
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
200/200 [=====] - 2s 11ms/step - loss: 0.7108 - accuracy
: 0.9294 - val_loss: 30.1557 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 6/100
200/200 [=====] - 2s 10ms/step - loss: 0.5712 - accuracy
: 0.9659 - val_loss: 1.7627 - val_accuracy: 0.6025 - lr: 1.5000e-04
Epoch 7/100
200/200 [=====] - 2s 10ms/step - loss: 0.4999 - accuracy
: 0.9862 - val_loss: 3.8233 - val_accuracy: 0.3162 - lr: 1.5000e-04
Epoch 8/100
200/200 [=====] - 2s 9ms/step - loss: 0.4667 - accuracy:
0.9844 - val_loss: 4.3739 - val_accuracy: 0.3475 - lr: 1.5000e-04
Epoch 9/100
200/200 [=====] - 2s 9ms/step - loss: 0.4379 - accuracy:
0.9834 - val_loss: 3.6591 - val_accuracy: 0.3462 - lr: 1.5000e-04
Epoch 10/100
200/200 [=====] - ETA: 0s - loss: 0.4017 - accuracy: 0.9
912
Epoch 10: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
200/200 [=====] - 2s 8ms/step - loss: 0.4017 - accuracy:
0.9912 - val_loss: 6.1254 - val_accuracy: 0.2862 - lr: 1.5000e-04
Epoch 11/100
200/200 [=====] - 2s 11ms/step - loss: 0.3751 - accuracy
: 0.9925 - val_loss: 0.5452 - val_accuracy: 0.9287 - lr: 7.5000e-05
Epoch 12/100
200/200 [=====] - 2s 10ms/step - loss: 0.3583 - accuracy
: 0.9934 - val_loss: 1.5310 - val_accuracy: 0.7800 - lr: 7.5000e-05
Epoch 13/100
200/200 [=====] - 2s 10ms/step - loss: 0.3392 - accuracy
: 0.9969 - val_loss: 2.2983 - val_accuracy: 0.5525 - lr: 7.5000e-05
Epoch 14/100
200/200 [=====] - 2s 10ms/step - loss: 0.3294 - accuracy
: 0.9959 - val_loss: 0.9615 - val_accuracy: 0.7700 - lr: 7.5000e-05
Epoch 15/100
196/200 [=====>.] - ETA: 0s - loss: 0.3198 - accuracy: 0.9
968
Epoch 15: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
200/200 [=====] - 2s 9ms/step - loss: 0.3193 - accuracy:
0.9969 - val_loss: 3.6489 - val_accuracy: 0.1887 - lr: 7.5000e-05

Epoch 16/100
200/200 [=====] - 2s 10ms/step - loss: 0.3052 - accuracy: 0.9978 - val_loss: 1.5907 - val_accuracy: 0.4938 - lr: 3.7500e-05

Epoch 17/100
200/200 [=====] - 2s 9ms/step - loss: 0.2966 - accuracy: 0.9984 - val_loss: 0.4419 - val_accuracy: 0.9488 - lr: 3.7500e-05

Epoch 18/100
200/200 [=====] - 2s 9ms/step - loss: 0.2910 - accuracy: 0.9978 - val_loss: 0.4737 - val_accuracy: 0.9325 - lr: 3.7500e-05

Epoch 19/100
200/200 [=====] - 2s 9ms/step - loss: 0.2836 - accuracy: 0.9978 - val_loss: 2.3173 - val_accuracy: 0.5362 - lr: 3.7500e-05

Epoch 20/100
200/200 [=====] - 2s 10ms/step - loss: 0.2748 - accuracy: 0.9994 - val_loss: 1.1023 - val_accuracy: 0.6837 - lr: 3.7500e-05

Epoch 21/100
200/200 [=====] - 2s 9ms/step - loss: 0.2711 - accuracy: 0.9991 - val_loss: 0.2816 - val_accuracy: 0.9950 - lr: 3.7500e-05

Epoch 22/100
200/200 [=====] - 2s 10ms/step - loss: 0.2606 - accuracy: 0.9984 - val_loss: 0.7869 - val_accuracy: 0.8087 - lr: 3.7500e-05

Epoch 23/100
200/200 [=====] - 2s 10ms/step - loss: 0.2578 - accuracy: 0.9978 - val_loss: 1.0584 - val_accuracy: 0.6938 - lr: 3.7500e-05

Epoch 24/100
200/200 [=====] - 2s 9ms/step - loss: 0.2468 - accuracy: 0.9991 - val_loss: 0.8891 - val_accuracy: 0.7663 - lr: 3.7500e-05

Epoch 25/100
195/200 [=====>.] - ETA: 0s - loss: 0.2411 - accuracy: 0.9990

Epoch 25: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
200/200 [=====] - 2s 11ms/step - loss: 0.2408 - accuracy: 0.9991 - val_loss: 2.6658 - val_accuracy: 0.4837 - lr: 3.7500e-05

Epoch 26/100
200/200 [=====] - 2s 9ms/step - loss: 0.2325 - accuracy: 0.9991 - val_loss: 0.2520 - val_accuracy: 0.9962 - lr: 1.8750e-05

Epoch 27/100
200/200 [=====] - 2s 9ms/step - loss: 0.2286 - accuracy: 0.9994 - val_loss: 0.3627 - val_accuracy: 0.9588 - lr: 1.8750e-05

Epoch 28/100
200/200 [=====] - 2s 10ms/step - loss: 0.2241 - accuracy: 0.9997 - val_loss: 0.2566 - val_accuracy: 0.9875 - lr: 1.8750e-05

Epoch 29/100
200/200 [=====] - 2s 9ms/step - loss: 0.2196 - accuracy: 0.9997 - val_loss: 0.2692 - val_accuracy: 0.9837 - lr: 1.8750e-05

Epoch 30/100
196/200 [=====>.] - ETA: 0s - loss: 0.2173 - accuracy: 0.9994

Epoch 30: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
200/200 [=====] - 2s 11ms/step - loss: 0.2173 - accuracy: 0.9994 - val_loss: 5.0274 - val_accuracy: 0.1787 - lr: 1.8750e-05

Epoch 31/100
200/200 [=====] - 2s 10ms/step - loss: 0.2126 - accuracy: 0.9997 - val_loss: 0.2192 - val_accuracy: 0.9975 - lr: 9.3750e-06

Epoch 32/100
200/200 [=====] - 2s 10ms/step - loss: 0.2102 - accuracy: 0.9997 - val_loss: 0.2151 - val_accuracy: 0.9950 - lr: 9.3750e-06

Epoch 33/100
200/200 [=====] - 2s 10ms/step - loss: 0.2083 - accuracy: 0.9994 - val_loss: 0.2465 - val_accuracy: 0.9912 - lr: 9.3750e-06

Epoch 34/100
200/200 [=====] - 2s 9ms/step - loss: 0.2043 - accuracy: 0.9994 - val_loss: 0.2164 - val_accuracy: 0.9925 - lr: 9.3750e-06

Epoch 35/100
200/200 [=====] - 2s 9ms/step - loss: 0.2027 - accuracy: 0.9997 - val_loss: 0.2881 - val_accuracy: 0.9737 - lr: 9.3750e-06

Epoch 36/100
196/200 [=====>.] - ETA: 0s - loss: 0.1995 - accuracy: 0.9997
Epoch 36: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
200/200 [=====] - 2s 10ms/step - loss: 0.1993 - accuracy: 0.9997 - val_loss: 0.2201 - val_accuracy: 0.9875 - lr: 9.3750e-06

Epoch 37/100
200/200 [=====] - 2s 9ms/step - loss: 0.1973 - accuracy: 1.0000 - val_loss: 0.2040 - val_accuracy: 0.9962 - lr: 4.6875e-06

Epoch 38/100
200/200 [=====] - 2s 10ms/step - loss: 0.1972 - accuracy: 0.9997 - val_loss: 0.1997 - val_accuracy: 0.9962 - lr: 4.6875e-06

Epoch 39/100
200/200 [=====] - 2s 9ms/step - loss: 0.1953 - accuracy: 1.0000 - val_loss: 0.2067 - val_accuracy: 0.9950 - lr: 4.6875e-06

Epoch 40/100
200/200 [=====] - 2s 11ms/step - loss: 0.1931 - accuracy: 1.0000 - val_loss: 0.2116 - val_accuracy: 0.9900 - lr: 4.6875e-06

Epoch 41/100
200/200 [=====] - 2s 9ms/step - loss: 0.1906 - accuracy: 1.0000 - val_loss: 0.2005 - val_accuracy: 0.9937 - lr: 4.6875e-06

Epoch 42/100
200/200 [=====] - 2s 9ms/step - loss: 0.1889 - accuracy: 1.0000 - val_loss: 0.1955 - val_accuracy: 0.9987 - lr: 4.6875e-06

Epoch 43/100
200/200 [=====] - 2s 9ms/step - loss: 0.1883 - accuracy: 1.0000 - val_loss: 0.2091 - val_accuracy: 0.9912 - lr: 4.6875e-06

Epoch 44/100
200/200 [=====] - 2s 9ms/step - loss: 0.1869 - accuracy: 0.9997 - val_loss: 0.2416 - val_accuracy: 0.9825 - lr: 4.6875e-06

Epoch 45/100
200/200 [=====] - 2s 11ms/step - loss: 0.1848 - accuracy: 1.0000 - val_loss: 0.1953 - val_accuracy: 0.9950 - lr: 4.6875e-06

Epoch 46/100
200/200 [=====] - 2s 10ms/step - loss: 0.1837 - accuracy: 0.9994 - val_loss: 0.2208 - val_accuracy: 0.9850 - lr: 4.6875e-06

Epoch 47/100
200/200 [=====] - 2s 10ms/step - loss: 0.1829 - accuracy: 1.0000 - val_loss: 0.1921 - val_accuracy: 0.9962 - lr: 4.6875e-06

Epoch 48/100
200/200 [=====] - 2s 11ms/step - loss: 0.1804 - accuracy: 0.9997 - val_loss: 0.2060 - val_accuracy: 0.9912 - lr: 4.6875e-06

Epoch 49/100
200/200 [=====] - 2s 9ms/step - loss: 0.1792 - accuracy: 1.0000 - val_loss: 0.2186 - val_accuracy: 0.9825 - lr: 4.6875e-06

Epoch 50/100
200/200 [=====] - 2s 10ms/step - loss: 0.1760 - accuracy: 1.0000 - val_loss: 0.1807 - val_accuracy: 0.9975 - lr: 4.6875e-06

Epoch 51/100
200/200 [=====] - 2s 9ms/step - loss: 0.1759 - accuracy: 0.9997 - val_loss: 0.1849 - val_accuracy: 0.9962 - lr: 4.6875e-06

Epoch 52/100
200/200 [=====] - 2s 10ms/step - loss: 0.1734 - accuracy: 1.0000 - val_loss: 0.1795 - val_accuracy: 0.9950 - lr: 4.6875e-06

Epoch 53/100
200/200 [=====] - 2s 10ms/step - loss: 0.1717 - accuracy
: 0.9997 - val_loss: 0.2064 - val_accuracy: 0.9862 - lr: 4.6875e-06
Epoch 54/100
200/200 [=====] - 2s 10ms/step - loss: 0.1703 - accuracy
: 1.0000 - val_loss: 0.1952 - val_accuracy: 0.9887 - lr: 4.6875e-06
[100, 16, 0.25]
17.5%
Epoch 1/100
188/188 [=====] - 3s 14ms/step - loss: 3.0833 - accuracy
: 0.1570 - val_loss: 3.9844 - val_accuracy: 0.0990 - lr: 3.0000e-04
Epoch 2/100
188/188 [=====] - 2s 13ms/step - loss: 2.4244 - accuracy
: 0.2917 - val_loss: 4.7729 - val_accuracy: 0.0930 - lr: 3.0000e-04
Epoch 3/100
188/188 [=====] - 2s 12ms/step - loss: 1.8399 - accuracy
: 0.5087 - val_loss: 5.0434 - val_accuracy: 0.2230 - lr: 3.0000e-04
Epoch 4/100
188/188 [=====] - 2s 11ms/step - loss: 1.2018 - accuracy
: 0.7597 - val_loss: 12.8710 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 5/100
188/188 [=====] - 2s 11ms/step - loss: 0.7966 - accuracy
: 0.9017 - val_loss: 3.3021 - val_accuracy: 0.2110 - lr: 3.0000e-04
Epoch 6/100
188/188 [=====] - 2s 13ms/step - loss: 0.6200 - accuracy
: 0.9520 - val_loss: 4.4729 - val_accuracy: 0.1920 - lr: 3.0000e-04
Epoch 7/100
188/188 [=====] - 2s 13ms/step - loss: 0.5095 - accuracy
: 0.9727 - val_loss: 38.8220 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 8/100
188/188 [=====] - 3s 13ms/step - loss: 0.4325 - accuracy
: 0.9863 - val_loss: 6.3622 - val_accuracy: 0.4410 - lr: 3.0000e-04
Epoch 9/100
188/188 [=====] - 2s 13ms/step - loss: 0.3900 - accuracy
: 0.9883 - val_loss: 2.4253 - val_accuracy: 0.4800 - lr: 3.0000e-04
Epoch 10/100
188/188 [=====] - 2s 12ms/step - loss: 0.3536 - accuracy
: 0.9913 - val_loss: 5.8908 - val_accuracy: 0.3220 - lr: 3.0000e-04
Epoch 11/100
188/188 [=====] - 2s 11ms/step - loss: 0.3291 - accuracy
: 0.9917 - val_loss: 8.1141 - val_accuracy: 0.4830 - lr: 3.0000e-04
Epoch 12/100
188/188 [=====] - 2s 13ms/step - loss: 0.2865 - accuracy
: 0.9947 - val_loss: 8.6883 - val_accuracy: 0.1800 - lr: 3.0000e-04
Epoch 13/100
185/188 [=====>.] - ETA: 0s - loss: 0.2689 - accuracy: 0.9
946
Epoch 13: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
188/188 [=====] - 3s 14ms/step - loss: 0.2686 - accuracy
: 0.9947 - val_loss: 8.2443 - val_accuracy: 0.3720 - lr: 3.0000e-04
Epoch 14/100
188/188 [=====] - 3s 13ms/step - loss: 0.2410 - accuracy
: 0.9987 - val_loss: 2.7231 - val_accuracy: 0.5960 - lr: 1.5000e-04
Epoch 15/100
188/188 [=====] - 2s 13ms/step - loss: 0.2230 - accuracy
: 0.9993 - val_loss: 10.3231 - val_accuracy: 0.0970 - lr: 1.5000e-04
Epoch 16/100
188/188 [=====] - 2s 13ms/step - loss: 0.2148 - accuracy
: 0.9987 - val_loss: 10.2740 - val_accuracy: 0.3120 - lr: 1.5000e-04
Epoch 17/100

186/188 [=====>.] - ETA: 0s - loss: 0.2061 - accuracy: 0.9983
Epoch 17: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
188/188 [=====] - 2s 12ms/step - loss: 0.2059 - accuracy: 0.9983 - val_loss: 3.2319 - val_accuracy: 0.3750 - lr: 1.5000e-04
Epoch 18/100
188/188 [=====] - 2s 11ms/step - loss: 0.1951 - accuracy: 0.9993 - val_loss: 1.3100 - val_accuracy: 0.6880 - lr: 7.5000e-05
Epoch 19/100
188/188 [=====] - 2s 13ms/step - loss: 0.1896 - accuracy: 0.9993 - val_loss: 0.8638 - val_accuracy: 0.7320 - lr: 7.5000e-05
Epoch 20/100
188/188 [=====] - 3s 14ms/step - loss: 0.1814 - accuracy: 0.9997 - val_loss: 1.2029 - val_accuracy: 0.7110 - lr: 7.5000e-05
Epoch 21/100
188/188 [=====] - 2s 13ms/step - loss: 0.1760 - accuracy: 0.9997 - val_loss: 3.4277 - val_accuracy: 0.4380 - lr: 7.5000e-05
Epoch 22/100
188/188 [=====] - 2s 12ms/step - loss: 0.1696 - accuracy: 0.9997 - val_loss: 4.9551 - val_accuracy: 0.2890 - lr: 7.5000e-05
Epoch 23/100
188/188 [=====] - 2s 11ms/step - loss: 0.1631 - accuracy: 0.9997 - val_loss: 0.2606 - val_accuracy: 0.9700 - lr: 7.5000e-05
Epoch 24/100
188/188 [=====] - 2s 9ms/step - loss: 0.1563 - accuracy: 1.0000 - val_loss: 0.4565 - val_accuracy: 0.8850 - lr: 7.5000e-05
Epoch 25/100
188/188 [=====] - 2s 11ms/step - loss: 0.1541 - accuracy: 0.9987 - val_loss: 0.2009 - val_accuracy: 0.9810 - lr: 7.5000e-05
Epoch 26/100
188/188 [=====] - 2s 10ms/step - loss: 0.1478 - accuracy: 0.9997 - val_loss: 1.8134 - val_accuracy: 0.6180 - lr: 7.5000e-05
Epoch 27/100
188/188 [=====] - 2s 10ms/step - loss: 0.1422 - accuracy: 0.9993 - val_loss: 4.8103 - val_accuracy: 0.2900 - lr: 7.5000e-05
Epoch 28/100
188/188 [=====] - 2s 11ms/step - loss: 0.1344 - accuracy: 0.9997 - val_loss: 13.4171 - val_accuracy: 0.2130 - lr: 7.5000e-05
Epoch 29/100
188/188 [=====] - ETA: 0s - loss: 0.1282 - accuracy: 1.0000
Epoch 29: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
188/188 [=====] - 2s 11ms/step - loss: 0.1282 - accuracy: 1.0000 - val_loss: 2.3822 - val_accuracy: 0.6000 - lr: 7.5000e-05
Epoch 30/100
188/188 [=====] - 2s 11ms/step - loss: 0.1233 - accuracy: 1.0000 - val_loss: 2.4947 - val_accuracy: 0.6840 - lr: 3.7500e-05
Epoch 31/100
188/188 [=====] - 2s 10ms/step - loss: 0.1216 - accuracy: 0.9997 - val_loss: 0.9483 - val_accuracy: 0.7840 - lr: 3.7500e-05
Epoch 32/100
188/188 [=====] - 2s 11ms/step - loss: 0.1171 - accuracy: 1.0000 - val_loss: 0.8866 - val_accuracy: 0.8480 - lr: 3.7500e-05
Epoch 33/100
188/188 [=====] - 2s 9ms/step - loss: 0.1146 - accuracy: 1.0000 - val_loss: 0.1258 - val_accuracy: 0.9980 - lr: 3.7500e-05
Epoch 34/100
188/188 [=====] - 2s 11ms/step - loss: 0.1105 - accuracy: 1.0000 - val_loss: 0.3472 - val_accuracy: 0.9130 - lr: 3.7500e-05
Epoch 35/100

188/188 [=====] - 2s 12ms/step - loss: 0.1065 - accuracy : 1.0000 - val_loss: 0.2292 - val_accuracy: 0.9690 - lr: 3.7500e-05
Epoch 36/100
188/188 [=====] - 2s 10ms/step - loss: 0.1043 - accuracy : 1.0000 - val_loss: 0.2900 - val_accuracy: 0.9200 - lr: 3.7500e-05
Epoch 37/100
184/188 [=====>.] - ETA: 0s - loss: 0.1009 - accuracy: 1.0000
Epoch 37: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
188/188 [=====] - 2s 11ms/step - loss: 0.1009 - accuracy : 1.0000 - val_loss: 0.1698 - val_accuracy: 0.9780 - lr: 3.7500e-05
Epoch 38/100
188/188 [=====] - 2s 12ms/step - loss: 0.0969 - accuracy : 1.0000 - val_loss: 0.1019 - val_accuracy: 0.9980 - lr: 1.8750e-05
Epoch 39/100
188/188 [=====] - 2s 12ms/step - loss: 0.0959 - accuracy : 1.0000 - val_loss: 0.1080 - val_accuracy: 0.9970 - lr: 1.8750e-05
Epoch 40/100
188/188 [=====] - 2s 12ms/step - loss: 0.0940 - accuracy : 1.0000 - val_loss: 0.3042 - val_accuracy: 0.9170 - lr: 1.8750e-05
Epoch 41/100
188/188 [=====] - 2s 13ms/step - loss: 0.0924 - accuracy : 1.0000 - val_loss: 0.2390 - val_accuracy: 0.9560 - lr: 1.8750e-05
Epoch 42/100
183/188 [=====>.] - ETA: 0s - loss: 0.0901 - accuracy: 1.0000
Epoch 42: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
188/188 [=====] - 2s 13ms/step - loss: 0.0901 - accuracy : 1.0000 - val_loss: 0.4888 - val_accuracy: 0.8870 - lr: 1.8750e-05
Epoch 43/100
188/188 [=====] - 2s 13ms/step - loss: 0.0882 - accuracy : 1.0000 - val_loss: 0.1213 - val_accuracy: 0.9890 - lr: 9.3750e-06
Epoch 44/100
188/188 [=====] - 2s 13ms/step - loss: 0.0879 - accuracy : 1.0000 - val_loss: 0.1184 - val_accuracy: 0.9890 - lr: 9.3750e-06
Epoch 45/100
188/188 [=====] - 3s 14ms/step - loss: 0.0870 - accuracy : 1.0000 - val_loss: 0.0984 - val_accuracy: 0.9940 - lr: 9.3750e-06
[100, 16, 0.30000000000000004]
20.0%
Epoch 1/100
175/175 [=====] - 3s 12ms/step - loss: 3.1118 - accuracy : 0.1536 - val_loss: 3.8326 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 2/100
175/175 [=====] - 2s 10ms/step - loss: 2.4905 - accuracy : 0.3039 - val_loss: 4.5472 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 3/100
175/175 [=====] - 2s 11ms/step - loss: 1.8774 - accuracy : 0.4804 - val_loss: 15.3001 - val_accuracy: 0.1292 - lr: 3.0000e-04
Epoch 4/100
175/175 [=====] - 2s 11ms/step - loss: 1.3328 - accuracy : 0.6804 - val_loss: 13.9633 - val_accuracy: 0.2017 - lr: 3.0000e-04
Epoch 5/100
174/175 [=====>.] - ETA: 0s - loss: 0.9371 - accuracy: 0.8337
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
175/175 [=====] - 2s 11ms/step - loss: 0.9374 - accuracy : 0.8332 - val_loss: 4.0643 - val_accuracy: 0.2975 - lr: 3.0000e-04
Epoch 6/100
175/175 [=====] - 2s 10ms/step - loss: 0.6786 - accuracy

: 0.9314 - val_loss: 4.7498 - val_accuracy: 0.3733 - lr: 1.5000e-04
Epoch 7/100
175/175 [=====] - 2s 11ms/step - loss: 0.5701 - accuracy
: 0.9639 - val_loss: 1.7051 - val_accuracy: 0.5392 - lr: 1.5000e-04
Epoch 8/100
175/175 [=====] - 2s 9ms/step - loss: 0.5119 - accuracy:
0.9725 - val_loss: 10.4612 - val_accuracy: 0.1775 - lr: 1.5000e-04
Epoch 9/100
175/175 [=====] - 2s 11ms/step - loss: 0.4580 - accuracy
: 0.9868 - val_loss: 4.8135 - val_accuracy: 0.4092 - lr: 1.5000e-04
Epoch 10/100
175/175 [=====] - 2s 9ms/step - loss: 0.4216 - accuracy:
0.9911 - val_loss: 9.2433 - val_accuracy: 0.1308 - lr: 1.5000e-04
Epoch 11/100
170/175 [=====>.] - ETA: 0s - loss: 0.3855 - accuracy: 0.9
952
Epoch 11: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
175/175 [=====] - 2s 10ms/step - loss: 0.3848 - accuracy
: 0.9950 - val_loss: 2.1213 - val_accuracy: 0.5542 - lr: 1.5000e-04
Epoch 12/100
175/175 [=====] - 2s 11ms/step - loss: 0.3586 - accuracy
: 0.9975 - val_loss: 1.1897 - val_accuracy: 0.6600 - lr: 7.5000e-05
Epoch 13/100
175/175 [=====] - 2s 10ms/step - loss: 0.3537 - accuracy
: 0.9957 - val_loss: 1.3870 - val_accuracy: 0.5642 - lr: 7.5000e-05
Epoch 14/100
175/175 [=====] - 2s 11ms/step - loss: 0.3382 - accuracy
: 0.9979 - val_loss: 2.2250 - val_accuracy: 0.4767 - lr: 7.5000e-05
Epoch 15/100
175/175 [=====] - 2s 11ms/step - loss: 0.3264 - accuracy
: 0.9989 - val_loss: 7.8163 - val_accuracy: 0.2892 - lr: 7.5000e-05
Epoch 16/100
175/175 [=====] - 2s 10ms/step - loss: 0.3151 - accuracy
: 0.9989 - val_loss: 0.4602 - val_accuracy: 0.9458 - lr: 7.5000e-05
Epoch 17/100
175/175 [=====] - 2s 11ms/step - loss: 0.3066 - accuracy
: 0.9989 - val_loss: 1.8550 - val_accuracy: 0.5333 - lr: 7.5000e-05
Epoch 18/100
175/175 [=====] - 2s 10ms/step - loss: 0.2934 - accuracy
: 0.9993 - val_loss: 5.2877 - val_accuracy: 0.2550 - lr: 7.5000e-05
Epoch 19/100
175/175 [=====] - 2s 9ms/step - loss: 0.2868 - accuracy:
0.9996 - val_loss: 2.9472 - val_accuracy: 0.5208 - lr: 7.5000e-05
Epoch 20/100
175/175 [=====] - 2s 11ms/step - loss: 0.2729 - accuracy
: 1.0000 - val_loss: 0.4419 - val_accuracy: 0.9508 - lr: 7.5000e-05
Epoch 21/100
175/175 [=====] - 2s 11ms/step - loss: 0.2634 - accuracy
: 0.9996 - val_loss: 1.7957 - val_accuracy: 0.5642 - lr: 7.5000e-05
Epoch 22/100
175/175 [=====] - 2s 10ms/step - loss: 0.2569 - accuracy
: 0.9996 - val_loss: 4.2458 - val_accuracy: 0.5550 - lr: 7.5000e-05
Epoch 23/100
175/175 [=====] - 2s 11ms/step - loss: 0.2453 - accuracy
: 1.0000 - val_loss: 6.0626 - val_accuracy: 0.3692 - lr: 7.5000e-05
Epoch 24/100
174/175 [=====>.] - ETA: 0s - loss: 0.2352 - accuracy: 0.9
993
Epoch 24: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
175/175 [=====] - 2s 11ms/step - loss: 0.2351 - accuracy

: 0.9993 - val_loss: 3.0459 - val_accuracy: 0.6150 - lr: 7.5000e-05
Epoch 25/100
175/175 [=====] - 2s 10ms/step - loss: 0.2277 - accuracy
: 1.0000 - val_loss: 0.8602 - val_accuracy: 0.8158 - lr: 3.7500e-05
Epoch 26/100
175/175 [=====] - 2s 12ms/step - loss: 0.2227 - accuracy
: 0.9996 - val_loss: 1.1960 - val_accuracy: 0.6767 - lr: 3.7500e-05
Epoch 27/100
175/175 [=====] - 2s 10ms/step - loss: 0.2156 - accuracy
: 1.0000 - val_loss: 2.1410 - val_accuracy: 0.5925 - lr: 3.7500e-05
Epoch 28/100
175/175 [=====] - 2s 9ms/step - loss: 0.2120 - accuracy:
1.0000 - val_loss: 0.4094 - val_accuracy: 0.9175 - lr: 3.7500e-05
Epoch 29/100
175/175 [=====] - 2s 11ms/step - loss: 0.2051 - accuracy
: 1.0000 - val_loss: 0.5504 - val_accuracy: 0.8842 - lr: 3.7500e-05
Epoch 30/100
175/175 [=====] - 2s 10ms/step - loss: 0.2010 - accuracy
: 1.0000 - val_loss: 1.5070 - val_accuracy: 0.6350 - lr: 3.7500e-05
Epoch 31/100
175/175 [=====] - 2s 10ms/step - loss: 0.1930 - accuracy
: 1.0000 - val_loss: 1.0108 - val_accuracy: 0.7750 - lr: 3.7500e-05
Epoch 32/100
175/175 [=====] - 2s 12ms/step - loss: 0.1897 - accuracy
: 0.9993 - val_loss: 0.3334 - val_accuracy: 0.9500 - lr: 3.7500e-05
[100, 16, 0.35000000000000003]
22.5%
Epoch 1/100
163/163 [=====] - 3s 16ms/step - loss: 3.0163 - accuracy
: 0.1647 - val_loss: 5.2218 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 2/100
163/163 [=====] - 2s 14ms/step - loss: 2.3907 - accuracy
: 0.2947 - val_loss: 4.9468 - val_accuracy: 0.1113 - lr: 3.0000e-04
Epoch 3/100
163/163 [=====] - 2s 13ms/step - loss: 1.9983 - accuracy
: 0.4413 - val_loss: 9.9297 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 4/100
163/163 [=====] - 2s 14ms/step - loss: 1.4843 - accuracy
: 0.6526 - val_loss: 11.3608 - val_accuracy: 0.1028 - lr: 3.0000e-04
Epoch 5/100
163/163 [=====] - 2s 14ms/step - loss: 1.0650 - accuracy
: 0.8068 - val_loss: 9.1105 - val_accuracy: 0.0928 - lr: 3.0000e-04
Epoch 6/100
159/163 [=====>.] - ETA: 0s - loss: 0.8006 - accuracy: 0.9
096
Epoch 6: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
163/163 [=====] - 2s 14ms/step - loss: 0.7981 - accuracy
: 0.9100 - val_loss: 10.8195 - val_accuracy: 0.2491 - lr: 3.0000e-04
Epoch 7/100
163/163 [=====] - 2s 14ms/step - loss: 0.6354 - accuracy
: 0.9565 - val_loss: 3.5484 - val_accuracy: 0.3697 - lr: 1.5000e-04
Epoch 8/100
163/163 [=====] - 2s 14ms/step - loss: 0.5668 - accuracy
: 0.9777 - val_loss: 3.2243 - val_accuracy: 0.4340 - lr: 1.5000e-04
Epoch 9/100
163/163 [=====] - 2s 13ms/step - loss: 0.5065 - accuracy
: 0.9808 - val_loss: 3.3885 - val_accuracy: 0.5689 - lr: 1.5000e-04
Epoch 10/100
163/163 [=====] - 2s 12ms/step - loss: 0.4688 - accuracy
: 0.9896 - val_loss: 7.1871 - val_accuracy: 0.1285 - lr: 1.5000e-04

Epoch 11/100
163/163 [=====] - 2s 12ms/step - loss: 0.4324 - accuracy : 0.9935 - val_loss: 7.2244 - val_accuracy: 0.4140 - lr: 1.5000e-04

Epoch 12/100
159/163 [=====>.] - ETA: 0s - loss: 0.4033 - accuracy: 0.9925

Epoch 12: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
163/163 [=====] - 2s 13ms/step - loss: 0.4038 - accuracy : 0.9923 - val_loss: 8.6593 - val_accuracy: 0.2305 - lr: 1.5000e-04

Epoch 13/100
163/163 [=====] - 2s 14ms/step - loss: 0.3784 - accuracy : 0.9958 - val_loss: 2.3373 - val_accuracy: 0.5011 - lr: 7.5000e-05

Epoch 14/100
163/163 [=====] - 2s 14ms/step - loss: 0.3614 - accuracy : 0.9981 - val_loss: 2.2858 - val_accuracy: 0.4461 - lr: 7.5000e-05

Epoch 15/100
163/163 [=====] - 2s 14ms/step - loss: 0.3491 - accuracy : 0.9977 - val_loss: 14.9389 - val_accuracy: 0.1378 - lr: 7.5000e-05

Epoch 16/100
163/163 [=====] - 2s 14ms/step - loss: 0.3382 - accuracy : 0.9977 - val_loss: 3.8751 - val_accuracy: 0.4168 - lr: 7.5000e-05

Epoch 17/100
163/163 [=====] - 2s 13ms/step - loss: 0.3222 - accuracy : 0.9985 - val_loss: 4.2038 - val_accuracy: 0.2976 - lr: 7.5000e-05

Epoch 18/100
162/163 [=====>.] - ETA: 0s - loss: 0.3159 - accuracy: 0.9965

Epoch 18: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
163/163 [=====] - 2s 12ms/step - loss: 0.3158 - accuracy : 0.9965 - val_loss: 2.4552 - val_accuracy: 0.5617 - lr: 7.5000e-05

Epoch 19/100
163/163 [=====] - 2s 14ms/step - loss: 0.3010 - accuracy : 0.9992 - val_loss: 0.4500 - val_accuracy: 0.9458 - lr: 3.7500e-05

Epoch 20/100
163/163 [=====] - 2s 13ms/step - loss: 0.2970 - accuracy : 0.9992 - val_loss: 0.3410 - val_accuracy: 0.9807 - lr: 3.7500e-05

Epoch 21/100
163/163 [=====] - 2s 13ms/step - loss: 0.2914 - accuracy : 0.9988 - val_loss: 3.4201 - val_accuracy: 0.3312 - lr: 3.7500e-05

Epoch 22/100
163/163 [=====] - 2s 14ms/step - loss: 0.2865 - accuracy : 1.0000 - val_loss: 1.7126 - val_accuracy: 0.6545 - lr: 3.7500e-05

Epoch 23/100
163/163 [=====] - 2s 13ms/step - loss: 0.2789 - accuracy : 0.9988 - val_loss: 0.6344 - val_accuracy: 0.8430 - lr: 3.7500e-05

Epoch 24/100
162/163 [=====>.] - ETA: 0s - loss: 0.2777 - accuracy: 0.9985

Epoch 24: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
163/163 [=====] - 2s 13ms/step - loss: 0.2787 - accuracy : 0.9981 - val_loss: 2.5399 - val_accuracy: 0.5460 - lr: 3.7500e-05

Epoch 25/100
163/163 [=====] - 2s 12ms/step - loss: 0.2696 - accuracy : 0.9996 - val_loss: 0.3791 - val_accuracy: 0.9650 - lr: 1.8750e-05

Epoch 26/100
163/163 [=====] - 2s 13ms/step - loss: 0.2640 - accuracy : 0.9996 - val_loss: 0.4330 - val_accuracy: 0.9358 - lr: 1.8750e-05

Epoch 27/100
163/163 [=====] - 2s 14ms/step - loss: 0.2621 - accuracy : 0.9988 - val_loss: 2.0012 - val_accuracy: 0.5139 - lr: 1.8750e-05

Epoch 28/100
163/163 [=====] - 2s 14ms/step - loss: 0.2586 - accuracy : 1.0000 - val_loss: 0.2885 - val_accuracy: 0.9843 - lr: 1.8750e-05

Epoch 29/100
163/163 [=====] - 2s 14ms/step - loss: 0.2563 - accuracy : 0.9992 - val_loss: 0.5714 - val_accuracy: 0.8722 - lr: 1.8750e-05

Epoch 30/100
163/163 [=====] - 2s 14ms/step - loss: 0.2518 - accuracy : 1.0000 - val_loss: 1.1656 - val_accuracy: 0.7445 - lr: 1.8750e-05

Epoch 31/100
163/163 [=====] - 2s 14ms/step - loss: 0.2483 - accuracy : 0.9992 - val_loss: 2.1890 - val_accuracy: 0.5924 - lr: 1.8750e-05

Epoch 32/100
159/163 [=====>.] - ETA: 0s - loss: 0.2445 - accuracy: 1.0000
Epoch 32: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
163/163 [=====] - 2s 12ms/step - loss: 0.2447 - accuracy : 1.0000 - val_loss: 0.8113 - val_accuracy: 0.7759 - lr: 1.8750e-05

Epoch 33/100
163/163 [=====] - 2s 14ms/step - loss: 0.2427 - accuracy : 0.9996 - val_loss: 0.2680 - val_accuracy: 0.9857 - lr: 9.3750e-06

Epoch 34/100
163/163 [=====] - 2s 13ms/step - loss: 0.2384 - accuracy : 0.9996 - val_loss: 0.2763 - val_accuracy: 0.9814 - lr: 9.3750e-06

Epoch 35/100
163/163 [=====] - 2s 14ms/step - loss: 0.2409 - accuracy : 0.9996 - val_loss: 0.3136 - val_accuracy: 0.9700 - lr: 9.3750e-06

Epoch 36/100
163/163 [=====] - 2s 13ms/step - loss: 0.2362 - accuracy : 1.0000 - val_loss: 1.2439 - val_accuracy: 0.7095 - lr: 9.3750e-06

Epoch 37/100
159/163 [=====>.] - ETA: 0s - loss: 0.2335 - accuracy: 0.9996
Epoch 37: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
163/163 [=====] - 2s 14ms/step - loss: 0.2333 - accuracy : 0.9996 - val_loss: 0.4474 - val_accuracy: 0.9265 - lr: 9.3750e-06

Epoch 38/100
163/163 [=====] - 2s 15ms/step - loss: 0.2314 - accuracy : 1.0000 - val_loss: 0.2515 - val_accuracy: 0.9864 - lr: 4.6875e-06

Epoch 39/100
163/163 [=====] - 2s 13ms/step - loss: 0.2306 - accuracy : 0.9996 - val_loss: 0.3607 - val_accuracy: 0.9543 - lr: 4.6875e-06

Epoch 40/100
163/163 [=====] - 2s 13ms/step - loss: 0.2300 - accuracy : 1.0000 - val_loss: 0.2557 - val_accuracy: 0.9886 - lr: 4.6875e-06

Epoch 41/100
163/163 [=====] - 2s 13ms/step - loss: 0.2264 - accuracy : 1.0000 - val_loss: 0.2467 - val_accuracy: 0.9872 - lr: 4.6875e-06

Epoch 42/100
163/163 [=====] - 2s 15ms/step - loss: 0.2263 - accuracy : 1.0000 - val_loss: 0.2714 - val_accuracy: 0.9800 - lr: 4.6875e-06

Epoch 43/100
163/163 [=====] - 3s 15ms/step - loss: 0.2254 - accuracy : 0.9996 - val_loss: 0.2550 - val_accuracy: 0.9886 - lr: 4.6875e-06

Epoch 44/100
163/163 [=====] - 2s 14ms/step - loss: 0.2224 - accuracy : 0.9996 - val_loss: 0.3759 - val_accuracy: 0.9429 - lr: 4.6875e-06

Epoch 45/100
163/163 [=====] - 2s 14ms/step - loss: 0.2212 - accuracy : 1.0000 - val_loss: 0.2410 - val_accuracy: 0.9893 - lr: 4.6875e-06

Epoch 46/100
163/163 [=====] - 2s 13ms/step - loss: 0.2195 - accuracy : 1.0000 - val_loss: 0.2401 - val_accuracy: 0.9872 - lr: 4.6875e-06

Epoch 47/100
163/163 [=====] - 2s 13ms/step - loss: 0.2174 - accuracy : 1.0000 - val_loss: 0.2520 - val_accuracy: 0.9822 - lr: 4.6875e-06

Epoch 48/100
163/163 [=====] - 2s 13ms/step - loss: 0.2164 - accuracy : 1.0000 - val_loss: 0.2636 - val_accuracy: 0.9807 - lr: 4.6875e-06

Epoch 49/100
163/163 [=====] - 2s 14ms/step - loss: 0.2154 - accuracy : 1.0000 - val_loss: 0.3484 - val_accuracy: 0.9429 - lr: 4.6875e-06

Epoch 50/100
161/163 [=====>.] - ETA: 0s - loss: 0.2148 - accuracy: 1.0000
Epoch 50: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
163/163 [=====] - 2s 14ms/step - loss: 0.2149 - accuracy : 1.0000 - val_loss: 0.3048 - val_accuracy: 0.9686 - lr: 4.6875e-06

Epoch 51/100
163/163 [=====] - 2s 13ms/step - loss: 0.2135 - accuracy : 1.0000 - val_loss: 0.2364 - val_accuracy: 0.9900 - lr: 2.3438e-06

Epoch 52/100
163/163 [=====] - 2s 14ms/step - loss: 0.2120 - accuracy : 0.9996 - val_loss: 0.2327 - val_accuracy: 0.9886 - lr: 2.3438e-06

Epoch 53/100
163/163 [=====] - 2s 13ms/step - loss: 0.2102 - accuracy : 1.0000 - val_loss: 0.2283 - val_accuracy: 0.9900 - lr: 2.3438e-06

Epoch 54/100
163/163 [=====] - 2s 13ms/step - loss: 0.2088 - accuracy : 1.0000 - val_loss: 0.2327 - val_accuracy: 0.9857 - lr: 2.3438e-06

Epoch 55/100
163/163 [=====] - 2s 13ms/step - loss: 0.2075 - accuracy : 1.0000 - val_loss: 0.2281 - val_accuracy: 0.9886 - lr: 2.3438e-06

Epoch 56/100
163/163 [=====] - 2s 13ms/step - loss: 0.2078 - accuracy : 0.9996 - val_loss: 0.2414 - val_accuracy: 0.9850 - lr: 2.3438e-06

Epoch 57/100
163/163 [=====] - 2s 14ms/step - loss: 0.2078 - accuracy : 0.9996 - val_loss: 0.2281 - val_accuracy: 0.9864 - lr: 2.3438e-06

Epoch 58/100
163/163 [=====] - 2s 13ms/step - loss: 0.2063 - accuracy : 1.0000 - val_loss: 0.2549 - val_accuracy: 0.9807 - lr: 2.3438e-06

Epoch 59/100
163/163 [=====] - 2s 14ms/step - loss: 0.2042 - accuracy : 0.9996 - val_loss: 0.2261 - val_accuracy: 0.9907 - lr: 2.3438e-06

Epoch 60/100
163/163 [=====] - 2s 14ms/step - loss: 0.2031 - accuracy : 1.0000 - val_loss: 0.2472 - val_accuracy: 0.9829 - lr: 2.3438e-06

Epoch 61/100
163/163 [=====] - 2s 12ms/step - loss: 0.2029 - accuracy : 1.0000 - val_loss: 0.2221 - val_accuracy: 0.9893 - lr: 2.3438e-06

Epoch 62/100
163/163 [=====] - 2s 13ms/step - loss: 0.2029 - accuracy : 1.0000 - val_loss: 0.2190 - val_accuracy: 0.9886 - lr: 2.3438e-06

Epoch 63/100
163/163 [=====] - 2s 14ms/step - loss: 0.2013 - accuracy : 1.0000 - val_loss: 0.2290 - val_accuracy: 0.9872 - lr: 2.3438e-06

Epoch 64/100
163/163 [=====] - 2s 14ms/step - loss: 0.1992 - accuracy : 1.0000 - val_loss: 0.2195 - val_accuracy: 0.9886 - lr: 2.3438e-06

Epoch 65/100
163/163 [=====] - 2s 14ms/step - loss: 0.1975 - accuracy : 1.0000 - val_loss: 0.2200 - val_accuracy: 0.9900 - lr: 2.3438e-06
Epoch 66/100
163/163 [=====] - 2s 14ms/step - loss: 0.1983 - accuracy : 0.9996 - val_loss: 0.2171 - val_accuracy: 0.9907 - lr: 2.3438e-06
Epoch 67/100
163/163 [=====] - 2s 14ms/step - loss: 0.1953 - accuracy : 1.0000 - val_loss: 0.2204 - val_accuracy: 0.9893 - lr: 2.3438e-06
Epoch 68/100
163/163 [=====] - 2s 12ms/step - loss: 0.1948 - accuracy : 1.0000 - val_loss: 0.2152 - val_accuracy: 0.9907 - lr: 2.3438e-06
Epoch 69/100
163/163 [=====] - 2s 12ms/step - loss: 0.1942 - accuracy : 1.0000 - val_loss: 0.2154 - val_accuracy: 0.9900 - lr: 2.3438e-06
Epoch 70/100
163/163 [=====] - 2s 12ms/step - loss: 0.1921 - accuracy : 1.0000 - val_loss: 0.2161 - val_accuracy: 0.9886 - lr: 2.3438e-06
Epoch 71/100
163/163 [=====] - 2s 14ms/step - loss: 0.1898 - accuracy : 1.0000 - val_loss: 0.2147 - val_accuracy: 0.9886 - lr: 2.3438e-06
[100, 16, 0.4]
25.0%
Epoch 1/100
150/150 [=====] - 3s 15ms/step - loss: 3.1269 - accuracy : 0.1488 - val_loss: 3.9113 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 2/100
150/150 [=====] - 2s 12ms/step - loss: 2.5369 - accuracy : 0.2812 - val_loss: 4.2892 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 3/100
150/150 [=====] - 2s 12ms/step - loss: 1.9199 - accuracy : 0.4775 - val_loss: 8.8005 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 4/100
150/150 [=====] - 2s 12ms/step - loss: 1.4756 - accuracy : 0.6308 - val_loss: 10.2418 - val_accuracy: 0.1200 - lr: 3.0000e-04
Epoch 5/100
147/150 [=====>.] - ETA: 0s - loss: 1.1453 - accuracy: 0.7713
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
150/150 [=====] - 2s 11ms/step - loss: 1.1432 - accuracy : 0.7725 - val_loss: 6.3013 - val_accuracy: 0.1219 - lr: 3.0000e-04
Epoch 6/100
150/150 [=====] - 2s 10ms/step - loss: 0.8493 - accuracy : 0.8929 - val_loss: 6.8539 - val_accuracy: 0.1181 - lr: 1.5000e-04
Epoch 7/100
150/150 [=====] - 2s 13ms/step - loss: 0.7249 - accuracy : 0.9346 - val_loss: 3.0297 - val_accuracy: 0.3619 - lr: 1.5000e-04
Epoch 8/100
150/150 [=====] - 2s 12ms/step - loss: 0.6419 - accuracy : 0.9542 - val_loss: 3.5792 - val_accuracy: 0.2606 - lr: 1.5000e-04
Epoch 9/100
150/150 [=====] - 2s 11ms/step - loss: 0.5793 - accuracy : 0.9712 - val_loss: 3.0129 - val_accuracy: 0.4706 - lr: 1.5000e-04
Epoch 10/100
150/150 [=====] - 2s 12ms/step - loss: 0.5330 - accuracy : 0.9754 - val_loss: 4.0221 - val_accuracy: 0.3706 - lr: 1.5000e-04
Epoch 11/100
150/150 [=====] - 2s 12ms/step - loss: 0.4832 - accuracy : 0.9850 - val_loss: 0.7966 - val_accuracy: 0.8363 - lr: 1.5000e-04
Epoch 12/100

150/150 [=====] - 2s 10ms/step - loss: 0.4499 - accuracy
: 0.9896 - val_loss: 2.2794 - val_accuracy: 0.4244 - lr: 1.5000e-04
Epoch 13/100
150/150 [=====] - 2s 12ms/step - loss: 0.4146 - accuracy
: 0.9912 - val_loss: 1.1350 - val_accuracy: 0.6600 - lr: 1.5000e-04
Epoch 14/100
150/150 [=====] - 2s 10ms/step - loss: 0.3875 - accuracy
: 0.9975 - val_loss: 2.1722 - val_accuracy: 0.4069 - lr: 1.5000e-04
Epoch 15/100
150/150 [=====] - 2s 10ms/step - loss: 0.3640 - accuracy
: 0.9958 - val_loss: 0.5250 - val_accuracy: 0.9406 - lr: 1.5000e-04
Epoch 16/100
150/150 [=====] - 2s 12ms/step - loss: 0.3504 - accuracy
: 0.9983 - val_loss: 4.6114 - val_accuracy: 0.2775 - lr: 1.5000e-04
Epoch 17/100
150/150 [=====] - 2s 11ms/step - loss: 0.3334 - accuracy
: 0.9967 - val_loss: 0.7365 - val_accuracy: 0.8313 - lr: 1.5000e-04
Epoch 18/100
150/150 [=====] - 2s 12ms/step - loss: 0.3148 - accuracy
: 0.9975 - val_loss: 1.0170 - val_accuracy: 0.7181 - lr: 1.5000e-04
Epoch 19/100
147/150 [=====>.] - ETA: 0s - loss: 0.3009 - accuracy: 0.9
979
Epoch 19: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
150/150 [=====] - 2s 12ms/step - loss: 0.3003 - accuracy
: 0.9979 - val_loss: 2.2698 - val_accuracy: 0.4231 - lr: 1.5000e-04
Epoch 20/100
150/150 [=====] - 2s 12ms/step - loss: 0.2839 - accuracy
: 0.9975 - val_loss: 1.0090 - val_accuracy: 0.7156 - lr: 7.5000e-05
Epoch 21/100
150/150 [=====] - 2s 12ms/step - loss: 0.2723 - accuracy
: 1.0000 - val_loss: 0.5376 - val_accuracy: 0.8875 - lr: 7.5000e-05
Epoch 22/100
150/150 [=====] - 2s 12ms/step - loss: 0.2640 - accuracy
: 0.9992 - val_loss: 1.0881 - val_accuracy: 0.6762 - lr: 7.5000e-05
Epoch 23/100
148/150 [=====>.] - ETA: 0s - loss: 0.2586 - accuracy: 0.9
992
Epoch 23: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
150/150 [=====] - 2s 10ms/step - loss: 0.2586 - accuracy
: 0.9992 - val_loss: 2.2219 - val_accuracy: 0.5319 - lr: 7.5000e-05
Epoch 24/100
150/150 [=====] - 2s 12ms/step - loss: 0.2525 - accuracy
: 0.9996 - val_loss: 0.2718 - val_accuracy: 0.9875 - lr: 3.7500e-05
Epoch 25/100
150/150 [=====] - 2s 12ms/step - loss: 0.2444 - accuracy
: 0.9996 - val_loss: 0.3027 - val_accuracy: 0.9800 - lr: 3.7500e-05
Epoch 26/100
150/150 [=====] - 2s 13ms/step - loss: 0.2397 - accuracy
: 1.0000 - val_loss: 0.2705 - val_accuracy: 0.9825 - lr: 3.7500e-05
Epoch 27/100
150/150 [=====] - 2s 13ms/step - loss: 0.2387 - accuracy
: 0.9996 - val_loss: 0.3598 - val_accuracy: 0.9719 - lr: 3.7500e-05
Epoch 28/100
150/150 [=====] - 2s 12ms/step - loss: 0.2321 - accuracy
: 1.0000 - val_loss: 0.3569 - val_accuracy: 0.9631 - lr: 3.7500e-05
Epoch 29/100
150/150 [=====] - 2s 12ms/step - loss: 0.2275 - accuracy
: 1.0000 - val_loss: 0.3406 - val_accuracy: 0.9656 - lr: 3.7500e-05
Epoch 30/100

148/150 [=====>.] - ETA: 0s - loss: 0.2226 - accuracy: 1.0000
Epoch 30: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
150/150 [=====] - 2s 13ms/step - loss: 0.2227 - accuracy: 1.0000 - val_loss: 0.4534 - val_accuracy: 0.9244 - lr: 3.7500e-05
Epoch 31/100
150/150 [=====] - 2s 13ms/step - loss: 0.2201 - accuracy: 0.9996 - val_loss: 0.2923 - val_accuracy: 0.9762 - lr: 1.8750e-05
Epoch 32/100
150/150 [=====] - 1s 10ms/step - loss: 0.2178 - accuracy: 0.9996 - val_loss: 0.2508 - val_accuracy: 0.9887 - lr: 1.8750e-05
Epoch 33/100
150/150 [=====] - 2s 11ms/step - loss: 0.2161 - accuracy: 0.9996 - val_loss: 0.2381 - val_accuracy: 0.9856 - lr: 1.8750e-05
Epoch 34/100
150/150 [=====] - 2s 12ms/step - loss: 0.2119 - accuracy: 1.0000 - val_loss: 0.2524 - val_accuracy: 0.9856 - lr: 1.8750e-05
Epoch 35/100
150/150 [=====] - 2s 11ms/step - loss: 0.2079 - accuracy: 1.0000 - val_loss: 0.2613 - val_accuracy: 0.9825 - lr: 1.8750e-05
Epoch 36/100
150/150 [=====] - 2s 11ms/step - loss: 0.2069 - accuracy: 1.0000 - val_loss: 0.2364 - val_accuracy: 0.9887 - lr: 1.8750e-05
Epoch 37/100
150/150 [=====] - 2s 12ms/step - loss: 0.2044 - accuracy: 1.0000 - val_loss: 0.6231 - val_accuracy: 0.8225 - lr: 1.8750e-05
Epoch 38/100
150/150 [=====] - 2s 13ms/step - loss: 0.1994 - accuracy: 1.0000 - val_loss: 0.2443 - val_accuracy: 0.9875 - lr: 1.8750e-05
Epoch 39/100
150/150 [=====] - 1s 10ms/step - loss: 0.1973 - accuracy: 1.0000 - val_loss: 0.2436 - val_accuracy: 0.9850 - lr: 1.8750e-05
Epoch 40/100
146/150 [=====>.] - ETA: 0s - loss: 0.1956 - accuracy: 1.0000
Epoch 40: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
150/150 [=====] - 2s 12ms/step - loss: 0.1955 - accuracy: 1.0000 - val_loss: 0.3086 - val_accuracy: 0.9625 - lr: 1.8750e-05
Epoch 41/100
150/150 [=====] - 2s 10ms/step - loss: 0.1927 - accuracy: 0.9996 - val_loss: 0.2525 - val_accuracy: 0.9825 - lr: 9.3750e-06
Epoch 42/100
150/150 [=====] - 1s 10ms/step - loss: 0.1902 - accuracy: 0.9996 - val_loss: 0.2074 - val_accuracy: 0.9912 - lr: 9.3750e-06
Epoch 43/100
150/150 [=====] - 2s 12ms/step - loss: 0.1904 - accuracy: 1.0000 - val_loss: 0.2125 - val_accuracy: 0.9900 - lr: 9.3750e-06
Epoch 44/100
150/150 [=====] - 2s 11ms/step - loss: 0.1864 - accuracy: 1.0000 - val_loss: 0.2163 - val_accuracy: 0.9831 - lr: 9.3750e-06
Epoch 45/100
150/150 [=====] - 2s 12ms/step - loss: 0.1845 - accuracy: 1.0000 - val_loss: 0.2442 - val_accuracy: 0.9812 - lr: 9.3750e-06
Epoch 46/100
148/150 [=====>.] - ETA: 0s - loss: 0.1846 - accuracy: 1.0000
Epoch 46: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
150/150 [=====] - 2s 12ms/step - loss: 0.1846 - accuracy: 1.0000 - val_loss: 0.2323 - val_accuracy: 0.9837 - lr: 9.3750e-06
Epoch 47/100

150/150 [=====] - 2s 12ms/step - loss: 0.1819 - accuracy
: 1.0000 - val_loss: 0.1989 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 48/100
150/150 [=====] - 2s 12ms/step - loss: 0.1825 - accuracy
: 0.9996 - val_loss: 0.2086 - val_accuracy: 0.9894 - lr: 4.6875e-06
Epoch 49/100
150/150 [=====] - 2s 11ms/step - loss: 0.1802 - accuracy
: 1.0000 - val_loss: 0.1961 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 50/100
150/150 [=====] - 2s 12ms/step - loss: 0.1780 - accuracy
: 1.0000 - val_loss: 0.1953 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 51/100
150/150 [=====] - 2s 10ms/step - loss: 0.1778 - accuracy
: 1.0000 - val_loss: 0.1951 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 52/100
150/150 [=====] - 2s 11ms/step - loss: 0.1780 - accuracy
: 1.0000 - val_loss: 0.2022 - val_accuracy: 0.9881 - lr: 4.6875e-06
Epoch 53/100
150/150 [=====] - 2s 12ms/step - loss: 0.1774 - accuracy
: 0.9996 - val_loss: 0.1929 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 54/100
150/150 [=====] - 2s 13ms/step - loss: 0.1757 - accuracy
: 1.0000 - val_loss: 0.1905 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 55/100
150/150 [=====] - 2s 12ms/step - loss: 0.1739 - accuracy
: 1.0000 - val_loss: 0.1941 - val_accuracy: 0.9894 - lr: 4.6875e-06
Epoch 56/100
150/150 [=====] - 2s 11ms/step - loss: 0.1730 - accuracy
: 1.0000 - val_loss: 0.1909 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 57/100
150/150 [=====] - 2s 13ms/step - loss: 0.1709 - accuracy
: 1.0000 - val_loss: 0.1890 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 58/100
150/150 [=====] - 2s 13ms/step - loss: 0.1696 - accuracy
: 1.0000 - val_loss: 0.1887 - val_accuracy: 0.9925 - lr: 4.6875e-06
Epoch 59/100
150/150 [=====] - 2s 11ms/step - loss: 0.1692 - accuracy
: 0.9992 - val_loss: 0.1932 - val_accuracy: 0.9919 - lr: 4.6875e-06
Epoch 60/100
150/150 [=====] - 2s 12ms/step - loss: 0.1680 - accuracy
: 1.0000 - val_loss: 0.1842 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 61/100
150/150 [=====] - 2s 11ms/step - loss: 0.1673 - accuracy
: 1.0000 - val_loss: 0.1844 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 62/100
150/150 [=====] - 2s 11ms/step - loss: 0.1655 - accuracy
: 1.0000 - val_loss: 0.1819 - val_accuracy: 0.9956 - lr: 4.6875e-06
Epoch 63/100
150/150 [=====] - 2s 12ms/step - loss: 0.1639 - accuracy
: 1.0000 - val_loss: 0.1825 - val_accuracy: 0.9919 - lr: 4.6875e-06
Epoch 64/100
150/150 [=====] - 2s 13ms/step - loss: 0.1630 - accuracy
: 1.0000 - val_loss: 0.1789 - val_accuracy: 0.9956 - lr: 4.6875e-06
Epoch 65/100
150/150 [=====] - 2s 11ms/step - loss: 0.1620 - accuracy
: 0.9996 - val_loss: 0.1962 - val_accuracy: 0.9875 - lr: 4.6875e-06
Epoch 66/100
150/150 [=====] - 2s 12ms/step - loss: 0.1599 - accuracy
: 1.0000 - val_loss: 0.1822 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 67/100

150/150 [=====] - 2s 12ms/step - loss: 0.1583 - accuracy
: 1.0000 - val_loss: 0.1771 - val_accuracy: 0.9944 - lr: 4.6875e-06
Epoch 68/100
150/150 [=====] - 2s 11ms/step - loss: 0.1582 - accuracy
: 1.0000 - val_loss: 0.1830 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 69/100
150/150 [=====] - 2s 12ms/step - loss: 0.1559 - accuracy
: 1.0000 - val_loss: 0.1742 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 70/100
150/150 [=====] - 2s 10ms/step - loss: 0.1547 - accuracy
: 1.0000 - val_loss: 0.1729 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 71/100
150/150 [=====] - 2s 13ms/step - loss: 0.1539 - accuracy
: 1.0000 - val_loss: 0.1763 - val_accuracy: 0.9919 - lr: 4.6875e-06
Epoch 72/100
150/150 [=====] - 2s 12ms/step - loss: 0.1515 - accuracy
: 1.0000 - val_loss: 0.1680 - val_accuracy: 0.9956 - lr: 4.6875e-06
Epoch 73/100
150/150 [=====] - 2s 11ms/step - loss: 0.1506 - accuracy
: 1.0000 - val_loss: 0.1684 - val_accuracy: 0.9956 - lr: 4.6875e-06
Epoch 74/100
150/150 [=====] - 2s 13ms/step - loss: 0.1500 - accuracy
: 0.9996 - val_loss: 0.1684 - val_accuracy: 0.9931 - lr: 4.6875e-06
[100, 24, 0.2]
27.5%
Epoch 1/100
134/134 [=====] - 3s 12ms/step - loss: 3.0071 - accuracy
: 0.1566 - val_loss: 7.7686 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 2/100
134/134 [=====] - 1s 11ms/step - loss: 2.3605 - accuracy
: 0.3006 - val_loss: 9.8858 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 3/100
134/134 [=====] - 1s 11ms/step - loss: 1.8355 - accuracy
: 0.5044 - val_loss: 12.4043 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 4/100
134/134 [=====] - 1s 9ms/step - loss: 1.2448 - accuracy:
0.7397 - val_loss: 17.9073 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 5/100
134/134 [=====] - ETA: 0s - loss: 0.8390 - accuracy: 0.8
906
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
134/134 [=====] - 1s 10ms/step - loss: 0.8390 - accuracy
: 0.8906 - val_loss: 17.2394 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 6/100
134/134 [=====] - 1s 11ms/step - loss: 0.6281 - accuracy
: 0.9578 - val_loss: 4.1699 - val_accuracy: 0.5013 - lr: 1.5000e-04
Epoch 7/100
134/134 [=====] - 2s 12ms/step - loss: 0.5600 - accuracy
: 0.9778 - val_loss: 3.6407 - val_accuracy: 0.2900 - lr: 1.5000e-04
Epoch 8/100
134/134 [=====] - 2s 12ms/step - loss: 0.5054 - accuracy
: 0.9862 - val_loss: 5.0346 - val_accuracy: 0.3250 - lr: 1.5000e-04
Epoch 9/100
134/134 [=====] - 2s 13ms/step - loss: 0.4615 - accuracy
: 0.9928 - val_loss: 7.2971 - val_accuracy: 0.3000 - lr: 1.5000e-04
Epoch 10/100
134/134 [=====] - 2s 13ms/step - loss: 0.4401 - accuracy
: 0.9909 - val_loss: 6.2370 - val_accuracy: 0.1787 - lr: 1.5000e-04
Epoch 11/100
134/134 [=====] - 2s 12ms/step - loss: 0.4073 - accuracy

```
: 0.9959 - val_loss: 3.3336 - val_accuracy: 0.5575 - lr: 1.5000e-04
Epoch 12/100
134/134 [=====] - 2s 12ms/step - loss: 0.3839 - accuracy
: 0.9959 - val_loss: 0.7396 - val_accuracy: 0.8637 - lr: 1.5000e-04
Epoch 13/100
134/134 [=====] - 2s 12ms/step - loss: 0.3568 - accuracy
: 0.9987 - val_loss: 2.2989 - val_accuracy: 0.5088 - lr: 1.5000e-04
Epoch 14/100
134/134 [=====] - 1s 9ms/step - loss: 0.3464 - accuracy:
0.9975 - val_loss: 2.2205 - val_accuracy: 0.4162 - lr: 1.5000e-04
Epoch 15/100
134/134 [=====] - 1s 10ms/step - loss: 0.3244 - accuracy
: 0.9981 - val_loss: 8.4678 - val_accuracy: 0.2450 - lr: 1.5000e-04
Epoch 16/100
134/134 [=====] - ETA: 0s - loss: 0.3096 - accuracy: 0.9
991
Epoch 16: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
134/134 [=====] - 2s 11ms/step - loss: 0.3096 - accuracy
: 0.9991 - val_loss: 7.5311 - val_accuracy: 0.1650 - lr: 1.5000e-04
Epoch 17/100
134/134 [=====] - 2s 13ms/step - loss: 0.2956 - accuracy
: 0.9991 - val_loss: 0.7776 - val_accuracy: 0.8138 - lr: 7.5000e-05
Epoch 18/100
134/134 [=====] - 2s 12ms/step - loss: 0.2852 - accuracy
: 0.9994 - val_loss: 1.1109 - val_accuracy: 0.7113 - lr: 7.5000e-05
Epoch 19/100
134/134 [=====] - 2s 13ms/step - loss: 0.2755 - accuracy
: 1.0000 - val_loss: 0.9407 - val_accuracy: 0.7912 - lr: 7.5000e-05
Epoch 20/100
134/134 [=====] - 2s 13ms/step - loss: 0.2691 - accuracy
: 0.9997 - val_loss: 0.6814 - val_accuracy: 0.8325 - lr: 7.5000e-05
Epoch 21/100
134/134 [=====] - 2s 11ms/step - loss: 0.2603 - accuracy
: 0.9994 - val_loss: 1.8516 - val_accuracy: 0.6737 - lr: 7.5000e-05
Epoch 22/100
134/134 [=====] - 2s 12ms/step - loss: 0.2535 - accuracy
: 0.9997 - val_loss: 1.2090 - val_accuracy: 0.6625 - lr: 7.5000e-05
Epoch 23/100
134/134 [=====] - 1s 11ms/step - loss: 0.2474 - accuracy
: 0.9997 - val_loss: 3.7470 - val_accuracy: 0.4863 - lr: 7.5000e-05
Epoch 24/100
133/134 [=====>.] - ETA: 0s - loss: 0.2390 - accuracy: 1.0
000
Epoch 24: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
134/134 [=====] - 1s 11ms/step - loss: 0.2389 - accuracy
: 1.0000 - val_loss: 2.3901 - val_accuracy: 0.5013 - lr: 7.5000e-05
[100, 24, 0.25]
30.0%
Epoch 1/100
125/125 [=====] - 2s 13ms/step - loss: 3.0483 - accuracy
: 0.1737 - val_loss: 5.3474 - val_accuracy: 0.1160 - lr: 3.0000e-04
Epoch 2/100
125/125 [=====] - 1s 10ms/step - loss: 2.4224 - accuracy
: 0.2953 - val_loss: 7.7933 - val_accuracy: 0.1160 - lr: 3.0000e-04
Epoch 3/100
125/125 [=====] - 1s 12ms/step - loss: 1.9167 - accuracy
: 0.4780 - val_loss: 8.0571 - val_accuracy: 0.1160 - lr: 3.0000e-04
Epoch 4/100
125/125 [=====] - 1s 10ms/step - loss: 1.4103 - accuracy
: 0.6787 - val_loss: 11.4006 - val_accuracy: 0.1310 - lr: 3.0000e-04
```

Epoch 5/100
122/125 [=====>.] - ETA: 0s - loss: 0.9985 - accuracy: 0.8303
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.000150000071246177.
125/125 [=====] - 1s 12ms/step - loss: 0.9963 - accuracy: 0.8313 - val_loss: 5.8943 - val_accuracy: 0.0940 - lr: 3.0000e-04
Epoch 6/100
125/125 [=====] - 1s 11ms/step - loss: 0.7640 - accuracy: 0.9227 - val_loss: 5.9100 - val_accuracy: 0.2480 - lr: 1.5000e-04
Epoch 7/100
125/125 [=====] - 1s 11ms/step - loss: 0.6468 - accuracy: 0.9560 - val_loss: 6.8714 - val_accuracy: 0.2360 - lr: 1.5000e-04
Epoch 8/100
125/125 [=====] - 1s 11ms/step - loss: 0.5970 - accuracy: 0.9610 - val_loss: 8.8258 - val_accuracy: 0.2830 - lr: 1.5000e-04
Epoch 9/100
124/125 [=====>.] - ETA: 0s - loss: 0.5430 - accuracy: 0.9761
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
125/125 [=====] - 1s 12ms/step - loss: 0.5425 - accuracy: 0.9763 - val_loss: 5.4106 - val_accuracy: 0.2770 - lr: 1.5000e-04
Epoch 10/100
125/125 [=====] - 1s 10ms/step - loss: 0.5040 - accuracy: 0.9830 - val_loss: 1.4541 - val_accuracy: 0.6020 - lr: 7.5000e-05
Epoch 11/100
125/125 [=====] - 1s 9ms/step - loss: 0.4832 - accuracy: 0.9870 - val_loss: 2.1851 - val_accuracy: 0.6190 - lr: 7.5000e-05
Epoch 12/100
125/125 [=====] - 1s 9ms/step - loss: 0.4560 - accuracy: 0.9937 - val_loss: 1.7287 - val_accuracy: 0.6410 - lr: 7.5000e-05
Epoch 13/100
125/125 [=====] - 1s 10ms/step - loss: 0.4385 - accuracy: 0.9943 - val_loss: 2.6956 - val_accuracy: 0.4390 - lr: 7.5000e-05
Epoch 14/100
121/125 [=====>.] - ETA: 0s - loss: 0.4292 - accuracy: 0.9928
Epoch 14: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
125/125 [=====] - 1s 12ms/step - loss: 0.4293 - accuracy: 0.9927 - val_loss: 2.3769 - val_accuracy: 0.5390 - lr: 7.5000e-05
Epoch 15/100
125/125 [=====] - 1s 10ms/step - loss: 0.4130 - accuracy: 0.9943 - val_loss: 2.4255 - val_accuracy: 0.4610 - lr: 3.7500e-05
Epoch 16/100
125/125 [=====] - 1s 11ms/step - loss: 0.4051 - accuracy: 0.9937 - val_loss: 0.6954 - val_accuracy: 0.8800 - lr: 3.7500e-05
Epoch 17/100
125/125 [=====] - 1s 10ms/step - loss: 0.4009 - accuracy: 0.9953 - val_loss: 0.8998 - val_accuracy: 0.8020 - lr: 3.7500e-05
Epoch 18/100
125/125 [=====] - 1s 10ms/step - loss: 0.3885 - accuracy: 0.9977 - val_loss: 0.7258 - val_accuracy: 0.8540 - lr: 3.7500e-05
Epoch 19/100
125/125 [=====] - 1s 8ms/step - loss: 0.3859 - accuracy: 0.9973 - val_loss: 0.7198 - val_accuracy: 0.8670 - lr: 3.7500e-05
Epoch 20/100
125/125 [=====] - 1s 11ms/step - loss: 0.3718 - accuracy: 0.9987 - val_loss: 0.6599 - val_accuracy: 0.8980 - lr: 3.7500e-05
Epoch 21/100
125/125 [=====] - 1s 10ms/step - loss: 0.3697 - accuracy: 0.9977 - val_loss: 0.9667 - val_accuracy: 0.7580 - lr: 3.7500e-05

Epoch 22/100
125/125 [=====] - 1s 12ms/step - loss: 0.3629 - accuracy : 0.9970 - val_loss: 0.7562 - val_accuracy: 0.8540 - lr: 3.7500e-05

Epoch 23/100
125/125 [=====] - 1s 9ms/step - loss: 0.3563 - accuracy: 0.9980 - val_loss: 1.5508 - val_accuracy: 0.5050 - lr: 3.7500e-05

Epoch 24/100
125/125 [=====] - 1s 9ms/step - loss: 0.3496 - accuracy: 0.9977 - val_loss: 0.4919 - val_accuracy: 0.9410 - lr: 3.7500e-05

Epoch 25/100
125/125 [=====] - 1s 10ms/step - loss: 0.3437 - accuracy : 0.9987 - val_loss: 2.4141 - val_accuracy: 0.5890 - lr: 3.7500e-05

Epoch 26/100
125/125 [=====] - 1s 11ms/step - loss: 0.3389 - accuracy : 0.9977 - val_loss: 0.6704 - val_accuracy: 0.8830 - lr: 3.7500e-05

Epoch 27/100
125/125 [=====] - 1s 10ms/step - loss: 0.3309 - accuracy : 0.9980 - val_loss: 1.4757 - val_accuracy: 0.6760 - lr: 3.7500e-05

Epoch 28/100
124/125 [=====>.] - ETA: 0s - loss: 0.3211 - accuracy: 0.9993

Epoch 28: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
125/125 [=====] - 1s 11ms/step - loss: 0.3210 - accuracy : 0.9993 - val_loss: 1.5486 - val_accuracy: 0.7030 - lr: 3.7500e-05

Epoch 29/100
125/125 [=====] - 1s 10ms/step - loss: 0.3167 - accuracy : 0.9993 - val_loss: 0.5123 - val_accuracy: 0.9370 - lr: 1.8750e-05

Epoch 30/100
125/125 [=====] - 1s 12ms/step - loss: 0.3133 - accuracy : 0.9997 - val_loss: 0.5462 - val_accuracy: 0.9110 - lr: 1.8750e-05

Epoch 31/100
125/125 [=====] - 1s 11ms/step - loss: 0.3092 - accuracy : 0.9987 - val_loss: 0.3660 - val_accuracy: 0.9800 - lr: 1.8750e-05

Epoch 32/100
125/125 [=====] - 1s 12ms/step - loss: 0.3057 - accuracy : 0.9990 - val_loss: 0.7165 - val_accuracy: 0.8340 - lr: 1.8750e-05

Epoch 33/100
125/125 [=====] - 1s 11ms/step - loss: 0.3007 - accuracy : 0.9990 - val_loss: 0.4027 - val_accuracy: 0.9610 - lr: 1.8750e-05

Epoch 34/100
125/125 [=====] - 1s 11ms/step - loss: 0.2968 - accuracy : 0.9997 - val_loss: 0.3823 - val_accuracy: 0.9730 - lr: 1.8750e-05

Epoch 35/100
125/125 [=====] - ETA: 0s - loss: 0.2934 - accuracy: 0.9997

Epoch 35: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
125/125 [=====] - 1s 10ms/step - loss: 0.2934 - accuracy : 0.9997 - val_loss: 0.4346 - val_accuracy: 0.9590 - lr: 1.8750e-05

Epoch 36/100
125/125 [=====] - 1s 10ms/step - loss: 0.2918 - accuracy : 1.0000 - val_loss: 0.4097 - val_accuracy: 0.9640 - lr: 9.3750e-06

Epoch 37/100
125/125 [=====] - 1s 10ms/step - loss: 0.2892 - accuracy : 0.9990 - val_loss: 0.3222 - val_accuracy: 0.9880 - lr: 9.3750e-06

Epoch 38/100
125/125 [=====] - 1s 11ms/step - loss: 0.2867 - accuracy : 0.9997 - val_loss: 0.3411 - val_accuracy: 0.9780 - lr: 9.3750e-06

Epoch 39/100
125/125 [=====] - 1s 11ms/step - loss: 0.2862 - accuracy : 0.9997 - val_loss: 0.5152 - val_accuracy: 0.9230 - lr: 9.3750e-06

Epoch 40/100
125/125 [=====] - 1s 11ms/step - loss: 0.2834 - accuracy : 0.9993 - val_loss: 0.4164 - val_accuracy: 0.9500 - lr: 9.3750e-06
Epoch 41/100
124/125 [=====>.] - ETA: 0s - loss: 0.2799 - accuracy: 1.0000
Epoch 41: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
125/125 [=====] - 1s 11ms/step - loss: 0.2798 - accuracy : 1.0000 - val_loss: 0.3434 - val_accuracy: 0.9790 - lr: 9.3750e-06
Epoch 42/100
125/125 [=====] - 1s 11ms/step - loss: 0.2798 - accuracy : 0.9997 - val_loss: 0.3377 - val_accuracy: 0.9820 - lr: 4.6875e-06
Epoch 43/100
125/125 [=====] - 1s 10ms/step - loss: 0.2777 - accuracy : 0.9997 - val_loss: 0.3628 - val_accuracy: 0.9760 - lr: 4.6875e-06
Epoch 44/100
125/125 [=====] - 1s 12ms/step - loss: 0.2765 - accuracy : 1.0000 - val_loss: 0.3096 - val_accuracy: 0.9890 - lr: 4.6875e-06
Epoch 45/100
125/125 [=====] - 1s 10ms/step - loss: 0.2769 - accuracy : 0.9993 - val_loss: 0.3231 - val_accuracy: 0.9830 - lr: 4.6875e-06
Epoch 46/100
125/125 [=====] - 1s 12ms/step - loss: 0.2743 - accuracy : 0.9997 - val_loss: 0.3301 - val_accuracy: 0.9860 - lr: 4.6875e-06
Epoch 47/100
125/125 [=====] - 1s 10ms/step - loss: 0.2746 - accuracy : 0.9993 - val_loss: 0.3202 - val_accuracy: 0.9850 - lr: 4.6875e-06
Epoch 48/100
121/125 [=====>.] - ETA: 0s - loss: 0.2709 - accuracy: 1.0000
Epoch 48: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
125/125 [=====] - 1s 10ms/step - loss: 0.2710 - accuracy : 1.0000 - val_loss: 0.3271 - val_accuracy: 0.9830 - lr: 4.6875e-06
Epoch 49/100
125/125 [=====] - 1s 9ms/step - loss: 0.2711 - accuracy : 0.9993 - val_loss: 0.3050 - val_accuracy: 0.9840 - lr: 2.3438e-06
Epoch 50/100
125/125 [=====] - 1s 11ms/step - loss: 0.2700 - accuracy : 1.0000 - val_loss: 0.3016 - val_accuracy: 0.9850 - lr: 2.3438e-06
Epoch 51/100
125/125 [=====] - 1s 10ms/step - loss: 0.2697 - accuracy : 0.9997 - val_loss: 0.2991 - val_accuracy: 0.9870 - lr: 2.3438e-06
Epoch 52/100
125/125 [=====] - 1s 10ms/step - loss: 0.2691 - accuracy : 1.0000 - val_loss: 0.2984 - val_accuracy: 0.9890 - lr: 2.3438e-06
Epoch 53/100
125/125 [=====] - 1s 11ms/step - loss: 0.2670 - accuracy : 0.9997 - val_loss: 0.3077 - val_accuracy: 0.9830 - lr: 2.3438e-06
Epoch 54/100
125/125 [=====] - 1s 11ms/step - loss: 0.2687 - accuracy : 1.0000 - val_loss: 0.3082 - val_accuracy: 0.9830 - lr: 2.3438e-06
Epoch 55/100
125/125 [=====] - 2s 12ms/step - loss: 0.2659 - accuracy : 1.0000 - val_loss: 0.3028 - val_accuracy: 0.9820 - lr: 2.3438e-06
Epoch 56/100
125/125 [=====] - 1s 11ms/step - loss: 0.2674 - accuracy : 0.9983 - val_loss: 0.2956 - val_accuracy: 0.9880 - lr: 2.3438e-06
[100, 24, 0.30000000000000004]
32.5%
Epoch 1/100

117/117 [=====] - 3s 15ms/step - loss: 3.0012 - accuracy
: 0.1579 - val_loss: 3.8964 - val_accuracy: 0.0983 - lr: 3.0000e-04
Epoch 2/100
117/117 [=====] - 1s 10ms/step - loss: 2.4470 - accuracy
: 0.2714 - val_loss: 4.1621 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 3/100
117/117 [=====] - 1s 9ms/step - loss: 2.1115 - accuracy:
0.3829 - val_loss: 4.2616 - val_accuracy: 0.1883 - lr: 3.0000e-04
Epoch 4/100
117/117 [=====] - 1s 12ms/step - loss: 1.5600 - accuracy
: 0.6039 - val_loss: 5.8878 - val_accuracy: 0.0975 - lr: 3.0000e-04
Epoch 5/100
116/117 [=====>.] - ETA: 0s - loss: 1.0888 - accuracy: 0.7
834
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
117/117 [=====] - 1s 12ms/step - loss: 1.0885 - accuracy
: 0.7832 - val_loss: 6.6533 - val_accuracy: 0.1308 - lr: 3.0000e-04
Epoch 6/100
117/117 [=====] - 1s 13ms/step - loss: 0.8019 - accuracy
: 0.8914 - val_loss: 3.2646 - val_accuracy: 0.2617 - lr: 1.5000e-04
Epoch 7/100
117/117 [=====] - 2s 13ms/step - loss: 0.6860 - accuracy
: 0.9325 - val_loss: 14.5203 - val_accuracy: 0.0992 - lr: 1.5000e-04
Epoch 8/100
117/117 [=====] - 1s 11ms/step - loss: 0.5899 - accuracy
: 0.9636 - val_loss: 8.7366 - val_accuracy: 0.2600 - lr: 1.5000e-04
Epoch 9/100
117/117 [=====] - 1s 12ms/step - loss: 0.5202 - accuracy
: 0.9764 - val_loss: 3.4949 - val_accuracy: 0.4942 - lr: 1.5000e-04
Epoch 10/100
114/117 [=====>.] - ETA: 0s - loss: 0.4822 - accuracy: 0.9
817
Epoch 10: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
117/117 [=====] - 2s 14ms/step - loss: 0.4809 - accuracy
: 0.9818 - val_loss: 6.6179 - val_accuracy: 0.2967 - lr: 1.5000e-04
Epoch 11/100
117/117 [=====] - 2s 14ms/step - loss: 0.4366 - accuracy
: 0.9925 - val_loss: 1.6364 - val_accuracy: 0.6675 - lr: 7.5000e-05
Epoch 12/100
117/117 [=====] - 2s 13ms/step - loss: 0.4163 - accuracy
: 0.9932 - val_loss: 1.9540 - val_accuracy: 0.4433 - lr: 7.5000e-05
Epoch 13/100
117/117 [=====] - 1s 12ms/step - loss: 0.3959 - accuracy
: 0.9975 - val_loss: 1.9811 - val_accuracy: 0.5800 - lr: 7.5000e-05
Epoch 14/100
117/117 [=====] - 1s 9ms/step - loss: 0.3825 - accuracy:
0.9968 - val_loss: 6.3567 - val_accuracy: 0.1908 - lr: 7.5000e-05
Epoch 15/100
112/117 [=====>..] - ETA: 0s - loss: 0.3695 - accuracy: 0.9
952
Epoch 15: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
117/117 [=====] - 1s 11ms/step - loss: 0.3693 - accuracy
: 0.9954 - val_loss: 3.6859 - val_accuracy: 0.4033 - lr: 7.5000e-05
Epoch 16/100
117/117 [=====] - 1s 13ms/step - loss: 0.3617 - accuracy
: 0.9971 - val_loss: 1.7988 - val_accuracy: 0.5325 - lr: 3.7500e-05
Epoch 17/100
117/117 [=====] - 1s 13ms/step - loss: 0.3514 - accuracy
: 0.9975 - val_loss: 0.4370 - val_accuracy: 0.9725 - lr: 3.7500e-05
Epoch 18/100

117/117 [=====] - 2s 13ms/step - loss: 0.3472 - accuracy
: 0.9982 - val_loss: 3.0969 - val_accuracy: 0.3708 - lr: 3.7500e-05
Epoch 19/100
117/117 [=====] - 1s 12ms/step - loss: 0.3405 - accuracy
: 0.9982 - val_loss: 0.5943 - val_accuracy: 0.8917 - lr: 3.7500e-05
Epoch 20/100
117/117 [=====] - 1s 12ms/step - loss: 0.3317 - accuracy
: 0.9989 - val_loss: 1.3506 - val_accuracy: 0.6375 - lr: 3.7500e-05
Epoch 21/100
114/117 [=====>.] - ETA: 0s - loss: 0.3288 - accuracy: 0.9
996
Epoch 21: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
117/117 [=====] - 2s 13ms/step - loss: 0.3288 - accuracy
: 0.9996 - val_loss: 1.0593 - val_accuracy: 0.7092 - lr: 3.7500e-05
Epoch 22/100
117/117 [=====] - 1s 12ms/step - loss: 0.3258 - accuracy
: 0.9982 - val_loss: 0.4217 - val_accuracy: 0.9725 - lr: 1.8750e-05
Epoch 23/100
117/117 [=====] - 1s 10ms/step - loss: 0.3178 - accuracy
: 0.9993 - val_loss: 0.6216 - val_accuracy: 0.8933 - lr: 1.8750e-05
Epoch 24/100
117/117 [=====] - 1s 12ms/step - loss: 0.3160 - accuracy
: 0.9986 - val_loss: 1.8301 - val_accuracy: 0.6633 - lr: 1.8750e-05
Epoch 25/100
117/117 [=====] - 1s 10ms/step - loss: 0.3135 - accuracy
: 0.9993 - val_loss: 0.3556 - val_accuracy: 0.9850 - lr: 1.8750e-05
Epoch 26/100
117/117 [=====] - 1s 10ms/step - loss: 0.3094 - accuracy
: 0.9993 - val_loss: 0.7400 - val_accuracy: 0.8292 - lr: 1.8750e-05
Epoch 27/100
117/117 [=====] - 1s 11ms/step - loss: 0.3069 - accuracy
: 0.9986 - val_loss: 0.5407 - val_accuracy: 0.9175 - lr: 1.8750e-05
Epoch 28/100
117/117 [=====] - 2s 14ms/step - loss: 0.3025 - accuracy
: 0.9996 - val_loss: 0.4786 - val_accuracy: 0.9400 - lr: 1.8750e-05
Epoch 29/100
111/117 [=====>..] - ETA: 0s - loss: 0.2994 - accuracy: 0.9
989
Epoch 29: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
117/117 [=====] - 2s 14ms/step - loss: 0.2995 - accuracy
: 0.9989 - val_loss: 0.5845 - val_accuracy: 0.8958 - lr: 1.8750e-05
Epoch 30/100
117/117 [=====] - 1s 12ms/step - loss: 0.2965 - accuracy
: 0.9996 - val_loss: 0.4014 - val_accuracy: 0.9642 - lr: 9.3750e-06
Epoch 31/100
117/117 [=====] - 1s 12ms/step - loss: 0.2962 - accuracy
: 0.9996 - val_loss: 0.3800 - val_accuracy: 0.9750 - lr: 9.3750e-06
Epoch 32/100
117/117 [=====] - 1s 12ms/step - loss: 0.2922 - accuracy
: 0.9996 - val_loss: 0.4095 - val_accuracy: 0.9667 - lr: 9.3750e-06
Epoch 33/100
117/117 [=====] - 2s 13ms/step - loss: 0.2911 - accuracy
: 0.9989 - val_loss: 0.3416 - val_accuracy: 0.9858 - lr: 9.3750e-06
Epoch 34/100
117/117 [=====] - 1s 13ms/step - loss: 0.2910 - accuracy
: 0.9993 - val_loss: 0.4274 - val_accuracy: 0.9533 - lr: 9.3750e-06
Epoch 35/100
117/117 [=====] - 1s 12ms/step - loss: 0.2853 - accuracy
: 1.0000 - val_loss: 0.3236 - val_accuracy: 0.9892 - lr: 9.3750e-06
Epoch 36/100

117/117 [=====] - 1s 11ms/step - loss: 0.2830 - accuracy : 1.0000 - val_loss: 0.4500 - val_accuracy: 0.9383 - lr: 9.3750e-06
Epoch 37/100
117/117 [=====] - 1s 10ms/step - loss: 0.2848 - accuracy : 0.9989 - val_loss: 0.3319 - val_accuracy: 0.9858 - lr: 9.3750e-06
Epoch 38/100
117/117 [=====] - 1s 12ms/step - loss: 0.2809 - accuracy : 1.0000 - val_loss: 0.3297 - val_accuracy: 0.9900 - lr: 9.3750e-06
Epoch 39/100
117/117 [=====] - ETA: 0s - loss: 0.2766 - accuracy: 0.9996
Epoch 39: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
117/117 [=====] - 1s 12ms/step - loss: 0.2766 - accuracy : 0.9996 - val_loss: 0.3376 - val_accuracy: 0.9750 - lr: 9.3750e-06
Epoch 40/100
117/117 [=====] - 2s 13ms/step - loss: 0.2775 - accuracy : 0.9993 - val_loss: 0.3137 - val_accuracy: 0.9883 - lr: 4.6875e-06
Epoch 41/100
117/117 [=====] - 1s 13ms/step - loss: 0.2751 - accuracy : 1.0000 - val_loss: 0.3083 - val_accuracy: 0.9917 - lr: 4.6875e-06
Epoch 42/100
117/117 [=====] - 1s 12ms/step - loss: 0.2734 - accuracy : 0.9996 - val_loss: 0.3095 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 43/100
117/117 [=====] - 2s 14ms/step - loss: 0.2742 - accuracy : 0.9996 - val_loss: 0.3151 - val_accuracy: 0.9867 - lr: 4.6875e-06
Epoch 44/100
117/117 [=====] - 1s 12ms/step - loss: 0.2696 - accuracy : 1.0000 - val_loss: 0.3027 - val_accuracy: 0.9917 - lr: 4.6875e-06
Epoch 45/100
117/117 [=====] - 1s 12ms/step - loss: 0.2703 - accuracy : 1.0000 - val_loss: 0.3170 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 46/100
117/117 [=====] - 2s 13ms/step - loss: 0.2685 - accuracy : 1.0000 - val_loss: 0.3111 - val_accuracy: 0.9883 - lr: 4.6875e-06
Epoch 47/100
117/117 [=====] - 1s 10ms/step - loss: 0.2691 - accuracy : 0.9996 - val_loss: 0.3038 - val_accuracy: 0.9892 - lr: 4.6875e-06
Epoch 48/100
117/117 [=====] - 1s 11ms/step - loss: 0.2651 - accuracy : 1.0000 - val_loss: 0.2953 - val_accuracy: 0.9908 - lr: 4.6875e-06
Epoch 49/100
117/117 [=====] - 1s 10ms/step - loss: 0.2651 - accuracy : 1.0000 - val_loss: 0.3682 - val_accuracy: 0.9700 - lr: 4.6875e-06
Epoch 50/100
117/117 [=====] - 2s 13ms/step - loss: 0.2628 - accuracy : 1.0000 - val_loss: 0.2932 - val_accuracy: 0.9892 - lr: 4.6875e-06
Epoch 51/100
117/117 [=====] - 2s 13ms/step - loss: 0.2646 - accuracy : 0.9989 - val_loss: 0.3189 - val_accuracy: 0.9850 - lr: 4.6875e-06
Epoch 52/100
117/117 [=====] - 1s 12ms/step - loss: 0.2633 - accuracy : 0.9993 - val_loss: 0.2872 - val_accuracy: 0.9917 - lr: 4.6875e-06
Epoch 53/100
117/117 [=====] - 2s 14ms/step - loss: 0.2607 - accuracy : 0.9996 - val_loss: 0.2884 - val_accuracy: 0.9925 - lr: 4.6875e-06
Epoch 54/100
117/117 [=====] - 2s 13ms/step - loss: 0.2595 - accuracy : 1.0000 - val_loss: 0.3796 - val_accuracy: 0.9625 - lr: 4.6875e-06
Epoch 55/100

117/117 [=====] - 1s 12ms/step - loss: 0.2583 - accuracy
: 0.9996 - val_loss: 0.2920 - val_accuracy: 0.9858 - lr: 4.6875e-06
Epoch 56/100
113/117 [=====>..] - ETA: 0s - loss: 0.2568 - accuracy: 1.0
000
Epoch 56: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
117/117 [=====] - 2s 14ms/step - loss: 0.2570 - accuracy
: 1.0000 - val_loss: 0.2963 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 57/100
117/117 [=====] - 1s 12ms/step - loss: 0.2545 - accuracy
: 1.0000 - val_loss: 0.2797 - val_accuracy: 0.9933 - lr: 2.3438e-06
Epoch 58/100
117/117 [=====] - 1s 12ms/step - loss: 0.2544 - accuracy
: 1.0000 - val_loss: 0.2788 - val_accuracy: 0.9908 - lr: 2.3438e-06
Epoch 59/100
117/117 [=====] - 1s 11ms/step - loss: 0.2548 - accuracy
: 0.9996 - val_loss: 0.2837 - val_accuracy: 0.9917 - lr: 2.3438e-06
Epoch 60/100
117/117 [=====] - 1s 10ms/step - loss: 0.2528 - accuracy
: 1.0000 - val_loss: 0.2824 - val_accuracy: 0.9883 - lr: 2.3438e-06
Epoch 61/100
117/117 [=====] - 1s 12ms/step - loss: 0.2513 - accuracy
: 1.0000 - val_loss: 0.2769 - val_accuracy: 0.9942 - lr: 2.3438e-06
Epoch 62/100
117/117 [=====] - 1s 12ms/step - loss: 0.2506 - accuracy
: 0.9996 - val_loss: 0.2772 - val_accuracy: 0.9950 - lr: 2.3438e-06
Epoch 63/100
117/117 [=====] - 2s 14ms/step - loss: 0.2504 - accuracy
: 1.0000 - val_loss: 0.2848 - val_accuracy: 0.9892 - lr: 2.3438e-06
Epoch 64/100
117/117 [=====] - 2s 13ms/step - loss: 0.2495 - accuracy
: 1.0000 - val_loss: 0.2732 - val_accuracy: 0.9933 - lr: 2.3438e-06
Epoch 65/100
117/117 [=====] - 1s 12ms/step - loss: 0.2499 - accuracy
: 0.9993 - val_loss: 0.2769 - val_accuracy: 0.9883 - lr: 2.3438e-06
Epoch 66/100
117/117 [=====] - 1s 11ms/step - loss: 0.2492 - accuracy
: 1.0000 - val_loss: 0.2729 - val_accuracy: 0.9942 - lr: 2.3438e-06
Epoch 67/100
117/117 [=====] - 2s 14ms/step - loss: 0.2480 - accuracy
: 1.0000 - val_loss: 0.2830 - val_accuracy: 0.9867 - lr: 2.3438e-06
Epoch 68/100
117/117 [=====] - 2s 14ms/step - loss: 0.2474 - accuracy
: 1.0000 - val_loss: 0.2725 - val_accuracy: 0.9900 - lr: 2.3438e-06
Epoch 69/100
117/117 [=====] - 1s 13ms/step - loss: 0.2475 - accuracy
: 1.0000 - val_loss: 0.2746 - val_accuracy: 0.9900 - lr: 2.3438e-06
Epoch 70/100
117/117 [=====] - 1s 9ms/step - loss: 0.2463 - accuracy
: 1.0000 - val_loss: 0.2675 - val_accuracy: 0.9942 - lr: 2.3438e-06
Epoch 71/100
117/117 [=====] - 1s 11ms/step - loss: 0.2428 - accuracy
: 1.0000 - val_loss: 0.2696 - val_accuracy: 0.9908 - lr: 2.3438e-06
Epoch 72/100
117/117 [=====] - 1s 12ms/step - loss: 0.2447 - accuracy
: 1.0000 - val_loss: 0.2701 - val_accuracy: 0.9892 - lr: 2.3438e-06
Epoch 73/100
117/117 [=====] - 1s 13ms/step - loss: 0.2426 - accuracy
: 1.0000 - val_loss: 0.2645 - val_accuracy: 0.9942 - lr: 2.3438e-06
Epoch 74/100

117/117 [=====] - 1s 11ms/step - loss: 0.2413 - accuracy
: 0.9996 - val_loss: 0.2696 - val_accuracy: 0.9917 - lr: 2.3438e-06
[100, 24, 0.35000000000000003]
35.0%
Epoch 1/100
109/109 [=====] - 2s 17ms/step - loss: 3.1045 - accuracy
: 0.1516 - val_loss: 3.7429 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 2/100
109/109 [=====] - 1s 14ms/step - loss: 2.5301 - accuracy
: 0.2720 - val_loss: 4.9942 - val_accuracy: 0.1135 - lr: 3.0000e-04
Epoch 3/100
109/109 [=====] - 1s 13ms/step - loss: 2.0003 - accuracy
: 0.4544 - val_loss: 7.7681 - val_accuracy: 0.0914 - lr: 3.0000e-04
Epoch 4/100
109/109 [=====] - 2s 15ms/step - loss: 1.5641 - accuracy
: 0.6033 - val_loss: 8.1822 - val_accuracy: 0.0928 - lr: 3.0000e-04
Epoch 5/100
106/109 [=====>.] - ETA: 0s - loss: 1.2125 - accuracy: 0.7
445
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
109/109 [=====] - 2s 14ms/step - loss: 1.2120 - accuracy
: 0.7430 - val_loss: 6.3641 - val_accuracy: 0.1028 - lr: 3.0000e-04
Epoch 6/100
109/109 [=====] - 1s 11ms/step - loss: 0.9517 - accuracy
: 0.8465 - val_loss: 11.4225 - val_accuracy: 0.0914 - lr: 1.5000e-04
Epoch 7/100
109/109 [=====] - 1s 9ms/step - loss: 0.8559 - accuracy:
0.8784 - val_loss: 3.4591 - val_accuracy: 0.1770 - lr: 1.5000e-04
Epoch 8/100
109/109 [=====] - 1s 13ms/step - loss: 0.7380 - accuracy
: 0.9219 - val_loss: 4.5363 - val_accuracy: 0.2848 - lr: 1.5000e-04
Epoch 9/100
109/109 [=====] - 1s 13ms/step - loss: 0.6725 - accuracy
: 0.9411 - val_loss: 3.2596 - val_accuracy: 0.2827 - lr: 1.5000e-04
Epoch 10/100
109/109 [=====] - 1s 12ms/step - loss: 0.5998 - accuracy
: 0.9615 - val_loss: 5.8992 - val_accuracy: 0.1385 - lr: 1.5000e-04
Epoch 11/100
109/109 [=====] - 2s 14ms/step - loss: 0.5440 - accuracy
: 0.9750 - val_loss: 3.8182 - val_accuracy: 0.2548 - lr: 1.5000e-04
Epoch 12/100
109/109 [=====] - 1s 14ms/step - loss: 0.5028 - accuracy
: 0.9842 - val_loss: 9.6817 - val_accuracy: 0.1991 - lr: 1.5000e-04
Epoch 13/100
106/109 [=====>.] - ETA: 0s - loss: 0.4716 - accuracy: 0.9
843
Epoch 13: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
109/109 [=====] - 1s 13ms/step - loss: 0.4723 - accuracy
: 0.9831 - val_loss: 3.4762 - val_accuracy: 0.5525 - lr: 1.5000e-04
Epoch 14/100
109/109 [=====] - 2s 14ms/step - loss: 0.4344 - accuracy
: 0.9935 - val_loss: 0.7483 - val_accuracy: 0.8473 - lr: 7.5000e-05
Epoch 15/100
109/109 [=====] - 2s 14ms/step - loss: 0.4152 - accuracy
: 0.9927 - val_loss: 1.1608 - val_accuracy: 0.7359 - lr: 7.5000e-05
Epoch 16/100
109/109 [=====] - 1s 13ms/step - loss: 0.4094 - accuracy
: 0.9942 - val_loss: 1.3627 - val_accuracy: 0.6974 - lr: 7.5000e-05
Epoch 17/100
109/109 [=====] - 1s 13ms/step - loss: 0.3923 - accuracy

: 0.9965 - val_loss: 1.0021 - val_accuracy: 0.7723 - lr: 7.5000e-05
Epoch 18/100
107/109 [=====>.] - ETA: 0s - loss: 0.3823 - accuracy: 0.9961
Epoch 18: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
109/109 [=====] - 1s 10ms/step - loss: 0.3822 - accuracy: 0.9962 - val_loss: 4.0507 - val_accuracy: 0.1256 - lr: 7.5000e-05
Epoch 19/100
109/109 [=====] - 1s 13ms/step - loss: 0.3749 - accuracy: 0.9969 - val_loss: 0.6620 - val_accuracy: 0.8722 - lr: 3.7500e-05
Epoch 20/100
109/109 [=====] - 1s 11ms/step - loss: 0.3659 - accuracy: 0.9981 - val_loss: 1.5836 - val_accuracy: 0.6353 - lr: 3.7500e-05
Epoch 21/100
109/109 [=====] - 2s 15ms/step - loss: 0.3572 - accuracy: 0.9996 - val_loss: 0.5420 - val_accuracy: 0.9243 - lr: 3.7500e-05
Epoch 22/100
109/109 [=====] - 2s 15ms/step - loss: 0.3495 - accuracy: 0.9996 - val_loss: 0.4779 - val_accuracy: 0.9507 - lr: 3.7500e-05
Epoch 23/100
109/109 [=====] - 2s 14ms/step - loss: 0.3462 - accuracy: 0.9985 - val_loss: 0.4304 - val_accuracy: 0.9729 - lr: 3.7500e-05
Epoch 24/100
109/109 [=====] - 1s 13ms/step - loss: 0.3445 - accuracy: 0.9996 - val_loss: 0.6645 - val_accuracy: 0.8565 - lr: 3.7500e-05
Epoch 25/100
109/109 [=====] - 2s 14ms/step - loss: 0.3355 - accuracy: 1.0000 - val_loss: 1.5463 - val_accuracy: 0.6310 - lr: 3.7500e-05
Epoch 26/100
109/109 [=====] - 2s 14ms/step - loss: 0.3355 - accuracy: 0.9977 - val_loss: 0.8820 - val_accuracy: 0.7816 - lr: 3.7500e-05
Epoch 27/100
106/109 [=====>.] - ETA: 0s - loss: 0.3285 - accuracy: 0.9972
Epoch 27: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
109/109 [=====] - 1s 13ms/step - loss: 0.3283 - accuracy: 0.9973 - val_loss: 0.4804 - val_accuracy: 0.9458 - lr: 3.7500e-05
Epoch 28/100
109/109 [=====] - 1s 11ms/step - loss: 0.3219 - accuracy: 0.9992 - val_loss: 0.6599 - val_accuracy: 0.8701 - lr: 1.8750e-05
Epoch 29/100
109/109 [=====] - 1s 10ms/step - loss: 0.3205 - accuracy: 0.9996 - val_loss: 0.4213 - val_accuracy: 0.9636 - lr: 1.8750e-05
Epoch 30/100
109/109 [=====] - 1s 12ms/step - loss: 0.3157 - accuracy: 0.9981 - val_loss: 0.3934 - val_accuracy: 0.9707 - lr: 1.8750e-05
Epoch 31/100
109/109 [=====] - 1s 13ms/step - loss: 0.3139 - accuracy: 0.9988 - val_loss: 0.4502 - val_accuracy: 0.9486 - lr: 1.8750e-05
Epoch 32/100
109/109 [=====] - 2s 14ms/step - loss: 0.3100 - accuracy: 0.9996 - val_loss: 0.7898 - val_accuracy: 0.8230 - lr: 1.8750e-05
Epoch 33/100
109/109 [=====] - 1s 13ms/step - loss: 0.3072 - accuracy: 0.9996 - val_loss: 0.4632 - val_accuracy: 0.9400 - lr: 1.8750e-05
Epoch 34/100
106/109 [=====>.] - ETA: 0s - loss: 0.3046 - accuracy: 1.0000
Epoch 34: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
109/109 [=====] - 2s 14ms/step - loss: 0.3045 - accuracy:

: 1.0000 - val_loss: 0.7774 - val_accuracy: 0.8337 - lr: 1.8750e-05
Epoch 35/100
109/109 [=====] - 1s 12ms/step - loss: 0.3048 - accuracy
: 0.9988 - val_loss: 0.4442 - val_accuracy: 0.9550 - lr: 9.3750e-06
[100, 24, 0.4]
37.5%
Epoch 1/100
100/100 [=====] - 2s 17ms/step - loss: 3.1006 - accuracy
: 0.1558 - val_loss: 4.4991 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 2/100
100/100 [=====] - 1s 12ms/step - loss: 2.5432 - accuracy
: 0.2821 - val_loss: 5.7383 - val_accuracy: 0.1025 - lr: 3.0000e-04
Epoch 3/100
100/100 [=====] - 1s 13ms/step - loss: 2.1148 - accuracy
: 0.4179 - val_loss: 8.4744 - val_accuracy: 0.1025 - lr: 3.0000e-04
Epoch 4/100
100/100 [=====] - 1s 11ms/step - loss: 1.6734 - accuracy
: 0.5683 - val_loss: 8.4863 - val_accuracy: 0.1363 - lr: 3.0000e-04
Epoch 5/100
93/100 [=====>...] - ETA: 0s - loss: 1.2793 - accuracy: 0.7
285
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
100/100 [=====] - 1s 11ms/step - loss: 1.2716 - accuracy
: 0.7300 - val_loss: 12.3017 - val_accuracy: 0.1231 - lr: 3.0000e-04
Epoch 6/100
100/100 [=====] - 1s 11ms/step - loss: 0.9817 - accuracy
: 0.8429 - val_loss: 9.1114 - val_accuracy: 0.2587 - lr: 1.5000e-04
Epoch 7/100
100/100 [=====] - 1s 12ms/step - loss: 0.8263 - accuracy
: 0.9054 - val_loss: 9.7614 - val_accuracy: 0.1656 - lr: 1.5000e-04
Epoch 8/100
100/100 [=====] - 1s 13ms/step - loss: 0.7310 - accuracy
: 0.9342 - val_loss: 9.1097 - val_accuracy: 0.3413 - lr: 1.5000e-04
Epoch 9/100
100/100 [=====] - 1s 12ms/step - loss: 0.6456 - accuracy
: 0.9583 - val_loss: 2.3412 - val_accuracy: 0.4556 - lr: 1.5000e-04
Epoch 10/100
100/100 [=====] - 1s 13ms/step - loss: 0.5926 - accuracy
: 0.9696 - val_loss: 4.1354 - val_accuracy: 0.2231 - lr: 1.5000e-04
Epoch 11/100
100/100 [=====] - 1s 12ms/step - loss: 0.5522 - accuracy
: 0.9758 - val_loss: 5.0383 - val_accuracy: 0.3831 - lr: 1.5000e-04
Epoch 12/100
100/100 [=====] - 1s 13ms/step - loss: 0.5127 - accuracy
: 0.9879 - val_loss: 4.6364 - val_accuracy: 0.3450 - lr: 1.5000e-04
Epoch 13/100
99/100 [=====>.] - ETA: 0s - loss: 0.4816 - accuracy: 0.9
882
Epoch 13: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
100/100 [=====] - 1s 13ms/step - loss: 0.4813 - accuracy
: 0.9883 - val_loss: 5.4562 - val_accuracy: 0.2419 - lr: 1.5000e-04
Epoch 14/100
100/100 [=====] - 1s 14ms/step - loss: 0.4541 - accuracy
: 0.9946 - val_loss: 1.8263 - val_accuracy: 0.5019 - lr: 7.5000e-05
Epoch 15/100
100/100 [=====] - 1s 11ms/step - loss: 0.4396 - accuracy
: 0.9946 - val_loss: 1.1909 - val_accuracy: 0.7462 - lr: 7.5000e-05
Epoch 16/100
100/100 [=====] - 1s 14ms/step - loss: 0.4248 - accuracy
: 0.9979 - val_loss: 1.1888 - val_accuracy: 0.7106 - lr: 7.5000e-05

Epoch 17/100
100/100 [=====] - 1s 10ms/step - loss: 0.4161 - accuracy : 0.9971 - val_loss: 0.7125 - val_accuracy: 0.8631 - lr: 7.5000e-05

Epoch 18/100
100/100 [=====] - 1s 11ms/step - loss: 0.4087 - accuracy : 0.9954 - val_loss: 4.3289 - val_accuracy: 0.4563 - lr: 7.5000e-05

Epoch 19/100
100/100 [=====] - 1s 10ms/step - loss: 0.3966 - accuracy : 0.9979 - val_loss: 0.5904 - val_accuracy: 0.9262 - lr: 7.5000e-05

Epoch 20/100
100/100 [=====] - 1s 13ms/step - loss: 0.3825 - accuracy : 0.9979 - val_loss: 1.6162 - val_accuracy: 0.4744 - lr: 7.5000e-05

Epoch 21/100
100/100 [=====] - 1s 13ms/step - loss: 0.3771 - accuracy : 0.9996 - val_loss: 1.0476 - val_accuracy: 0.7275 - lr: 7.5000e-05

Epoch 22/100
100/100 [=====] - 1s 12ms/step - loss: 0.3681 - accuracy : 0.9979 - val_loss: 2.0285 - val_accuracy: 0.5425 - lr: 7.5000e-05

Epoch 23/100
97/100 [=====>.] - ETA: 0s - loss: 0.3578 - accuracy: 0.9991

Epoch 23: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
100/100 [=====] - 1s 14ms/step - loss: 0.3583 - accuracy : 0.9992 - val_loss: 1.2831 - val_accuracy: 0.6794 - lr: 7.5000e-05

Epoch 24/100
100/100 [=====] - 1s 13ms/step - loss: 0.3511 - accuracy : 0.9975 - val_loss: 2.0210 - val_accuracy: 0.5925 - lr: 3.7500e-05

Epoch 25/100
100/100 [=====] - 1s 14ms/step - loss: 0.3420 - accuracy : 0.9987 - val_loss: 0.6664 - val_accuracy: 0.8531 - lr: 3.7500e-05

Epoch 26/100
100/100 [=====] - 1s 12ms/step - loss: 0.3402 - accuracy : 1.0000 - val_loss: 0.9386 - val_accuracy: 0.7700 - lr: 3.7500e-05

Epoch 27/100
97/100 [=====>.] - ETA: 0s - loss: 0.3364 - accuracy: 0.9983

Epoch 27: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
100/100 [=====] - 1s 14ms/step - loss: 0.3360 - accuracy : 0.9983 - val_loss: 0.8558 - val_accuracy: 0.7906 - lr: 3.7500e-05

Epoch 28/100
100/100 [=====] - 1s 12ms/step - loss: 0.3294 - accuracy : 0.9992 - val_loss: 0.3809 - val_accuracy: 0.9775 - lr: 1.8750e-05

Epoch 29/100
100/100 [=====] - 1s 14ms/step - loss: 0.3313 - accuracy : 1.0000 - val_loss: 0.3859 - val_accuracy: 0.9775 - lr: 1.8750e-05

Epoch 30/100
100/100 [=====] - 1s 11ms/step - loss: 0.3263 - accuracy : 0.9992 - val_loss: 0.5694 - val_accuracy: 0.9019 - lr: 1.8750e-05

Epoch 31/100
100/100 [=====] - 1s 11ms/step - loss: 0.3239 - accuracy : 0.9983 - val_loss: 0.3567 - val_accuracy: 0.9887 - lr: 1.8750e-05

Epoch 32/100
100/100 [=====] - 1s 11ms/step - loss: 0.3193 - accuracy : 1.0000 - val_loss: 0.3946 - val_accuracy: 0.9756 - lr: 1.8750e-05

Epoch 33/100
100/100 [=====] - 1s 12ms/step - loss: 0.3146 - accuracy : 0.9996 - val_loss: 0.7156 - val_accuracy: 0.8400 - lr: 1.8750e-05

Epoch 34/100
100/100 [=====] - 1s 12ms/step - loss: 0.3140 - accuracy : 1.0000 - val_loss: 0.6570 - val_accuracy: 0.8631 - lr: 1.8750e-05

Epoch 35/100
96/100 [=====>..] - ETA: 0s - loss: 0.3122 - accuracy: 0.9996
Epoch 35: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
100/100 [=====] - 1s 13ms/step - loss: 0.3118 - accuracy: 0.9996 - val_loss: 0.3908 - val_accuracy: 0.9744 - lr: 1.8750e-05
Epoch 36/100
100/100 [=====] - 1s 13ms/step - loss: 0.3090 - accuracy: 0.9996 - val_loss: 0.3428 - val_accuracy: 0.9875 - lr: 9.3750e-06
Epoch 37/100
100/100 [=====] - 1s 12ms/step - loss: 0.3079 - accuracy: 0.9996 - val_loss: 0.3838 - val_accuracy: 0.9787 - lr: 9.3750e-06
Epoch 38/100
100/100 [=====] - 1s 14ms/step - loss: 0.3074 - accuracy: 1.0000 - val_loss: 0.3316 - val_accuracy: 0.9925 - lr: 9.3750e-06
Epoch 39/100
100/100 [=====] - 1s 12ms/step - loss: 0.3061 - accuracy: 0.9996 - val_loss: 0.3337 - val_accuracy: 0.9900 - lr: 9.3750e-06
Epoch 40/100
100/100 [=====] - 1s 14ms/step - loss: 0.3008 - accuracy: 1.0000 - val_loss: 0.3594 - val_accuracy: 0.9837 - lr: 9.3750e-06
Epoch 41/100
100/100 [=====] - 1s 12ms/step - loss: 0.3030 - accuracy: 0.9996 - val_loss: 0.3530 - val_accuracy: 0.9844 - lr: 9.3750e-06
Epoch 42/100
99/100 [=====>..] - ETA: 0s - loss: 0.2999 - accuracy: 0.9996
Epoch 42: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
100/100 [=====] - 1s 12ms/step - loss: 0.2999 - accuracy: 0.9996 - val_loss: 0.4264 - val_accuracy: 0.9525 - lr: 9.3750e-06
Epoch 43/100
100/100 [=====] - 1s 11ms/step - loss: 0.2970 - accuracy: 0.9996 - val_loss: 0.3288 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 44/100
100/100 [=====] - 1s 12ms/step - loss: 0.2993 - accuracy: 0.9987 - val_loss: 0.3926 - val_accuracy: 0.9719 - lr: 4.6875e-06
Epoch 45/100
100/100 [=====] - 1s 11ms/step - loss: 0.2972 - accuracy: 0.9992 - val_loss: 0.3196 - val_accuracy: 0.9912 - lr: 4.6875e-06
Epoch 46/100
100/100 [=====] - 1s 12ms/step - loss: 0.2948 - accuracy: 1.0000 - val_loss: 0.3249 - val_accuracy: 0.9931 - lr: 4.6875e-06
Epoch 47/100
100/100 [=====] - 1s 13ms/step - loss: 0.2926 - accuracy: 1.0000 - val_loss: 0.3613 - val_accuracy: 0.9762 - lr: 4.6875e-06
Epoch 48/100
100/100 [=====] - 1s 13ms/step - loss: 0.2934 - accuracy: 1.0000 - val_loss: 0.3218 - val_accuracy: 0.9931 - lr: 4.6875e-06
Epoch 49/100
100/100 [=====] - 1s 13ms/step - loss: 0.2921 - accuracy: 1.0000 - val_loss: 0.3174 - val_accuracy: 0.9906 - lr: 4.6875e-06
Epoch 50/100
100/100 [=====] - 1s 12ms/step - loss: 0.2897 - accuracy: 0.9996 - val_loss: 0.3145 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 51/100
100/100 [=====] - 1s 13ms/step - loss: 0.2902 - accuracy: 1.0000 - val_loss: 0.3760 - val_accuracy: 0.9725 - lr: 4.6875e-06
Epoch 52/100
100/100 [=====] - 1s 12ms/step - loss: 0.2879 - accuracy: 1.0000 - val_loss: 0.3210 - val_accuracy: 0.9894 - lr: 4.6875e-06

Epoch 53/100
100/100 [=====] - 1s 14ms/step - loss: 0.2886 - accuracy : 1.0000 - val_loss: 0.3146 - val_accuracy: 0.9919 - lr: 4.6875e-06
Epoch 54/100
96/100 [=====>..] - ETA: 0s - loss: 0.2857 - accuracy: 0.9996
Epoch 54: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
100/100 [=====] - 1s 12ms/step - loss: 0.2856 - accuracy : 0.9996 - val_loss: 0.3312 - val_accuracy: 0.9862 - lr: 4.6875e-06
Epoch 55/100
100/100 [=====] - 1s 14ms/step - loss: 0.2844 - accuracy : 1.0000 - val_loss: 0.3393 - val_accuracy: 0.9831 - lr: 2.3438e-06
Epoch 56/100
100/100 [=====] - 1s 10ms/step - loss: 0.2857 - accuracy : 0.9996 - val_loss: 0.3299 - val_accuracy: 0.9869 - lr: 2.3438e-06
Epoch 57/100
100/100 [=====] - 1s 10ms/step - loss: 0.2878 - accuracy : 1.0000 - val_loss: 0.3077 - val_accuracy: 0.9925 - lr: 2.3438e-06
Epoch 58/100
100/100 [=====] - 1s 10ms/step - loss: 0.2869 - accuracy : 0.9996 - val_loss: 0.3133 - val_accuracy: 0.9925 - lr: 2.3438e-06
[100, 32, 0.2]
40.0%
Epoch 1/100
100/100 [=====] - 2s 13ms/step - loss: 3.0347 - accuracy : 0.1656 - val_loss: 3.6080 - val_accuracy: 0.1075 - lr: 3.0000e-04
Epoch 2/100
100/100 [=====] - 1s 9ms/step - loss: 2.5266 - accuracy: 0.2853 - val_loss: 3.6635 - val_accuracy: 0.1075 - lr: 3.0000e-04
Epoch 3/100
100/100 [=====] - 1s 10ms/step - loss: 2.1064 - accuracy : 0.4112 - val_loss: 4.4125 - val_accuracy: 0.1475 - lr: 3.0000e-04
Epoch 4/100
100/100 [=====] - 1s 10ms/step - loss: 1.6619 - accuracy : 0.5866 - val_loss: 6.0132 - val_accuracy: 0.0925 - lr: 3.0000e-04
Epoch 5/100
98/100 [=====>..] - ETA: 0s - loss: 1.2367 - accuracy: 0.7605
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
100/100 [=====] - 1s 9ms/step - loss: 1.2352 - accuracy: 0.7616 - val_loss: 5.1795 - val_accuracy: 0.0900 - lr: 3.0000e-04
Epoch 6/100
100/100 [=====] - 1s 10ms/step - loss: 0.9409 - accuracy : 0.8706 - val_loss: 3.7806 - val_accuracy: 0.1425 - lr: 1.5000e-04
Epoch 7/100
100/100 [=====] - 1s 10ms/step - loss: 0.7804 - accuracy : 0.9272 - val_loss: 3.2575 - val_accuracy: 0.2375 - lr: 1.5000e-04
Epoch 8/100
100/100 [=====] - 1s 11ms/step - loss: 0.6912 - accuracy : 0.9566 - val_loss: 1.7980 - val_accuracy: 0.5813 - lr: 1.5000e-04
Epoch 9/100
100/100 [=====] - 1s 9ms/step - loss: 0.6192 - accuracy: 0.9734 - val_loss: 2.0654 - val_accuracy: 0.4812 - lr: 1.5000e-04
Epoch 10/100
100/100 [=====] - 1s 9ms/step - loss: 0.5672 - accuracy: 0.9806 - val_loss: 2.1786 - val_accuracy: 0.4750 - lr: 1.5000e-04
Epoch 11/100
100/100 [=====] - 1s 9ms/step - loss: 0.5273 - accuracy: 0.9856 - val_loss: 1.7024 - val_accuracy: 0.6300 - lr: 1.5000e-04
Epoch 12/100

100/100 [=====] - 1s 9ms/step - loss: 0.4857 - accuracy: 0.9897 - val_loss: 3.5243 - val_accuracy: 0.4787 - lr: 1.5000e-04
Epoch 13/100
100/100 [=====] - 1s 9ms/step - loss: 0.4530 - accuracy: 0.9950 - val_loss: 3.8208 - val_accuracy: 0.4825 - lr: 1.5000e-04
Epoch 14/100
100/100 [=====] - 1s 8ms/step - loss: 0.4361 - accuracy: 0.9928 - val_loss: 1.5269 - val_accuracy: 0.5625 - lr: 1.5000e-04
Epoch 15/100
100/100 [=====] - 1s 9ms/step - loss: 0.4100 - accuracy: 0.9956 - val_loss: 1.8918 - val_accuracy: 0.6050 - lr: 1.5000e-04
Epoch 16/100
100/100 [=====] - 1s 10ms/step - loss: 0.3950 - accuracy: 0.9978 - val_loss: 0.9539 - val_accuracy: 0.7538 - lr: 1.5000e-04
Epoch 17/100
100/100 [=====] - 1s 10ms/step - loss: 0.3765 - accuracy: 0.9972 - val_loss: 3.9946 - val_accuracy: 0.3325 - lr: 1.5000e-04
Epoch 18/100
100/100 [=====] - 1s 11ms/step - loss: 0.3596 - accuracy: 0.9972 - val_loss: 0.8568 - val_accuracy: 0.7912 - lr: 1.5000e-04
Epoch 19/100
100/100 [=====] - 1s 10ms/step - loss: 0.3404 - accuracy: 0.9987 - val_loss: 3.6685 - val_accuracy: 0.4225 - lr: 1.5000e-04
Epoch 20/100
100/100 [=====] - 1s 9ms/step - loss: 0.3287 - accuracy: 0.9966 - val_loss: 3.4303 - val_accuracy: 0.1650 - lr: 1.5000e-04
Epoch 21/100
100/100 [=====] - 1s 10ms/step - loss: 0.3125 - accuracy: 0.9987 - val_loss: 2.4498 - val_accuracy: 0.3913 - lr: 1.5000e-04
Epoch 22/100
99/100 [=====>.] - ETA: 0s - loss: 0.2990 - accuracy: 0.9991
Epoch 22: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
100/100 [=====] - 1s 9ms/step - loss: 0.2989 - accuracy: 0.9991 - val_loss: 3.7348 - val_accuracy: 0.3512 - lr: 1.5000e-04
Epoch 23/100
100/100 [=====] - 1s 10ms/step - loss: 0.2851 - accuracy: 0.9997 - val_loss: 0.3373 - val_accuracy: 0.9850 - lr: 7.5000e-05
Epoch 24/100
100/100 [=====] - 1s 10ms/step - loss: 0.2793 - accuracy: 0.9997 - val_loss: 1.1026 - val_accuracy: 0.6913 - lr: 7.5000e-05
Epoch 25/100
100/100 [=====] - 1s 9ms/step - loss: 0.2737 - accuracy: 0.9997 - val_loss: 2.0350 - val_accuracy: 0.5962 - lr: 7.5000e-05
Epoch 26/100
100/100 [=====] - 1s 9ms/step - loss: 0.2668 - accuracy: 0.9997 - val_loss: 2.3824 - val_accuracy: 0.4050 - lr: 7.5000e-05
Epoch 27/100
95/100 [=====>..] - ETA: 0s - loss: 0.2605 - accuracy: 0.9997
Epoch 27: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
100/100 [=====] - 1s 9ms/step - loss: 0.2603 - accuracy: 0.9997 - val_loss: 3.0010 - val_accuracy: 0.4112 - lr: 7.5000e-05
Epoch 28/100
100/100 [=====] - 1s 9ms/step - loss: 0.2544 - accuracy: 0.9997 - val_loss: 1.2281 - val_accuracy: 0.6087 - lr: 3.7500e-05
Epoch 29/100
100/100 [=====] - 1s 10ms/step - loss: 0.2515 - accuracy: 1.0000 - val_loss: 0.3089 - val_accuracy: 0.9812 - lr: 3.7500e-05
Epoch 30/100

100/100 [=====] - 1s 8ms/step - loss: 0.2491 - accuracy: 1.0000 - val_loss: 0.2919 - val_accuracy: 0.9787 - lr: 3.7500e-05
Epoch 31/100
100/100 [=====] - 1s 9ms/step - loss: 0.2471 - accuracy: 1.0000 - val_loss: 0.9709 - val_accuracy: 0.7063 - lr: 3.7500e-05
Epoch 32/100
100/100 [=====] - 1s 9ms/step - loss: 0.2411 - accuracy: 1.0000 - val_loss: 0.2752 - val_accuracy: 0.9900 - lr: 3.7500e-05
Epoch 33/100
100/100 [=====] - 1s 10ms/step - loss: 0.2381 - accuracy: 1.0000 - val_loss: 0.6615 - val_accuracy: 0.8650 - lr: 3.7500e-05
Epoch 34/100
100/100 [=====] - 1s 9ms/step - loss: 0.2343 - accuracy: 1.0000 - val_loss: 0.2838 - val_accuracy: 0.9837 - lr: 3.7500e-05
Epoch 35/100
100/100 [=====] - 1s 11ms/step - loss: 0.2330 - accuracy: 1.0000 - val_loss: 0.3287 - val_accuracy: 0.9775 - lr: 3.7500e-05
Epoch 36/100
99/100 [=====>.] - ETA: 0s - loss: 0.2286 - accuracy: 1.0000
Epoch 36: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
100/100 [=====] - 1s 10ms/step - loss: 0.2285 - accuracy: 1.0000 - val_loss: 0.3687 - val_accuracy: 0.9475 - lr: 3.7500e-05
Epoch 37/100
100/100 [=====] - 1s 11ms/step - loss: 0.2266 - accuracy: 1.0000 - val_loss: 0.2736 - val_accuracy: 0.9825 - lr: 1.8750e-05
Epoch 38/100
100/100 [=====] - 1s 9ms/step - loss: 0.2240 - accuracy: 1.0000 - val_loss: 0.7509 - val_accuracy: 0.7763 - lr: 1.8750e-05
Epoch 39/100
100/100 [=====] - 1s 9ms/step - loss: 0.2220 - accuracy: 1.0000 - val_loss: 0.2504 - val_accuracy: 0.9887 - lr: 1.8750e-05
Epoch 40/100
100/100 [=====] - 1s 9ms/step - loss: 0.2192 - accuracy: 1.0000 - val_loss: 0.2426 - val_accuracy: 0.9937 - lr: 1.8750e-05
Epoch 41/100
100/100 [=====] - 1s 9ms/step - loss: 0.2173 - accuracy: 1.0000 - val_loss: 0.3140 - val_accuracy: 0.9775 - lr: 1.8750e-05
Epoch 42/100
100/100 [=====] - 1s 11ms/step - loss: 0.2162 - accuracy: 1.0000 - val_loss: 0.2424 - val_accuracy: 0.9862 - lr: 1.8750e-05
Epoch 43/100
100/100 [=====] - 1s 9ms/step - loss: 0.2141 - accuracy: 1.0000 - val_loss: 0.2419 - val_accuracy: 0.9900 - lr: 1.8750e-05
Epoch 44/100
100/100 [=====] - 1s 9ms/step - loss: 0.2124 - accuracy: 1.0000 - val_loss: 0.5306 - val_accuracy: 0.8850 - lr: 1.8750e-05
Epoch 45/100
100/100 [=====] - 1s 10ms/step - loss: 0.2104 - accuracy: 1.0000 - val_loss: 0.4584 - val_accuracy: 0.9187 - lr: 1.8750e-05
Epoch 46/100
100/100 [=====] - 1s 9ms/step - loss: 0.2068 - accuracy: 0.9997 - val_loss: 0.3194 - val_accuracy: 0.9725 - lr: 1.8750e-05
Epoch 47/100
100/100 [=====] - ETA: 0s - loss: 0.2060 - accuracy: 1.0000
Epoch 47: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
100/100 [=====] - 1s 9ms/step - loss: 0.2060 - accuracy: 1.0000 - val_loss: 0.2929 - val_accuracy: 0.9737 - lr: 1.8750e-05
Epoch 48/100

100/100 [=====] - 1s 9ms/step - loss: 0.2035 - accuracy: 1.0000 - val_loss: 0.2748 - val_accuracy: 0.9875 - lr: 9.3750e-06
Epoch 49/100
100/100 [=====] - 1s 9ms/step - loss: 0.2024 - accuracy: 1.0000 - val_loss: 0.2259 - val_accuracy: 0.9887 - lr: 9.3750e-06
Epoch 50/100
100/100 [=====] - 1s 9ms/step - loss: 0.2018 - accuracy: 0.9997 - val_loss: 0.2292 - val_accuracy: 0.9912 - lr: 9.3750e-06
Epoch 51/100
100/100 [=====] - 1s 10ms/step - loss: 0.1999 - accuracy: 1.0000 - val_loss: 0.2215 - val_accuracy: 0.9862 - lr: 9.3750e-06
Epoch 52/100
100/100 [=====] - 1s 11ms/step - loss: 0.1989 - accuracy: 1.0000 - val_loss: 0.2152 - val_accuracy: 0.9912 - lr: 9.3750e-06
[100, 32, 0.25]
42.5%
Epoch 1/100
94/94 [=====] - 2s 14ms/step - loss: 3.0895 - accuracy: 0.1607 - val_loss: 3.7678 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 2/100
94/94 [=====] - 1s 12ms/step - loss: 2.4441 - accuracy: 0.3063 - val_loss: 6.4428 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 3/100
94/94 [=====] - 1s 13ms/step - loss: 1.9242 - accuracy: 0.4693 - val_loss: 8.6052 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 4/100
94/94 [=====] - 1s 12ms/step - loss: 1.4362 - accuracy: 0.6667 - val_loss: 11.4600 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 5/100
93/94 [=====>.] - ETA: 0s - loss: 1.0170 - accuracy: 0.8441
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
94/94 [=====] - 1s 11ms/step - loss: 1.0174 - accuracy: 0.8433 - val_loss: 11.5291 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 6/100
94/94 [=====] - 1s 11ms/step - loss: 0.7749 - accuracy: 0.9307 - val_loss: 12.5895 - val_accuracy: 0.1020 - lr: 1.5000e-04
Epoch 7/100
94/94 [=====] - 1s 13ms/step - loss: 0.6663 - accuracy: 0.9557 - val_loss: 10.8433 - val_accuracy: 0.0950 - lr: 1.5000e-04
Epoch 8/100
94/94 [=====] - 1s 10ms/step - loss: 0.5982 - accuracy: 0.9677 - val_loss: 4.4980 - val_accuracy: 0.1330 - lr: 1.5000e-04
Epoch 9/100
91/94 [=====>.] - ETA: 0s - loss: 0.5414 - accuracy: 0.9870
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
94/94 [=====] - 1s 9ms/step - loss: 0.5414 - accuracy: 0.9870 - val_loss: 5.9703 - val_accuracy: 0.2280 - lr: 1.5000e-04
Epoch 10/100
94/94 [=====] - 1s 9ms/step - loss: 0.5051 - accuracy: 0.9887 - val_loss: 2.9464 - val_accuracy: 0.3180 - lr: 7.5000e-05
Epoch 11/100
94/94 [=====] - 1s 10ms/step - loss: 0.4831 - accuracy: 0.9893 - val_loss: 1.4428 - val_accuracy: 0.6640 - lr: 7.5000e-05
Epoch 12/100
94/94 [=====] - 1s 13ms/step - loss: 0.4627 - accuracy: 0.9940 - val_loss: 0.7735 - val_accuracy: 0.8380 - lr: 7.5000e-05
Epoch 13/100
94/94 [=====] - 1s 11ms/step - loss: 0.4453 - accuracy:

0.9953 - val_loss: 2.2654 - val_accuracy: 0.5730 - lr: 7.5000e-05
Epoch 14/100
94/94 [=====] - 1s 12ms/step - loss: 0.4364 - accuracy:
0.9947 - val_loss: 2.6647 - val_accuracy: 0.4440 - lr: 7.5000e-05
Epoch 15/100
94/94 [=====] - 1s 11ms/step - loss: 0.4223 - accuracy:
0.9940 - val_loss: 3.3627 - val_accuracy: 0.4380 - lr: 7.5000e-05
Epoch 16/100
90/94 [=====>..] - ETA: 0s - loss: 0.4094 - accuracy: 0.997
2
Epoch 16: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
94/94 [=====] - 1s 11ms/step - loss: 0.4088 - accuracy:
0.9970 - val_loss: 2.1586 - val_accuracy: 0.3650 - lr: 7.5000e-05
Epoch 17/100
94/94 [=====] - 1s 13ms/step - loss: 0.3931 - accuracy:
0.9970 - val_loss: 0.8862 - val_accuracy: 0.8070 - lr: 3.7500e-05
Epoch 18/100
94/94 [=====] - 1s 11ms/step - loss: 0.3865 - accuracy:
0.9993 - val_loss: 0.6406 - val_accuracy: 0.9310 - lr: 3.7500e-05
Epoch 19/100
94/94 [=====] - 1s 12ms/step - loss: 0.3855 - accuracy:
0.9977 - val_loss: 0.6723 - val_accuracy: 0.8660 - lr: 3.7500e-05
Epoch 20/100
94/94 [=====] - 1s 10ms/step - loss: 0.3760 - accuracy:
0.9973 - val_loss: 0.5181 - val_accuracy: 0.9620 - lr: 3.7500e-05
Epoch 21/100
94/94 [=====] - 1s 9ms/step - loss: 0.3707 - accuracy: 0
.9977 - val_loss: 1.1723 - val_accuracy: 0.6690 - lr: 3.7500e-05
Epoch 22/100
94/94 [=====] - 1s 10ms/step - loss: 0.3644 - accuracy:
0.9993 - val_loss: 0.4810 - val_accuracy: 0.9700 - lr: 3.7500e-05
Epoch 23/100
94/94 [=====] - 1s 13ms/step - loss: 0.3585 - accuracy:
0.9993 - val_loss: 1.4217 - val_accuracy: 0.6450 - lr: 3.7500e-05
Epoch 24/100
94/94 [=====] - 1s 9ms/step - loss: 0.3547 - accuracy: 0
.9983 - val_loss: 0.6170 - val_accuracy: 0.9140 - lr: 3.7500e-05
Epoch 25/100
94/94 [=====] - 1s 9ms/step - loss: 0.3470 - accuracy: 1
.0000 - val_loss: 0.6957 - val_accuracy: 0.8800 - lr: 3.7500e-05
Epoch 26/100
93/94 [=====>..] - ETA: 0s - loss: 0.3424 - accuracy: 0.998
3
Epoch 26: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
94/94 [=====] - 1s 10ms/step - loss: 0.3423 - accuracy:
0.9983 - val_loss: 0.5024 - val_accuracy: 0.9470 - lr: 3.7500e-05
Epoch 27/100
94/94 [=====] - 1s 11ms/step - loss: 0.3377 - accuracy:
0.9993 - val_loss: 0.4560 - val_accuracy: 0.9700 - lr: 1.8750e-05
Epoch 28/100
94/94 [=====] - 1s 11ms/step - loss: 0.3319 - accuracy:
0.9997 - val_loss: 0.4301 - val_accuracy: 0.9740 - lr: 1.8750e-05
Epoch 29/100
94/94 [=====] - 1s 12ms/step - loss: 0.3311 - accuracy:
1.0000 - val_loss: 0.3773 - val_accuracy: 0.9850 - lr: 1.8750e-05
Epoch 30/100
94/94 [=====] - 1s 12ms/step - loss: 0.3291 - accuracy:
0.9987 - val_loss: 0.3996 - val_accuracy: 0.9730 - lr: 1.8750e-05
Epoch 31/100
94/94 [=====] - 1s 11ms/step - loss: 0.3248 - accuracy:

0.9993 - val_loss: 0.6918 - val_accuracy: 0.8670 - lr: 1.8750e-05
Epoch 32/100
94/94 [=====] - 1s 12ms/step - loss: 0.3226 - accuracy:
1.0000 - val_loss: 1.0596 - val_accuracy: 0.6630 - lr: 1.8750e-05
Epoch 33/100
92/94 [=====>.] - ETA: 0s - loss: 0.3181 - accuracy: 0.999
7
Epoch 33: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
94/94 [=====] - 1s 11ms/step - loss: 0.3182 - accuracy:
0.9997 - val_loss: 0.3774 - val_accuracy: 0.9790 - lr: 1.8750e-05
Epoch 34/100
94/94 [=====] - 1s 11ms/step - loss: 0.3179 - accuracy:
0.9997 - val_loss: 0.3682 - val_accuracy: 0.9720 - lr: 9.3750e-06
Epoch 35/100
94/94 [=====] - 1s 12ms/step - loss: 0.3152 - accuracy:
0.9997 - val_loss: 0.3928 - val_accuracy: 0.9740 - lr: 9.3750e-06
Epoch 36/100
94/94 [=====] - 1s 13ms/step - loss: 0.3129 - accuracy:
0.9993 - val_loss: 0.4369 - val_accuracy: 0.9640 - lr: 9.3750e-06
Epoch 37/100
94/94 [=====] - 1s 12ms/step - loss: 0.3111 - accuracy:
0.9990 - val_loss: 0.5015 - val_accuracy: 0.9380 - lr: 9.3750e-06
Epoch 38/100
89/94 [=====>..] - ETA: 0s - loss: 0.3120 - accuracy: 1.000
0
Epoch 38: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
94/94 [=====] - 1s 12ms/step - loss: 0.3118 - accuracy:
1.0000 - val_loss: 0.4275 - val_accuracy: 0.9640 - lr: 9.3750e-06
Epoch 39/100
94/94 [=====] - 1s 9ms/step - loss: 0.3073 - accuracy: 0
.9993 - val_loss: 0.3409 - val_accuracy: 0.9830 - lr: 4.6875e-06
Epoch 40/100
94/94 [=====] - 1s 10ms/step - loss: 0.3073 - accuracy:
0.9997 - val_loss: 0.3430 - val_accuracy: 0.9830 - lr: 4.6875e-06
Epoch 41/100
94/94 [=====] - 1s 10ms/step - loss: 0.3064 - accuracy:
0.9997 - val_loss: 0.3484 - val_accuracy: 0.9820 - lr: 4.6875e-06
[100, 32, 0.30000000000000004]
45.0%
Epoch 1/100
88/88 [=====] - 2s 17ms/step - loss: 3.1158 - accuracy:
0.1650 - val_loss: 3.5055 - val_accuracy: 0.1033 - lr: 3.0000e-04
Epoch 2/100
88/88 [=====] - 1s 12ms/step - loss: 2.4872 - accuracy:
0.2993 - val_loss: 4.4237 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 3/100
88/88 [=====] - 1s 13ms/step - loss: 2.0644 - accuracy:
0.4275 - val_loss: 7.1084 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 4/100
88/88 [=====] - 1s 12ms/step - loss: 1.5560 - accuracy:
0.6268 - val_loss: 9.6457 - val_accuracy: 0.1308 - lr: 3.0000e-04
Epoch 5/100
88/88 [=====] - ETA: 0s - loss: 1.1375 - accuracy: 0.790
0
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
88/88 [=====] - 1s 13ms/step - loss: 1.1375 - accuracy:
0.7900 - val_loss: 12.8895 - val_accuracy: 0.1508 - lr: 3.0000e-04
Epoch 6/100
88/88 [=====] - 1s 12ms/step - loss: 0.8472 - accuracy:
0.9036 - val_loss: 12.2577 - val_accuracy: 0.1108 - lr: 1.5000e-04

Epoch 7/100
88/88 [=====] - 1s 12ms/step - loss: 0.7312 - accuracy: 0.9454 - val_loss: 11.5390 - val_accuracy: 0.0983 - lr: 1.5000e-04
Epoch 8/100
88/88 [=====] - 1s 12ms/step - loss: 0.6546 - accuracy: 0.9646 - val_loss: 7.7958 - val_accuracy: 0.1667 - lr: 1.5000e-04
Epoch 9/100
83/88 [=====>..] - ETA: 0s - loss: 0.6001 - accuracy: 0.9752
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
88/88 [=====] - 1s 13ms/step - loss: 0.6011 - accuracy: 0.9743 - val_loss: 9.5860 - val_accuracy: 0.0983 - lr: 1.5000e-04
Epoch 10/100
88/88 [=====] - 1s 11ms/step - loss: 0.5523 - accuracy: 0.9825 - val_loss: 4.6882 - val_accuracy: 0.1617 - lr: 7.5000e-05
Epoch 11/100
88/88 [=====] - 1s 12ms/step - loss: 0.5289 - accuracy: 0.9854 - val_loss: 1.6501 - val_accuracy: 0.5717 - lr: 7.5000e-05
Epoch 12/100
88/88 [=====] - 1s 9ms/step - loss: 0.5154 - accuracy: 0.9893 - val_loss: 1.4378 - val_accuracy: 0.5483 - lr: 7.5000e-05
Epoch 13/100
88/88 [=====] - 1s 10ms/step - loss: 0.4972 - accuracy: 0.9914 - val_loss: 1.0713 - val_accuracy: 0.7500 - lr: 7.5000e-05
Epoch 14/100
88/88 [=====] - 1s 10ms/step - loss: 0.4796 - accuracy: 0.9946 - val_loss: 0.8423 - val_accuracy: 0.8583 - lr: 7.5000e-05
Epoch 15/100
88/88 [=====] - 1s 11ms/step - loss: 0.4632 - accuracy: 0.9943 - val_loss: 6.0639 - val_accuracy: 0.2900 - lr: 7.5000e-05
Epoch 16/100
88/88 [=====] - 1s 14ms/step - loss: 0.4521 - accuracy: 0.9954 - val_loss: 7.5163 - val_accuracy: 0.4400 - lr: 7.5000e-05
Epoch 17/100
88/88 [=====] - 1s 11ms/step - loss: 0.4403 - accuracy: 0.9946 - val_loss: 0.5520 - val_accuracy: 0.9600 - lr: 7.5000e-05
Epoch 18/100
88/88 [=====] - 1s 12ms/step - loss: 0.4249 - accuracy: 0.9964 - val_loss: 0.6341 - val_accuracy: 0.9225 - lr: 7.5000e-05
Epoch 19/100
88/88 [=====] - 1s 14ms/step - loss: 0.4171 - accuracy: 0.9950 - val_loss: 1.5609 - val_accuracy: 0.6300 - lr: 7.5000e-05
Epoch 20/100
88/88 [=====] - 1s 12ms/step - loss: 0.4081 - accuracy: 0.9964 - val_loss: 2.1699 - val_accuracy: 0.5183 - lr: 7.5000e-05
Epoch 21/100
88/88 [=====] - 1s 14ms/step - loss: 0.3952 - accuracy: 0.9989 - val_loss: 0.5178 - val_accuracy: 0.9575 - lr: 7.5000e-05
Epoch 22/100
88/88 [=====] - 1s 11ms/step - loss: 0.3861 - accuracy: 0.9993 - val_loss: 0.5160 - val_accuracy: 0.9558 - lr: 7.5000e-05
Epoch 23/100
88/88 [=====] - 1s 13ms/step - loss: 0.3768 - accuracy: 0.9975 - val_loss: 0.5353 - val_accuracy: 0.9375 - lr: 7.5000e-05
Epoch 24/100
88/88 [=====] - 1s 13ms/step - loss: 0.3692 - accuracy: 0.9986 - val_loss: 2.1926 - val_accuracy: 0.5717 - lr: 7.5000e-05
Epoch 25/100
88/88 [=====] - 1s 12ms/step - loss: 0.3599 - accuracy: 0.9979 - val_loss: 0.5607 - val_accuracy: 0.9067 - lr: 7.5000e-05

Epoch 26/100
88/88 [=====] - ETA: 0s - loss: 0.3452 - accuracy: 0.9989
Epoch 26: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
88/88 [=====] - 1s 12ms/step - loss: 0.3452 - accuracy: 0.9989 - val_loss: 0.8815 - val_accuracy: 0.7725 - lr: 7.5000e-05
Epoch 27/100
88/88 [=====] - 1s 10ms/step - loss: 0.3406 - accuracy: 0.9996 - val_loss: 0.4280 - val_accuracy: 0.9650 - lr: 3.7500e-05
Epoch 28/100
88/88 [=====] - 1s 10ms/step - loss: 0.3357 - accuracy: 0.9993 - val_loss: 0.7642 - val_accuracy: 0.8200 - lr: 3.7500e-05
Epoch 29/100
88/88 [=====] - 1s 10ms/step - loss: 0.3312 - accuracy: 0.9996 - val_loss: 0.3788 - val_accuracy: 0.9825 - lr: 3.7500e-05
Epoch 30/100
88/88 [=====] - 1s 10ms/step - loss: 0.3291 - accuracy: 0.9996 - val_loss: 0.4913 - val_accuracy: 0.9392 - lr: 3.7500e-05
Epoch 31/100
88/88 [=====] - 1s 13ms/step - loss: 0.3225 - accuracy: 0.9996 - val_loss: 0.3711 - val_accuracy: 0.9800 - lr: 3.7500e-05
Epoch 32/100
88/88 [=====] - 1s 12ms/step - loss: 0.3160 - accuracy: 1.0000 - val_loss: 0.7848 - val_accuracy: 0.8042 - lr: 3.7500e-05
Epoch 33/100
88/88 [=====] - 1s 12ms/step - loss: 0.3135 - accuracy: 0.9996 - val_loss: 0.3796 - val_accuracy: 0.9775 - lr: 3.7500e-05
Epoch 34/100
88/88 [=====] - 1s 11ms/step - loss: 0.3075 - accuracy: 0.9996 - val_loss: 0.4370 - val_accuracy: 0.9658 - lr: 3.7500e-05
Epoch 35/100
88/88 [=====] - 1s 13ms/step - loss: 0.3044 - accuracy: 1.0000 - val_loss: 0.3558 - val_accuracy: 0.9792 - lr: 3.7500e-05
Epoch 36/100
88/88 [=====] - 1s 12ms/step - loss: 0.3002 - accuracy: 1.0000 - val_loss: 0.3843 - val_accuracy: 0.9708 - lr: 3.7500e-05
Epoch 37/100
88/88 [=====] - 1s 12ms/step - loss: 0.2943 - accuracy: 0.9996 - val_loss: 0.3793 - val_accuracy: 0.9717 - lr: 3.7500e-05
Epoch 38/100
88/88 [=====] - 1s 11ms/step - loss: 0.2905 - accuracy: 0.9993 - val_loss: 1.2275 - val_accuracy: 0.6233 - lr: 3.7500e-05
Epoch 39/100
87/88 [=====>.] - ETA: 0s - loss: 0.2856 - accuracy: 0.9993
Epoch 39: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
88/88 [=====] - 1s 13ms/step - loss: 0.2856 - accuracy: 0.9993 - val_loss: 0.7988 - val_accuracy: 0.7733 - lr: 3.7500e-05
Epoch 40/100
88/88 [=====] - 1s 14ms/step - loss: 0.2818 - accuracy: 0.9996 - val_loss: 0.3463 - val_accuracy: 0.9817 - lr: 1.8750e-05
Epoch 41/100
88/88 [=====] - 1s 12ms/step - loss: 0.2798 - accuracy: 1.0000 - val_loss: 0.3604 - val_accuracy: 0.9750 - lr: 1.8750e-05
[100, 32, 0.35000000000000003]
47.5%
Epoch 1/100
82/82 [=====] - 2s 17ms/step - loss: 3.1415 - accuracy: 0.1624 - val_loss: 4.2603 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 2/100

82/82 [=====] - 1s 10ms/step - loss: 2.6156 - accuracy: 0.2697 - val_loss: 4.3055 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 3/100
82/82 [=====] - 1s 10ms/step - loss: 2.2686 - accuracy: 0.3655 - val_loss: 3.9804 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 4/100
82/82 [=====] - 1s 13ms/step - loss: 1.9210 - accuracy: 0.4875 - val_loss: 3.5801 - val_accuracy: 0.1071 - lr: 3.0000e-04
Epoch 5/100
82/82 [=====] - 1s 14ms/step - loss: 1.6167 - accuracy: 0.5972 - val_loss: 6.0842 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 6/100
82/82 [=====] - 1s 13ms/step - loss: 1.3435 - accuracy: 0.7076 - val_loss: 6.2620 - val_accuracy: 0.1884 - lr: 3.0000e-04
Epoch 7/100
82/82 [=====] - 1s 14ms/step - loss: 1.1286 - accuracy: 0.7930 - val_loss: 11.5799 - val_accuracy: 0.1021 - lr: 3.0000e-04
Epoch 8/100
79/82 [=====>..] - ETA: 0s - loss: 0.9434 - accuracy: 0.8584
Epoch 8: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
82/82 [=====] - 1s 13ms/step - loss: 0.9403 - accuracy: 0.8599 - val_loss: 6.9774 - val_accuracy: 0.2427 - lr: 3.0000e-04
Epoch 9/100
82/82 [=====] - 1s 13ms/step - loss: 0.7723 - accuracy: 0.9230 - val_loss: 3.4223 - val_accuracy: 0.2712 - lr: 1.5000e-04
Epoch 10/100
82/82 [=====] - 1s 14ms/step - loss: 0.6742 - accuracy: 0.9584 - val_loss: 3.1539 - val_accuracy: 0.3333 - lr: 1.5000e-04
Epoch 11/100
82/82 [=====] - 1s 12ms/step - loss: 0.6208 - accuracy: 0.9688 - val_loss: 6.0679 - val_accuracy: 0.1406 - lr: 1.5000e-04
Epoch 12/100
82/82 [=====] - 1s 13ms/step - loss: 0.5701 - accuracy: 0.9792 - val_loss: 9.4981 - val_accuracy: 0.2020 - lr: 1.5000e-04
Epoch 13/100
82/82 [=====] - 1s 13ms/step - loss: 0.5303 - accuracy: 0.9842 - val_loss: 0.9490 - val_accuracy: 0.7937 - lr: 1.5000e-04
Epoch 14/100
82/82 [=====] - 1s 14ms/step - loss: 0.4918 - accuracy: 0.9946 - val_loss: 3.5589 - val_accuracy: 0.2991 - lr: 1.5000e-04
Epoch 15/100
82/82 [=====] - 1s 12ms/step - loss: 0.4753 - accuracy: 0.9919 - val_loss: 1.6601 - val_accuracy: 0.4325 - lr: 1.5000e-04
Epoch 16/100
82/82 [=====] - 1s 12ms/step - loss: 0.4513 - accuracy: 0.9942 - val_loss: 1.6960 - val_accuracy: 0.5396 - lr: 1.5000e-04
Epoch 17/100
76/82 [=====>...] - ETA: 0s - loss: 0.4307 - accuracy: 0.9959
Epoch 17: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
82/82 [=====] - 1s 10ms/step - loss: 0.4304 - accuracy: 0.9958 - val_loss: 2.0662 - val_accuracy: 0.4954 - lr: 1.5000e-04
Epoch 18/100
82/82 [=====] - 1s 11ms/step - loss: 0.4133 - accuracy: 0.9981 - val_loss: 0.7802 - val_accuracy: 0.8237 - lr: 7.5000e-05
Epoch 19/100
82/82 [=====] - 1s 15ms/step - loss: 0.4029 - accuracy: 0.9973 - val_loss: 0.6472 - val_accuracy: 0.9122 - lr: 7.5000e-05
Epoch 20/100

82/82 [=====] - 1s 13ms/step - loss: 0.3975 - accuracy: 0.9969 - val_loss: 1.0807 - val_accuracy: 0.7395 - lr: 7.5000e-05
Epoch 21/100
82/82 [=====] - 1s 13ms/step - loss: 0.3905 - accuracy: 0.9981 - val_loss: 0.4688 - val_accuracy: 0.9736 - lr: 7.5000e-05
Epoch 22/100
82/82 [=====] - 1s 13ms/step - loss: 0.3884 - accuracy: 0.9981 - val_loss: 1.5076 - val_accuracy: 0.6610 - lr: 7.5000e-05
Epoch 23/100
82/82 [=====] - 1s 14ms/step - loss: 0.3743 - accuracy: 0.9992 - val_loss: 0.5873 - val_accuracy: 0.9172 - lr: 7.5000e-05
Epoch 24/100
82/82 [=====] - 1s 12ms/step - loss: 0.3678 - accuracy: 0.9988 - val_loss: 0.6770 - val_accuracy: 0.8879 - lr: 7.5000e-05
Epoch 25/100
81/82 [=====>.] - ETA: 0s - loss: 0.3582 - accuracy: 0.9996
Epoch 25: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
82/82 [=====] - 1s 14ms/step - loss: 0.3583 - accuracy: 0.9996 - val_loss: 0.5129 - val_accuracy: 0.9500 - lr: 7.5000e-05
Epoch 26/100
82/82 [=====] - 1s 13ms/step - loss: 0.3504 - accuracy: 0.9996 - val_loss: 0.4954 - val_accuracy: 0.9500 - lr: 3.7500e-05
Epoch 27/100
82/82 [=====] - 1s 14ms/step - loss: 0.3477 - accuracy: 0.9985 - val_loss: 0.4009 - val_accuracy: 0.9814 - lr: 3.7500e-05
Epoch 28/100
82/82 [=====] - 1s 13ms/step - loss: 0.3445 - accuracy: 0.9988 - val_loss: 0.5775 - val_accuracy: 0.9300 - lr: 3.7500e-05
Epoch 29/100
82/82 [=====] - 1s 13ms/step - loss: 0.3383 - accuracy: 0.9996 - val_loss: 0.7931 - val_accuracy: 0.8258 - lr: 3.7500e-05
Epoch 30/100
82/82 [=====] - 1s 12ms/step - loss: 0.3398 - accuracy: 0.9985 - val_loss: 0.6595 - val_accuracy: 0.9015 - lr: 3.7500e-05
Epoch 31/100
82/82 [=====] - 1s 11ms/step - loss: 0.3321 - accuracy: 0.9992 - val_loss: 0.3680 - val_accuracy: 0.9864 - lr: 3.7500e-05
Epoch 32/100
82/82 [=====] - 1s 10ms/step - loss: 0.3293 - accuracy: 0.9992 - val_loss: 0.4352 - val_accuracy: 0.9665 - lr: 3.7500e-05
Epoch 33/100
82/82 [=====] - 1s 10ms/step - loss: 0.3261 - accuracy: 0.9996 - val_loss: 0.4625 - val_accuracy: 0.9522 - lr: 3.7500e-05
Epoch 34/100
82/82 [=====] - 1s 11ms/step - loss: 0.3221 - accuracy: 0.9988 - val_loss: 0.4610 - val_accuracy: 0.9507 - lr: 3.7500e-05
Epoch 35/100
82/82 [=====] - ETA: 0s - loss: 0.3163 - accuracy: 0.9996
Epoch 35: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
82/82 [=====] - 1s 11ms/step - loss: 0.3163 - accuracy: 0.9996 - val_loss: 0.4292 - val_accuracy: 0.9679 - lr: 3.7500e-05
Epoch 36/100
82/82 [=====] - 1s 12ms/step - loss: 0.3139 - accuracy: 0.9992 - val_loss: 0.3521 - val_accuracy: 0.9850 - lr: 1.8750e-05
Epoch 37/100
82/82 [=====] - 1s 10ms/step - loss: 0.3098 - accuracy: 1.0000 - val_loss: 0.3560 - val_accuracy: 0.9822 - lr: 1.8750e-05
Epoch 38/100

82/82 [=====] - 1s 10ms/step - loss: 0.3103 - accuracy: 0.9992 - val_loss: 0.3848 - val_accuracy: 0.9736 - lr: 1.8750e-05
Epoch 39/100
82/82 [=====] - 1s 11ms/step - loss: 0.3070 - accuracy: 0.9992 - val_loss: 0.7367 - val_accuracy: 0.8444 - lr: 1.8750e-05
Epoch 40/100
80/82 [=====>.] - ETA: 0s - loss: 0.3038 - accuracy: 0.9992
Epoch 40: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
82/82 [=====] - 1s 10ms/step - loss: 0.3038 - accuracy: 0.9992 - val_loss: 0.4272 - val_accuracy: 0.9636 - lr: 1.8750e-05
Epoch 41/100
82/82 [=====] - 1s 10ms/step - loss: 0.3028 - accuracy: 0.9996 - val_loss: 0.3463 - val_accuracy: 0.9850 - lr: 9.3750e-06
Epoch 42/100
82/82 [=====] - 1s 12ms/step - loss: 0.3001 - accuracy: 1.0000 - val_loss: 0.3556 - val_accuracy: 0.9829 - lr: 9.3750e-06
Epoch 43/100
82/82 [=====] - 1s 12ms/step - loss: 0.3002 - accuracy: 0.9992 - val_loss: 0.3297 - val_accuracy: 0.9864 - lr: 9.3750e-06
[100, 32, 0.4]
50.0%
Epoch 1/100
75/75 [=====] - 2s 15ms/step - loss: 3.1834 - accuracy: 0.1321 - val_loss: 4.0131 - val_accuracy: 0.0875 - lr: 3.0000e-04
Epoch 2/100
75/75 [=====] - 1s 10ms/step - loss: 2.6902 - accuracy: 0.2267 - val_loss: 5.0041 - val_accuracy: 0.0875 - lr: 3.0000e-04
Epoch 3/100
75/75 [=====] - 1s 12ms/step - loss: 2.3497 - accuracy: 0.3413 - val_loss: 5.8699 - val_accuracy: 0.0875 - lr: 3.0000e-04
Epoch 4/100
75/75 [=====] - 1s 10ms/step - loss: 1.9097 - accuracy: 0.4758 - val_loss: 7.3591 - val_accuracy: 0.1569 - lr: 3.0000e-04
Epoch 5/100
70/75 [=====>..] - ETA: 0s - loss: 1.4862 - accuracy: 0.6420
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
75/75 [=====] - 1s 10ms/step - loss: 1.4779 - accuracy: 0.6450 - val_loss: 9.5995 - val_accuracy: 0.0913 - lr: 3.0000e-04
Epoch 6/100
75/75 [=====] - 1s 10ms/step - loss: 1.1391 - accuracy: 0.7917 - val_loss: 11.3592 - val_accuracy: 0.0913 - lr: 1.5000e-04
Epoch 7/100
75/75 [=====] - 1s 10ms/step - loss: 0.9858 - accuracy: 0.8425 - val_loss: 10.5438 - val_accuracy: 0.0913 - lr: 1.5000e-04
Epoch 8/100
75/75 [=====] - 1s 10ms/step - loss: 0.8523 - accuracy: 0.8946 - val_loss: 10.2460 - val_accuracy: 0.0925 - lr: 1.5000e-04
Epoch 9/100
72/75 [=====>..] - ETA: 0s - loss: 0.7806 - accuracy: 0.9084
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
75/75 [=====] - 1s 10ms/step - loss: 0.7805 - accuracy: 0.9100 - val_loss: 13.3911 - val_accuracy: 0.0956 - lr: 1.5000e-04
Epoch 10/100
75/75 [=====] - 1s 12ms/step - loss: 0.7188 - accuracy: 0.9375 - val_loss: 4.3247 - val_accuracy: 0.2706 - lr: 7.5000e-05
Epoch 11/100
75/75 [=====] - 1s 11ms/step - loss: 0.6733 - accuracy:

0.9517 - val_loss: 3.1689 - val_accuracy: 0.2663 - lr: 7.5000e-05
Epoch 12/100
75/75 [=====] - 1s 10ms/step - loss: 0.6422 - accuracy:
0.9642 - val_loss: 3.2742 - val_accuracy: 0.4325 - lr: 7.5000e-05
Epoch 13/100
75/75 [=====] - 1s 11ms/step - loss: 0.5976 - accuracy:
0.9750 - val_loss: 3.4503 - val_accuracy: 0.4187 - lr: 7.5000e-05
Epoch 14/100
75/75 [=====] - 1s 10ms/step - loss: 0.5807 - accuracy:
0.9737 - val_loss: 1.9251 - val_accuracy: 0.5150 - lr: 7.5000e-05
Epoch 15/100
75/75 [=====] - 1s 12ms/step - loss: 0.5604 - accuracy:
0.9783 - val_loss: 3.7506 - val_accuracy: 0.3481 - lr: 7.5000e-05
Epoch 16/100
75/75 [=====] - 1s 10ms/step - loss: 0.5429 - accuracy:
0.9779 - val_loss: 2.3255 - val_accuracy: 0.5231 - lr: 7.5000e-05
Epoch 17/100
75/75 [=====] - 1s 10ms/step - loss: 0.5185 - accuracy:
0.9879 - val_loss: 2.0875 - val_accuracy: 0.4737 - lr: 7.5000e-05
Epoch 18/100
75/75 [=====] - ETA: 0s - loss: 0.5068 - accuracy: 0.985
4
Epoch 18: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
75/75 [=====] - 1s 11ms/step - loss: 0.5068 - accuracy:
0.9854 - val_loss: 2.4608 - val_accuracy: 0.3756 - lr: 7.5000e-05
Epoch 19/100
75/75 [=====] - 1s 10ms/step - loss: 0.4876 - accuracy:
0.9883 - val_loss: 2.9261 - val_accuracy: 0.2894 - lr: 3.7500e-05
Epoch 20/100
75/75 [=====] - 1s 10ms/step - loss: 0.4719 - accuracy:
0.9921 - val_loss: 0.7743 - val_accuracy: 0.8712 - lr: 3.7500e-05
Epoch 21/100
75/75 [=====] - 1s 12ms/step - loss: 0.4677 - accuracy:
0.9900 - val_loss: 1.8232 - val_accuracy: 0.5806 - lr: 3.7500e-05
Epoch 22/100
75/75 [=====] - 1s 10ms/step - loss: 0.4591 - accuracy:
0.9925 - val_loss: 1.3808 - val_accuracy: 0.5944 - lr: 3.7500e-05
Epoch 23/100
75/75 [=====] - 1s 10ms/step - loss: 0.4510 - accuracy:
0.9921 - val_loss: 0.7510 - val_accuracy: 0.8737 - lr: 3.7500e-05
Epoch 24/100
75/75 [=====] - 1s 10ms/step - loss: 0.4466 - accuracy:
0.9925 - val_loss: 2.9411 - val_accuracy: 0.3537 - lr: 3.7500e-05
Epoch 25/100
75/75 [=====] - 1s 10ms/step - loss: 0.4399 - accuracy:
0.9912 - val_loss: 1.0019 - val_accuracy: 0.7700 - lr: 3.7500e-05
Epoch 26/100
75/75 [=====] - 1s 10ms/step - loss: 0.4346 - accuracy:
0.9937 - val_loss: 0.5941 - val_accuracy: 0.9388 - lr: 3.7500e-05
Epoch 27/100
75/75 [=====] - 1s 12ms/step - loss: 0.4244 - accuracy:
0.9958 - val_loss: 2.1109 - val_accuracy: 0.4244 - lr: 3.7500e-05
Epoch 28/100
75/75 [=====] - 1s 10ms/step - loss: 0.4171 - accuracy:
0.9946 - val_loss: 0.6185 - val_accuracy: 0.9231 - lr: 3.7500e-05
Epoch 29/100
75/75 [=====] - 1s 11ms/step - loss: 0.4103 - accuracy:
0.9967 - val_loss: 1.0579 - val_accuracy: 0.7194 - lr: 3.7500e-05
Epoch 30/100
71/75 [=====>..] - ETA: 0s - loss: 0.4103 - accuracy: 0.994

3

Epoch 30: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.

75/75 [=====] - 1s 12ms/step - loss: 0.4094 - accuracy: 0.9946 - val_loss: 0.7706 - val_accuracy: 0.8606 - lr: 3.7500e-05

Epoch 31/100

75/75 [=====] - 1s 13ms/step - loss: 0.4015 - accuracy: 0.9975 - val_loss: 0.6192 - val_accuracy: 0.9137 - lr: 1.8750e-05

Epoch 32/100

75/75 [=====] - 1s 12ms/step - loss: 0.3998 - accuracy: 0.9950 - val_loss: 0.6124 - val_accuracy: 0.9200 - lr: 1.8750e-05

Epoch 33/100

75/75 [=====] - 1s 12ms/step - loss: 0.3858 - accuracy: 0.9983 - val_loss: 0.6761 - val_accuracy: 0.9000 - lr: 1.8750e-05

Epoch 34/100

72/75 [=====>..] - ETA: 0s - loss: 0.3871 - accuracy: 0.9961

Epoch 34: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.

75/75 [=====] - 1s 11ms/step - loss: 0.3870 - accuracy: 0.9958 - val_loss: 0.6458 - val_accuracy: 0.9075 - lr: 1.8750e-05

Epoch 35/100

75/75 [=====] - 1s 12ms/step - loss: 0.3844 - accuracy: 0.9971 - val_loss: 0.4974 - val_accuracy: 0.9506 - lr: 9.3750e-06

Epoch 36/100

75/75 [=====] - 1s 12ms/step - loss: 0.3808 - accuracy: 0.9979 - val_loss: 0.4944 - val_accuracy: 0.9531 - lr: 9.3750e-06

Epoch 37/100

75/75 [=====] - 1s 14ms/step - loss: 0.3836 - accuracy: 0.9971 - val_loss: 0.4941 - val_accuracy: 0.9525 - lr: 9.3750e-06

Epoch 38/100

75/75 [=====] - 1s 11ms/step - loss: 0.3832 - accuracy: 0.9987 - val_loss: 1.1835 - val_accuracy: 0.6956 - lr: 9.3750e-06

Epoch 39/100

75/75 [=====] - 1s 12ms/step - loss: 0.3770 - accuracy: 0.9987 - val_loss: 0.6172 - val_accuracy: 0.9131 - lr: 9.3750e-06

Epoch 40/100

75/75 [=====] - 1s 11ms/step - loss: 0.3774 - accuracy: 0.9967 - val_loss: 0.5245 - val_accuracy: 0.9450 - lr: 9.3750e-06

Epoch 41/100

75/75 [=====] - ETA: 0s - loss: 0.3725 - accuracy: 0.9983

Epoch 41: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.

75/75 [=====] - 1s 11ms/step - loss: 0.3725 - accuracy: 0.9983 - val_loss: 0.5214 - val_accuracy: 0.9450 - lr: 9.3750e-06

Epoch 42/100

75/75 [=====] - 1s 11ms/step - loss: 0.3723 - accuracy: 0.9987 - val_loss: 0.4668 - val_accuracy: 0.9581 - lr: 4.6875e-06

Epoch 43/100

75/75 [=====] - 1s 11ms/step - loss: 0.3700 - accuracy: 0.9971 - val_loss: 0.4689 - val_accuracy: 0.9575 - lr: 4.6875e-06

Epoch 44/100

75/75 [=====] - 1s 12ms/step - loss: 0.3704 - accuracy: 0.9983 - val_loss: 0.4891 - val_accuracy: 0.9494 - lr: 4.6875e-06

Epoch 45/100

75/75 [=====] - 1s 10ms/step - loss: 0.3694 - accuracy: 0.9987 - val_loss: 0.4783 - val_accuracy: 0.9544 - lr: 4.6875e-06

Epoch 46/100

68/75 [=====>...] - ETA: 0s - loss: 0.3710 - accuracy: 0.9982

Epoch 46: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.

75/75 [=====] - 1s 10ms/step - loss: 0.3722 - accuracy:

0.9983 - val_loss: 0.4886 - val_accuracy: 0.9506 - lr: 4.6875e-06
Epoch 47/100
75/75 [=====] - 1s 11ms/step - loss: 0.3705 - accuracy:
0.9996 - val_loss: 0.4654 - val_accuracy: 0.9581 - lr: 2.3438e-06
Epoch 48/100
75/75 [=====] - 1s 12ms/step - loss: 0.3718 - accuracy:
0.9979 - val_loss: 0.4658 - val_accuracy: 0.9563 - lr: 2.3438e-06
Epoch 49/100
75/75 [=====] - 1s 11ms/step - loss: 0.3688 - accuracy:
0.9992 - val_loss: 0.4622 - val_accuracy: 0.9575 - lr: 2.3438e-06
Epoch 50/100
75/75 [=====] - 1s 12ms/step - loss: 0.3673 - accuracy:
0.9983 - val_loss: 0.4669 - val_accuracy: 0.9588 - lr: 2.3438e-06
Epoch 51/100
75/75 [=====] - 1s 11ms/step - loss: 0.3655 - accuracy:
0.9987 - val_loss: 0.4643 - val_accuracy: 0.9588 - lr: 2.3438e-06
Epoch 52/100
75/75 [=====] - 1s 12ms/step - loss: 0.3680 - accuracy:
0.9962 - val_loss: 0.4650 - val_accuracy: 0.9606 - lr: 2.3438e-06
Epoch 53/100
75/75 [=====] - 1s 13ms/step - loss: 0.3646 - accuracy:
0.9983 - val_loss: 0.4582 - val_accuracy: 0.9613 - lr: 2.3438e-06
Epoch 54/100
75/75 [=====] - 1s 11ms/step - loss: 0.3616 - accuracy:
0.9992 - val_loss: 0.4576 - val_accuracy: 0.9588 - lr: 2.3438e-06
Epoch 55/100
75/75 [=====] - 1s 11ms/step - loss: 0.3653 - accuracy:
0.9975 - val_loss: 0.4717 - val_accuracy: 0.9581 - lr: 2.3438e-06
Epoch 56/100
75/75 [=====] - 1s 12ms/step - loss: 0.3658 - accuracy:
0.9962 - val_loss: 0.4586 - val_accuracy: 0.9631 - lr: 2.3438e-06
Epoch 57/100
75/75 [=====] - 1s 13ms/step - loss: 0.3624 - accuracy:
0.9979 - val_loss: 0.4554 - val_accuracy: 0.9606 - lr: 2.3438e-06
Epoch 58/100
75/75 [=====] - 1s 10ms/step - loss: 0.3620 - accuracy:
0.9979 - val_loss: 0.5061 - val_accuracy: 0.9475 - lr: 2.3438e-06
Epoch 59/100
75/75 [=====] - 1s 12ms/step - loss: 0.3633 - accuracy:
0.9975 - val_loss: 0.4709 - val_accuracy: 0.9550 - lr: 2.3438e-06
Epoch 60/100
75/75 [=====] - 1s 12ms/step - loss: 0.3617 - accuracy:
0.9979 - val_loss: 0.4578 - val_accuracy: 0.9588 - lr: 2.3438e-06
Epoch 61/100
75/75 [=====] - ETA: 0s - loss: 0.3633 - accuracy: 0.997
9
Epoch 61: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
75/75 [=====] - 1s 11ms/step - loss: 0.3633 - accuracy:
0.9979 - val_loss: 0.4694 - val_accuracy: 0.9581 - lr: 2.3438e-06
Epoch 62/100
75/75 [=====] - 1s 10ms/step - loss: 0.3586 - accuracy:
0.9996 - val_loss: 0.4627 - val_accuracy: 0.9606 - lr: 1.1719e-06
Epoch 63/100
75/75 [=====] - 1s 11ms/step - loss: 0.3591 - accuracy:
0.9979 - val_loss: 0.4609 - val_accuracy: 0.9594 - lr: 1.1719e-06
Epoch 64/100
75/75 [=====] - 1s 11ms/step - loss: 0.3603 - accuracy:
0.9992 - val_loss: 0.4554 - val_accuracy: 0.9581 - lr: 1.1719e-06
Epoch 65/100
75/75 [=====] - 1s 10ms/step - loss: 0.3590 - accuracy:

0.9996 - val_loss: 0.4528 - val_accuracy: 0.9606 - lr: 1.1719e-06
Epoch 66/100
75/75 [=====] - 1s 12ms/step - loss: 0.3573 - accuracy:
0.9975 - val_loss: 0.4529 - val_accuracy: 0.9581 - lr: 1.1719e-06
Epoch 67/100
75/75 [=====] - 1s 11ms/step - loss: 0.3595 - accuracy:
0.9983 - val_loss: 0.4523 - val_accuracy: 0.9619 - lr: 1.1719e-06
Epoch 68/100
75/75 [=====] - 1s 11ms/step - loss: 0.3585 - accuracy:
0.9992 - val_loss: 0.4534 - val_accuracy: 0.9613 - lr: 1.1719e-06
[100, 40, 0.2]
52.5%
Epoch 1/100
80/80 [=====] - 2s 16ms/step - loss: 3.0921 - accuracy:
0.1713 - val_loss: 3.4911 - val_accuracy: 0.0988 - lr: 3.0000e-04
Epoch 2/100
80/80 [=====] - 1s 10ms/step - loss: 2.5099 - accuracy:
0.2916 - val_loss: 3.1655 - val_accuracy: 0.0988 - lr: 3.0000e-04
Epoch 3/100
80/80 [=====] - 1s 9ms/step - loss: 2.0796 - accuracy: 0
.4187 - val_loss: 3.2662 - val_accuracy: 0.0938 - lr: 3.0000e-04
Epoch 4/100
80/80 [=====] - 1s 10ms/step - loss: 1.6432 - accuracy:
0.5872 - val_loss: 5.0054 - val_accuracy: 0.0938 - lr: 3.0000e-04
Epoch 5/100
80/80 [=====] - 1s 10ms/step - loss: 1.2806 - accuracy:
0.7387 - val_loss: 7.2581 - val_accuracy: 0.0938 - lr: 3.0000e-04
Epoch 6/100
80/80 [=====] - ETA: 0s - loss: 0.9494 - accuracy: 0.883
1
Epoch 6: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
80/80 [=====] - 1s 9ms/step - loss: 0.9494 - accuracy: 0
.8831 - val_loss: 8.6983 - val_accuracy: 0.0938 - lr: 3.0000e-04
Epoch 7/100
80/80 [=====] - 1s 10ms/step - loss: 0.7508 - accuracy:
0.9391 - val_loss: 7.0023 - val_accuracy: 0.1950 - lr: 1.5000e-04
Epoch 8/100
80/80 [=====] - 1s 10ms/step - loss: 0.6710 - accuracy:
0.9631 - val_loss: 4.0424 - val_accuracy: 0.1762 - lr: 1.5000e-04
Epoch 9/100
80/80 [=====] - 1s 10ms/step - loss: 0.6114 - accuracy:
0.9778 - val_loss: 3.8715 - val_accuracy: 0.1937 - lr: 1.5000e-04
Epoch 10/100
80/80 [=====] - 1s 10ms/step - loss: 0.5650 - accuracy:
0.9841 - val_loss: 2.1095 - val_accuracy: 0.4200 - lr: 1.5000e-04
Epoch 11/100
80/80 [=====] - 1s 10ms/step - loss: 0.5340 - accuracy:
0.9828 - val_loss: 0.8665 - val_accuracy: 0.8300 - lr: 1.5000e-04
Epoch 12/100
80/80 [=====] - 1s 10ms/step - loss: 0.5066 - accuracy:
0.9875 - val_loss: 3.1439 - val_accuracy: 0.3225 - lr: 1.5000e-04
Epoch 13/100
80/80 [=====] - 1s 9ms/step - loss: 0.4795 - accuracy: 0
.9922 - val_loss: 3.6217 - val_accuracy: 0.4050 - lr: 1.5000e-04
Epoch 14/100
80/80 [=====] - 1s 9ms/step - loss: 0.4517 - accuracy: 0
.9953 - val_loss: 1.7582 - val_accuracy: 0.6125 - lr: 1.5000e-04
Epoch 15/100
76/80 [=====>..] - ETA: 0s - loss: 0.4294 - accuracy: 0.995
7

Epoch 15: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
80/80 [=====] - 1s 9ms/step - loss: 0.4297 - accuracy: 0.9956 - val_loss: 1.5612 - val_accuracy: 0.5550 - lr: 1.5000e-04
Epoch 16/100
80/80 [=====] - 1s 10ms/step - loss: 0.4085 - accuracy: 0.9978 - val_loss: 0.6378 - val_accuracy: 0.9225 - lr: 7.5000e-05
Epoch 17/100
80/80 [=====] - 1s 10ms/step - loss: 0.4004 - accuracy: 0.9978 - val_loss: 0.8695 - val_accuracy: 0.7937 - lr: 7.5000e-05
Epoch 18/100
80/80 [=====] - 1s 10ms/step - loss: 0.3896 - accuracy: 0.9978 - val_loss: 1.2341 - val_accuracy: 0.5888 - lr: 7.5000e-05
Epoch 19/100
80/80 [=====] - 1s 10ms/step - loss: 0.3833 - accuracy: 0.9978 - val_loss: 0.8931 - val_accuracy: 0.7575 - lr: 7.5000e-05
Epoch 20/100
76/80 [=====>..] - ETA: 0s - loss: 0.3702 - accuracy: 0.9987
Epoch 20: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
80/80 [=====] - 1s 10ms/step - loss: 0.3699 - accuracy: 0.9987 - val_loss: 1.5731 - val_accuracy: 0.5863 - lr: 7.5000e-05
Epoch 21/100
80/80 [=====] - 1s 10ms/step - loss: 0.3632 - accuracy: 0.9997 - val_loss: 0.4944 - val_accuracy: 0.9613 - lr: 3.7500e-05
Epoch 22/100
80/80 [=====] - 1s 11ms/step - loss: 0.3600 - accuracy: 0.9994 - val_loss: 0.4206 - val_accuracy: 0.9737 - lr: 3.7500e-05
Epoch 23/100
80/80 [=====] - 1s 10ms/step - loss: 0.3536 - accuracy: 0.9984 - val_loss: 0.8089 - val_accuracy: 0.8213 - lr: 3.7500e-05
Epoch 24/100
80/80 [=====] - 1s 10ms/step - loss: 0.3521 - accuracy: 0.9987 - val_loss: 0.6616 - val_accuracy: 0.8825 - lr: 3.7500e-05
Epoch 25/100
80/80 [=====] - 1s 10ms/step - loss: 0.3484 - accuracy: 0.9994 - val_loss: 1.2922 - val_accuracy: 0.6112 - lr: 3.7500e-05
Epoch 26/100
80/80 [=====] - 1s 11ms/step - loss: 0.3407 - accuracy: 1.0000 - val_loss: 0.4027 - val_accuracy: 0.9762 - lr: 3.7500e-05
Epoch 27/100
80/80 [=====] - 1s 10ms/step - loss: 0.3384 - accuracy: 0.9997 - val_loss: 1.8564 - val_accuracy: 0.4750 - lr: 3.7500e-05
Epoch 28/100
80/80 [=====] - 1s 10ms/step - loss: 0.3361 - accuracy: 0.9987 - val_loss: 0.4182 - val_accuracy: 0.9775 - lr: 3.7500e-05
Epoch 29/100
80/80 [=====] - 1s 10ms/step - loss: 0.3296 - accuracy: 0.9994 - val_loss: 0.6569 - val_accuracy: 0.8863 - lr: 3.7500e-05
Epoch 30/100
80/80 [=====] - 1s 10ms/step - loss: 0.3251 - accuracy: 0.9997 - val_loss: 0.3929 - val_accuracy: 0.9762 - lr: 3.7500e-05
Epoch 31/100
80/80 [=====] - 1s 10ms/step - loss: 0.3225 - accuracy: 0.9994 - val_loss: 0.7177 - val_accuracy: 0.8388 - lr: 3.7500e-05
Epoch 32/100
80/80 [=====] - 1s 10ms/step - loss: 0.3183 - accuracy: 0.9994 - val_loss: 0.3782 - val_accuracy: 0.9825 - lr: 3.7500e-05
Epoch 33/100
80/80 [=====] - 1s 9ms/step - loss: 0.3129 - accuracy: 0.9997 - val_loss: 0.5807 - val_accuracy: 0.9150 - lr: 3.7500e-05

Epoch 34/100
80/80 [=====] - 1s 9ms/step - loss: 0.3075 - accuracy: 1.0000 - val_loss: 0.4492 - val_accuracy: 0.9538 - lr: 3.7500e-05

Epoch 35/100
80/80 [=====] - 1s 9ms/step - loss: 0.3053 - accuracy: 1.0000 - val_loss: 0.4982 - val_accuracy: 0.9413 - lr: 3.7500e-05

Epoch 36/100
77/80 [=====>..] - ETA: 0s - loss: 0.3000 - accuracy: 0.9997

Epoch 36: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
80/80 [=====] - 1s 10ms/step - loss: 0.2999 - accuracy: 0.9997 - val_loss: 0.4496 - val_accuracy: 0.9625 - lr: 3.7500e-05

Epoch 37/100
80/80 [=====] - 1s 10ms/step - loss: 0.2986 - accuracy: 1.0000 - val_loss: 0.3487 - val_accuracy: 0.9762 - lr: 1.8750e-05

Epoch 38/100
80/80 [=====] - 1s 10ms/step - loss: 0.2934 - accuracy: 0.9997 - val_loss: 0.3397 - val_accuracy: 0.9825 - lr: 1.8750e-05

Epoch 39/100
80/80 [=====] - 1s 10ms/step - loss: 0.2926 - accuracy: 1.0000 - val_loss: 0.3610 - val_accuracy: 0.9750 - lr: 1.8750e-05

Epoch 40/100
80/80 [=====] - 1s 11ms/step - loss: 0.2897 - accuracy: 1.0000 - val_loss: 0.3335 - val_accuracy: 0.9837 - lr: 1.8750e-05

Epoch 41/100
80/80 [=====] - 1s 10ms/step - loss: 0.2883 - accuracy: 1.0000 - val_loss: 0.3807 - val_accuracy: 0.9750 - lr: 1.8750e-05

Epoch 42/100
80/80 [=====] - 1s 10ms/step - loss: 0.2856 - accuracy: 0.9997 - val_loss: 0.3382 - val_accuracy: 0.9775 - lr: 1.8750e-05

Epoch 43/100
80/80 [=====] - 1s 10ms/step - loss: 0.2822 - accuracy: 1.0000 - val_loss: 0.3448 - val_accuracy: 0.9725 - lr: 1.8750e-05

Epoch 44/100
76/80 [=====>..] - ETA: 0s - loss: 0.2810 - accuracy: 0.9997

Epoch 44: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
80/80 [=====] - 1s 10ms/step - loss: 0.2809 - accuracy: 0.9997 - val_loss: 0.3566 - val_accuracy: 0.9775 - lr: 1.8750e-05

Epoch 45/100
80/80 [=====] - 1s 9ms/step - loss: 0.2774 - accuracy: 1.0000 - val_loss: 0.3170 - val_accuracy: 0.9850 - lr: 9.3750e-06

Epoch 46/100
80/80 [=====] - 1s 10ms/step - loss: 0.2779 - accuracy: 0.9997 - val_loss: 0.3301 - val_accuracy: 0.9775 - lr: 9.3750e-06

Epoch 47/100
80/80 [=====] - 1s 10ms/step - loss: 0.2762 - accuracy: 1.0000 - val_loss: 0.3134 - val_accuracy: 0.9825 - lr: 9.3750e-06

Epoch 48/100
80/80 [=====] - 1s 10ms/step - loss: 0.2748 - accuracy: 1.0000 - val_loss: 0.3291 - val_accuracy: 0.9812 - lr: 9.3750e-06

Epoch 49/100
80/80 [=====] - 1s 10ms/step - loss: 0.2733 - accuracy: 0.9997 - val_loss: 0.3888 - val_accuracy: 0.9775 - lr: 9.3750e-06

Epoch 50/100
80/80 [=====] - 1s 10ms/step - loss: 0.2735 - accuracy: 0.9997 - val_loss: 0.3192 - val_accuracy: 0.9812 - lr: 9.3750e-06

Epoch 51/100
80/80 [=====] - 1s 10ms/step - loss: 0.2705 - accuracy: 1.0000 - val_loss: 0.3116 - val_accuracy: 0.9787 - lr: 9.3750e-06

Epoch 52/100
80/80 [=====] - 1s 10ms/step - loss: 0.2694 - accuracy: 0.9997 - val_loss: 0.3117 - val_accuracy: 0.9812 - lr: 9.3750e-06
Epoch 53/100
80/80 [=====] - 1s 9ms/step - loss: 0.2673 - accuracy: 1.0000 - val_loss: 0.3025 - val_accuracy: 0.9800 - lr: 9.3750e-06
Epoch 54/100
80/80 [=====] - 1s 9ms/step - loss: 0.2654 - accuracy: 1.0000 - val_loss: 0.3160 - val_accuracy: 0.9812 - lr: 9.3750e-06
Epoch 55/100
80/80 [=====] - 1s 10ms/step - loss: 0.2642 - accuracy: 1.0000 - val_loss: 0.3008 - val_accuracy: 0.9837 - lr: 9.3750e-06
Epoch 56/100
80/80 [=====] - 1s 10ms/step - loss: 0.2632 - accuracy: 1.0000 - val_loss: 0.3386 - val_accuracy: 0.9775 - lr: 9.3750e-06
Epoch 57/100
80/80 [=====] - 1s 10ms/step - loss: 0.2610 - accuracy: 1.0000 - val_loss: 0.3086 - val_accuracy: 0.9787 - lr: 9.3750e-06
[100, 40, 0.25]
55.0%
Epoch 1/100
75/75 [=====] - 2s 16ms/step - loss: 3.1138 - accuracy: 0.1570 - val_loss: 3.6246 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 2/100
75/75 [=====] - 1s 10ms/step - loss: 2.5491 - accuracy: 0.2873 - val_loss: 4.7652 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 3/100
75/75 [=====] - 1s 10ms/step - loss: 2.2453 - accuracy: 0.3510 - val_loss: 6.2006 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 4/100
75/75 [=====] - 1s 9ms/step - loss: 1.8855 - accuracy: 0.4950 - val_loss: 8.3785 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 5/100
75/75 [=====] - ETA: 0s - loss: 1.5133 - accuracy: 0.6433
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
75/75 [=====] - 1s 10ms/step - loss: 1.5133 - accuracy: 0.6433 - val_loss: 10.1729 - val_accuracy: 0.1340 - lr: 3.0000e-04
Epoch 6/100
75/75 [=====] - 1s 10ms/step - loss: 1.1749 - accuracy: 0.7770 - val_loss: 10.0957 - val_accuracy: 0.1850 - lr: 1.5000e-04
Epoch 7/100
75/75 [=====] - 1s 11ms/step - loss: 0.9858 - accuracy: 0.8523 - val_loss: 8.5979 - val_accuracy: 0.2650 - lr: 1.5000e-04
Epoch 8/100
75/75 [=====] - 1s 10ms/step - loss: 0.8510 - accuracy: 0.9020 - val_loss: 8.0785 - val_accuracy: 0.2480 - lr: 1.5000e-04
Epoch 9/100
74/75 [=====>.] - ETA: 0s - loss: 0.7546 - accuracy: 0.9365
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
75/75 [=====] - 1s 11ms/step - loss: 0.7552 - accuracy: 0.9357 - val_loss: 9.0535 - val_accuracy: 0.1460 - lr: 1.5000e-04
Epoch 10/100
75/75 [=====] - 1s 10ms/step - loss: 0.6772 - accuracy: 0.9660 - val_loss: 6.0070 - val_accuracy: 0.3550 - lr: 7.5000e-05
Epoch 11/100
75/75 [=====] - 1s 10ms/step - loss: 0.6276 - accuracy: 0.9757 - val_loss: 5.9209 - val_accuracy: 0.3920 - lr: 7.5000e-05
Epoch 12/100

75/75 [=====] - 1s 10ms/step - loss: 0.6112 - accuracy: 0.9757 - val_loss: 3.4280 - val_accuracy: 0.4960 - lr: 7.5000e-05
Epoch 13/100
75/75 [=====] - 1s 10ms/step - loss: 0.5846 - accuracy: 0.9833 - val_loss: 1.6943 - val_accuracy: 0.5720 - lr: 7.5000e-05
Epoch 14/100
75/75 [=====] - 1s 10ms/step - loss: 0.5637 - accuracy: 0.9847 - val_loss: 2.2741 - val_accuracy: 0.5000 - lr: 7.5000e-05
Epoch 15/100
75/75 [=====] - 1s 10ms/step - loss: 0.5433 - accuracy: 0.9847 - val_loss: 0.7552 - val_accuracy: 0.9080 - lr: 7.5000e-05
Epoch 16/100
75/75 [=====] - 1s 10ms/step - loss: 0.5235 - accuracy: 0.9910 - val_loss: 1.4885 - val_accuracy: 0.6570 - lr: 7.5000e-05
Epoch 17/100
75/75 [=====] - 1s 10ms/step - loss: 0.5017 - accuracy: 0.9940 - val_loss: 1.2994 - val_accuracy: 0.6640 - lr: 7.5000e-05
Epoch 18/100
75/75 [=====] - 1s 10ms/step - loss: 0.4934 - accuracy: 0.9923 - val_loss: 3.6517 - val_accuracy: 0.3420 - lr: 7.5000e-05
Epoch 19/100
73/75 [=====>.] - ETA: 0s - loss: 0.4768 - accuracy: 0.9945
Epoch 19: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
75/75 [=====] - 1s 11ms/step - loss: 0.4766 - accuracy: 0.9947 - val_loss: 1.6843 - val_accuracy: 0.6800 - lr: 7.5000e-05
Epoch 20/100
75/75 [=====] - 1s 11ms/step - loss: 0.4652 - accuracy: 0.9947 - val_loss: 2.4398 - val_accuracy: 0.5420 - lr: 3.7500e-05
Epoch 21/100
75/75 [=====] - 1s 10ms/step - loss: 0.4590 - accuracy: 0.9953 - val_loss: 1.0867 - val_accuracy: 0.7450 - lr: 3.7500e-05
Epoch 22/100
75/75 [=====] - 1s 11ms/step - loss: 0.4513 - accuracy: 0.9980 - val_loss: 0.6374 - val_accuracy: 0.9240 - lr: 3.7500e-05
Epoch 23/100
75/75 [=====] - 1s 10ms/step - loss: 0.4479 - accuracy: 0.9980 - val_loss: 0.7017 - val_accuracy: 0.9070 - lr: 3.7500e-05
Epoch 24/100
75/75 [=====] - 1s 11ms/step - loss: 0.4370 - accuracy: 0.9973 - val_loss: 2.1879 - val_accuracy: 0.5570 - lr: 3.7500e-05
Epoch 25/100
75/75 [=====] - 1s 10ms/step - loss: 0.4319 - accuracy: 0.9987 - val_loss: 0.8441 - val_accuracy: 0.8440 - lr: 3.7500e-05
Epoch 26/100
75/75 [=====] - 1s 10ms/step - loss: 0.4302 - accuracy: 0.9980 - val_loss: 0.6012 - val_accuracy: 0.9400 - lr: 3.7500e-05
Epoch 27/100
75/75 [=====] - 1s 10ms/step - loss: 0.4246 - accuracy: 0.9987 - val_loss: 0.5582 - val_accuracy: 0.9620 - lr: 3.7500e-05
Epoch 28/100
75/75 [=====] - 1s 10ms/step - loss: 0.4168 - accuracy: 0.9980 - val_loss: 0.7203 - val_accuracy: 0.8770 - lr: 3.7500e-05
Epoch 29/100
75/75 [=====] - 1s 11ms/step - loss: 0.4140 - accuracy: 0.9983 - val_loss: 0.7736 - val_accuracy: 0.8600 - lr: 3.7500e-05
Epoch 30/100
75/75 [=====] - 1s 10ms/step - loss: 0.4061 - accuracy: 0.9990 - val_loss: 2.8514 - val_accuracy: 0.4690 - lr: 3.7500e-05
Epoch 31/100

71/75 [=====>..] - ETA: 0s - loss: 0.4017 - accuracy: 0.9996
Epoch 31: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
75/75 [=====] - 1s 10ms/step - loss: 0.4009 - accuracy: 0.9997 - val_loss: 1.7768 - val_accuracy: 0.5090 - lr: 3.7500e-05
Epoch 32/100
75/75 [=====] - 1s 10ms/step - loss: 0.3970 - accuracy: 0.9993 - val_loss: 0.6085 - val_accuracy: 0.9250 - lr: 1.8750e-05
Epoch 33/100
75/75 [=====] - 1s 11ms/step - loss: 0.3912 - accuracy: 0.9993 - val_loss: 0.4619 - val_accuracy: 0.9840 - lr: 1.8750e-05
Epoch 34/100
75/75 [=====] - 1s 11ms/step - loss: 0.3919 - accuracy: 0.9993 - val_loss: 0.5668 - val_accuracy: 0.9410 - lr: 1.8750e-05
Epoch 35/100
75/75 [=====] - 1s 10ms/step - loss: 0.3881 - accuracy: 0.9990 - val_loss: 0.6587 - val_accuracy: 0.8970 - lr: 1.8750e-05
Epoch 36/100
75/75 [=====] - 1s 10ms/step - loss: 0.3850 - accuracy: 0.9993 - val_loss: 0.4818 - val_accuracy: 0.9680 - lr: 1.8750e-05
Epoch 37/100
70/75 [=====>..] - ETA: 0s - loss: 0.3830 - accuracy: 0.9993
Epoch 37: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
75/75 [=====] - 1s 9ms/step - loss: 0.3830 - accuracy: 0.9993 - val_loss: 0.4671 - val_accuracy: 0.9710 - lr: 1.8750e-05
Epoch 38/100
75/75 [=====] - 1s 9ms/step - loss: 0.3797 - accuracy: 0.9990 - val_loss: 0.4291 - val_accuracy: 0.9780 - lr: 9.3750e-06
Epoch 39/100
75/75 [=====] - 1s 9ms/step - loss: 0.3806 - accuracy: 0.9990 - val_loss: 0.4299 - val_accuracy: 0.9780 - lr: 9.3750e-06
Epoch 40/100
75/75 [=====] - 1s 9ms/step - loss: 0.3778 - accuracy: 0.9993 - val_loss: 0.4766 - val_accuracy: 0.9690 - lr: 9.3750e-06
Epoch 41/100
75/75 [=====] - 1s 10ms/step - loss: 0.3790 - accuracy: 0.9990 - val_loss: 0.4667 - val_accuracy: 0.9730 - lr: 9.3750e-06
Epoch 42/100
75/75 [=====] - 1s 9ms/step - loss: 0.3747 - accuracy: 0.9990 - val_loss: 0.4247 - val_accuracy: 0.9830 - lr: 9.3750e-06
Epoch 43/100
75/75 [=====] - 1s 11ms/step - loss: 0.3731 - accuracy: 0.9997 - val_loss: 0.4345 - val_accuracy: 0.9800 - lr: 9.3750e-06
Epoch 44/100
75/75 [=====] - 1s 10ms/step - loss: 0.3707 - accuracy: 0.9997 - val_loss: 0.4293 - val_accuracy: 0.9780 - lr: 9.3750e-06
Epoch 45/100
75/75 [=====] - 1s 10ms/step - loss: 0.3717 - accuracy: 0.9990 - val_loss: 0.4118 - val_accuracy: 0.9870 - lr: 9.3750e-06
Epoch 46/100
75/75 [=====] - 1s 11ms/step - loss: 0.3663 - accuracy: 0.9997 - val_loss: 0.4038 - val_accuracy: 0.9870 - lr: 9.3750e-06
Epoch 47/100
75/75 [=====] - 1s 10ms/step - loss: 0.3659 - accuracy: 1.0000 - val_loss: 0.4468 - val_accuracy: 0.9750 - lr: 9.3750e-06
Epoch 48/100
75/75 [=====] - 1s 10ms/step - loss: 0.3660 - accuracy: 0.9993 - val_loss: 0.5223 - val_accuracy: 0.9490 - lr: 9.3750e-06
Epoch 49/100

75/75 [=====] - 1s 10ms/step - loss: 0.3646 - accuracy: 0.9997 - val_loss: 0.4140 - val_accuracy: 0.9800 - lr: 9.3750e-06
Epoch 50/100
75/75 [=====] - 1s 10ms/step - loss: 0.3614 - accuracy: 0.9993 - val_loss: 0.3916 - val_accuracy: 0.9880 - lr: 9.3750e-06
Epoch 51/100
75/75 [=====] - 1s 11ms/step - loss: 0.3611 - accuracy: 0.9997 - val_loss: 0.4354 - val_accuracy: 0.9690 - lr: 9.3750e-06
Epoch 52/100
75/75 [=====] - 1s 11ms/step - loss: 0.3587 - accuracy: 1.0000 - val_loss: 0.3993 - val_accuracy: 0.9910 - lr: 9.3750e-06
Epoch 53/100
75/75 [=====] - 1s 10ms/step - loss: 0.3583 - accuracy: 1.0000 - val_loss: 0.4530 - val_accuracy: 0.9690 - lr: 9.3750e-06
Epoch 54/100
71/75 [=====>..] - ETA: 0s - loss: 0.3552 - accuracy: 0.9982
Epoch 54: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
75/75 [=====] - 1s 10ms/step - loss: 0.3553 - accuracy: 0.9983 - val_loss: 0.3989 - val_accuracy: 0.9880 - lr: 9.3750e-06
Epoch 55/100
75/75 [=====] - 1s 10ms/step - loss: 0.3541 - accuracy: 0.9997 - val_loss: 0.4028 - val_accuracy: 0.9800 - lr: 4.6875e-06
Epoch 56/100
75/75 [=====] - 1s 10ms/step - loss: 0.3506 - accuracy: 0.9997 - val_loss: 0.3828 - val_accuracy: 0.9910 - lr: 4.6875e-06
Epoch 57/100
75/75 [=====] - 1s 10ms/step - loss: 0.3508 - accuracy: 0.9997 - val_loss: 0.3840 - val_accuracy: 0.9890 - lr: 4.6875e-06
Epoch 58/100
75/75 [=====] - 1s 10ms/step - loss: 0.3491 - accuracy: 1.0000 - val_loss: 0.3851 - val_accuracy: 0.9850 - lr: 4.6875e-06
Epoch 59/100
75/75 [=====] - 1s 9ms/step - loss: 0.3509 - accuracy: 1.0000 - val_loss: 0.3852 - val_accuracy: 0.9880 - lr: 4.6875e-06
Epoch 60/100
73/75 [=====>..] - ETA: 0s - loss: 0.3478 - accuracy: 0.9997
Epoch 60: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
75/75 [=====] - 1s 10ms/step - loss: 0.3478 - accuracy: 0.9997 - val_loss: 0.3833 - val_accuracy: 0.9900 - lr: 4.6875e-06
Epoch 61/100
75/75 [=====] - 1s 10ms/step - loss: 0.3474 - accuracy: 1.0000 - val_loss: 0.3753 - val_accuracy: 0.9920 - lr: 2.3438e-06
Epoch 62/100
75/75 [=====] - 1s 10ms/step - loss: 0.3457 - accuracy: 1.0000 - val_loss: 0.3754 - val_accuracy: 0.9920 - lr: 2.3438e-06
Epoch 63/100
75/75 [=====] - 1s 10ms/step - loss: 0.3465 - accuracy: 1.0000 - val_loss: 0.3850 - val_accuracy: 0.9860 - lr: 2.3438e-06
Epoch 64/100
75/75 [=====] - 1s 10ms/step - loss: 0.3446 - accuracy: 0.9997 - val_loss: 0.3868 - val_accuracy: 0.9890 - lr: 2.3438e-06
Epoch 65/100
70/75 [=====>..] - ETA: 0s - loss: 0.3449 - accuracy: 0.9993
Epoch 65: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
75/75 [=====] - 1s 10ms/step - loss: 0.3451 - accuracy: 0.9993 - val_loss: 0.3759 - val_accuracy: 0.9900 - lr: 2.3438e-06
Epoch 66/100

75/75 [=====] - 1s 10ms/step - loss: 0.3425 - accuracy: 0.9993 - val_loss: 0.3743 - val_accuracy: 0.9870 - lr: 1.1719e-06
Epoch 67/100
75/75 [=====] - 1s 10ms/step - loss: 0.3422 - accuracy: 1.0000 - val_loss: 0.3731 - val_accuracy: 0.9900 - lr: 1.1719e-06
Epoch 68/100
75/75 [=====] - 1s 11ms/step - loss: 0.3431 - accuracy: 0.9993 - val_loss: 0.3730 - val_accuracy: 0.9880 - lr: 1.1719e-06
Epoch 69/100
75/75 [=====] - 1s 11ms/step - loss: 0.3446 - accuracy: 0.9990 - val_loss: 0.3734 - val_accuracy: 0.9890 - lr: 1.1719e-06
Epoch 70/100
75/75 [=====] - 1s 10ms/step - loss: 0.3445 - accuracy: 0.9993 - val_loss: 0.3744 - val_accuracy: 0.9890 - lr: 1.1719e-06
Epoch 71/100
75/75 [=====] - ETA: 0s - loss: 0.3433 - accuracy: 0.9997
Epoch 71: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.
75/75 [=====] - 1s 10ms/step - loss: 0.3433 - accuracy: 0.9997 - val_loss: 0.3749 - val_accuracy: 0.9890 - lr: 1.1719e-06
Epoch 72/100
75/75 [=====] - 1s 11ms/step - loss: 0.3426 - accuracy: 1.0000 - val_loss: 0.3750 - val_accuracy: 0.9880 - lr: 5.8594e-07
Epoch 73/100
75/75 [=====] - 1s 10ms/step - loss: 0.3436 - accuracy: 1.0000 - val_loss: 0.3740 - val_accuracy: 0.9880 - lr: 5.8594e-07
[100, 40, 0.30000000000000004]
57.5%
Epoch 1/100
70/70 [=====] - 2s 17ms/step - loss: 3.1541 - accuracy: 0.1686 - val_loss: 3.7124 - val_accuracy: 0.1017 - lr: 3.0000e-04
Epoch 2/100
70/70 [=====] - 1s 10ms/step - loss: 2.4903 - accuracy: 0.2904 - val_loss: 4.9365 - val_accuracy: 0.0983 - lr: 3.0000e-04
Epoch 3/100
70/70 [=====] - 1s 9ms/step - loss: 2.1194 - accuracy: 0.4164 - val_loss: 6.3150 - val_accuracy: 0.0983 - lr: 3.0000e-04
Epoch 4/100
70/70 [=====] - 1s 10ms/step - loss: 1.6877 - accuracy: 0.5864 - val_loss: 8.6979 - val_accuracy: 0.1183 - lr: 3.0000e-04
Epoch 5/100
64/70 [=====>...] - ETA: 0s - loss: 1.2999 - accuracy: 0.7414
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
70/70 [=====] - 1s 9ms/step - loss: 1.2824 - accuracy: 0.7475 - val_loss: 12.2959 - val_accuracy: 0.1008 - lr: 3.0000e-04
Epoch 6/100
70/70 [=====] - 1s 10ms/step - loss: 0.9600 - accuracy: 0.8750 - val_loss: 10.8904 - val_accuracy: 0.1400 - lr: 1.5000e-04
Epoch 7/100
70/70 [=====] - 1s 11ms/step - loss: 0.8247 - accuracy: 0.9207 - val_loss: 11.8367 - val_accuracy: 0.1950 - lr: 1.5000e-04
Epoch 8/100
70/70 [=====] - 1s 10ms/step - loss: 0.7223 - accuracy: 0.9532 - val_loss: 8.9077 - val_accuracy: 0.1025 - lr: 1.5000e-04
Epoch 9/100
69/70 [=====>.] - ETA: 0s - loss: 0.6528 - accuracy: 0.9714
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
70/70 [=====] - 1s 11ms/step - loss: 0.6523 - accuracy:

0.9718 - val_loss: 11.6138 - val_accuracy: 0.2300 - lr: 1.5000e-04
Epoch 10/100
70/70 [=====] - 1s 11ms/step - loss: 0.6078 - accuracy:
0.9800 - val_loss: 6.6073 - val_accuracy: 0.1983 - lr: 7.5000e-05
Epoch 11/100
70/70 [=====] - 1s 10ms/step - loss: 0.5870 - accuracy:
0.9850 - val_loss: 2.0648 - val_accuracy: 0.4183 - lr: 7.5000e-05
Epoch 12/100
70/70 [=====] - 1s 10ms/step - loss: 0.5630 - accuracy:
0.9868 - val_loss: 1.9032 - val_accuracy: 0.6267 - lr: 7.5000e-05
Epoch 13/100
70/70 [=====] - 1s 10ms/step - loss: 0.5510 - accuracy:
0.9871 - val_loss: 1.0147 - val_accuracy: 0.7992 - lr: 7.5000e-05
Epoch 14/100
70/70 [=====] - 1s 10ms/step - loss: 0.5336 - accuracy:
0.9893 - val_loss: 7.0057 - val_accuracy: 0.1075 - lr: 7.5000e-05
Epoch 15/100
70/70 [=====] - 1s 11ms/step - loss: 0.5116 - accuracy:
0.9946 - val_loss: 1.4580 - val_accuracy: 0.6192 - lr: 7.5000e-05
Epoch 16/100
70/70 [=====] - 1s 11ms/step - loss: 0.5043 - accuracy:
0.9929 - val_loss: 5.8528 - val_accuracy: 0.3083 - lr: 7.5000e-05
Epoch 17/100
68/70 [=====>.] - ETA: 0s - loss: 0.4888 - accuracy: 0.994
5
Epoch 17: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
70/70 [=====] - 1s 11ms/step - loss: 0.4896 - accuracy:
0.9943 - val_loss: 10.8517 - val_accuracy: 0.1017 - lr: 7.5000e-05
Epoch 18/100
70/70 [=====] - 1s 11ms/step - loss: 0.4776 - accuracy:
0.9943 - val_loss: 0.6236 - val_accuracy: 0.9467 - lr: 3.7500e-05
Epoch 19/100
70/70 [=====] - 1s 10ms/step - loss: 0.4722 - accuracy:
0.9957 - val_loss: 1.1441 - val_accuracy: 0.7200 - lr: 3.7500e-05
Epoch 20/100
70/70 [=====] - 1s 10ms/step - loss: 0.4660 - accuracy:
0.9964 - val_loss: 0.8341 - val_accuracy: 0.8533 - lr: 3.7500e-05
Epoch 21/100
70/70 [=====] - 1s 11ms/step - loss: 0.4590 - accuracy:
0.9979 - val_loss: 1.7174 - val_accuracy: 0.5767 - lr: 3.7500e-05
Epoch 22/100
69/70 [=====>.] - ETA: 0s - loss: 0.4543 - accuracy: 0.996
4
Epoch 22: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
70/70 [=====] - 1s 11ms/step - loss: 0.4541 - accuracy:
0.9964 - val_loss: 6.1107 - val_accuracy: 0.1275 - lr: 3.7500e-05
Epoch 23/100
70/70 [=====] - 1s 11ms/step - loss: 0.4480 - accuracy:
0.9982 - val_loss: 0.5639 - val_accuracy: 0.9625 - lr: 1.8750e-05
Epoch 24/100
70/70 [=====] - 1s 10ms/step - loss: 0.4453 - accuracy:
0.9986 - val_loss: 0.5655 - val_accuracy: 0.9608 - lr: 1.8750e-05
Epoch 25/100
70/70 [=====] - 1s 10ms/step - loss: 0.4434 - accuracy:
0.9968 - val_loss: 0.5592 - val_accuracy: 0.9575 - lr: 1.8750e-05
Epoch 26/100
70/70 [=====] - 1s 10ms/step - loss: 0.4413 - accuracy:
0.9986 - val_loss: 0.5367 - val_accuracy: 0.9617 - lr: 1.8750e-05
Epoch 27/100
70/70 [=====] - 1s 10ms/step - loss: 0.4370 - accuracy:

0.9989 - val_loss: 0.5276 - val_accuracy: 0.9683 - lr: 1.8750e-05
Epoch 28/100
70/70 [=====] - 1s 10ms/step - loss: 0.4357 - accuracy:
0.9971 - val_loss: 0.5524 - val_accuracy: 0.9508 - lr: 1.8750e-05
Epoch 29/100
70/70 [=====] - 1s 10ms/step - loss: 0.4293 - accuracy:
0.9971 - val_loss: 0.6217 - val_accuracy: 0.9317 - lr: 1.8750e-05
Epoch 30/100
70/70 [=====] - 1s 10ms/step - loss: 0.4299 - accuracy:
0.9982 - val_loss: 1.4955 - val_accuracy: 0.6325 - lr: 1.8750e-05
Epoch 31/100
70/70 [=====] - 1s 11ms/step - loss: 0.4263 - accuracy:
0.9975 - val_loss: 0.5039 - val_accuracy: 0.9667 - lr: 1.8750e-05
Epoch 32/100
70/70 [=====] - 1s 11ms/step - loss: 0.4207 - accuracy:
0.9989 - val_loss: 0.4802 - val_accuracy: 0.9742 - lr: 1.8750e-05
Epoch 33/100
70/70 [=====] - 1s 10ms/step - loss: 0.4204 - accuracy:
0.9989 - val_loss: 0.8025 - val_accuracy: 0.8525 - lr: 1.8750e-05
Epoch 34/100
70/70 [=====] - 1s 10ms/step - loss: 0.4163 - accuracy:
0.9979 - val_loss: 0.4874 - val_accuracy: 0.9700 - lr: 1.8750e-05
Epoch 35/100
70/70 [=====] - 1s 11ms/step - loss: 0.4128 - accuracy:
0.9982 - val_loss: 0.4935 - val_accuracy: 0.9667 - lr: 1.8750e-05
Epoch 36/100
65/70 [=====>...] - ETA: 0s - loss: 0.4105 - accuracy: 0.998
5
Epoch 36: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
70/70 [=====] - 1s 10ms/step - loss: 0.4106 - accuracy:
0.9986 - val_loss: 2.3907 - val_accuracy: 0.4233 - lr: 1.8750e-05
Epoch 37/100
70/70 [=====] - 1s 10ms/step - loss: 0.4073 - accuracy:
0.9986 - val_loss: 0.4830 - val_accuracy: 0.9692 - lr: 9.3750e-06
Epoch 38/100
70/70 [=====] - 1s 10ms/step - loss: 0.4072 - accuracy:
0.9982 - val_loss: 0.4948 - val_accuracy: 0.9725 - lr: 9.3750e-06
Epoch 39/100
70/70 [=====] - 1s 11ms/step - loss: 0.4016 - accuracy:
0.9989 - val_loss: 0.5318 - val_accuracy: 0.9542 - lr: 9.3750e-06
Epoch 40/100
70/70 [=====] - 1s 11ms/step - loss: 0.4031 - accuracy:
0.9986 - val_loss: 0.4611 - val_accuracy: 0.9792 - lr: 9.3750e-06
Epoch 41/100
70/70 [=====] - 1s 11ms/step - loss: 0.4021 - accuracy:
0.9989 - val_loss: 0.7217 - val_accuracy: 0.8800 - lr: 9.3750e-06
Epoch 42/100
70/70 [=====] - 1s 11ms/step - loss: 0.3989 - accuracy:
1.0000 - val_loss: 0.6135 - val_accuracy: 0.9258 - lr: 9.3750e-06
Epoch 43/100
70/70 [=====] - 1s 11ms/step - loss: 0.3992 - accuracy:
0.9993 - val_loss: 0.5126 - val_accuracy: 0.9683 - lr: 9.3750e-06
Epoch 44/100
68/70 [=====>.] - ETA: 0s - loss: 0.3942 - accuracy: 1.000
0
Epoch 44: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
70/70 [=====] - 1s 11ms/step - loss: 0.3940 - accuracy:
1.0000 - val_loss: 0.5319 - val_accuracy: 0.9483 - lr: 9.3750e-06
Epoch 45/100
70/70 [=====] - 1s 11ms/step - loss: 0.3940 - accuracy:

0.9986 - val_loss: 0.4624 - val_accuracy: 0.9742 - lr: 4.6875e-06
Epoch 46/100
70/70 [=====] - 1s 10ms/step - loss: 0.3934 - accuracy:
1.0000 - val_loss: 0.4796 - val_accuracy: 0.9667 - lr: 4.6875e-06
Epoch 47/100
70/70 [=====] - 1s 10ms/step - loss: 0.3919 - accuracy:
0.9996 - val_loss: 0.4460 - val_accuracy: 0.9808 - lr: 4.6875e-06
Epoch 48/100
70/70 [=====] - 1s 10ms/step - loss: 0.3885 - accuracy:
0.9993 - val_loss: 0.4679 - val_accuracy: 0.9742 - lr: 4.6875e-06
Epoch 49/100
70/70 [=====] - 1s 10ms/step - loss: 0.3918 - accuracy:
1.0000 - val_loss: 0.4529 - val_accuracy: 0.9758 - lr: 4.6875e-06
Epoch 50/100
70/70 [=====] - 1s 11ms/step - loss: 0.3869 - accuracy:
1.0000 - val_loss: 0.4818 - val_accuracy: 0.9642 - lr: 4.6875e-06
Epoch 51/100
70/70 [=====] - 1s 10ms/step - loss: 0.3869 - accuracy:
1.0000 - val_loss: 0.4443 - val_accuracy: 0.9783 - lr: 4.6875e-06
Epoch 52/100
70/70 [=====] - 1s 10ms/step - loss: 0.3886 - accuracy:
0.9986 - val_loss: 0.4426 - val_accuracy: 0.9775 - lr: 4.6875e-06
Epoch 53/100
70/70 [=====] - 1s 11ms/step - loss: 0.3857 - accuracy:
0.9996 - val_loss: 0.4433 - val_accuracy: 0.9783 - lr: 4.6875e-06
Epoch 54/100
70/70 [=====] - 1s 12ms/step - loss: 0.3841 - accuracy:
0.9993 - val_loss: 0.5994 - val_accuracy: 0.9308 - lr: 4.6875e-06
Epoch 55/100
70/70 [=====] - 1s 11ms/step - loss: 0.3811 - accuracy:
0.9996 - val_loss: 0.4377 - val_accuracy: 0.9808 - lr: 4.6875e-06
Epoch 56/100
70/70 [=====] - 1s 11ms/step - loss: 0.3820 - accuracy:
0.9996 - val_loss: 0.7051 - val_accuracy: 0.8833 - lr: 4.6875e-06
Epoch 57/100
70/70 [=====] - 1s 11ms/step - loss: 0.3803 - accuracy:
0.9996 - val_loss: 0.4355 - val_accuracy: 0.9808 - lr: 4.6875e-06
Epoch 58/100
70/70 [=====] - 1s 11ms/step - loss: 0.3784 - accuracy:
0.9989 - val_loss: 0.4500 - val_accuracy: 0.9733 - lr: 4.6875e-06
Epoch 59/100
70/70 [=====] - 1s 10ms/step - loss: 0.3784 - accuracy:
0.9993 - val_loss: 0.4401 - val_accuracy: 0.9792 - lr: 4.6875e-06
[100, 40, 0.35000000000000003]
60.0%
Epoch 1/100
65/65 [=====] - 2s 19ms/step - loss: 3.1186 - accuracy:
0.1454 - val_loss: 3.2718 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 2/100
65/65 [=====] - 1s 13ms/step - loss: 2.5670 - accuracy:
0.2447 - val_loss: 3.9343 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 3/100
65/65 [=====] - 1s 12ms/step - loss: 2.1913 - accuracy:
0.3582 - val_loss: 6.3093 - val_accuracy: 0.1028 - lr: 3.0000e-04
Epoch 4/100
65/65 [=====] - 1s 13ms/step - loss: 1.8168 - accuracy:
0.5048 - val_loss: 8.4073 - val_accuracy: 0.1028 - lr: 3.0000e-04
Epoch 5/100
64/65 [=====>.] - ETA: 0s - loss: 1.3837 - accuracy: 0.676
2

Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
65/65 [=====] - 1s 14ms/step - loss: 1.3799 - accuracy: 0.6764 - val_loss: 12.7230 - val_accuracy: 0.1028 - lr: 3.0000e-04
Epoch 6/100
65/65 [=====] - 1s 11ms/step - loss: 1.0362 - accuracy: 0.8311 - val_loss: 13.5029 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 7/100
65/65 [=====] - 1s 11ms/step - loss: 0.8628 - accuracy: 0.9015 - val_loss: 13.9569 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 8/100
65/65 [=====] - 1s 11ms/step - loss: 0.7541 - accuracy: 0.9381 - val_loss: 14.1189 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 9/100
64/65 [=====>.] - ETA: 0s - loss: 0.6870 - accuracy: 0.9469
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
65/65 [=====] - 1s 11ms/step - loss: 0.6866 - accuracy: 0.9469 - val_loss: 12.6106 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 10/100
65/65 [=====] - 1s 13ms/step - loss: 0.6211 - accuracy: 0.9681 - val_loss: 9.6741 - val_accuracy: 0.1970 - lr: 7.5000e-05
Epoch 11/100
65/65 [=====] - 1s 13ms/step - loss: 0.5949 - accuracy: 0.9773 - val_loss: 6.9798 - val_accuracy: 0.2584 - lr: 7.5000e-05
Epoch 12/100
65/65 [=====] - 1s 13ms/step - loss: 0.5644 - accuracy: 0.9800 - val_loss: 6.5420 - val_accuracy: 0.2548 - lr: 7.5000e-05
Epoch 13/100
65/65 [=====] - 1s 15ms/step - loss: 0.5444 - accuracy: 0.9861 - val_loss: 2.9794 - val_accuracy: 0.3919 - lr: 7.5000e-05
Epoch 14/100
65/65 [=====] - 1s 13ms/step - loss: 0.5273 - accuracy: 0.9861 - val_loss: 1.7492 - val_accuracy: 0.4789 - lr: 7.5000e-05
Epoch 15/100
65/65 [=====] - 1s 12ms/step - loss: 0.5066 - accuracy: 0.9888 - val_loss: 1.4555 - val_accuracy: 0.5760 - lr: 7.5000e-05
Epoch 16/100
65/65 [=====] - 1s 12ms/step - loss: 0.4829 - accuracy: 0.9931 - val_loss: 5.5978 - val_accuracy: 0.2548 - lr: 7.5000e-05
Epoch 17/100
65/65 [=====] - 1s 13ms/step - loss: 0.4708 - accuracy: 0.9938 - val_loss: 7.3744 - val_accuracy: 0.1635 - lr: 7.5000e-05
Epoch 18/100
65/65 [=====] - 1s 13ms/step - loss: 0.4498 - accuracy: 0.9946 - val_loss: 4.6718 - val_accuracy: 0.2784 - lr: 7.5000e-05
Epoch 19/100
65/65 [=====] - 1s 13ms/step - loss: 0.4357 - accuracy: 0.9950 - val_loss: 0.6193 - val_accuracy: 0.9322 - lr: 7.5000e-05
Epoch 20/100
65/65 [=====] - 1s 16ms/step - loss: 0.4247 - accuracy: 0.9965 - val_loss: 3.0325 - val_accuracy: 0.4775 - lr: 7.5000e-05
Epoch 21/100
65/65 [=====] - 1s 13ms/step - loss: 0.4101 - accuracy: 0.9969 - val_loss: 0.5761 - val_accuracy: 0.9386 - lr: 7.5000e-05
Epoch 22/100
65/65 [=====] - 1s 12ms/step - loss: 0.4018 - accuracy: 0.9965 - val_loss: 0.5910 - val_accuracy: 0.9315 - lr: 7.5000e-05
Epoch 23/100
65/65 [=====] - 1s 14ms/step - loss: 0.3876 - accuracy: 0.9981 - val_loss: 5.1432 - val_accuracy: 0.2277 - lr: 7.5000e-05

Epoch 24/100
65/65 [=====] - 1s 12ms/step - loss: 0.3820 - accuracy: 0.9985 - val_loss: 4.0445 - val_accuracy: 0.4268 - lr: 7.5000e-05
Epoch 25/100
60/65 [=====>...] - ETA: 0s - loss: 0.3684 - accuracy: 0.9979
Epoch 25: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
65/65 [=====] - 1s 11ms/step - loss: 0.3675 - accuracy: 0.9981 - val_loss: 3.0437 - val_accuracy: 0.2712 - lr: 7.5000e-05
Epoch 26/100
65/65 [=====] - 1s 11ms/step - loss: 0.3639 - accuracy: 0.9985 - val_loss: 0.5236 - val_accuracy: 0.9429 - lr: 3.7500e-05
Epoch 27/100
65/65 [=====] - 1s 11ms/step - loss: 0.3569 - accuracy: 0.9988 - val_loss: 0.6125 - val_accuracy: 0.8958 - lr: 3.7500e-05
Epoch 28/100
65/65 [=====] - 1s 11ms/step - loss: 0.3499 - accuracy: 0.9981 - val_loss: 1.8928 - val_accuracy: 0.5632 - lr: 3.7500e-05
Epoch 29/100
65/65 [=====] - 1s 11ms/step - loss: 0.3446 - accuracy: 0.9992 - val_loss: 2.8626 - val_accuracy: 0.3776 - lr: 3.7500e-05
Epoch 30/100
62/65 [=====>..] - ETA: 0s - loss: 0.3428 - accuracy: 0.9992
Epoch 30: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
65/65 [=====] - 1s 13ms/step - loss: 0.3430 - accuracy: 0.9992 - val_loss: 1.0167 - val_accuracy: 0.7573 - lr: 3.7500e-05
Epoch 31/100
65/65 [=====] - 1s 13ms/step - loss: 0.3375 - accuracy: 0.9988 - val_loss: 0.4991 - val_accuracy: 0.9400 - lr: 1.8750e-05
Epoch 32/100
65/65 [=====] - 1s 12ms/step - loss: 0.3346 - accuracy: 0.9992 - val_loss: 0.4847 - val_accuracy: 0.9486 - lr: 1.8750e-05
Epoch 33/100
65/65 [=====] - 1s 14ms/step - loss: 0.3352 - accuracy: 0.9996 - val_loss: 0.4347 - val_accuracy: 0.9679 - lr: 1.8750e-05
Epoch 34/100
65/65 [=====] - 1s 17ms/step - loss: 0.3316 - accuracy: 0.9992 - val_loss: 0.4665 - val_accuracy: 0.9579 - lr: 1.8750e-05
Epoch 35/100
65/65 [=====] - 1s 13ms/step - loss: 0.3290 - accuracy: 1.0000 - val_loss: 0.5164 - val_accuracy: 0.9358 - lr: 1.8750e-05
Epoch 36/100
65/65 [=====] - 1s 11ms/step - loss: 0.3267 - accuracy: 1.0000 - val_loss: 0.4805 - val_accuracy: 0.9486 - lr: 1.8750e-05
Epoch 37/100
65/65 [=====] - ETA: 0s - loss: 0.3226 - accuracy: 1.0000
Epoch 37: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
65/65 [=====] - 1s 12ms/step - loss: 0.3226 - accuracy: 1.0000 - val_loss: 0.4655 - val_accuracy: 0.9458 - lr: 1.8750e-05
Epoch 38/100
65/65 [=====] - 1s 12ms/step - loss: 0.3229 - accuracy: 0.9996 - val_loss: 0.4032 - val_accuracy: 0.9764 - lr: 9.3750e-06
Epoch 39/100
65/65 [=====] - 1s 13ms/step - loss: 0.3206 - accuracy: 1.0000 - val_loss: 0.3889 - val_accuracy: 0.9807 - lr: 9.3750e-06
Epoch 40/100
65/65 [=====] - 1s 14ms/step - loss: 0.3213 - accuracy: 0.9992 - val_loss: 0.4307 - val_accuracy: 0.9650 - lr: 9.3750e-06

Epoch 41/100
65/65 [=====] - 1s 11ms/step - loss: 0.3202 - accuracy: 1.0000 - val_loss: 0.7563 - val_accuracy: 0.8280 - lr: 9.3750e-06
Epoch 42/100
65/65 [=====] - 1s 13ms/step - loss: 0.3174 - accuracy: 1.0000 - val_loss: 0.4427 - val_accuracy: 0.9600 - lr: 9.3750e-06
Epoch 43/100
65/65 [=====] - 1s 15ms/step - loss: 0.3155 - accuracy: 1.0000 - val_loss: 0.3813 - val_accuracy: 0.9764 - lr: 9.3750e-06
Epoch 44/100
65/65 [=====] - 1s 13ms/step - loss: 0.3169 - accuracy: 0.9992 - val_loss: 0.3819 - val_accuracy: 0.9757 - lr: 9.3750e-06
Epoch 45/100
65/65 [=====] - 1s 10ms/step - loss: 0.3163 - accuracy: 1.0000 - val_loss: 0.4749 - val_accuracy: 0.9415 - lr: 9.3750e-06
Epoch 46/100
65/65 [=====] - 1s 10ms/step - loss: 0.3119 - accuracy: 1.0000 - val_loss: 0.3852 - val_accuracy: 0.9736 - lr: 9.3750e-06
Epoch 47/100
63/65 [=====>.] - ETA: 0s - loss: 0.3131 - accuracy: 0.9996
Epoch 47: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
65/65 [=====] - 1s 10ms/step - loss: 0.3132 - accuracy: 0.9996 - val_loss: 0.4415 - val_accuracy: 0.9529 - lr: 9.3750e-06
Epoch 48/100
65/65 [=====] - 1s 11ms/step - loss: 0.3087 - accuracy: 1.0000 - val_loss: 0.4049 - val_accuracy: 0.9707 - lr: 4.6875e-06
Epoch 49/100
65/65 [=====] - 1s 12ms/step - loss: 0.3107 - accuracy: 0.9996 - val_loss: 0.3864 - val_accuracy: 0.9779 - lr: 4.6875e-06
Epoch 50/100
65/65 [=====] - 1s 14ms/step - loss: 0.3070 - accuracy: 1.0000 - val_loss: 0.3986 - val_accuracy: 0.9693 - lr: 4.6875e-06
Epoch 51/100
65/65 [=====] - 1s 13ms/step - loss: 0.3076 - accuracy: 0.9996 - val_loss: 0.3736 - val_accuracy: 0.9786 - lr: 4.6875e-06
[100, 40, 0.4]
62.5%
Epoch 1/100
60/60 [=====] - 2s 19ms/step - loss: 3.2006 - accuracy: 0.1612 - val_loss: 5.2673 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 2/100
60/60 [=====] - 1s 12ms/step - loss: 2.6277 - accuracy: 0.2646 - val_loss: 6.9135 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 3/100
60/60 [=====] - 1s 12ms/step - loss: 2.3677 - accuracy: 0.3204 - val_loss: 6.0689 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 4/100
60/60 [=====] - 1s 11ms/step - loss: 2.1033 - accuracy: 0.4200 - val_loss: 7.6859 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 5/100
55/60 [=====>...] - ETA: 0s - loss: 1.7377 - accuracy: 0.5632
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
60/60 [=====] - 1s 10ms/step - loss: 1.7193 - accuracy: 0.5671 - val_loss: 9.5043 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 6/100
60/60 [=====] - 1s 11ms/step - loss: 1.3665 - accuracy: 0.6975 - val_loss: 10.1028 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 7/100

60/60 [=====] - 1s 11ms/step - loss: 1.1898 - accuracy: 0.7775 - val_loss: 8.2710 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 8/100
60/60 [=====] - 1s 11ms/step - loss: 1.0410 - accuracy: 0.8267 - val_loss: 8.7589 - val_accuracy: 0.1956 - lr: 1.5000e-04
Epoch 9/100
59/60 [=====>.] - ETA: 0s - loss: 0.9516 - accuracy: 0.8589
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
60/60 [=====] - 1s 11ms/step - loss: 0.9526 - accuracy: 0.8583 - val_loss: 7.0167 - val_accuracy: 0.1200 - lr: 1.5000e-04
Epoch 10/100
60/60 [=====] - 1s 11ms/step - loss: 0.8382 - accuracy: 0.9092 - val_loss: 8.3235 - val_accuracy: 0.2019 - lr: 7.5000e-05
Epoch 11/100
60/60 [=====] - 1s 11ms/step - loss: 0.7943 - accuracy: 0.9100 - val_loss: 4.7884 - val_accuracy: 0.1281 - lr: 7.5000e-05
Epoch 12/100
60/60 [=====] - 1s 11ms/step - loss: 0.7667 - accuracy: 0.9308 - val_loss: 5.5211 - val_accuracy: 0.1706 - lr: 7.5000e-05
Epoch 13/100
60/60 [=====] - 1s 12ms/step - loss: 0.7354 - accuracy: 0.9379 - val_loss: 4.7254 - val_accuracy: 0.1825 - lr: 7.5000e-05
Epoch 14/100
60/60 [=====] - 1s 11ms/step - loss: 0.7020 - accuracy: 0.9504 - val_loss: 3.5490 - val_accuracy: 0.4988 - lr: 7.5000e-05
Epoch 15/100
60/60 [=====] - 1s 11ms/step - loss: 0.6751 - accuracy: 0.9558 - val_loss: 5.6879 - val_accuracy: 0.2144 - lr: 7.5000e-05
Epoch 16/100
60/60 [=====] - 1s 10ms/step - loss: 0.6487 - accuracy: 0.9642 - val_loss: 9.9037 - val_accuracy: 0.1138 - lr: 7.5000e-05
Epoch 17/100
60/60 [=====] - 1s 11ms/step - loss: 0.6311 - accuracy: 0.9625 - val_loss: 3.6759 - val_accuracy: 0.2550 - lr: 7.5000e-05
Epoch 18/100
55/60 [=====>...] - ETA: 0s - loss: 0.6079 - accuracy: 0.9709
Epoch 18: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
60/60 [=====] - 1s 12ms/step - loss: 0.6068 - accuracy: 0.9712 - val_loss: 3.6400 - val_accuracy: 0.5069 - lr: 7.5000e-05
Epoch 19/100
60/60 [=====] - 1s 12ms/step - loss: 0.5903 - accuracy: 0.9704 - val_loss: 3.1940 - val_accuracy: 0.1713 - lr: 3.7500e-05
Epoch 20/100
60/60 [=====] - 1s 11ms/step - loss: 0.5843 - accuracy: 0.9762 - val_loss: 2.6477 - val_accuracy: 0.3119 - lr: 3.7500e-05
Epoch 21/100
60/60 [=====] - 1s 12ms/step - loss: 0.5698 - accuracy: 0.9808 - val_loss: 4.1123 - val_accuracy: 0.3650 - lr: 3.7500e-05
Epoch 22/100
60/60 [=====] - 1s 12ms/step - loss: 0.5551 - accuracy: 0.9837 - val_loss: 1.7814 - val_accuracy: 0.5344 - lr: 3.7500e-05
Epoch 23/100
60/60 [=====] - 1s 12ms/step - loss: 0.5433 - accuracy: 0.9871 - val_loss: 4.9383 - val_accuracy: 0.1956 - lr: 3.7500e-05
Epoch 24/100
60/60 [=====] - 1s 12ms/step - loss: 0.5337 - accuracy: 0.9892 - val_loss: 3.1289 - val_accuracy: 0.3025 - lr: 3.7500e-05
Epoch 25/100

60/60 [=====] - 1s 11ms/step - loss: 0.5312 - accuracy: 0.9871 - val_loss: 3.8550 - val_accuracy: 0.2706 - lr: 3.7500e-05
Epoch 26/100
60/60 [=====] - 1s 13ms/step - loss: 0.5187 - accuracy: 0.9904 - val_loss: 0.9724 - val_accuracy: 0.7981 - lr: 3.7500e-05
Epoch 27/100
60/60 [=====] - 1s 11ms/step - loss: 0.5124 - accuracy: 0.9900 - val_loss: 2.6862 - val_accuracy: 0.4013 - lr: 3.7500e-05
Epoch 28/100
60/60 [=====] - 1s 11ms/step - loss: 0.5098 - accuracy: 0.9904 - val_loss: 0.8887 - val_accuracy: 0.8331 - lr: 3.7500e-05
Epoch 29/100
60/60 [=====] - 1s 11ms/step - loss: 0.4974 - accuracy: 0.9921 - val_loss: 0.8698 - val_accuracy: 0.8300 - lr: 3.7500e-05
Epoch 30/100
60/60 [=====] - 1s 12ms/step - loss: 0.4917 - accuracy: 0.9900 - val_loss: 2.6477 - val_accuracy: 0.5138 - lr: 3.7500e-05
Epoch 31/100
60/60 [=====] - 1s 12ms/step - loss: 0.4878 - accuracy: 0.9917 - val_loss: 2.3812 - val_accuracy: 0.5369 - lr: 3.7500e-05
Epoch 32/100
60/60 [=====] - 1s 12ms/step - loss: 0.4821 - accuracy: 0.9908 - val_loss: 3.1731 - val_accuracy: 0.3125 - lr: 3.7500e-05
Epoch 33/100
57/60 [=====>..] - ETA: 0s - loss: 0.4737 - accuracy: 0.9943
Epoch 33: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
60/60 [=====] - 1s 12ms/step - loss: 0.4734 - accuracy: 0.9946 - val_loss: 2.0489 - val_accuracy: 0.5806 - lr: 3.7500e-05
Epoch 34/100
60/60 [=====] - 1s 11ms/step - loss: 0.4670 - accuracy: 0.9962 - val_loss: 1.7127 - val_accuracy: 0.5694 - lr: 1.8750e-05
Epoch 35/100
60/60 [=====] - 1s 11ms/step - loss: 0.4652 - accuracy: 0.9933 - val_loss: 0.6987 - val_accuracy: 0.9044 - lr: 1.8750e-05
Epoch 36/100
60/60 [=====] - 1s 11ms/step - loss: 0.4624 - accuracy: 0.9937 - val_loss: 2.0300 - val_accuracy: 0.4575 - lr: 1.8750e-05
Epoch 37/100
60/60 [=====] - 1s 11ms/step - loss: 0.4526 - accuracy: 0.9962 - val_loss: 0.5959 - val_accuracy: 0.9413 - lr: 1.8750e-05
Epoch 38/100
60/60 [=====] - 1s 11ms/step - loss: 0.4587 - accuracy: 0.9954 - val_loss: 0.9378 - val_accuracy: 0.8062 - lr: 1.8750e-05
Epoch 39/100
60/60 [=====] - 1s 11ms/step - loss: 0.4503 - accuracy: 0.9962 - val_loss: 0.8350 - val_accuracy: 0.8456 - lr: 1.8750e-05
Epoch 40/100
60/60 [=====] - 1s 10ms/step - loss: 0.4457 - accuracy: 0.9954 - val_loss: 3.0214 - val_accuracy: 0.2706 - lr: 1.8750e-05
Epoch 41/100
57/60 [=====>..] - ETA: 0s - loss: 0.4451 - accuracy: 0.9969
Epoch 41: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
60/60 [=====] - 1s 11ms/step - loss: 0.4477 - accuracy: 0.9954 - val_loss: 0.7575 - val_accuracy: 0.8719 - lr: 1.8750e-05
Epoch 42/100
60/60 [=====] - 1s 11ms/step - loss: 0.4396 - accuracy: 0.9954 - val_loss: 0.8337 - val_accuracy: 0.8438 - lr: 9.3750e-06
Epoch 43/100

60/60 [=====] - 1s 11ms/step - loss: 0.4373 - accuracy: 0.9958 - val_loss: 1.5357 - val_accuracy: 0.5931 - lr: 9.3750e-06
Epoch 44/100
60/60 [=====] - 1s 12ms/step - loss: 0.4413 - accuracy: 0.9971 - val_loss: 0.6077 - val_accuracy: 0.9256 - lr: 9.3750e-06
Epoch 45/100
57/60 [=====>..] - ETA: 0s - loss: 0.4359 - accuracy: 0.9974
Epoch 45: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
60/60 [=====] - 1s 12ms/step - loss: 0.4348 - accuracy: 0.9975 - val_loss: 0.6839 - val_accuracy: 0.8944 - lr: 9.3750e-06
Epoch 46/100
60/60 [=====] - 1s 12ms/step - loss: 0.4369 - accuracy: 0.9962 - val_loss: 0.8509 - val_accuracy: 0.8169 - lr: 4.6875e-06
Epoch 47/100
60/60 [=====] - 1s 12ms/step - loss: 0.4301 - accuracy: 0.9979 - val_loss: 0.6123 - val_accuracy: 0.9262 - lr: 4.6875e-06
Epoch 48/100
60/60 [=====] - 1s 12ms/step - loss: 0.4297 - accuracy: 0.9967 - val_loss: 0.7691 - val_accuracy: 0.8531 - lr: 4.6875e-06
Epoch 49/100
60/60 [=====] - 1s 12ms/step - loss: 0.4356 - accuracy: 0.9954 - val_loss: 0.5665 - val_accuracy: 0.9438 - lr: 4.6875e-06
Epoch 50/100
60/60 [=====] - 1s 11ms/step - loss: 0.4262 - accuracy: 0.9979 - val_loss: 0.5848 - val_accuracy: 0.9362 - lr: 4.6875e-06
Epoch 51/100
60/60 [=====] - 1s 12ms/step - loss: 0.4306 - accuracy: 0.9983 - val_loss: 0.6530 - val_accuracy: 0.9081 - lr: 4.6875e-06
Epoch 52/100
60/60 [=====] - 1s 12ms/step - loss: 0.4290 - accuracy: 0.9983 - val_loss: 0.5541 - val_accuracy: 0.9475 - lr: 4.6875e-06
Epoch 53/100
60/60 [=====] - 1s 12ms/step - loss: 0.4278 - accuracy: 0.9979 - val_loss: 0.5897 - val_accuracy: 0.9400 - lr: 4.6875e-06
Epoch 54/100
60/60 [=====] - 1s 11ms/step - loss: 0.4272 - accuracy: 0.9975 - val_loss: 0.5982 - val_accuracy: 0.9275 - lr: 4.6875e-06
Epoch 55/100
60/60 [=====] - 1s 12ms/step - loss: 0.4266 - accuracy: 0.9971 - val_loss: 0.6470 - val_accuracy: 0.9081 - lr: 4.6875e-06
Epoch 56/100
55/60 [=====>...] - ETA: 0s - loss: 0.4212 - accuracy: 0.9977
Epoch 56: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
60/60 [=====] - 1s 12ms/step - loss: 0.4204 - accuracy: 0.9979 - val_loss: 0.5987 - val_accuracy: 0.9306 - lr: 4.6875e-06
Epoch 57/100
60/60 [=====] - 1s 11ms/step - loss: 0.4291 - accuracy: 0.9975 - val_loss: 0.5512 - val_accuracy: 0.9494 - lr: 2.3438e-06
Epoch 58/100
60/60 [=====] - 1s 12ms/step - loss: 0.4233 - accuracy: 0.9975 - val_loss: 0.5413 - val_accuracy: 0.9488 - lr: 2.3438e-06
Epoch 59/100
60/60 [=====] - 1s 12ms/step - loss: 0.4292 - accuracy: 0.9992 - val_loss: 0.5888 - val_accuracy: 0.9337 - lr: 2.3438e-06
Epoch 60/100
60/60 [=====] - 1s 11ms/step - loss: 0.4189 - accuracy: 0.9987 - val_loss: 0.5366 - val_accuracy: 0.9488 - lr: 2.3438e-06
Epoch 61/100

60/60 [=====] - 1s 10ms/step - loss: 0.4220 - accuracy: 0.9987 - val_loss: 0.5365 - val_accuracy: 0.9544 - lr: 2.3438e-06
Epoch 62/100
60/60 [=====] - 1s 11ms/step - loss: 0.4231 - accuracy: 0.9983 - val_loss: 0.5383 - val_accuracy: 0.9475 - lr: 2.3438e-06
Epoch 63/100
60/60 [=====] - 1s 10ms/step - loss: 0.4235 - accuracy: 0.9979 - val_loss: 0.5796 - val_accuracy: 0.9344 - lr: 2.3438e-06
Epoch 64/100
59/60 [=====>.] - ETA: 0s - loss: 0.4225 - accuracy: 0.9970
Epoch 64: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
60/60 [=====] - 1s 11ms/step - loss: 0.4223 - accuracy: 0.9971 - val_loss: 0.5618 - val_accuracy: 0.9388 - lr: 2.3438e-06
Epoch 65/100
60/60 [=====] - 1s 11ms/step - loss: 0.4175 - accuracy: 0.9992 - val_loss: 0.5440 - val_accuracy: 0.9488 - lr: 1.1719e-06
Epoch 66/100
60/60 [=====] - 1s 11ms/step - loss: 0.4206 - accuracy: 0.9975 - val_loss: 0.5367 - val_accuracy: 0.9519 - lr: 1.1719e-06
Epoch 67/100
60/60 [=====] - 1s 12ms/step - loss: 0.4200 - accuracy: 0.9987 - val_loss: 0.5538 - val_accuracy: 0.9438 - lr: 1.1719e-06
Epoch 68/100
60/60 [=====] - ETA: 0s - loss: 0.4199 - accuracy: 0.9979
Epoch 68: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.
60/60 [=====] - 1s 11ms/step - loss: 0.4199 - accuracy: 0.9979 - val_loss: 0.5482 - val_accuracy: 0.9463 - lr: 1.1719e-06
Epoch 69/100
60/60 [=====] - 1s 11ms/step - loss: 0.4228 - accuracy: 0.9975 - val_loss: 0.5371 - val_accuracy: 0.9475 - lr: 5.8594e-07
Epoch 70/100
60/60 [=====] - 1s 12ms/step - loss: 0.4159 - accuracy: 0.9979 - val_loss: 0.5374 - val_accuracy: 0.9481 - lr: 5.8594e-07
Epoch 71/100
60/60 [=====] - 1s 12ms/step - loss: 0.4215 - accuracy: 0.9975 - val_loss: 0.5362 - val_accuracy: 0.9456 - lr: 5.8594e-07
Epoch 72/100
60/60 [=====] - 1s 12ms/step - loss: 0.4180 - accuracy: 0.9983 - val_loss: 0.5379 - val_accuracy: 0.9481 - lr: 5.8594e-07
Epoch 73/100
60/60 [=====] - 1s 12ms/step - loss: 0.4216 - accuracy: 0.9975 - val_loss: 0.5304 - val_accuracy: 0.9506 - lr: 5.8594e-07
[100, 48, 0.2]
65.0%
Epoch 1/100
67/67 [=====] - 2s 16ms/step - loss: 3.0640 - accuracy: 0.1538 - val_loss: 3.7431 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 2/100
67/67 [=====] - 1s 11ms/step - loss: 2.4744 - accuracy: 0.2778 - val_loss: 6.6365 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 3/100
67/67 [=====] - 1s 12ms/step - loss: 2.0590 - accuracy: 0.4303 - val_loss: 8.2342 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 4/100
67/67 [=====] - 1s 13ms/step - loss: 1.6074 - accuracy: 0.6191 - val_loss: 8.7012 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 5/100
62/67 [=====>...] - ETA: 0s - loss: 1.2576 - accuracy: 0.752

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Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.

67/67 [=====] - 1s 12ms/step - loss: 1.2520 - accuracy: 0.7541 - val_loss: 9.3396 - val_accuracy: 0.1088 - lr: 3.0000e-04

Epoch 6/100

67/67 [=====] - 1s 12ms/step - loss: 0.9765 - accuracy: 0.8594 - val_loss: 10.2176 - val_accuracy: 0.2013 - lr: 1.5000e-04

Epoch 7/100

67/67 [=====] - 1s 11ms/step - loss: 0.8415 - accuracy: 0.9097 - val_loss: 10.2135 - val_accuracy: 0.0962 - lr: 1.5000e-04

Epoch 8/100

67/67 [=====] - 1s 10ms/step - loss: 0.7660 - accuracy: 0.9294 - val_loss: 8.9586 - val_accuracy: 0.0950 - lr: 1.5000e-04

Epoch 9/100

63/67 [=====>..] - ETA: 0s - loss: 0.6956 - accuracy: 0.948

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Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.

67/67 [=====] - 1s 10ms/step - loss: 0.6943 - accuracy: 0.9497 - val_loss: 7.4893 - val_accuracy: 0.1325 - lr: 1.5000e-04

Epoch 10/100

67/67 [=====] - 1s 10ms/step - loss: 0.6285 - accuracy: 0.9716 - val_loss: 6.7813 - val_accuracy: 0.1838 - lr: 7.5000e-05

Epoch 11/100

67/67 [=====] - 1s 10ms/step - loss: 0.6052 - accuracy: 0.9744 - val_loss: 3.8377 - val_accuracy: 0.2400 - lr: 7.5000e-05

Epoch 12/100

67/67 [=====] - 1s 12ms/step - loss: 0.5842 - accuracy: 0.9819 - val_loss: 2.2600 - val_accuracy: 0.4787 - lr: 7.5000e-05

Epoch 13/100

67/67 [=====] - 1s 12ms/step - loss: 0.5577 - accuracy: 0.9847 - val_loss: 2.5893 - val_accuracy: 0.4162 - lr: 7.5000e-05

Epoch 14/100

67/67 [=====] - 1s 13ms/step - loss: 0.5427 - accuracy: 0.9859 - val_loss: 1.2989 - val_accuracy: 0.6562 - lr: 7.5000e-05

Epoch 15/100

67/67 [=====] - 1s 11ms/step - loss: 0.5150 - accuracy: 0.9900 - val_loss: 3.5889 - val_accuracy: 0.3300 - lr: 7.5000e-05

Epoch 16/100

67/67 [=====] - 1s 11ms/step - loss: 0.5013 - accuracy: 0.9909 - val_loss: 6.6648 - val_accuracy: 0.1887 - lr: 7.5000e-05

Epoch 17/100

67/67 [=====] - 1s 13ms/step - loss: 0.4862 - accuracy: 0.9906 - val_loss: 3.5297 - val_accuracy: 0.3175 - lr: 7.5000e-05

Epoch 18/100

64/67 [=====>..] - ETA: 0s - loss: 0.4700 - accuracy: 0.994

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Epoch 18: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.

67/67 [=====] - 1s 12ms/step - loss: 0.4701 - accuracy: 0.9947 - val_loss: 4.6118 - val_accuracy: 0.1950 - lr: 7.5000e-05

Epoch 19/100

67/67 [=====] - 1s 12ms/step - loss: 0.4634 - accuracy: 0.9931 - val_loss: 0.8762 - val_accuracy: 0.8288 - lr: 3.7500e-05

Epoch 20/100

67/67 [=====] - 1s 12ms/step - loss: 0.4535 - accuracy: 0.9919 - val_loss: 4.5550 - val_accuracy: 0.1963 - lr: 3.7500e-05

Epoch 21/100

67/67 [=====] - 1s 12ms/step - loss: 0.4468 - accuracy: 0.9934 - val_loss: 1.6109 - val_accuracy: 0.5713 - lr: 3.7500e-05

Epoch 22/100

67/67 [=====] - 1s 12ms/step - loss: 0.4398 - accuracy:

0.9944 - val_loss: 1.0064 - val_accuracy: 0.8200 - lr: 3.7500e-05
Epoch 23/100
67/67 [=====] - ETA: 0s - loss: 0.4300 - accuracy: 0.9969
Epoch 23: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
67/67 [=====] - 1s 12ms/step - loss: 0.4300 - accuracy: 0.9969 - val_loss: 2.3497 - val_accuracy: 0.4300 - lr: 3.7500e-05
Epoch 24/100
67/67 [=====] - 1s 12ms/step - loss: 0.4253 - accuracy: 0.9966 - val_loss: 0.9522 - val_accuracy: 0.7837 - lr: 1.8750e-05
Epoch 25/100
67/67 [=====] - 1s 13ms/step - loss: 0.4226 - accuracy: 0.9950 - val_loss: 0.5113 - val_accuracy: 0.9638 - lr: 1.8750e-05
Epoch 26/100
67/67 [=====] - 1s 12ms/step - loss: 0.4171 - accuracy: 0.9959 - val_loss: 0.9380 - val_accuracy: 0.8050 - lr: 1.8750e-05
Epoch 27/100
67/67 [=====] - 1s 12ms/step - loss: 0.4173 - accuracy: 0.9956 - val_loss: 0.7182 - val_accuracy: 0.8788 - lr: 1.8750e-05
Epoch 28/100
67/67 [=====] - 1s 10ms/step - loss: 0.4104 - accuracy: 0.9966 - val_loss: 1.9788 - val_accuracy: 0.4800 - lr: 1.8750e-05
Epoch 29/100
63/67 [=====>..] - ETA: 0s - loss: 0.4116 - accuracy: 0.9967
Epoch 29: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
67/67 [=====] - 1s 10ms/step - loss: 0.4114 - accuracy: 0.9969 - val_loss: 0.8346 - val_accuracy: 0.8425 - lr: 1.8750e-05
Epoch 30/100
67/67 [=====] - 1s 10ms/step - loss: 0.4073 - accuracy: 0.9975 - val_loss: 0.5682 - val_accuracy: 0.9500 - lr: 9.3750e-06
Epoch 31/100
67/67 [=====] - 1s 11ms/step - loss: 0.4041 - accuracy: 0.9972 - val_loss: 0.5532 - val_accuracy: 0.9375 - lr: 9.3750e-06
Epoch 32/100
67/67 [=====] - 1s 11ms/step - loss: 0.3990 - accuracy: 0.9978 - val_loss: 1.2419 - val_accuracy: 0.6950 - lr: 9.3750e-06
Epoch 33/100
66/67 [=====>.] - ETA: 0s - loss: 0.4021 - accuracy: 0.9965
Epoch 33: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
67/67 [=====] - 1s 12ms/step - loss: 0.4019 - accuracy: 0.9966 - val_loss: 0.5830 - val_accuracy: 0.9325 - lr: 9.3750e-06
Epoch 34/100
67/67 [=====] - 1s 12ms/step - loss: 0.3992 - accuracy: 0.9978 - val_loss: 0.5003 - val_accuracy: 0.9625 - lr: 4.6875e-06
Epoch 35/100
67/67 [=====] - 1s 13ms/step - loss: 0.4006 - accuracy: 0.9972 - val_loss: 0.4505 - val_accuracy: 0.9750 - lr: 4.6875e-06
Epoch 36/100
67/67 [=====] - 1s 12ms/step - loss: 0.3981 - accuracy: 0.9969 - val_loss: 0.6011 - val_accuracy: 0.9300 - lr: 4.6875e-06
Epoch 37/100
67/67 [=====] - 1s 12ms/step - loss: 0.3974 - accuracy: 0.9972 - val_loss: 0.4966 - val_accuracy: 0.9638 - lr: 4.6875e-06
Epoch 38/100
67/67 [=====] - 1s 12ms/step - loss: 0.3938 - accuracy: 0.9984 - val_loss: 0.4561 - val_accuracy: 0.9812 - lr: 4.6875e-06
Epoch 39/100
63/67 [=====>..] - ETA: 0s - loss: 0.3935 - accuracy: 0.997

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Epoch 39: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.

67/67 [=====] - 1s 12ms/step - loss: 0.3946 - accuracy: 0.9975 - val_loss: 0.4606 - val_accuracy: 0.9775 - lr: 4.6875e-06

Epoch 40/100

67/67 [=====] - 1s 12ms/step - loss: 0.3921 - accuracy: 0.9975 - val_loss: 0.4639 - val_accuracy: 0.9775 - lr: 2.3438e-06

Epoch 41/100

67/67 [=====] - 1s 13ms/step - loss: 0.3941 - accuracy: 0.9978 - val_loss: 0.4412 - val_accuracy: 0.9825 - lr: 2.3438e-06

Epoch 42/100

67/67 [=====] - 1s 11ms/step - loss: 0.3943 - accuracy: 0.9978 - val_loss: 0.4513 - val_accuracy: 0.9800 - lr: 2.3438e-06

Epoch 43/100

67/67 [=====] - 1s 12ms/step - loss: 0.3914 - accuracy: 0.9978 - val_loss: 0.4499 - val_accuracy: 0.9837 - lr: 2.3438e-06

Epoch 44/100

67/67 [=====] - 1s 11ms/step - loss: 0.3915 - accuracy: 0.9969 - val_loss: 0.4388 - val_accuracy: 0.9787 - lr: 2.3438e-06

Epoch 45/100

67/67 [=====] - 1s 11ms/step - loss: 0.3892 - accuracy: 0.9987 - val_loss: 0.4698 - val_accuracy: 0.9775 - lr: 2.3438e-06

Epoch 46/100

67/67 [=====] - 1s 12ms/step - loss: 0.3905 - accuracy: 0.9981 - val_loss: 0.4518 - val_accuracy: 0.9737 - lr: 2.3438e-06

Epoch 47/100

67/67 [=====] - 1s 11ms/step - loss: 0.3914 - accuracy: 0.9962 - val_loss: 0.4404 - val_accuracy: 0.9787 - lr: 2.3438e-06

Epoch 48/100

66/67 [=====>.] - ETA: 0s - loss: 0.3889 - accuracy: 0.9981

Epoch 48: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.

67/67 [=====] - 1s 11ms/step - loss: 0.3887 - accuracy: 0.9981 - val_loss: 0.4435 - val_accuracy: 0.9837 - lr: 2.3438e-06

Epoch 49/100

67/67 [=====] - 1s 10ms/step - loss: 0.3893 - accuracy: 0.9981 - val_loss: 0.4352 - val_accuracy: 0.9812 - lr: 1.1719e-06

Epoch 50/100

67/67 [=====] - 1s 10ms/step - loss: 0.3872 - accuracy: 0.9981 - val_loss: 0.4365 - val_accuracy: 0.9825 - lr: 1.1719e-06

Epoch 51/100

67/67 [=====] - 1s 10ms/step - loss: 0.3855 - accuracy: 0.9981 - val_loss: 0.4364 - val_accuracy: 0.9775 - lr: 1.1719e-06

Epoch 52/100

67/67 [=====] - 1s 10ms/step - loss: 0.3874 - accuracy: 0.9991 - val_loss: 0.4375 - val_accuracy: 0.9775 - lr: 1.1719e-06

Epoch 53/100

65/67 [=====>.] - ETA: 0s - loss: 0.3867 - accuracy: 0.9981

Epoch 53: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.

67/67 [=====] - 1s 11ms/step - loss: 0.3869 - accuracy: 0.9981 - val_loss: 0.4391 - val_accuracy: 0.9850 - lr: 1.1719e-06

Epoch 54/100

67/67 [=====] - 1s 12ms/step - loss: 0.3858 - accuracy: 0.9984 - val_loss: 0.4354 - val_accuracy: 0.9825 - lr: 5.8594e-07

Epoch 55/100

67/67 [=====] - 1s 12ms/step - loss: 0.3872 - accuracy: 0.9984 - val_loss: 0.4357 - val_accuracy: 0.9825 - lr: 5.8594e-07

Epoch 56/100

67/67 [=====] - 1s 12ms/step - loss: 0.3885 - accuracy:

0.9978 - val_loss: 0.4348 - val_accuracy: 0.9825 - lr: 5.8594e-07
Epoch 57/100
67/67 [=====] - 1s 13ms/step - loss: 0.3880 - accuracy:
0.9975 - val_loss: 0.4335 - val_accuracy: 0.9825 - lr: 5.8594e-07
Epoch 58/100
67/67 [=====] - 1s 12ms/step - loss: 0.3895 - accuracy:
0.9966 - val_loss: 0.4372 - val_accuracy: 0.9825 - lr: 5.8594e-07
Epoch 59/100
67/67 [=====] - 1s 12ms/step - loss: 0.3877 - accuracy:
0.9978 - val_loss: 0.4326 - val_accuracy: 0.9800 - lr: 5.8594e-07
Epoch 60/100
67/67 [=====] - 1s 13ms/step - loss: 0.3855 - accuracy:
0.9981 - val_loss: 0.4346 - val_accuracy: 0.9775 - lr: 5.8594e-07
Epoch 61/100
67/67 [=====] - 1s 12ms/step - loss: 0.3867 - accuracy:
0.9972 - val_loss: 0.4336 - val_accuracy: 0.9812 - lr: 5.8594e-07
Epoch 62/100
67/67 [=====] - 1s 12ms/step - loss: 0.3865 - accuracy:
0.9987 - val_loss: 0.4322 - val_accuracy: 0.9787 - lr: 5.8594e-07
Epoch 63/100
67/67 [=====] - 1s 12ms/step - loss: 0.3869 - accuracy:
0.9975 - val_loss: 0.4322 - val_accuracy: 0.9825 - lr: 5.8594e-07
Epoch 64/100
67/67 [=====] - 1s 13ms/step - loss: 0.3830 - accuracy:
0.9987 - val_loss: 0.4338 - val_accuracy: 0.9787 - lr: 5.8594e-07
Epoch 65/100
67/67 [=====] - 1s 13ms/step - loss: 0.3858 - accuracy:
0.9972 - val_loss: 0.4352 - val_accuracy: 0.9775 - lr: 5.8594e-07
[100, 48, 0.25]
67.5%
Epoch 1/100
63/63 [=====] - 2s 20ms/step - loss: 3.1985 - accuracy:
0.1417 - val_loss: 3.6958 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 2/100
63/63 [=====] - 1s 9ms/step - loss: 2.6168 - accuracy: 0
.2647 - val_loss: 4.7594 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 3/100
63/63 [=====] - 1s 9ms/step - loss: 2.2128 - accuracy: 0
.3853 - val_loss: 7.0856 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 4/100
63/63 [=====] - 1s 9ms/step - loss: 1.7368 - accuracy: 0
.5673 - val_loss: 7.8714 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 5/100
60/63 [=====>..] - ETA: 0s - loss: 1.3630 - accuracy: 0.703
1
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
63/63 [=====] - 1s 10ms/step - loss: 1.3505 - accuracy:
0.7097 - val_loss: 9.1674 - val_accuracy: 0.0940 - lr: 3.0000e-04
Epoch 6/100
63/63 [=====] - 1s 10ms/step - loss: 1.0495 - accuracy:
0.8380 - val_loss: 10.3376 - val_accuracy: 0.0930 - lr: 1.5000e-04
Epoch 7/100
63/63 [=====] - 1s 12ms/step - loss: 0.9259 - accuracy:
0.8743 - val_loss: 10.4943 - val_accuracy: 0.0930 - lr: 1.5000e-04
Epoch 8/100
63/63 [=====] - 1s 11ms/step - loss: 0.7940 - accuracy:
0.9220 - val_loss: 9.5924 - val_accuracy: 0.0930 - lr: 1.5000e-04
Epoch 9/100
59/63 [=====>..] - ETA: 0s - loss: 0.7336 - accuracy: 0.941
4

Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
63/63 [=====] - 1s 12ms/step - loss: 0.7300 - accuracy: 0.9427 - val_loss: 8.0616 - val_accuracy: 0.0930 - lr: 1.5000e-04
Epoch 10/100
63/63 [=====] - 1s 12ms/step - loss: 0.6706 - accuracy: 0.9567 - val_loss: 6.5994 - val_accuracy: 0.1810 - lr: 7.5000e-05
Epoch 11/100
63/63 [=====] - 1s 11ms/step - loss: 0.6347 - accuracy: 0.9690 - val_loss: 3.7894 - val_accuracy: 0.3100 - lr: 7.5000e-05
Epoch 12/100
63/63 [=====] - 1s 12ms/step - loss: 0.6103 - accuracy: 0.9717 - val_loss: 1.4723 - val_accuracy: 0.5610 - lr: 7.5000e-05
Epoch 13/100
63/63 [=====] - 1s 12ms/step - loss: 0.5910 - accuracy: 0.9747 - val_loss: 1.2330 - val_accuracy: 0.7300 - lr: 7.5000e-05
Epoch 14/100
63/63 [=====] - 1s 12ms/step - loss: 0.5726 - accuracy: 0.9783 - val_loss: 1.5111 - val_accuracy: 0.7070 - lr: 7.5000e-05
Epoch 15/100
63/63 [=====] - 1s 12ms/step - loss: 0.5552 - accuracy: 0.9823 - val_loss: 0.8567 - val_accuracy: 0.8540 - lr: 7.5000e-05
Epoch 16/100
63/63 [=====] - 1s 12ms/step - loss: 0.5354 - accuracy: 0.9870 - val_loss: 2.5311 - val_accuracy: 0.5480 - lr: 7.5000e-05
Epoch 17/100
63/63 [=====] - 1s 13ms/step - loss: 0.5179 - accuracy: 0.9903 - val_loss: 2.9018 - val_accuracy: 0.4390 - lr: 7.5000e-05
Epoch 18/100
63/63 [=====] - 1s 12ms/step - loss: 0.5043 - accuracy: 0.9873 - val_loss: 2.4981 - val_accuracy: 0.4260 - lr: 7.5000e-05
Epoch 19/100
63/63 [=====] - ETA: 0s - loss: 0.4897 - accuracy: 0.9910
Epoch 19: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
63/63 [=====] - 1s 12ms/step - loss: 0.4897 - accuracy: 0.9910 - val_loss: 2.5899 - val_accuracy: 0.5200 - lr: 7.5000e-05
Epoch 20/100
63/63 [=====] - 1s 13ms/step - loss: 0.4712 - accuracy: 0.9957 - val_loss: 1.8450 - val_accuracy: 0.4750 - lr: 3.7500e-05
Epoch 21/100
63/63 [=====] - 1s 12ms/step - loss: 0.4700 - accuracy: 0.9933 - val_loss: 0.6723 - val_accuracy: 0.9280 - lr: 3.7500e-05
Epoch 22/100
63/63 [=====] - 1s 12ms/step - loss: 0.4568 - accuracy: 0.9950 - val_loss: 0.8995 - val_accuracy: 0.7920 - lr: 3.7500e-05
Epoch 23/100
63/63 [=====] - 1s 12ms/step - loss: 0.4535 - accuracy: 0.9947 - val_loss: 0.5674 - val_accuracy: 0.9590 - lr: 3.7500e-05
Epoch 24/100
63/63 [=====] - 1s 11ms/step - loss: 0.4466 - accuracy: 0.9940 - val_loss: 0.7044 - val_accuracy: 0.8950 - lr: 3.7500e-05
Epoch 25/100
63/63 [=====] - 1s 10ms/step - loss: 0.4428 - accuracy: 0.9970 - val_loss: 0.6665 - val_accuracy: 0.9120 - lr: 3.7500e-05
Epoch 26/100
63/63 [=====] - 1s 10ms/step - loss: 0.4350 - accuracy: 0.9957 - val_loss: 0.8649 - val_accuracy: 0.8080 - lr: 3.7500e-05
Epoch 27/100
63/63 [=====] - ETA: 0s - loss: 0.4294 - accuracy: 0.9960

Epoch 27: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
63/63 [=====] - 1s 11ms/step - loss: 0.4294 - accuracy: 0.9960 - val_loss: 0.6651 - val_accuracy: 0.8940 - lr: 3.7500e-05
Epoch 28/100
63/63 [=====] - 1s 12ms/step - loss: 0.4208 - accuracy: 0.9963 - val_loss: 0.5883 - val_accuracy: 0.9460 - lr: 1.8750e-05
Epoch 29/100
63/63 [=====] - 1s 12ms/step - loss: 0.4173 - accuracy: 0.9973 - val_loss: 0.8221 - val_accuracy: 0.8320 - lr: 1.8750e-05
Epoch 30/100
63/63 [=====] - 1s 13ms/step - loss: 0.4161 - accuracy: 0.9967 - val_loss: 0.5115 - val_accuracy: 0.9630 - lr: 1.8750e-05
Epoch 31/100
63/63 [=====] - 1s 12ms/step - loss: 0.4133 - accuracy: 0.9983 - val_loss: 0.5686 - val_accuracy: 0.9560 - lr: 1.8750e-05
Epoch 32/100
63/63 [=====] - 1s 13ms/step - loss: 0.4093 - accuracy: 0.9973 - val_loss: 1.3420 - val_accuracy: 0.6100 - lr: 1.8750e-05
Epoch 33/100
63/63 [=====] - 1s 11ms/step - loss: 0.4070 - accuracy: 0.9977 - val_loss: 0.5707 - val_accuracy: 0.9520 - lr: 1.8750e-05
Epoch 34/100
63/63 [=====] - ETA: 0s - loss: 0.4043 - accuracy: 0.9980
Epoch 34: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
63/63 [=====] - 1s 12ms/step - loss: 0.4043 - accuracy: 0.9980 - val_loss: 0.6158 - val_accuracy: 0.9130 - lr: 1.8750e-05
Epoch 35/100
63/63 [=====] - 1s 12ms/step - loss: 0.3981 - accuracy: 0.9997 - val_loss: 0.4993 - val_accuracy: 0.9590 - lr: 9.3750e-06
Epoch 36/100
63/63 [=====] - 1s 13ms/step - loss: 0.3977 - accuracy: 0.9987 - val_loss: 0.4651 - val_accuracy: 0.9760 - lr: 9.3750e-06
Epoch 37/100
63/63 [=====] - 1s 12ms/step - loss: 0.3966 - accuracy: 0.9983 - val_loss: 0.6621 - val_accuracy: 0.9130 - lr: 9.3750e-06
Epoch 38/100
63/63 [=====] - 1s 12ms/step - loss: 0.3963 - accuracy: 0.9997 - val_loss: 0.6134 - val_accuracy: 0.9350 - lr: 9.3750e-06
Epoch 39/100
63/63 [=====] - 1s 13ms/step - loss: 0.3968 - accuracy: 0.9977 - val_loss: 0.4821 - val_accuracy: 0.9690 - lr: 9.3750e-06
Epoch 40/100
63/63 [=====] - 1s 13ms/step - loss: 0.3952 - accuracy: 0.9997 - val_loss: 0.4573 - val_accuracy: 0.9750 - lr: 9.3750e-06
Epoch 41/100
63/63 [=====] - 1s 12ms/step - loss: 0.3925 - accuracy: 0.9980 - val_loss: 0.7067 - val_accuracy: 0.8800 - lr: 9.3750e-06
Epoch 42/100
63/63 [=====] - 1s 12ms/step - loss: 0.3900 - accuracy: 0.9987 - val_loss: 0.5174 - val_accuracy: 0.9480 - lr: 9.3750e-06
Epoch 43/100
63/63 [=====] - 1s 13ms/step - loss: 0.3891 - accuracy: 0.9993 - val_loss: 0.4647 - val_accuracy: 0.9760 - lr: 9.3750e-06
Epoch 44/100
63/63 [=====] - 1s 12ms/step - loss: 0.3866 - accuracy: 0.9987 - val_loss: 0.4441 - val_accuracy: 0.9800 - lr: 9.3750e-06
Epoch 45/100
63/63 [=====] - 1s 11ms/step - loss: 0.3812 - accuracy: 0.9990 - val_loss: 0.4403 - val_accuracy: 0.9810 - lr: 9.3750e-06

Epoch 46/100
63/63 [=====] - 1s 11ms/step - loss: 0.3827 - accuracy: 0.9987 - val_loss: 0.4613 - val_accuracy: 0.9740 - lr: 9.3750e-06
Epoch 47/100
63/63 [=====] - 1s 10ms/step - loss: 0.3805 - accuracy: 0.9990 - val_loss: 0.5439 - val_accuracy: 0.9420 - lr: 9.3750e-06
Epoch 48/100
63/63 [=====] - 1s 10ms/step - loss: 0.3808 - accuracy: 0.9983 - val_loss: 0.4806 - val_accuracy: 0.9640 - lr: 9.3750e-06
Epoch 49/100
61/63 [=====>.] - ETA: 0s - loss: 0.3780 - accuracy: 0.9986
Epoch 49: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
63/63 [=====] - 1s 13ms/step - loss: 0.3781 - accuracy: 0.9987 - val_loss: 0.4778 - val_accuracy: 0.9650 - lr: 9.3750e-06
Epoch 50/100
63/63 [=====] - 1s 12ms/step - loss: 0.3771 - accuracy: 0.9983 - val_loss: 0.4547 - val_accuracy: 0.9740 - lr: 4.6875e-06
Epoch 51/100
63/63 [=====] - 1s 13ms/step - loss: 0.3760 - accuracy: 0.9997 - val_loss: 0.4620 - val_accuracy: 0.9670 - lr: 4.6875e-06
Epoch 52/100
63/63 [=====] - 1s 12ms/step - loss: 0.3765 - accuracy: 0.9980 - val_loss: 0.5084 - val_accuracy: 0.9580 - lr: 4.6875e-06
Epoch 53/100
63/63 [=====] - ETA: 0s - loss: 0.3768 - accuracy: 0.9987
Epoch 53: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
63/63 [=====] - 1s 13ms/step - loss: 0.3768 - accuracy: 0.9987 - val_loss: 0.4463 - val_accuracy: 0.9730 - lr: 4.6875e-06
Epoch 54/100
63/63 [=====] - 1s 13ms/step - loss: 0.3754 - accuracy: 0.9993 - val_loss: 0.4423 - val_accuracy: 0.9720 - lr: 2.3438e-06
Epoch 55/100
63/63 [=====] - 1s 12ms/step - loss: 0.3702 - accuracy: 0.9987 - val_loss: 0.4329 - val_accuracy: 0.9760 - lr: 2.3438e-06
Epoch 56/100
63/63 [=====] - 1s 12ms/step - loss: 0.3735 - accuracy: 0.9993 - val_loss: 0.4286 - val_accuracy: 0.9770 - lr: 2.3438e-06
Epoch 57/100
63/63 [=====] - 1s 12ms/step - loss: 0.3705 - accuracy: 0.9990 - val_loss: 0.4349 - val_accuracy: 0.9790 - lr: 2.3438e-06
[100, 48, 0.30000000000000004]
70.0%
Epoch 1/100
59/59 [=====] - 2s 19ms/step - loss: 3.1378 - accuracy: 0.1407 - val_loss: 3.5084 - val_accuracy: 0.1017 - lr: 3.0000e-04
Epoch 2/100
59/59 [=====] - 1s 11ms/step - loss: 2.5980 - accuracy: 0.2554 - val_loss: 5.0542 - val_accuracy: 0.1017 - lr: 3.0000e-04
Epoch 3/100
59/59 [=====] - 1s 10ms/step - loss: 2.2607 - accuracy: 0.3596 - val_loss: 5.6912 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 4/100
59/59 [=====] - 1s 11ms/step - loss: 1.8662 - accuracy: 0.4821 - val_loss: 6.9079 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 5/100
56/59 [=====>..] - ETA: 0s - loss: 1.5639 - accuracy: 0.6042
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.

59/59 [=====] - 1s 11ms/step - loss: 1.5682 - accuracy: 0.6029 - val_loss: 7.6529 - val_accuracy: 0.1050 - lr: 3.0000e-04
Epoch 6/100
59/59 [=====] - 1s 12ms/step - loss: 1.2899 - accuracy: 0.7275 - val_loss: 7.5982 - val_accuracy: 0.0992 - lr: 1.5000e-04
Epoch 7/100
59/59 [=====] - 1s 11ms/step - loss: 1.1565 - accuracy: 0.7736 - val_loss: 6.8948 - val_accuracy: 0.1308 - lr: 1.5000e-04
Epoch 8/100
59/59 [=====] - 1s 11ms/step - loss: 1.0607 - accuracy: 0.8136 - val_loss: 6.7388 - val_accuracy: 0.1333 - lr: 1.5000e-04
Epoch 9/100
55/59 [=====>...] - ETA: 0s - loss: 0.9431 - accuracy: 0.8598
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.50000356230885e-05.
59/59 [=====] - 1s 10ms/step - loss: 0.9410 - accuracy: 0.8614 - val_loss: 6.2987 - val_accuracy: 0.1367 - lr: 1.5000e-04
Epoch 10/100
59/59 [=====] - 1s 10ms/step - loss: 0.8716 - accuracy: 0.8854 - val_loss: 3.7148 - val_accuracy: 0.2617 - lr: 7.5000e-05
Epoch 11/100
59/59 [=====] - 1s 10ms/step - loss: 0.8251 - accuracy: 0.9011 - val_loss: 3.3677 - val_accuracy: 0.3300 - lr: 7.5000e-05
Epoch 12/100
59/59 [=====] - 1s 10ms/step - loss: 0.7983 - accuracy: 0.9075 - val_loss: 2.7351 - val_accuracy: 0.3650 - lr: 7.5000e-05
Epoch 13/100
59/59 [=====] - 1s 12ms/step - loss: 0.7629 - accuracy: 0.9221 - val_loss: 2.2021 - val_accuracy: 0.4358 - lr: 7.5000e-05
Epoch 14/100
59/59 [=====] - 1s 12ms/step - loss: 0.7351 - accuracy: 0.9279 - val_loss: 1.8668 - val_accuracy: 0.3992 - lr: 7.5000e-05
Epoch 15/100
59/59 [=====] - 1s 11ms/step - loss: 0.6995 - accuracy: 0.9418 - val_loss: 1.5158 - val_accuracy: 0.6133 - lr: 7.5000e-05
Epoch 16/100
59/59 [=====] - 1s 12ms/step - loss: 0.6679 - accuracy: 0.9507 - val_loss: 0.9364 - val_accuracy: 0.8283 - lr: 7.5000e-05
Epoch 17/100
59/59 [=====] - 1s 12ms/step - loss: 0.6465 - accuracy: 0.9564 - val_loss: 2.7558 - val_accuracy: 0.4892 - lr: 7.5000e-05
Epoch 18/100
59/59 [=====] - 1s 11ms/step - loss: 0.6277 - accuracy: 0.9571 - val_loss: 2.4451 - val_accuracy: 0.4692 - lr: 7.5000e-05
Epoch 19/100
59/59 [=====] - 1s 13ms/step - loss: 0.5945 - accuracy: 0.9704 - val_loss: 2.4767 - val_accuracy: 0.4892 - lr: 7.5000e-05
Epoch 20/100
57/59 [=====>..] - ETA: 0s - loss: 0.5837 - accuracy: 0.9697
Epoch 20: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
59/59 [=====] - 1s 12ms/step - loss: 0.5835 - accuracy: 0.9696 - val_loss: 1.3433 - val_accuracy: 0.6192 - lr: 7.5000e-05
Epoch 21/100
59/59 [=====] - 1s 12ms/step - loss: 0.5594 - accuracy: 0.9739 - val_loss: 1.2096 - val_accuracy: 0.6783 - lr: 3.7500e-05
Epoch 22/100
59/59 [=====] - 1s 11ms/step - loss: 0.5437 - accuracy: 0.9800 - val_loss: 1.4851 - val_accuracy: 0.6142 - lr: 3.7500e-05
Epoch 23/100

59/59 [=====] - 1s 11ms/step - loss: 0.5362 - accuracy: 0.9804 - val_loss: 0.7151 - val_accuracy: 0.9058 - lr: 3.7500e-05
Epoch 24/100
59/59 [=====] - 1s 12ms/step - loss: 0.5265 - accuracy: 0.9825 - val_loss: 0.7536 - val_accuracy: 0.8992 - lr: 3.7500e-05
Epoch 25/100
59/59 [=====] - 1s 12ms/step - loss: 0.5073 - accuracy: 0.9864 - val_loss: 0.7974 - val_accuracy: 0.8842 - lr: 3.7500e-05
Epoch 26/100
59/59 [=====] - 1s 11ms/step - loss: 0.5029 - accuracy: 0.9868 - val_loss: 0.9804 - val_accuracy: 0.7633 - lr: 3.7500e-05
Epoch 27/100
55/59 [=====>...] - ETA: 0s - loss: 0.4972 - accuracy: 0.9883
Epoch 27: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
59/59 [=====] - 1s 12ms/step - loss: 0.4963 - accuracy: 0.9889 - val_loss: 1.0750 - val_accuracy: 0.7708 - lr: 3.7500e-05
Epoch 28/100
59/59 [=====] - 1s 11ms/step - loss: 0.4845 - accuracy: 0.9925 - val_loss: 0.7091 - val_accuracy: 0.9092 - lr: 1.8750e-05
Epoch 29/100
59/59 [=====] - 1s 11ms/step - loss: 0.4812 - accuracy: 0.9896 - val_loss: 0.6741 - val_accuracy: 0.9208 - lr: 1.8750e-05
Epoch 30/100
59/59 [=====] - 1s 12ms/step - loss: 0.4743 - accuracy: 0.9896 - val_loss: 0.6238 - val_accuracy: 0.9300 - lr: 1.8750e-05
Epoch 31/100
59/59 [=====] - 1s 11ms/step - loss: 0.4737 - accuracy: 0.9911 - val_loss: 0.6422 - val_accuracy: 0.9283 - lr: 1.8750e-05
Epoch 32/100
59/59 [=====] - 1s 10ms/step - loss: 0.4695 - accuracy: 0.9914 - val_loss: 0.6388 - val_accuracy: 0.9250 - lr: 1.8750e-05
Epoch 33/100
59/59 [=====] - 1s 10ms/step - loss: 0.4658 - accuracy: 0.9904 - val_loss: 0.6733 - val_accuracy: 0.9142 - lr: 1.8750e-05
Epoch 34/100
59/59 [=====] - ETA: 0s - loss: 0.4614 - accuracy: 0.9921
Epoch 34: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
59/59 [=====] - 1s 10ms/step - loss: 0.4614 - accuracy: 0.9921 - val_loss: 0.6666 - val_accuracy: 0.9133 - lr: 1.8750e-05
Epoch 35/100
59/59 [=====] - 1s 10ms/step - loss: 0.4601 - accuracy: 0.9946 - val_loss: 0.5825 - val_accuracy: 0.9375 - lr: 9.3750e-06
Epoch 36/100
59/59 [=====] - 1s 11ms/step - loss: 0.4531 - accuracy: 0.9957 - val_loss: 0.6246 - val_accuracy: 0.9317 - lr: 9.3750e-06
Epoch 37/100
59/59 [=====] - 1s 12ms/step - loss: 0.4477 - accuracy: 0.9946 - val_loss: 0.5786 - val_accuracy: 0.9375 - lr: 9.3750e-06
Epoch 38/100
59/59 [=====] - 1s 11ms/step - loss: 0.4466 - accuracy: 0.9961 - val_loss: 0.5977 - val_accuracy: 0.9342 - lr: 9.3750e-06
Epoch 39/100
59/59 [=====] - 1s 13ms/step - loss: 0.4454 - accuracy: 0.9950 - val_loss: 0.5845 - val_accuracy: 0.9392 - lr: 9.3750e-06
Epoch 40/100
59/59 [=====] - 1s 12ms/step - loss: 0.4463 - accuracy: 0.9943 - val_loss: 0.5843 - val_accuracy: 0.9342 - lr: 9.3750e-06
Epoch 41/100

55/59 [=====>...] - ETA: 0s - loss: 0.4425 - accuracy: 0.995
1
Epoch 41: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
59/59 [=====] - 1s 12ms/step - loss: 0.4424 - accuracy:
0.9954 - val_loss: 0.5803 - val_accuracy: 0.9358 - lr: 9.3750e-06
Epoch 42/100
59/59 [=====] - 1s 11ms/step - loss: 0.4408 - accuracy:
0.9943 - val_loss: 0.5633 - val_accuracy: 0.9450 - lr: 4.6875e-06
Epoch 43/100
59/59 [=====] - 1s 12ms/step - loss: 0.4399 - accuracy:
0.9946 - val_loss: 0.5733 - val_accuracy: 0.9408 - lr: 4.6875e-06
Epoch 44/100
59/59 [=====] - 1s 13ms/step - loss: 0.4382 - accuracy:
0.9957 - val_loss: 0.5631 - val_accuracy: 0.9450 - lr: 4.6875e-06
Epoch 45/100
59/59 [=====] - 1s 12ms/step - loss: 0.4405 - accuracy:
0.9954 - val_loss: 0.5643 - val_accuracy: 0.9450 - lr: 4.6875e-06
Epoch 46/100
59/59 [=====] - 1s 12ms/step - loss: 0.4355 - accuracy:
0.9957 - val_loss: 0.5631 - val_accuracy: 0.9433 - lr: 4.6875e-06
Epoch 47/100
59/59 [=====] - 1s 12ms/step - loss: 0.4377 - accuracy:
0.9936 - val_loss: 0.5763 - val_accuracy: 0.9400 - lr: 4.6875e-06
Epoch 48/100
56/59 [=====>...] - ETA: 0s - loss: 0.4349 - accuracy: 0.995
2
Epoch 48: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
59/59 [=====] - 1s 12ms/step - loss: 0.4346 - accuracy:
0.9954 - val_loss: 0.5674 - val_accuracy: 0.9433 - lr: 4.6875e-06
Epoch 49/100
59/59 [=====] - 1s 11ms/step - loss: 0.4328 - accuracy:
0.9964 - val_loss: 0.5565 - val_accuracy: 0.9442 - lr: 2.3438e-06
Epoch 50/100
59/59 [=====] - 1s 12ms/step - loss: 0.4337 - accuracy:
0.9961 - val_loss: 0.5542 - val_accuracy: 0.9442 - lr: 2.3438e-06
Epoch 51/100
59/59 [=====] - 1s 13ms/step - loss: 0.4358 - accuracy:
0.9946 - val_loss: 0.5556 - val_accuracy: 0.9442 - lr: 2.3438e-06
Epoch 52/100
59/59 [=====] - 1s 12ms/step - loss: 0.4316 - accuracy:
0.9964 - val_loss: 0.5542 - val_accuracy: 0.9442 - lr: 2.3438e-06
Epoch 53/100
59/59 [=====] - 1s 11ms/step - loss: 0.4326 - accuracy:
0.9939 - val_loss: 0.5548 - val_accuracy: 0.9417 - lr: 2.3438e-06
Epoch 54/100
59/59 [=====] - 1s 12ms/step - loss: 0.4304 - accuracy:
0.9964 - val_loss: 0.5530 - val_accuracy: 0.9433 - lr: 2.3438e-06
[100, 48, 0.35000000000000003]
72.5%
Epoch 1/100
55/55 [=====] - 2s 19ms/step - loss: 3.1193 - accuracy:
0.1454 - val_loss: 3.6906 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 2/100
55/55 [=====] - 1s 11ms/step - loss: 2.4469 - accuracy:
0.2974 - val_loss: 5.1848 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 3/100
55/55 [=====] - 1s 11ms/step - loss: 2.0296 - accuracy:
0.4348 - val_loss: 5.5931 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 4/100
55/55 [=====] - 1s 13ms/step - loss: 1.6707 - accuracy:

0.5721 - val_loss: 5.9439 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 5/100
52/55 [=====>..] - ETA: 0s - loss: 1.3647 - accuracy: 0.7019
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
55/55 [=====] - 1s 13ms/step - loss: 1.3645 - accuracy: 0.7010 - val_loss: 7.1794 - val_accuracy: 0.1028 - lr: 3.0000e-04
Epoch 6/100
55/55 [=====] - 1s 14ms/step - loss: 1.0874 - accuracy: 0.8015 - val_loss: 8.5504 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 7/100
55/55 [=====] - 1s 13ms/step - loss: 0.9720 - accuracy: 0.8519 - val_loss: 9.4700 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 8/100
55/55 [=====] - 1s 13ms/step - loss: 0.8835 - accuracy: 0.8788 - val_loss: 9.3035 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 9/100
52/55 [=====>..] - ETA: 0s - loss: 0.8033 - accuracy: 0.8970
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
55/55 [=====] - 1s 14ms/step - loss: 0.8042 - accuracy: 0.8965 - val_loss: 8.4527 - val_accuracy: 0.1028 - lr: 1.5000e-04
Epoch 10/100
55/55 [=====] - 1s 14ms/step - loss: 0.7249 - accuracy: 0.9300 - val_loss: 6.1596 - val_accuracy: 0.2305 - lr: 7.5000e-05
Epoch 11/100
55/55 [=====] - 1s 12ms/step - loss: 0.6757 - accuracy: 0.9438 - val_loss: 5.4651 - val_accuracy: 0.1263 - lr: 7.5000e-05
Epoch 12/100
55/55 [=====] - 1s 14ms/step - loss: 0.6487 - accuracy: 0.9500 - val_loss: 2.5255 - val_accuracy: 0.4604 - lr: 7.5000e-05
Epoch 13/100
55/55 [=====] - 1s 14ms/step - loss: 0.6213 - accuracy: 0.9638 - val_loss: 4.0343 - val_accuracy: 0.1927 - lr: 7.5000e-05
Epoch 14/100
55/55 [=====] - 1s 13ms/step - loss: 0.5963 - accuracy: 0.9684 - val_loss: 3.7221 - val_accuracy: 0.2884 - lr: 7.5000e-05
Epoch 15/100
55/55 [=====] - 1s 14ms/step - loss: 0.5701 - accuracy: 0.9746 - val_loss: 1.2668 - val_accuracy: 0.6403 - lr: 7.5000e-05
Epoch 16/100
55/55 [=====] - 1s 14ms/step - loss: 0.5584 - accuracy: 0.9758 - val_loss: 4.1930 - val_accuracy: 0.2227 - lr: 7.5000e-05
Epoch 17/100
55/55 [=====] - 1s 14ms/step - loss: 0.5318 - accuracy: 0.9811 - val_loss: 5.5771 - val_accuracy: 0.1570 - lr: 7.5000e-05
Epoch 18/100
55/55 [=====] - 1s 15ms/step - loss: 0.5241 - accuracy: 0.9835 - val_loss: 1.4517 - val_accuracy: 0.6552 - lr: 7.5000e-05
Epoch 19/100
55/55 [=====] - ETA: 0s - loss: 0.5004 - accuracy: 0.9842
Epoch 19: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
55/55 [=====] - 1s 15ms/step - loss: 0.5004 - accuracy: 0.9842 - val_loss: 1.7629 - val_accuracy: 0.5617 - lr: 7.5000e-05
Epoch 20/100
55/55 [=====] - 1s 14ms/step - loss: 0.4787 - accuracy: 0.9881 - val_loss: 0.7423 - val_accuracy: 0.9065 - lr: 3.7500e-05
Epoch 21/100
55/55 [=====] - 1s 12ms/step - loss: 0.4748 - accuracy:

0.9873 - val_loss: 2.4269 - val_accuracy: 0.4083 - lr: 3.7500e-05
Epoch 22/100
55/55 [=====] - 1s 12ms/step - loss: 0.4650 - accuracy:
0.9908 - val_loss: 0.8539 - val_accuracy: 0.8380 - lr: 3.7500e-05
Epoch 23/100
55/55 [=====] - 1s 11ms/step - loss: 0.4561 - accuracy:
0.9931 - val_loss: 0.7072 - val_accuracy: 0.9022 - lr: 3.7500e-05
Epoch 24/100
55/55 [=====] - 1s 11ms/step - loss: 0.4508 - accuracy:
0.9912 - val_loss: 0.9524 - val_accuracy: 0.7637 - lr: 3.7500e-05
Epoch 25/100
55/55 [=====] - 1s 11ms/step - loss: 0.4375 - accuracy:
0.9931 - val_loss: 0.9153 - val_accuracy: 0.8116 - lr: 3.7500e-05
Epoch 26/100
55/55 [=====] - 1s 13ms/step - loss: 0.4305 - accuracy:
0.9950 - val_loss: 0.7769 - val_accuracy: 0.8737 - lr: 3.7500e-05
Epoch 27/100
52/55 [=====>..] - ETA: 0s - loss: 0.4266 - accuracy: 0.996
4
Epoch 27: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
55/55 [=====] - 1s 12ms/step - loss: 0.4270 - accuracy:
0.9965 - val_loss: 1.5818 - val_accuracy: 0.6181 - lr: 3.7500e-05
Epoch 28/100
55/55 [=====] - 1s 13ms/step - loss: 0.4194 - accuracy:
0.9958 - val_loss: 0.5946 - val_accuracy: 0.9343 - lr: 1.8750e-05
Epoch 29/100
55/55 [=====] - 1s 13ms/step - loss: 0.4090 - accuracy:
0.9973 - val_loss: 0.5739 - val_accuracy: 0.9358 - lr: 1.8750e-05
Epoch 30/100
55/55 [=====] - 1s 14ms/step - loss: 0.4072 - accuracy:
0.9962 - val_loss: 0.5407 - val_accuracy: 0.9429 - lr: 1.8750e-05
Epoch 31/100
55/55 [=====] - 1s 14ms/step - loss: 0.4036 - accuracy:
0.9958 - val_loss: 0.8193 - val_accuracy: 0.8458 - lr: 1.8750e-05
Epoch 32/100
55/55 [=====] - 1s 14ms/step - loss: 0.4026 - accuracy:
0.9962 - val_loss: 0.5545 - val_accuracy: 0.9293 - lr: 1.8750e-05
Epoch 33/100
55/55 [=====] - 1s 13ms/step - loss: 0.3990 - accuracy:
0.9965 - val_loss: 0.6323 - val_accuracy: 0.9151 - lr: 1.8750e-05
Epoch 34/100
54/55 [=====>..] - ETA: 0s - loss: 0.3924 - accuracy: 0.998
1
Epoch 34: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
55/55 [=====] - 1s 13ms/step - loss: 0.3926 - accuracy:
0.9981 - val_loss: 1.0868 - val_accuracy: 0.7102 - lr: 1.8750e-05
Epoch 35/100
55/55 [=====] - 1s 14ms/step - loss: 0.3941 - accuracy:
0.9958 - val_loss: 0.5668 - val_accuracy: 0.9365 - lr: 9.3750e-06
Epoch 36/100
55/55 [=====] - 1s 14ms/step - loss: 0.3885 - accuracy:
0.9988 - val_loss: 0.5969 - val_accuracy: 0.9258 - lr: 9.3750e-06
Epoch 37/100
55/55 [=====] - 1s 14ms/step - loss: 0.3874 - accuracy:
0.9981 - val_loss: 0.5841 - val_accuracy: 0.9322 - lr: 9.3750e-06
Epoch 38/100
52/55 [=====>..] - ETA: 0s - loss: 0.3856 - accuracy: 0.998
8
Epoch 38: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
55/55 [=====] - 1s 14ms/step - loss: 0.3861 - accuracy:

0.9988 - val_loss: 0.6297 - val_accuracy: 0.9072 - lr: 9.3750e-06
Epoch 39/100
55/55 [=====] - 1s 14ms/step - loss: 0.3859 - accuracy:
0.9981 - val_loss: 0.5242 - val_accuracy: 0.9443 - lr: 4.6875e-06
Epoch 40/100
55/55 [=====] - 1s 13ms/step - loss: 0.3795 - accuracy:
0.9992 - val_loss: 0.5811 - val_accuracy: 0.9193 - lr: 4.6875e-06
Epoch 41/100
55/55 [=====] - 1s 14ms/step - loss: 0.3841 - accuracy:
0.9992 - val_loss: 0.5151 - val_accuracy: 0.9450 - lr: 4.6875e-06
Epoch 42/100
55/55 [=====] - 1s 13ms/step - loss: 0.3847 - accuracy:
0.9965 - val_loss: 0.5045 - val_accuracy: 0.9515 - lr: 4.6875e-06
Epoch 43/100
55/55 [=====] - 1s 12ms/step - loss: 0.3843 - accuracy:
0.9969 - val_loss: 0.5056 - val_accuracy: 0.9536 - lr: 4.6875e-06
Epoch 44/100
55/55 [=====] - 1s 12ms/step - loss: 0.3816 - accuracy:
0.9981 - val_loss: 0.5137 - val_accuracy: 0.9507 - lr: 4.6875e-06
Epoch 45/100
55/55 [=====] - 1s 12ms/step - loss: 0.3817 - accuracy:
0.9977 - val_loss: 0.5010 - val_accuracy: 0.9500 - lr: 4.6875e-06
Epoch 46/100
55/55 [=====] - 1s 11ms/step - loss: 0.3820 - accuracy:
0.9977 - val_loss: 0.5105 - val_accuracy: 0.9465 - lr: 4.6875e-06
Epoch 47/100
55/55 [=====] - 1s 11ms/step - loss: 0.3782 - accuracy:
0.9977 - val_loss: 0.5029 - val_accuracy: 0.9479 - lr: 4.6875e-06
Epoch 48/100
55/55 [=====] - 1s 13ms/step - loss: 0.3777 - accuracy:
0.9977 - val_loss: 0.4969 - val_accuracy: 0.9500 - lr: 4.6875e-06
Epoch 49/100
55/55 [=====] - 1s 12ms/step - loss: 0.3756 - accuracy:
0.9985 - val_loss: 0.5017 - val_accuracy: 0.9493 - lr: 4.6875e-06
Epoch 50/100
55/55 [=====] - 1s 13ms/step - loss: 0.3721 - accuracy:
0.9985 - val_loss: 0.5104 - val_accuracy: 0.9450 - lr: 4.6875e-06
Epoch 51/100
55/55 [=====] - 1s 13ms/step - loss: 0.3746 - accuracy:
0.9981 - val_loss: 0.4985 - val_accuracy: 0.9500 - lr: 4.6875e-06
Epoch 52/100
55/55 [=====] - 1s 14ms/step - loss: 0.3746 - accuracy:
0.9973 - val_loss: 0.4961 - val_accuracy: 0.9465 - lr: 4.6875e-06
Epoch 53/100
55/55 [=====] - 1s 13ms/step - loss: 0.3716 - accuracy:
0.9985 - val_loss: 0.5010 - val_accuracy: 0.9472 - lr: 4.6875e-06
Epoch 54/100
55/55 [=====] - 1s 13ms/step - loss: 0.3680 - accuracy:
0.9988 - val_loss: 0.4891 - val_accuracy: 0.9522 - lr: 4.6875e-06
Epoch 55/100
55/55 [=====] - 1s 12ms/step - loss: 0.3695 - accuracy:
0.9973 - val_loss: 0.4861 - val_accuracy: 0.9515 - lr: 4.6875e-06
[100, 48, 0.4]
75.0%
Epoch 1/100
50/50 [=====] - 2s 22ms/step - loss: 3.1312 - accuracy:
0.1454 - val_loss: 3.3270 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 2/100
50/50 [=====] - 1s 14ms/step - loss: 2.6322 - accuracy:
0.2438 - val_loss: 4.4003 - val_accuracy: 0.1081 - lr: 3.0000e-04

Epoch 3/100
50/50 [=====] - 1s 12ms/step - loss: 2.2834 - accuracy: 0.3483 - val_loss: 5.3917 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 4/100
50/50 [=====] - 1s 13ms/step - loss: 1.8839 - accuracy: 0.4946 - val_loss: 7.0495 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 5/100
45/50 [=====>...] - ETA: 0s - loss: 1.5571 - accuracy: 0.6106
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
50/50 [=====] - 1s 13ms/step - loss: 1.5413 - accuracy: 0.6187 - val_loss: 7.6750 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 6/100
50/50 [=====] - 1s 13ms/step - loss: 1.2685 - accuracy: 0.7437 - val_loss: 9.0945 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 7/100
50/50 [=====] - 1s 14ms/step - loss: 1.1516 - accuracy: 0.7887 - val_loss: 10.2379 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 8/100
50/50 [=====] - 1s 13ms/step - loss: 1.0191 - accuracy: 0.8454 - val_loss: 8.8999 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 9/100
45/50 [=====>...] - ETA: 0s - loss: 0.9026 - accuracy: 0.8838
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
50/50 [=====] - 1s 13ms/step - loss: 0.8999 - accuracy: 0.8842 - val_loss: 9.0158 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 10/100
50/50 [=====] - 1s 12ms/step - loss: 0.8167 - accuracy: 0.9175 - val_loss: 8.8598 - val_accuracy: 0.1106 - lr: 7.5000e-05
Epoch 11/100
50/50 [=====] - 1s 12ms/step - loss: 0.7657 - accuracy: 0.9392 - val_loss: 6.7514 - val_accuracy: 0.2444 - lr: 7.5000e-05
Epoch 12/100
50/50 [=====] - 1s 11ms/step - loss: 0.7303 - accuracy: 0.9517 - val_loss: 5.8387 - val_accuracy: 0.2069 - lr: 7.5000e-05
Epoch 13/100
48/50 [=====>...] - ETA: 0s - loss: 0.6890 - accuracy: 0.9588
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
50/50 [=====] - 1s 11ms/step - loss: 0.6901 - accuracy: 0.9575 - val_loss: 4.9683 - val_accuracy: 0.2325 - lr: 7.5000e-05
Epoch 14/100
50/50 [=====] - 1s 12ms/step - loss: 0.6563 - accuracy: 0.9671 - val_loss: 3.2164 - val_accuracy: 0.2819 - lr: 3.7500e-05
Epoch 15/100
50/50 [=====] - 1s 12ms/step - loss: 0.6463 - accuracy: 0.9683 - val_loss: 2.5102 - val_accuracy: 0.4425 - lr: 3.7500e-05
Epoch 16/100
50/50 [=====] - 1s 13ms/step - loss: 0.6304 - accuracy: 0.9704 - val_loss: 2.2309 - val_accuracy: 0.4106 - lr: 3.7500e-05
Epoch 17/100
50/50 [=====] - 1s 13ms/step - loss: 0.6108 - accuracy: 0.9758 - val_loss: 2.5566 - val_accuracy: 0.4913 - lr: 3.7500e-05
Epoch 18/100
50/50 [=====] - 1s 14ms/step - loss: 0.6002 - accuracy: 0.9779 - val_loss: 1.5996 - val_accuracy: 0.6031 - lr: 3.7500e-05
Epoch 19/100
50/50 [=====] - 1s 13ms/step - loss: 0.5824 - accuracy: 0.9846 - val_loss: 1.7956 - val_accuracy: 0.4712 - lr: 3.7500e-05

Epoch 20/100
50/50 [=====] - 1s 13ms/step - loss: 0.5758 - accuracy: 0.9800 - val_loss: 1.8478 - val_accuracy: 0.6294 - lr: 3.7500e-05
Epoch 21/100
50/50 [=====] - 1s 13ms/step - loss: 0.5581 - accuracy: 0.9821 - val_loss: 1.8177 - val_accuracy: 0.5669 - lr: 3.7500e-05
Epoch 22/100
49/50 [=====>.] - ETA: 0s - loss: 0.5541 - accuracy: 0.9864
Epoch 22: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
50/50 [=====] - 1s 13ms/step - loss: 0.5537 - accuracy: 0.9867 - val_loss: 1.9794 - val_accuracy: 0.6319 - lr: 3.7500e-05
Epoch 23/100
50/50 [=====] - 1s 14ms/step - loss: 0.5412 - accuracy: 0.9892 - val_loss: 0.7480 - val_accuracy: 0.9050 - lr: 1.8750e-05
Epoch 24/100
50/50 [=====] - 1s 12ms/step - loss: 0.5380 - accuracy: 0.9883 - val_loss: 0.8735 - val_accuracy: 0.8531 - lr: 1.8750e-05
Epoch 25/100
50/50 [=====] - 1s 13ms/step - loss: 0.5350 - accuracy: 0.9879 - val_loss: 0.9698 - val_accuracy: 0.8094 - lr: 1.8750e-05
Epoch 26/100
50/50 [=====] - 1s 15ms/step - loss: 0.5298 - accuracy: 0.9892 - val_loss: 4.6671 - val_accuracy: 0.2969 - lr: 1.8750e-05
Epoch 27/100
49/50 [=====>.] - ETA: 0s - loss: 0.5295 - accuracy: 0.9860
Epoch 27: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
50/50 [=====] - 1s 13ms/step - loss: 0.5287 - accuracy: 0.9858 - val_loss: 0.7648 - val_accuracy: 0.9137 - lr: 1.8750e-05
Epoch 28/100
50/50 [=====] - 1s 13ms/step - loss: 0.5163 - accuracy: 0.9925 - val_loss: 1.3098 - val_accuracy: 0.6587 - lr: 9.3750e-06
Epoch 29/100
50/50 [=====] - 1s 14ms/step - loss: 0.5153 - accuracy: 0.9921 - val_loss: 1.0035 - val_accuracy: 0.7987 - lr: 9.3750e-06
Epoch 30/100
50/50 [=====] - 1s 13ms/step - loss: 0.5136 - accuracy: 0.9908 - val_loss: 1.3860 - val_accuracy: 0.6800 - lr: 9.3750e-06
Epoch 31/100
49/50 [=====>.] - ETA: 0s - loss: 0.5083 - accuracy: 0.9940
Epoch 31: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
50/50 [=====] - 1s 13ms/step - loss: 0.5089 - accuracy: 0.9937 - val_loss: 0.8931 - val_accuracy: 0.8275 - lr: 9.3750e-06
Epoch 32/100
50/50 [=====] - 1s 13ms/step - loss: 0.5099 - accuracy: 0.9946 - val_loss: 0.6439 - val_accuracy: 0.9425 - lr: 4.6875e-06
Epoch 33/100
50/50 [=====] - 1s 13ms/step - loss: 0.5062 - accuracy: 0.9921 - val_loss: 0.6657 - val_accuracy: 0.9337 - lr: 4.6875e-06
Epoch 34/100
50/50 [=====] - 1s 13ms/step - loss: 0.5067 - accuracy: 0.9929 - val_loss: 0.7138 - val_accuracy: 0.9144 - lr: 4.6875e-06
Epoch 35/100
50/50 [=====] - 1s 12ms/step - loss: 0.5024 - accuracy: 0.9954 - val_loss: 0.6412 - val_accuracy: 0.9463 - lr: 4.6875e-06
Epoch 36/100
50/50 [=====] - 1s 11ms/step - loss: 0.5031 - accuracy: 0.9925 - val_loss: 0.6710 - val_accuracy: 0.9331 - lr: 4.6875e-06

Epoch 37/100
50/50 [=====] - 1s 12ms/step - loss: 0.5037 - accuracy: 0.9912 - val_loss: 0.6902 - val_accuracy: 0.9269 - lr: 4.6875e-06
Epoch 38/100
50/50 [=====] - 1s 11ms/step - loss: 0.5035 - accuracy: 0.9925 - val_loss: 0.6767 - val_accuracy: 0.9331 - lr: 4.6875e-06
Epoch 39/100
50/50 [=====] - 1s 12ms/step - loss: 0.4975 - accuracy: 0.9950 - val_loss: 0.6362 - val_accuracy: 0.9431 - lr: 4.6875e-06
Epoch 40/100
50/50 [=====] - 1s 12ms/step - loss: 0.4965 - accuracy: 0.9942 - val_loss: 0.6920 - val_accuracy: 0.9144 - lr: 4.6875e-06
Epoch 41/100
50/50 [=====] - 1s 13ms/step - loss: 0.4946 - accuracy: 0.9946 - val_loss: 0.6444 - val_accuracy: 0.9356 - lr: 4.6875e-06
Epoch 42/100
50/50 [=====] - 1s 12ms/step - loss: 0.4955 - accuracy: 0.9929 - val_loss: 0.7918 - val_accuracy: 0.8775 - lr: 4.6875e-06
Epoch 43/100
50/50 [=====] - ETA: 0s - loss: 0.4944 - accuracy: 0.9925
Epoch 43: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
50/50 [=====] - 1s 13ms/step - loss: 0.4944 - accuracy: 0.9925 - val_loss: 0.6801 - val_accuracy: 0.9231 - lr: 4.6875e-06
Epoch 44/100
50/50 [=====] - 1s 13ms/step - loss: 0.4939 - accuracy: 0.9946 - val_loss: 0.6773 - val_accuracy: 0.9281 - lr: 2.3438e-06
Epoch 45/100
50/50 [=====] - 1s 13ms/step - loss: 0.4972 - accuracy: 0.9929 - val_loss: 0.6655 - val_accuracy: 0.9300 - lr: 2.3438e-06
Epoch 46/100
50/50 [=====] - 1s 12ms/step - loss: 0.4918 - accuracy: 0.9937 - val_loss: 0.6335 - val_accuracy: 0.9350 - lr: 2.3438e-06
Epoch 47/100
50/50 [=====] - 1s 13ms/step - loss: 0.4954 - accuracy: 0.9937 - val_loss: 0.6245 - val_accuracy: 0.9494 - lr: 2.3438e-06
Epoch 48/100
50/50 [=====] - 1s 13ms/step - loss: 0.4900 - accuracy: 0.9958 - val_loss: 0.6209 - val_accuracy: 0.9519 - lr: 2.3438e-06
Epoch 49/100
50/50 [=====] - 1s 12ms/step - loss: 0.4897 - accuracy: 0.9942 - val_loss: 0.6522 - val_accuracy: 0.9356 - lr: 2.3438e-06
Epoch 50/100
50/50 [=====] - 1s 12ms/step - loss: 0.4857 - accuracy: 0.9950 - val_loss: 0.6233 - val_accuracy: 0.9438 - lr: 2.3438e-06
Epoch 51/100
50/50 [=====] - 1s 13ms/step - loss: 0.4898 - accuracy: 0.9950 - val_loss: 0.6142 - val_accuracy: 0.9481 - lr: 2.3438e-06
Epoch 52/100
50/50 [=====] - 1s 14ms/step - loss: 0.4883 - accuracy: 0.9954 - val_loss: 0.6219 - val_accuracy: 0.9475 - lr: 2.3438e-06
Epoch 53/100
50/50 [=====] - 1s 13ms/step - loss: 0.4947 - accuracy: 0.9921 - val_loss: 0.6148 - val_accuracy: 0.9475 - lr: 2.3438e-06
Epoch 54/100
50/50 [=====] - 1s 13ms/step - loss: 0.4869 - accuracy: 0.9958 - val_loss: 0.6162 - val_accuracy: 0.9456 - lr: 2.3438e-06
Epoch 55/100
48/50 [=====>..] - ETA: 0s - loss: 0.4882 - accuracy: 0.9952

Epoch 55: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
50/50 [=====] - 1s 13ms/step - loss: 0.4876 - accuracy: 0.9954 - val_loss: 0.6267 - val_accuracy: 0.9394 - lr: 2.3438e-06
Epoch 56/100
50/50 [=====] - 1s 14ms/step - loss: 0.4854 - accuracy: 0.9925 - val_loss: 0.6168 - val_accuracy: 0.9488 - lr: 1.1719e-06
Epoch 57/100
50/50 [=====] - 1s 13ms/step - loss: 0.4817 - accuracy: 0.9979 - val_loss: 0.6149 - val_accuracy: 0.9494 - lr: 1.1719e-06
Epoch 58/100
50/50 [=====] - 1s 13ms/step - loss: 0.4822 - accuracy: 0.9962 - val_loss: 0.6193 - val_accuracy: 0.9438 - lr: 1.1719e-06
Epoch 59/100
50/50 [=====] - 1s 14ms/step - loss: 0.4821 - accuracy: 0.9946 - val_loss: 0.6090 - val_accuracy: 0.9525 - lr: 1.1719e-06
Epoch 60/100
50/50 [=====] - 1s 12ms/step - loss: 0.4831 - accuracy: 0.9967 - val_loss: 0.6071 - val_accuracy: 0.9506 - lr: 1.1719e-06
Epoch 61/100
50/50 [=====] - 1s 12ms/step - loss: 0.4841 - accuracy: 0.9962 - val_loss: 0.6258 - val_accuracy: 0.9400 - lr: 1.1719e-06
Epoch 62/100
50/50 [=====] - 1s 12ms/step - loss: 0.4812 - accuracy: 0.9946 - val_loss: 0.6172 - val_accuracy: 0.9438 - lr: 1.1719e-06
Epoch 63/100
50/50 [=====] - 1s 12ms/step - loss: 0.4819 - accuracy: 0.9962 - val_loss: 0.6086 - val_accuracy: 0.9513 - lr: 1.1719e-06
Epoch 64/100
46/50 [=====>...] - ETA: 0s - loss: 0.4860 - accuracy: 0.9937
Epoch 64: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.
50/50 [=====] - 1s 12ms/step - loss: 0.4856 - accuracy: 0.9942 - val_loss: 0.6080 - val_accuracy: 0.9494 - lr: 1.1719e-06
Epoch 65/100
50/50 [=====] - 1s 12ms/step - loss: 0.4829 - accuracy: 0.9954 - val_loss: 0.6090 - val_accuracy: 0.9513 - lr: 5.8594e-07
Epoch 66/100
50/50 [=====] - 1s 13ms/step - loss: 0.4862 - accuracy: 0.9967 - val_loss: 0.6378 - val_accuracy: 0.9369 - lr: 5.8594e-07
Epoch 67/100
50/50 [=====] - 1s 12ms/step - loss: 0.4761 - accuracy: 0.9962 - val_loss: 0.6294 - val_accuracy: 0.9413 - lr: 5.8594e-07
Epoch 68/100
49/50 [=====>.] - ETA: 0s - loss: 0.4844 - accuracy: 0.9966
Epoch 68: ReduceLROnPlateau reducing learning rate to 2.9296876391526894e-07.
50/50 [=====] - 1s 13ms/step - loss: 0.4843 - accuracy: 0.9967 - val_loss: 0.6111 - val_accuracy: 0.9456 - lr: 5.8594e-07
Epoch 69/100
50/50 [=====] - 1s 12ms/step - loss: 0.4798 - accuracy: 0.9962 - val_loss: 0.6084 - val_accuracy: 0.9481 - lr: 2.9297e-07
Epoch 70/100
50/50 [=====] - 1s 11ms/step - loss: 0.4814 - accuracy: 0.9954 - val_loss: 0.6057 - val_accuracy: 0.9519 - lr: 2.9297e-07
Epoch 71/100
50/50 [=====] - 1s 12ms/step - loss: 0.4843 - accuracy: 0.9946 - val_loss: 0.6061 - val_accuracy: 0.9513 - lr: 2.9297e-07
[100, 56, 0.2]
77.5%
Epoch 1/100

58/58 [=====] - 2s 16ms/step - loss: 3.0983 - accuracy: 0.1328 - val_loss: 3.1286 - val_accuracy: 0.0925 - lr: 3.0000e-04
Epoch 2/100
58/58 [=====] - 1s 12ms/step - loss: 2.6005 - accuracy: 0.2516 - val_loss: 4.0074 - val_accuracy: 0.0900 - lr: 3.0000e-04
Epoch 3/100
58/58 [=====] - 1s 14ms/step - loss: 2.2737 - accuracy: 0.3491 - val_loss: 5.4622 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 4/100
58/58 [=====] - 1s 13ms/step - loss: 1.9190 - accuracy: 0.4866 - val_loss: 6.9851 - val_accuracy: 0.1088 - lr: 3.0000e-04
Epoch 5/100
54/58 [=====>...] - ETA: 0s - loss: 1.5660 - accuracy: 0.6382
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
58/58 [=====] - 1s 12ms/step - loss: 1.5565 - accuracy: 0.6422 - val_loss: 9.2969 - val_accuracy: 0.0975 - lr: 3.0000e-04
Epoch 6/100
58/58 [=====] - 1s 11ms/step - loss: 1.2774 - accuracy: 0.7437 - val_loss: 10.3087 - val_accuracy: 0.0900 - lr: 1.5000e-04
Epoch 7/100
58/58 [=====] - 1s 13ms/step - loss: 1.1316 - accuracy: 0.8122 - val_loss: 11.0343 - val_accuracy: 0.0900 - lr: 1.5000e-04
Epoch 8/100
58/58 [=====] - 1s 13ms/step - loss: 1.0071 - accuracy: 0.8519 - val_loss: 12.5777 - val_accuracy: 0.0950 - lr: 1.5000e-04
Epoch 9/100
58/58 [=====] - ETA: 0s - loss: 0.9158 - accuracy: 0.8763
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
58/58 [=====] - 1s 12ms/step - loss: 0.9158 - accuracy: 0.8763 - val_loss: 10.3481 - val_accuracy: 0.1825 - lr: 1.5000e-04
Epoch 10/100
58/58 [=====] - 1s 11ms/step - loss: 0.8249 - accuracy: 0.9100 - val_loss: 6.9816 - val_accuracy: 0.1813 - lr: 7.5000e-05
Epoch 11/100
58/58 [=====] - 1s 11ms/step - loss: 0.7771 - accuracy: 0.9275 - val_loss: 4.2432 - val_accuracy: 0.1838 - lr: 7.5000e-05
Epoch 12/100
58/58 [=====] - 1s 10ms/step - loss: 0.7377 - accuracy: 0.9344 - val_loss: 4.6506 - val_accuracy: 0.3262 - lr: 7.5000e-05
Epoch 13/100
58/58 [=====] - 1s 10ms/step - loss: 0.6973 - accuracy: 0.9413 - val_loss: 1.6742 - val_accuracy: 0.4550 - lr: 7.5000e-05
Epoch 14/100
58/58 [=====] - 1s 10ms/step - loss: 0.6645 - accuracy: 0.9578 - val_loss: 2.1904 - val_accuracy: 0.3887 - lr: 7.5000e-05
Epoch 15/100
58/58 [=====] - 1s 10ms/step - loss: 0.6268 - accuracy: 0.9659 - val_loss: 5.7657 - val_accuracy: 0.2637 - lr: 7.5000e-05
Epoch 16/100
58/58 [=====] - 1s 12ms/step - loss: 0.6022 - accuracy: 0.9753 - val_loss: 1.3943 - val_accuracy: 0.6587 - lr: 7.5000e-05
Epoch 17/100
58/58 [=====] - 1s 14ms/step - loss: 0.5703 - accuracy: 0.9812 - val_loss: 1.5110 - val_accuracy: 0.5550 - lr: 7.5000e-05
Epoch 18/100
58/58 [=====] - 1s 11ms/step - loss: 0.5487 - accuracy: 0.9856 - val_loss: 1.0980 - val_accuracy: 0.8238 - lr: 7.5000e-05
Epoch 19/100

58/58 [=====] - 1s 12ms/step - loss: 0.5292 - accuracy: 0.9837 - val_loss: 1.8450 - val_accuracy: 0.5013 - lr: 7.5000e-05
Epoch 20/100
58/58 [=====] - 1s 12ms/step - loss: 0.5093 - accuracy: 0.9900 - val_loss: 1.1547 - val_accuracy: 0.7362 - lr: 7.5000e-05
Epoch 21/100
58/58 [=====] - 1s 11ms/step - loss: 0.4834 - accuracy: 0.9937 - val_loss: 1.3739 - val_accuracy: 0.6125 - lr: 7.5000e-05
Epoch 22/100
58/58 [=====] - ETA: 0s - loss: 0.4712 - accuracy: 0.9941
1
Epoch 22: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
58/58 [=====] - 1s 12ms/step - loss: 0.4712 - accuracy: 0.9941 - val_loss: 4.0181 - val_accuracy: 0.4050 - lr: 7.5000e-05
Epoch 23/100
58/58 [=====] - 1s 12ms/step - loss: 0.4604 - accuracy: 0.9944 - val_loss: 0.7664 - val_accuracy: 0.8850 - lr: 3.7500e-05
Epoch 24/100
58/58 [=====] - 1s 12ms/step - loss: 0.4487 - accuracy: 0.9966 - val_loss: 0.7015 - val_accuracy: 0.9175 - lr: 3.7500e-05
Epoch 25/100
58/58 [=====] - 1s 11ms/step - loss: 0.4385 - accuracy: 0.9984 - val_loss: 0.5469 - val_accuracy: 0.9588 - lr: 3.7500e-05
Epoch 26/100
58/58 [=====] - 1s 12ms/step - loss: 0.4379 - accuracy: 0.9956 - val_loss: 1.3971 - val_accuracy: 0.6237 - lr: 3.7500e-05
Epoch 27/100
58/58 [=====] - 1s 11ms/step - loss: 0.4361 - accuracy: 0.9950 - val_loss: 0.6737 - val_accuracy: 0.9125 - lr: 3.7500e-05
Epoch 28/100
58/58 [=====] - 1s 12ms/step - loss: 0.4253 - accuracy: 0.9981 - val_loss: 3.3092 - val_accuracy: 0.4013 - lr: 3.7500e-05
Epoch 29/100
55/58 [=====>..] - ETA: 0s - loss: 0.4175 - accuracy: 0.9974
4
Epoch 29: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
58/58 [=====] - 1s 12ms/step - loss: 0.4179 - accuracy: 0.9972 - val_loss: 0.8184 - val_accuracy: 0.8413 - lr: 3.7500e-05
Epoch 30/100
58/58 [=====] - 1s 12ms/step - loss: 0.4099 - accuracy: 0.9972 - val_loss: 0.5224 - val_accuracy: 0.9488 - lr: 1.8750e-05
Epoch 31/100
58/58 [=====] - 1s 12ms/step - loss: 0.4073 - accuracy: 0.9994 - val_loss: 0.6622 - val_accuracy: 0.9000 - lr: 1.8750e-05
Epoch 32/100
58/58 [=====] - 1s 11ms/step - loss: 0.4027 - accuracy: 0.9994 - val_loss: 0.6389 - val_accuracy: 0.9337 - lr: 1.8750e-05
Epoch 33/100
58/58 [=====] - 1s 12ms/step - loss: 0.4019 - accuracy: 0.9978 - val_loss: 0.7706 - val_accuracy: 0.8487 - lr: 1.8750e-05
Epoch 34/100
58/58 [=====] - 1s 13ms/step - loss: 0.3960 - accuracy: 0.9972 - val_loss: 0.4937 - val_accuracy: 0.9700 - lr: 1.8750e-05
Epoch 35/100
58/58 [=====] - 1s 11ms/step - loss: 0.3930 - accuracy: 0.9987 - val_loss: 0.5654 - val_accuracy: 0.9488 - lr: 1.8750e-05
Epoch 36/100
58/58 [=====] - 1s 10ms/step - loss: 0.3974 - accuracy: 0.9987 - val_loss: 0.6426 - val_accuracy: 0.9087 - lr: 1.8750e-05
Epoch 37/100

58/58 [=====] - 1s 11ms/step - loss: 0.3896 - accuracy: 0.9991 - val_loss: 0.5089 - val_accuracy: 0.9575 - lr: 1.8750e-05
Epoch 38/100
56/58 [=====>..] - ETA: 0s - loss: 0.3893 - accuracy: 0.9987
Epoch 38: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
58/58 [=====] - 1s 10ms/step - loss: 0.3900 - accuracy: 0.9987 - val_loss: 1.0281 - val_accuracy: 0.7513 - lr: 1.8750e-05
Epoch 39/100
58/58 [=====] - 1s 11ms/step - loss: 0.3861 - accuracy: 0.9978 - val_loss: 0.4712 - val_accuracy: 0.9712 - lr: 9.3750e-06
Epoch 40/100
58/58 [=====] - 1s 13ms/step - loss: 0.3847 - accuracy: 0.9972 - val_loss: 0.5186 - val_accuracy: 0.9575 - lr: 9.3750e-06
Epoch 41/100
58/58 [=====] - 1s 12ms/step - loss: 0.3809 - accuracy: 0.9987 - val_loss: 0.4848 - val_accuracy: 0.9625 - lr: 9.3750e-06
Epoch 42/100
58/58 [=====] - 1s 12ms/step - loss: 0.3804 - accuracy: 0.9991 - val_loss: 0.4492 - val_accuracy: 0.9700 - lr: 9.3750e-06
Epoch 43/100
58/58 [=====] - 1s 13ms/step - loss: 0.3793 - accuracy: 0.9994 - val_loss: 0.5064 - val_accuracy: 0.9500 - lr: 9.3750e-06
Epoch 44/100
58/58 [=====] - 1s 14ms/step - loss: 0.3763 - accuracy: 0.9991 - val_loss: 0.4386 - val_accuracy: 0.9663 - lr: 9.3750e-06
Epoch 45/100
58/58 [=====] - 1s 13ms/step - loss: 0.3759 - accuracy: 0.9991 - val_loss: 0.4501 - val_accuracy: 0.9675 - lr: 9.3750e-06
Epoch 46/100
58/58 [=====] - 1s 13ms/step - loss: 0.3752 - accuracy: 0.9978 - val_loss: 0.5391 - val_accuracy: 0.9488 - lr: 9.3750e-06
Epoch 47/100
58/58 [=====] - 1s 12ms/step - loss: 0.3750 - accuracy: 0.9994 - val_loss: 0.4561 - val_accuracy: 0.9625 - lr: 9.3750e-06
Epoch 48/100
58/58 [=====] - 1s 12ms/step - loss: 0.3723 - accuracy: 0.9987 - val_loss: 0.4322 - val_accuracy: 0.9650 - lr: 9.3750e-06
Epoch 49/100
58/58 [=====] - 1s 12ms/step - loss: 0.3698 - accuracy: 0.9994 - val_loss: 0.4528 - val_accuracy: 0.9700 - lr: 9.3750e-06
Epoch 50/100
58/58 [=====] - 1s 13ms/step - loss: 0.3690 - accuracy: 0.9991 - val_loss: 0.4436 - val_accuracy: 0.9700 - lr: 9.3750e-06
Epoch 51/100
58/58 [=====] - 1s 11ms/step - loss: 0.3665 - accuracy: 0.9991 - val_loss: 0.5535 - val_accuracy: 0.9500 - lr: 9.3750e-06
[100, 56, 0.25]
80.0%
Epoch 1/100
54/54 [=====] - 2s 21ms/step - loss: 3.1960 - accuracy: 0.1477 - val_loss: 3.7794 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 2/100
54/54 [=====] - 1s 13ms/step - loss: 2.6363 - accuracy: 0.2597 - val_loss: 4.9406 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 3/100
54/54 [=====] - 1s 12ms/step - loss: 2.2791 - accuracy: 0.3400 - val_loss: 6.9417 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 4/100
54/54 [=====] - 1s 13ms/step - loss: 1.9232 - accuracy:

0.4623 - val_loss: 8.9778 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 5/100
52/54 [=====>..] - ETA: 0s - loss: 1.5530 - accuracy: 0.6174
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
54/54 [=====] - 1s 10ms/step - loss: 1.5495 - accuracy: 0.6173 - val_loss: 10.0623 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 6/100
54/54 [=====] - 1s 10ms/step - loss: 1.2054 - accuracy: 0.7633 - val_loss: 11.4110 - val_accuracy: 0.1020 - lr: 1.5000e-04
Epoch 7/100
54/54 [=====] - 1s 10ms/step - loss: 1.0308 - accuracy: 0.8287 - val_loss: 12.2440 - val_accuracy: 0.1160 - lr: 1.5000e-04
Epoch 8/100
54/54 [=====] - 1s 10ms/step - loss: 0.8581 - accuracy: 0.9010 - val_loss: 12.7809 - val_accuracy: 0.1160 - lr: 1.5000e-04
Epoch 9/100
51/54 [=====>..] - ETA: 0s - loss: 0.7478 - accuracy: 0.9317
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
54/54 [=====] - 1s 11ms/step - loss: 0.7468 - accuracy: 0.9317 - val_loss: 13.1961 - val_accuracy: 0.1160 - lr: 1.5000e-04
Epoch 10/100
54/54 [=====] - 1s 11ms/step - loss: 0.6687 - accuracy: 0.9587 - val_loss: 8.0679 - val_accuracy: 0.1890 - lr: 7.5000e-05
Epoch 11/100
54/54 [=====] - 1s 12ms/step - loss: 0.6248 - accuracy: 0.9720 - val_loss: 5.8620 - val_accuracy: 0.2910 - lr: 7.5000e-05
Epoch 12/100
54/54 [=====] - 1s 13ms/step - loss: 0.6032 - accuracy: 0.9770 - val_loss: 5.5597 - val_accuracy: 0.3160 - lr: 7.5000e-05
Epoch 13/100
51/54 [=====>..] - ETA: 0s - loss: 0.5769 - accuracy: 0.9804
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
54/54 [=====] - 1s 13ms/step - loss: 0.5765 - accuracy: 0.9803 - val_loss: 4.4614 - val_accuracy: 0.2660 - lr: 7.5000e-05
Epoch 14/100
54/54 [=====] - 1s 11ms/step - loss: 0.5584 - accuracy: 0.9870 - val_loss: 4.1630 - val_accuracy: 0.3770 - lr: 3.7500e-05
Epoch 15/100
54/54 [=====] - 1s 13ms/step - loss: 0.5475 - accuracy: 0.9850 - val_loss: 1.3084 - val_accuracy: 0.6810 - lr: 3.7500e-05
Epoch 16/100
54/54 [=====] - 1s 13ms/step - loss: 0.5371 - accuracy: 0.9890 - val_loss: 1.1141 - val_accuracy: 0.7290 - lr: 3.7500e-05
Epoch 17/100
54/54 [=====] - 1s 13ms/step - loss: 0.5267 - accuracy: 0.9863 - val_loss: 1.4480 - val_accuracy: 0.6240 - lr: 3.7500e-05
Epoch 18/100
54/54 [=====] - 1s 13ms/step - loss: 0.5158 - accuracy: 0.9910 - val_loss: 0.9653 - val_accuracy: 0.8050 - lr: 3.7500e-05
Epoch 19/100
54/54 [=====] - 1s 13ms/step - loss: 0.5054 - accuracy: 0.9933 - val_loss: 1.4490 - val_accuracy: 0.5970 - lr: 3.7500e-05
Epoch 20/100
54/54 [=====] - 1s 13ms/step - loss: 0.5006 - accuracy: 0.9933 - val_loss: 1.5398 - val_accuracy: 0.5750 - lr: 3.7500e-05
Epoch 21/100
54/54 [=====] - 1s 12ms/step - loss: 0.4895 - accuracy:

0.9937 - val_loss: 1.7646 - val_accuracy: 0.5200 - lr: 3.7500e-05
Epoch 22/100
54/54 [=====] - 1s 12ms/step - loss: 0.4863 - accuracy:
0.9950 - val_loss: 0.6762 - val_accuracy: 0.9240 - lr: 3.7500e-05
Epoch 23/100
54/54 [=====] - 1s 12ms/step - loss: 0.4751 - accuracy:
0.9967 - val_loss: 1.2809 - val_accuracy: 0.6680 - lr: 3.7500e-05
Epoch 24/100
54/54 [=====] - 1s 13ms/step - loss: 0.4681 - accuracy:
0.9953 - val_loss: 1.0599 - val_accuracy: 0.7660 - lr: 3.7500e-05
Epoch 25/100
54/54 [=====] - 1s 13ms/step - loss: 0.4639 - accuracy:
0.9937 - val_loss: 0.7858 - val_accuracy: 0.8580 - lr: 3.7500e-05
Epoch 26/100
52/54 [=====>..] - ETA: 0s - loss: 0.4550 - accuracy: 0.995
9
Epoch 26: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
54/54 [=====] - 1s 14ms/step - loss: 0.4549 - accuracy:
0.9960 - val_loss: 1.2988 - val_accuracy: 0.6520 - lr: 3.7500e-05
Epoch 27/100
54/54 [=====] - 1s 14ms/step - loss: 0.4514 - accuracy:
0.9970 - val_loss: 0.6562 - val_accuracy: 0.9250 - lr: 1.8750e-05
Epoch 28/100
54/54 [=====] - 1s 13ms/step - loss: 0.4453 - accuracy:
0.9970 - val_loss: 0.5527 - val_accuracy: 0.9460 - lr: 1.8750e-05
Epoch 29/100
54/54 [=====] - 1s 12ms/step - loss: 0.4418 - accuracy:
0.9967 - val_loss: 0.9037 - val_accuracy: 0.8220 - lr: 1.8750e-05
Epoch 30/100
54/54 [=====] - 1s 11ms/step - loss: 0.4393 - accuracy:
0.9973 - val_loss: 0.5610 - val_accuracy: 0.9500 - lr: 1.8750e-05
Epoch 31/100
54/54 [=====] - 1s 11ms/step - loss: 0.4348 - accuracy:
0.9980 - val_loss: 0.7782 - val_accuracy: 0.8530 - lr: 1.8750e-05
Epoch 32/100
50/54 [=====>...] - ETA: 0s - loss: 0.4347 - accuracy: 0.997
9
Epoch 32: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
54/54 [=====] - 1s 11ms/step - loss: 0.4346 - accuracy:
0.9980 - val_loss: 1.0701 - val_accuracy: 0.7510 - lr: 1.8750e-05
Epoch 33/100
54/54 [=====] - 1s 11ms/step - loss: 0.4279 - accuracy:
0.9983 - val_loss: 0.6862 - val_accuracy: 0.9040 - lr: 9.3750e-06
Epoch 34/100
54/54 [=====] - 1s 12ms/step - loss: 0.4246 - accuracy:
0.9990 - val_loss: 0.5035 - val_accuracy: 0.9610 - lr: 9.3750e-06
Epoch 35/100
54/54 [=====] - 1s 12ms/step - loss: 0.4244 - accuracy:
0.9990 - val_loss: 0.5641 - val_accuracy: 0.9460 - lr: 9.3750e-06
Epoch 36/100
54/54 [=====] - 1s 13ms/step - loss: 0.4248 - accuracy:
0.9977 - val_loss: 0.5608 - val_accuracy: 0.9450 - lr: 9.3750e-06
Epoch 37/100
54/54 [=====] - 1s 12ms/step - loss: 0.4230 - accuracy:
0.9973 - val_loss: 0.5128 - val_accuracy: 0.9600 - lr: 9.3750e-06
Epoch 38/100
52/54 [=====>..] - ETA: 0s - loss: 0.4202 - accuracy: 0.996
6
Epoch 38: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
54/54 [=====] - 1s 14ms/step - loss: 0.4203 - accuracy:

0.9967 - val_loss: 0.5285 - val_accuracy: 0.9580 - lr: 9.3750e-06
Epoch 39/100
54/54 [=====] - 1s 11ms/step - loss: 0.4198 - accuracy:
0.9993 - val_loss: 0.5020 - val_accuracy: 0.9630 - lr: 4.6875e-06
Epoch 40/100
54/54 [=====] - 1s 13ms/step - loss: 0.4181 - accuracy:
0.9990 - val_loss: 0.5054 - val_accuracy: 0.9640 - lr: 4.6875e-06
Epoch 41/100
54/54 [=====] - 1s 14ms/step - loss: 0.4176 - accuracy:
0.9987 - val_loss: 0.4938 - val_accuracy: 0.9660 - lr: 4.6875e-06
Epoch 42/100
54/54 [=====] - 1s 12ms/step - loss: 0.4205 - accuracy:
0.9967 - val_loss: 0.6270 - val_accuracy: 0.9260 - lr: 4.6875e-06
Epoch 43/100
54/54 [=====] - 1s 13ms/step - loss: 0.4161 - accuracy:
0.9983 - val_loss: 0.5219 - val_accuracy: 0.9590 - lr: 4.6875e-06
Epoch 44/100
54/54 [=====] - 1s 13ms/step - loss: 0.4154 - accuracy:
0.9977 - val_loss: 0.4890 - val_accuracy: 0.9650 - lr: 4.6875e-06
Epoch 45/100
54/54 [=====] - 1s 12ms/step - loss: 0.4168 - accuracy:
0.9983 - val_loss: 0.4920 - val_accuracy: 0.9650 - lr: 4.6875e-06
Epoch 46/100
54/54 [=====] - 1s 13ms/step - loss: 0.4135 - accuracy:
0.9993 - val_loss: 0.5011 - val_accuracy: 0.9610 - lr: 4.6875e-06
Epoch 47/100
54/54 [=====] - 1s 13ms/step - loss: 0.4123 - accuracy:
0.9987 - val_loss: 0.4880 - val_accuracy: 0.9650 - lr: 4.6875e-06
Epoch 48/100
54/54 [=====] - 1s 13ms/step - loss: 0.4132 - accuracy:
0.9977 - val_loss: 0.5022 - val_accuracy: 0.9630 - lr: 4.6875e-06
Epoch 49/100
54/54 [=====] - 1s 13ms/step - loss: 0.4142 - accuracy:
0.9983 - val_loss: 0.4916 - val_accuracy: 0.9640 - lr: 4.6875e-06
Epoch 50/100
54/54 [=====] - 1s 13ms/step - loss: 0.4100 - accuracy:
0.9983 - val_loss: 0.5119 - val_accuracy: 0.9600 - lr: 4.6875e-06
Epoch 51/100
54/54 [=====] - 1s 13ms/step - loss: 0.4093 - accuracy:
0.9997 - val_loss: 0.4869 - val_accuracy: 0.9660 - lr: 4.6875e-06
Epoch 52/100
54/54 [=====] - 1s 13ms/step - loss: 0.4085 - accuracy:
0.9987 - val_loss: 0.5013 - val_accuracy: 0.9630 - lr: 4.6875e-06
Epoch 53/100
54/54 [=====] - 1s 11ms/step - loss: 0.4072 - accuracy:
0.9987 - val_loss: 0.4956 - val_accuracy: 0.9630 - lr: 4.6875e-06
[100, 56, 0.30000000000000004]
82.5%
Epoch 1/100
50/50 [=====] - 2s 19ms/step - loss: 3.1154 - accuracy:
0.1679 - val_loss: 3.2438 - val_accuracy: 0.1017 - lr: 3.0000e-04
Epoch 2/100
50/50 [=====] - 0s 10ms/step - loss: 2.5246 - accuracy:
0.2643 - val_loss: 4.0595 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 3/100
50/50 [=====] - 1s 11ms/step - loss: 2.2042 - accuracy:
0.3671 - val_loss: 5.4292 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 4/100
50/50 [=====] - 1s 12ms/step - loss: 1.9282 - accuracy:
0.4804 - val_loss: 7.2358 - val_accuracy: 0.1133 - lr: 3.0000e-04

Epoch 5/100
46/50 [=====>...] - ETA: 0s - loss: 1.5856 - accuracy: 0.6056
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.000150000071246177.
50/50 [=====] - 1s 12ms/step - loss: 1.5769 - accuracy: 0.6114 - val_loss: 9.6185 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 6/100
50/50 [=====] - 1s 11ms/step - loss: 1.2390 - accuracy: 0.7589 - val_loss: 12.5336 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 7/100
50/50 [=====] - 1s 12ms/step - loss: 1.0709 - accuracy: 0.8279 - val_loss: 14.4932 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 8/100
50/50 [=====] - 1s 11ms/step - loss: 0.9346 - accuracy: 0.8718 - val_loss: 15.1118 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 9/100
49/50 [=====>.] - ETA: 0s - loss: 0.8371 - accuracy: 0.9060
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
50/50 [=====] - 1s 11ms/step - loss: 0.8381 - accuracy: 0.9050 - val_loss: 14.5750 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 10/100
50/50 [=====] - 1s 11ms/step - loss: 0.7565 - accuracy: 0.9318 - val_loss: 15.5378 - val_accuracy: 0.1133 - lr: 7.5000e-05
Epoch 11/100
50/50 [=====] - 1s 12ms/step - loss: 0.7070 - accuracy: 0.9468 - val_loss: 12.0764 - val_accuracy: 0.1133 - lr: 7.5000e-05
Epoch 12/100
50/50 [=====] - 1s 12ms/step - loss: 0.6891 - accuracy: 0.9554 - val_loss: 10.0400 - val_accuracy: 0.1175 - lr: 7.5000e-05
Epoch 13/100
46/50 [=====>...] - ETA: 0s - loss: 0.6482 - accuracy: 0.9701
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
50/50 [=====] - 1s 12ms/step - loss: 0.6491 - accuracy: 0.9693 - val_loss: 9.3078 - val_accuracy: 0.1875 - lr: 7.5000e-05
Epoch 14/100
50/50 [=====] - 1s 12ms/step - loss: 0.6209 - accuracy: 0.9721 - val_loss: 6.0943 - val_accuracy: 0.1608 - lr: 3.7500e-05
Epoch 15/100
50/50 [=====] - 1s 12ms/step - loss: 0.6128 - accuracy: 0.9764 - val_loss: 3.9822 - val_accuracy: 0.2425 - lr: 3.7500e-05
Epoch 16/100
50/50 [=====] - 1s 13ms/step - loss: 0.6043 - accuracy: 0.9796 - val_loss: 2.4055 - val_accuracy: 0.3142 - lr: 3.7500e-05
Epoch 17/100
50/50 [=====] - 1s 12ms/step - loss: 0.5909 - accuracy: 0.9807 - val_loss: 1.3655 - val_accuracy: 0.6275 - lr: 3.7500e-05
Epoch 18/100
50/50 [=====] - 1s 11ms/step - loss: 0.5766 - accuracy: 0.9854 - val_loss: 2.0913 - val_accuracy: 0.4592 - lr: 3.7500e-05
Epoch 19/100
50/50 [=====] - 1s 12ms/step - loss: 0.5670 - accuracy: 0.9868 - val_loss: 0.8797 - val_accuracy: 0.8442 - lr: 3.7500e-05
Epoch 20/100
50/50 [=====] - 1s 12ms/step - loss: 0.5553 - accuracy: 0.9854 - val_loss: 0.9658 - val_accuracy: 0.8067 - lr: 3.7500e-05
Epoch 21/100
50/50 [=====] - 1s 12ms/step - loss: 0.5448 - accuracy: 0.9850 - val_loss: 0.8867 - val_accuracy: 0.8150 - lr: 3.7500e-05

Epoch 22/100
50/50 [=====] - 1s 11ms/step - loss: 0.5342 - accuracy: 0.9911 - val_loss: 1.2857 - val_accuracy: 0.6667 - lr: 3.7500e-05
Epoch 23/100
45/50 [=====>...] - ETA: 0s - loss: 0.5300 - accuracy: 0.9877
Epoch 23: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
50/50 [=====] - 1s 11ms/step - loss: 0.5292 - accuracy: 0.9882 - val_loss: 0.9528 - val_accuracy: 0.7842 - lr: 3.7500e-05
Epoch 24/100
50/50 [=====] - 1s 12ms/step - loss: 0.5196 - accuracy: 0.9921 - val_loss: 0.6372 - val_accuracy: 0.9442 - lr: 1.8750e-05
Epoch 25/100
50/50 [=====] - 1s 10ms/step - loss: 0.5172 - accuracy: 0.9886 - val_loss: 0.6479 - val_accuracy: 0.9400 - lr: 1.8750e-05
Epoch 26/100
50/50 [=====] - 1s 11ms/step - loss: 0.5096 - accuracy: 0.9932 - val_loss: 0.8085 - val_accuracy: 0.8933 - lr: 1.8750e-05
Epoch 27/100
50/50 [=====] - 1s 11ms/step - loss: 0.5107 - accuracy: 0.9914 - val_loss: 0.6759 - val_accuracy: 0.9358 - lr: 1.8750e-05
Epoch 28/100
47/50 [=====>..] - ETA: 0s - loss: 0.4990 - accuracy: 0.9958
Epoch 28: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
50/50 [=====] - 1s 11ms/step - loss: 0.4988 - accuracy: 0.9961 - val_loss: 0.7093 - val_accuracy: 0.9075 - lr: 1.8750e-05
Epoch 29/100
50/50 [=====] - 1s 11ms/step - loss: 0.5011 - accuracy: 0.9939 - val_loss: 0.6144 - val_accuracy: 0.9500 - lr: 9.3750e-06
Epoch 30/100
50/50 [=====] - 1s 11ms/step - loss: 0.4965 - accuracy: 0.9921 - val_loss: 0.7017 - val_accuracy: 0.9167 - lr: 9.3750e-06
Epoch 31/100
50/50 [=====] - 1s 12ms/step - loss: 0.4953 - accuracy: 0.9943 - val_loss: 0.6902 - val_accuracy: 0.9283 - lr: 9.3750e-06
Epoch 32/100
50/50 [=====] - 1s 12ms/step - loss: 0.4854 - accuracy: 0.9939 - val_loss: 0.6105 - val_accuracy: 0.9433 - lr: 9.3750e-06
Epoch 33/100
50/50 [=====] - 1s 12ms/step - loss: 0.4873 - accuracy: 0.9957 - val_loss: 0.6443 - val_accuracy: 0.9300 - lr: 9.3750e-06
Epoch 34/100
50/50 [=====] - 1s 12ms/step - loss: 0.4911 - accuracy: 0.9943 - val_loss: 0.6953 - val_accuracy: 0.9142 - lr: 9.3750e-06
Epoch 35/100
50/50 [=====] - 1s 11ms/step - loss: 0.4884 - accuracy: 0.9961 - val_loss: 0.6380 - val_accuracy: 0.9408 - lr: 9.3750e-06
Epoch 36/100
50/50 [=====] - 1s 12ms/step - loss: 0.4864 - accuracy: 0.9943 - val_loss: 0.5931 - val_accuracy: 0.9542 - lr: 9.3750e-06
Epoch 37/100
50/50 [=====] - 1s 12ms/step - loss: 0.4853 - accuracy: 0.9939 - val_loss: 0.5970 - val_accuracy: 0.9542 - lr: 9.3750e-06
Epoch 38/100
50/50 [=====] - 1s 12ms/step - loss: 0.4800 - accuracy: 0.9961 - val_loss: 0.6147 - val_accuracy: 0.9325 - lr: 9.3750e-06
Epoch 39/100
50/50 [=====] - 1s 12ms/step - loss: 0.4772 - accuracy: 0.9921 - val_loss: 0.5911 - val_accuracy: 0.9525 - lr: 9.3750e-06

Epoch 40/100
50/50 [=====] - 1s 11ms/step - loss: 0.4793 - accuracy: 0.9943 - val_loss: 0.6246 - val_accuracy: 0.9475 - lr: 9.3750e-06
Epoch 41/100
50/50 [=====] - 1s 12ms/step - loss: 0.4738 - accuracy: 0.9936 - val_loss: 0.5859 - val_accuracy: 0.9558 - lr: 9.3750e-06
Epoch 42/100
50/50 [=====] - 1s 11ms/step - loss: 0.4736 - accuracy: 0.9961 - val_loss: 0.6114 - val_accuracy: 0.9383 - lr: 9.3750e-06
Epoch 43/100
50/50 [=====] - 1s 12ms/step - loss: 0.4700 - accuracy: 0.9961 - val_loss: 0.6563 - val_accuracy: 0.9225 - lr: 9.3750e-06
Epoch 44/100
50/50 [=====] - 1s 12ms/step - loss: 0.4696 - accuracy: 0.9957 - val_loss: 0.6265 - val_accuracy: 0.9300 - lr: 9.3750e-06
Epoch 45/100
50/50 [=====] - 1s 12ms/step - loss: 0.4689 - accuracy: 0.9943 - val_loss: 0.5688 - val_accuracy: 0.9550 - lr: 9.3750e-06
Epoch 46/100
50/50 [=====] - 1s 12ms/step - loss: 0.4616 - accuracy: 0.9964 - val_loss: 0.5881 - val_accuracy: 0.9550 - lr: 9.3750e-06
Epoch 47/100
50/50 [=====] - 1s 11ms/step - loss: 0.4620 - accuracy: 0.9950 - val_loss: 0.6276 - val_accuracy: 0.9408 - lr: 9.3750e-06
Epoch 48/100
50/50 [=====] - 1s 12ms/step - loss: 0.4556 - accuracy: 0.9961 - val_loss: 0.5704 - val_accuracy: 0.9550 - lr: 9.3750e-06
Epoch 49/100
46/50 [=====>...] - ETA: 0s - loss: 0.4575 - accuracy: 0.9969
Epoch 49: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
50/50 [=====] - 1s 13ms/step - loss: 0.4572 - accuracy: 0.9968 - val_loss: 0.5825 - val_accuracy: 0.9550 - lr: 9.3750e-06
Epoch 50/100
50/50 [=====] - 1s 12ms/step - loss: 0.4582 - accuracy: 0.9964 - val_loss: 0.5636 - val_accuracy: 0.9558 - lr: 4.6875e-06
Epoch 51/100
50/50 [=====] - 1s 13ms/step - loss: 0.4565 - accuracy: 0.9943 - val_loss: 0.5505 - val_accuracy: 0.9575 - lr: 4.6875e-06
Epoch 52/100
50/50 [=====] - 1s 11ms/step - loss: 0.4544 - accuracy: 0.9954 - val_loss: 0.5629 - val_accuracy: 0.9567 - lr: 4.6875e-06
Epoch 53/100
50/50 [=====] - 1s 11ms/step - loss: 0.4550 - accuracy: 0.9971 - val_loss: 0.5844 - val_accuracy: 0.9542 - lr: 4.6875e-06
Epoch 54/100
50/50 [=====] - 1s 10ms/step - loss: 0.4501 - accuracy: 0.9968 - val_loss: 0.5552 - val_accuracy: 0.9567 - lr: 4.6875e-06
Epoch 55/100
46/50 [=====>...] - ETA: 0s - loss: 0.4499 - accuracy: 0.9965
Epoch 55: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
50/50 [=====] - 1s 11ms/step - loss: 0.4495 - accuracy: 0.9968 - val_loss: 0.5580 - val_accuracy: 0.9567 - lr: 4.6875e-06
Epoch 56/100
50/50 [=====] - 1s 11ms/step - loss: 0.4473 - accuracy: 0.9961 - val_loss: 0.5411 - val_accuracy: 0.9633 - lr: 2.3438e-06
Epoch 57/100
50/50 [=====] - 1s 11ms/step - loss: 0.4472 - accuracy: 0.9986 - val_loss: 0.5455 - val_accuracy: 0.9608 - lr: 2.3438e-06

Epoch 58/100
50/50 [=====] - 1s 13ms/step - loss: 0.4522 - accuracy: 0.9954 - val_loss: 0.5401 - val_accuracy: 0.9608 - lr: 2.3438e-06
Epoch 59/100
50/50 [=====] - 1s 12ms/step - loss: 0.4490 - accuracy: 0.9964 - val_loss: 0.5399 - val_accuracy: 0.9617 - lr: 2.3438e-06
Epoch 60/100
50/50 [=====] - 1s 12ms/step - loss: 0.4472 - accuracy: 0.9968 - val_loss: 0.5375 - val_accuracy: 0.9608 - lr: 2.3438e-06
Epoch 61/100
50/50 [=====] - 1s 12ms/step - loss: 0.4469 - accuracy: 0.9982 - val_loss: 0.5444 - val_accuracy: 0.9600 - lr: 2.3438e-06
Epoch 62/100
50/50 [=====] - 1s 12ms/step - loss: 0.4479 - accuracy: 0.9964 - val_loss: 0.5344 - val_accuracy: 0.9650 - lr: 2.3438e-06
Epoch 63/100
50/50 [=====] - 1s 12ms/step - loss: 0.4435 - accuracy: 0.9986 - val_loss: 0.5430 - val_accuracy: 0.9592 - lr: 2.3438e-06
Epoch 64/100
50/50 [=====] - 1s 11ms/step - loss: 0.4438 - accuracy: 0.9979 - val_loss: 0.5318 - val_accuracy: 0.9633 - lr: 2.3438e-06
Epoch 65/100
50/50 [=====] - 1s 12ms/step - loss: 0.4447 - accuracy: 0.9957 - val_loss: 0.5302 - val_accuracy: 0.9617 - lr: 2.3438e-06
Epoch 66/100
50/50 [=====] - 1s 12ms/step - loss: 0.4432 - accuracy: 0.9975 - val_loss: 0.5308 - val_accuracy: 0.9658 - lr: 2.3438e-06
Epoch 67/100
50/50 [=====] - 1s 12ms/step - loss: 0.4411 - accuracy: 0.9979 - val_loss: 0.5325 - val_accuracy: 0.9642 - lr: 2.3438e-06
Epoch 68/100
50/50 [=====] - 1s 13ms/step - loss: 0.4422 - accuracy: 0.9971 - val_loss: 0.5332 - val_accuracy: 0.9633 - lr: 2.3438e-06
Epoch 69/100
50/50 [=====] - 1s 12ms/step - loss: 0.4429 - accuracy: 0.9982 - val_loss: 0.5287 - val_accuracy: 0.9642 - lr: 2.3438e-06
Epoch 70/100
50/50 [=====] - 1s 12ms/step - loss: 0.4418 - accuracy: 0.9979 - val_loss: 0.5259 - val_accuracy: 0.9642 - lr: 2.3438e-06
Epoch 71/100
50/50 [=====] - 1s 13ms/step - loss: 0.4401 - accuracy: 0.9957 - val_loss: 0.5285 - val_accuracy: 0.9633 - lr: 2.3438e-06
Epoch 72/100
50/50 [=====] - 1s 13ms/step - loss: 0.4377 - accuracy: 0.9975 - val_loss: 0.5260 - val_accuracy: 0.9642 - lr: 2.3438e-06
Epoch 73/100
50/50 [=====] - 1s 12ms/step - loss: 0.4393 - accuracy: 0.9954 - val_loss: 0.5258 - val_accuracy: 0.9617 - lr: 2.3438e-06
Epoch 74/100
45/50 [=====>...] - ETA: 0s - loss: 0.4395 - accuracy: 0.9964
Epoch 74: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
50/50 [=====] - 1s 12ms/step - loss: 0.4415 - accuracy: 0.9961 - val_loss: 0.5443 - val_accuracy: 0.9558 - lr: 2.3438e-06
Epoch 75/100
50/50 [=====] - 1s 12ms/step - loss: 0.4381 - accuracy: 0.9961 - val_loss: 0.5301 - val_accuracy: 0.9592 - lr: 1.1719e-06
Epoch 76/100
50/50 [=====] - 1s 11ms/step - loss: 0.4414 - accuracy: 0.9968 - val_loss: 0.5268 - val_accuracy: 0.9617 - lr: 1.1719e-06

Epoch 77/100
50/50 [=====] - 1s 11ms/step - loss: 0.4365 - accuracy: 0.9986 - val_loss: 0.5247 - val_accuracy: 0.9633 - lr: 1.1719e-06
Epoch 78/100
50/50 [=====] - 1s 13ms/step - loss: 0.4350 - accuracy: 0.9975 - val_loss: 0.5245 - val_accuracy: 0.9625 - lr: 1.1719e-06
[100, 56, 0.35000000000000003]
85.0%
Epoch 1/100
47/47 [=====] - 2s 24ms/step - loss: 3.2236 - accuracy: 0.1504 - val_loss: 3.3366 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 2/100
47/47 [=====] - 1s 11ms/step - loss: 2.6083 - accuracy: 0.2682 - val_loss: 4.7036 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 3/100
47/47 [=====] - 1s 12ms/step - loss: 2.3526 - accuracy: 0.3324 - val_loss: 6.3380 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 4/100
47/47 [=====] - 1s 13ms/step - loss: 2.0182 - accuracy: 0.4540 - val_loss: 7.5446 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 5/100
47/47 [=====] - ETA: 0s - loss: 1.6523 - accuracy: 0.6060
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
47/47 [=====] - 1s 13ms/step - loss: 1.6523 - accuracy: 0.6060 - val_loss: 8.1439 - val_accuracy: 0.0999 - lr: 3.0000e-04
Epoch 6/100
47/47 [=====] - 1s 13ms/step - loss: 1.3504 - accuracy: 0.7091 - val_loss: 9.1078 - val_accuracy: 0.0999 - lr: 1.5000e-04
Epoch 7/100
47/47 [=====] - 1s 13ms/step - loss: 1.2011 - accuracy: 0.7749 - val_loss: 10.5700 - val_accuracy: 0.0999 - lr: 1.5000e-04
Epoch 8/100
47/47 [=====] - 1s 14ms/step - loss: 1.0572 - accuracy: 0.8284 - val_loss: 12.1170 - val_accuracy: 0.0999 - lr: 1.5000e-04
Epoch 9/100
44/47 [=====>..] - ETA: 0s - loss: 0.9497 - accuracy: 0.8770
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
47/47 [=====] - 1s 15ms/step - loss: 0.9467 - accuracy: 0.8780 - val_loss: 14.4142 - val_accuracy: 0.0999 - lr: 1.5000e-04
Epoch 10/100
47/47 [=====] - 1s 15ms/step - loss: 0.8649 - accuracy: 0.9034 - val_loss: 14.5560 - val_accuracy: 0.0999 - lr: 7.5000e-05
Epoch 11/100
47/47 [=====] - 1s 14ms/step - loss: 0.8158 - accuracy: 0.9219 - val_loss: 14.6925 - val_accuracy: 0.0999 - lr: 7.5000e-05
Epoch 12/100
47/47 [=====] - 1s 14ms/step - loss: 0.7764 - accuracy: 0.9384 - val_loss: 10.9385 - val_accuracy: 0.0999 - lr: 7.5000e-05
Epoch 13/100
46/47 [=====>.] - ETA: 0s - loss: 0.7473 - accuracy: 0.9414
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
47/47 [=====] - 1s 14ms/step - loss: 0.7479 - accuracy: 0.9411 - val_loss: 12.0369 - val_accuracy: 0.0999 - lr: 7.5000e-05
[100, 56, 0.4]
87.5%
Epoch 1/100
43/43 [=====] - 2s 23ms/step - loss: 3.1835 - accuracy:

0.1358 - val_loss: 3.0602 - val_accuracy: 0.0875 - lr: 3.0000e-04
Epoch 2/100
43/43 [=====] - 1s 15ms/step - loss: 2.6504 - accuracy:
0.2504 - val_loss: 4.2638 - val_accuracy: 0.1025 - lr: 3.0000e-04
Epoch 3/100
43/43 [=====] - 1s 16ms/step - loss: 2.3354 - accuracy:
0.3317 - val_loss: 5.9655 - val_accuracy: 0.1025 - lr: 3.0000e-04
Epoch 4/100
43/43 [=====] - 1s 15ms/step - loss: 2.0485 - accuracy:
0.4183 - val_loss: 8.6443 - val_accuracy: 0.1025 - lr: 3.0000e-04
Epoch 5/100
39/43 [=====>...] - ETA: 0s - loss: 1.7327 - accuracy: 0.549
5
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
43/43 [=====] - 1s 14ms/step - loss: 1.7066 - accuracy:
0.5600 - val_loss: 10.3057 - val_accuracy: 0.1025 - lr: 3.0000e-04
Epoch 6/100
43/43 [=====] - 1s 13ms/step - loss: 1.3923 - accuracy:
0.7171 - val_loss: 10.8887 - val_accuracy: 0.1025 - lr: 1.5000e-04
Epoch 7/100
43/43 [=====] - 1s 16ms/step - loss: 1.2130 - accuracy:
0.7871 - val_loss: 11.2399 - val_accuracy: 0.1025 - lr: 1.5000e-04
Epoch 8/100
43/43 [=====] - 1s 15ms/step - loss: 1.0637 - accuracy:
0.8425 - val_loss: 11.2723 - val_accuracy: 0.1025 - lr: 1.5000e-04
Epoch 9/100
43/43 [=====] - ETA: 0s - loss: 0.9534 - accuracy: 0.877
1
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
43/43 [=====] - 1s 12ms/step - loss: 0.9534 - accuracy:
0.8771 - val_loss: 10.9947 - val_accuracy: 0.1025 - lr: 1.5000e-04
Epoch 10/100
43/43 [=====] - 1s 13ms/step - loss: 0.8688 - accuracy:
0.9092 - val_loss: 10.5206 - val_accuracy: 0.1025 - lr: 7.5000e-05
Epoch 11/100
43/43 [=====] - 1s 12ms/step - loss: 0.8105 - accuracy:
0.9279 - val_loss: 10.0203 - val_accuracy: 0.1131 - lr: 7.5000e-05
Epoch 12/100
43/43 [=====] - 1s 12ms/step - loss: 0.7815 - accuracy:
0.9408 - val_loss: 7.2319 - val_accuracy: 0.2631 - lr: 7.5000e-05
Epoch 13/100
38/43 [=====>....] - ETA: 0s - loss: 0.7520 - accuracy: 0.944
5
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
43/43 [=====] - 1s 12ms/step - loss: 0.7552 - accuracy:
0.9417 - val_loss: 5.5952 - val_accuracy: 0.0913 - lr: 7.5000e-05
Epoch 14/100
43/43 [=====] - 1s 14ms/step - loss: 0.7160 - accuracy:
0.9563 - val_loss: 6.6991 - val_accuracy: 0.2169 - lr: 3.7500e-05
Epoch 15/100
43/43 [=====] - 1s 12ms/step - loss: 0.6929 - accuracy:
0.9667 - val_loss: 4.4828 - val_accuracy: 0.2619 - lr: 3.7500e-05
Epoch 16/100
43/43 [=====] - 1s 14ms/step - loss: 0.6834 - accuracy:
0.9671 - val_loss: 4.0032 - val_accuracy: 0.2819 - lr: 3.7500e-05
Epoch 17/100
43/43 [=====] - 1s 15ms/step - loss: 0.6654 - accuracy:
0.9663 - val_loss: 2.1451 - val_accuracy: 0.3506 - lr: 3.7500e-05
Epoch 18/100
43/43 [=====] - 1s 16ms/step - loss: 0.6506 - accuracy:

0.9742 - val_loss: 1.5321 - val_accuracy: 0.5819 - lr: 3.7500e-05
Epoch 19/100
43/43 [=====] - 1s 16ms/step - loss: 0.6362 - accuracy:
0.9804 - val_loss: 7.2838 - val_accuracy: 0.1000 - lr: 3.7500e-05
Epoch 20/100
43/43 [=====] - 1s 14ms/step - loss: 0.6249 - accuracy:
0.9783 - val_loss: 2.9766 - val_accuracy: 0.2906 - lr: 3.7500e-05
Epoch 21/100
43/43 [=====] - 1s 15ms/step - loss: 0.6105 - accuracy:
0.9837 - val_loss: 0.9034 - val_accuracy: 0.8612 - lr: 3.7500e-05
Epoch 22/100
43/43 [=====] - 1s 15ms/step - loss: 0.6094 - accuracy:
0.9775 - val_loss: 1.5637 - val_accuracy: 0.5512 - lr: 3.7500e-05
Epoch 23/100
43/43 [=====] - 1s 15ms/step - loss: 0.6017 - accuracy:
0.9800 - val_loss: 3.5135 - val_accuracy: 0.2431 - lr: 3.7500e-05
Epoch 24/100
43/43 [=====] - 1s 16ms/step - loss: 0.5889 - accuracy:
0.9800 - val_loss: 0.8957 - val_accuracy: 0.8494 - lr: 3.7500e-05
Epoch 25/100
43/43 [=====] - 1s 14ms/step - loss: 0.5748 - accuracy:
0.9842 - val_loss: 2.7679 - val_accuracy: 0.3869 - lr: 3.7500e-05
Epoch 26/100
43/43 [=====] - 1s 14ms/step - loss: 0.5681 - accuracy:
0.9879 - val_loss: 4.0837 - val_accuracy: 0.1912 - lr: 3.7500e-05
Epoch 27/100
43/43 [=====] - 1s 15ms/step - loss: 0.5582 - accuracy:
0.9875 - val_loss: 0.7360 - val_accuracy: 0.9212 - lr: 3.7500e-05
Epoch 28/100
43/43 [=====] - 1s 14ms/step - loss: 0.5445 - accuracy:
0.9929 - val_loss: 1.9217 - val_accuracy: 0.4387 - lr: 3.7500e-05
Epoch 29/100
43/43 [=====] - 1s 15ms/step - loss: 0.5365 - accuracy:
0.9900 - val_loss: 0.7674 - val_accuracy: 0.8950 - lr: 3.7500e-05
Epoch 30/100
43/43 [=====] - 1s 15ms/step - loss: 0.5310 - accuracy:
0.9883 - val_loss: 1.5438 - val_accuracy: 0.6000 - lr: 3.7500e-05
Epoch 31/100
42/43 [=====>.] - ETA: 0s - loss: 0.5162 - accuracy: 0.994
5
Epoch 31: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
43/43 [=====] - 1s 14ms/step - loss: 0.5171 - accuracy:
0.9942 - val_loss: 1.5968 - val_accuracy: 0.6144 - lr: 3.7500e-05
Epoch 32/100
43/43 [=====] - 1s 15ms/step - loss: 0.5115 - accuracy:
0.9917 - val_loss: 1.0255 - val_accuracy: 0.7581 - lr: 1.8750e-05
Epoch 33/100
43/43 [=====] - 1s 15ms/step - loss: 0.5050 - accuracy:
0.9929 - val_loss: 0.6631 - val_accuracy: 0.9463 - lr: 1.8750e-05
Epoch 34/100
43/43 [=====] - 1s 15ms/step - loss: 0.5002 - accuracy:
0.9954 - val_loss: 0.8490 - val_accuracy: 0.8600 - lr: 1.8750e-05
Epoch 35/100
43/43 [=====] - 1s 13ms/step - loss: 0.5026 - accuracy:
0.9937 - val_loss: 2.5436 - val_accuracy: 0.3875 - lr: 1.8750e-05
Epoch 36/100
43/43 [=====] - 1s 13ms/step - loss: 0.4990 - accuracy:
0.9921 - val_loss: 1.9414 - val_accuracy: 0.5219 - lr: 1.8750e-05
Epoch 37/100
43/43 [=====] - ETA: 0s - loss: 0.4987 - accuracy: 0.991

2

Epoch 37: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.

43/43 [=====] - 1s 12ms/step - loss: 0.4987 - accuracy: 0.9912 - val_loss: 0.7249 - val_accuracy: 0.9131 - lr: 1.8750e-05

Epoch 38/100

43/43 [=====] - 1s 12ms/step - loss: 0.4843 - accuracy: 0.9967 - val_loss: 0.7087 - val_accuracy: 0.9112 - lr: 9.3750e-06

Epoch 39/100

43/43 [=====] - 1s 12ms/step - loss: 0.4862 - accuracy: 0.9933 - val_loss: 0.7341 - val_accuracy: 0.9013 - lr: 9.3750e-06

Epoch 40/100

43/43 [=====] - 1s 13ms/step - loss: 0.4879 - accuracy: 0.9950 - val_loss: 0.8308 - val_accuracy: 0.8631 - lr: 9.3750e-06

Epoch 41/100

43/43 [=====] - ETA: 0s - loss: 0.4819 - accuracy: 0.996

2

Epoch 41: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.

43/43 [=====] - 1s 15ms/step - loss: 0.4819 - accuracy: 0.9962 - val_loss: 0.6690 - val_accuracy: 0.9250 - lr: 9.3750e-06

Epoch 42/100

43/43 [=====] - 1s 16ms/step - loss: 0.4832 - accuracy: 0.9967 - val_loss: 0.6153 - val_accuracy: 0.9419 - lr: 4.6875e-06

Epoch 43/100

43/43 [=====] - 1s 14ms/step - loss: 0.4796 - accuracy: 0.9962 - val_loss: 0.6113 - val_accuracy: 0.9494 - lr: 4.6875e-06

Epoch 44/100

43/43 [=====] - 1s 14ms/step - loss: 0.4824 - accuracy: 0.9950 - val_loss: 0.8265 - val_accuracy: 0.8562 - lr: 4.6875e-06

Epoch 45/100

43/43 [=====] - 1s 14ms/step - loss: 0.4827 - accuracy: 0.9942 - val_loss: 0.6124 - val_accuracy: 0.9513 - lr: 4.6875e-06

Epoch 46/100

43/43 [=====] - 1s 14ms/step - loss: 0.4770 - accuracy: 0.9954 - val_loss: 0.5975 - val_accuracy: 0.9538 - lr: 4.6875e-06

Epoch 47/100

43/43 [=====] - 1s 14ms/step - loss: 0.4774 - accuracy: 0.9950 - val_loss: 0.6142 - val_accuracy: 0.9506 - lr: 4.6875e-06

Epoch 48/100

43/43 [=====] - 1s 13ms/step - loss: 0.4792 - accuracy: 0.9971 - val_loss: 0.5829 - val_accuracy: 0.9531 - lr: 4.6875e-06

Epoch 49/100

43/43 [=====] - 1s 13ms/step - loss: 0.4757 - accuracy: 0.9946 - val_loss: 0.7899 - val_accuracy: 0.8600 - lr: 4.6875e-06

Epoch 50/100

43/43 [=====] - 1s 14ms/step - loss: 0.4702 - accuracy: 0.9962 - val_loss: 0.5868 - val_accuracy: 0.9563 - lr: 4.6875e-06

Epoch 51/100

43/43 [=====] - 1s 14ms/step - loss: 0.4711 - accuracy: 0.9958 - val_loss: 0.5944 - val_accuracy: 0.9506 - lr: 4.6875e-06

Epoch 52/100

40/43 [=====>...] - ETA: 0s - loss: 0.4706 - accuracy: 0.996

4

Epoch 52: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.

43/43 [=====] - 1s 14ms/step - loss: 0.4692 - accuracy: 0.9967 - val_loss: 0.6203 - val_accuracy: 0.9425 - lr: 4.6875e-06

Epoch 53/100

43/43 [=====] - 1s 15ms/step - loss: 0.4703 - accuracy: 0.9950 - val_loss: 0.5933 - val_accuracy: 0.9525 - lr: 2.3438e-06

Epoch 54/100

43/43 [=====] - 1s 15ms/step - loss: 0.4714 - accuracy:

0.9954 - val_loss: 0.6121 - val_accuracy: 0.9444 - lr: 2.3438e-06
Epoch 55/100
43/43 [=====] - 1s 14ms/step - loss: 0.4685 - accuracy:
0.9967 - val_loss: 0.6324 - val_accuracy: 0.9419 - lr: 2.3438e-06
Epoch 56/100
43/43 [=====] - 1s 15ms/step - loss: 0.4690 - accuracy:
0.9967 - val_loss: 0.5772 - val_accuracy: 0.9544 - lr: 2.3438e-06
Epoch 57/100
43/43 [=====] - 1s 16ms/step - loss: 0.4704 - accuracy:
0.9958 - val_loss: 0.5773 - val_accuracy: 0.9581 - lr: 2.3438e-06
Epoch 58/100
43/43 [=====] - 1s 15ms/step - loss: 0.4685 - accuracy:
0.9958 - val_loss: 0.6299 - val_accuracy: 0.9406 - lr: 2.3438e-06
Epoch 59/100
43/43 [=====] - 1s 16ms/step - loss: 0.4695 - accuracy:
0.9958 - val_loss: 0.6054 - val_accuracy: 0.9431 - lr: 2.3438e-06
Epoch 60/100
43/43 [=====] - ETA: 0s - loss: 0.4685 - accuracy: 0.995
8
Epoch 60: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
43/43 [=====] - 1s 15ms/step - loss: 0.4685 - accuracy:
0.9958 - val_loss: 0.5884 - val_accuracy: 0.9506 - lr: 2.3438e-06
Epoch 61/100
43/43 [=====] - 1s 13ms/step - loss: 0.4626 - accuracy:
0.9979 - val_loss: 0.5802 - val_accuracy: 0.9531 - lr: 1.1719e-06
Epoch 62/100
43/43 [=====] - 1s 12ms/step - loss: 0.4701 - accuracy:
0.9950 - val_loss: 0.5756 - val_accuracy: 0.9519 - lr: 1.1719e-06
Epoch 63/100
43/43 [=====] - 1s 12ms/step - loss: 0.4665 - accuracy:
0.9958 - val_loss: 0.5751 - val_accuracy: 0.9538 - lr: 1.1719e-06
Epoch 64/100
43/43 [=====] - 1s 12ms/step - loss: 0.4649 - accuracy:
0.9942 - val_loss: 0.5802 - val_accuracy: 0.9506 - lr: 1.1719e-06
Epoch 65/100
43/43 [=====] - 1s 12ms/step - loss: 0.4660 - accuracy:
0.9958 - val_loss: 0.5814 - val_accuracy: 0.9506 - lr: 1.1719e-06
Epoch 66/100
43/43 [=====] - 1s 12ms/step - loss: 0.4648 - accuracy:
0.9971 - val_loss: 0.5788 - val_accuracy: 0.9544 - lr: 1.1719e-06
Epoch 67/100
43/43 [=====] - 1s 15ms/step - loss: 0.4668 - accuracy:
0.9967 - val_loss: 0.5720 - val_accuracy: 0.9588 - lr: 1.1719e-06
Epoch 68/100
43/43 [=====] - 1s 14ms/step - loss: 0.4689 - accuracy:
0.9946 - val_loss: 0.5684 - val_accuracy: 0.9600 - lr: 1.1719e-06
Epoch 69/100
43/43 [=====] - 1s 14ms/step - loss: 0.4681 - accuracy:
0.9967 - val_loss: 0.5694 - val_accuracy: 0.9575 - lr: 1.1719e-06
Epoch 70/100
43/43 [=====] - 1s 14ms/step - loss: 0.4606 - accuracy:
0.9946 - val_loss: 0.5696 - val_accuracy: 0.9544 - lr: 1.1719e-06
Epoch 71/100
43/43 [=====] - 1s 14ms/step - loss: 0.4617 - accuracy:
0.9975 - val_loss: 0.5696 - val_accuracy: 0.9575 - lr: 1.1719e-06
Epoch 72/100
43/43 [=====] - ETA: 0s - loss: 0.4653 - accuracy: 0.995
8
Epoch 72: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.
43/43 [=====] - 1s 14ms/step - loss: 0.4653 - accuracy:

0.9958 - val_loss: 0.5761 - val_accuracy: 0.9569 - lr: 1.1719e-06
Epoch 73/100
43/43 [=====] - 1s 15ms/step - loss: 0.4619 - accuracy:
0.9967 - val_loss: 0.5676 - val_accuracy: 0.9600 - lr: 5.8594e-07
Epoch 74/100
43/43 [=====] - 1s 15ms/step - loss: 0.4682 - accuracy:
0.9950 - val_loss: 0.5668 - val_accuracy: 0.9569 - lr: 5.8594e-07
Epoch 75/100
43/43 [=====] - 1s 14ms/step - loss: 0.4603 - accuracy:
0.9979 - val_loss: 0.5677 - val_accuracy: 0.9575 - lr: 5.8594e-07
Epoch 76/100
43/43 [=====] - 1s 15ms/step - loss: 0.4632 - accuracy:
0.9962 - val_loss: 0.5671 - val_accuracy: 0.9544 - lr: 5.8594e-07
Epoch 77/100
43/43 [=====] - 1s 16ms/step - loss: 0.4581 - accuracy:
0.9950 - val_loss: 0.5661 - val_accuracy: 0.9606 - lr: 5.8594e-07
Epoch 78/100
43/43 [=====] - 1s 14ms/step - loss: 0.4628 - accuracy:
0.9962 - val_loss: 0.5699 - val_accuracy: 0.9544 - lr: 5.8594e-07
Epoch 79/100
43/43 [=====] - 1s 14ms/step - loss: 0.4618 - accuracy:
0.9967 - val_loss: 0.5750 - val_accuracy: 0.9531 - lr: 5.8594e-07
Epoch 80/100
43/43 [=====] - 1s 14ms/step - loss: 0.4598 - accuracy:
0.9975 - val_loss: 0.5664 - val_accuracy: 0.9538 - lr: 5.8594e-07
Epoch 81/100
43/43 [=====] - ETA: 0s - loss: 0.4603 - accuracy: 0.995
8
Epoch 81: ReduceLROnPlateau reducing learning rate to 2.9296876391526894e-07.
43/43 [=====] - 1s 16ms/step - loss: 0.4603 - accuracy:
0.9958 - val_loss: 0.5665 - val_accuracy: 0.9556 - lr: 5.8594e-07
Epoch 82/100
43/43 [=====] - 1s 14ms/step - loss: 0.4631 - accuracy:
0.9958 - val_loss: 0.5661 - val_accuracy: 0.9563 - lr: 2.9297e-07
Epoch 83/100
43/43 [=====] - 1s 12ms/step - loss: 0.4646 - accuracy:
0.9954 - val_loss: 0.5653 - val_accuracy: 0.9569 - lr: 2.9297e-07
Epoch 84/100
43/43 [=====] - 1s 12ms/step - loss: 0.4612 - accuracy:
0.9971 - val_loss: 0.5667 - val_accuracy: 0.9550 - lr: 2.9297e-07
Epoch 85/100
43/43 [=====] - 1s 12ms/step - loss: 0.4595 - accuracy:
0.9971 - val_loss: 0.5681 - val_accuracy: 0.9531 - lr: 2.9297e-07
Epoch 86/100
43/43 [=====] - 1s 13ms/step - loss: 0.4634 - accuracy:
0.9962 - val_loss: 0.5647 - val_accuracy: 0.9569 - lr: 2.9297e-07
Epoch 87/100
43/43 [=====] - 1s 16ms/step - loss: 0.4630 - accuracy:
0.9958 - val_loss: 0.5650 - val_accuracy: 0.9544 - lr: 2.9297e-07
Epoch 88/100
43/43 [=====] - 1s 12ms/step - loss: 0.4627 - accuracy:
0.9950 - val_loss: 0.5638 - val_accuracy: 0.9575 - lr: 2.9297e-07
Epoch 89/100
43/43 [=====] - 1s 12ms/step - loss: 0.4574 - accuracy:
0.9962 - val_loss: 0.5638 - val_accuracy: 0.9575 - lr: 2.9297e-07
[100, 64, 0.2]
90.0%
Epoch 1/100
50/50 [=====] - 1s 17ms/step - loss: 3.1404 - accuracy:
0.1534 - val_loss: 3.7519 - val_accuracy: 0.0950 - lr: 3.0000e-04

Epoch 2/100
50/50 [=====] - 1s 10ms/step - loss: 2.5896 - accuracy: 0.2741 - val_loss: 5.5444 - val_accuracy: 0.0950 - lr: 3.0000e-04
Epoch 3/100
50/50 [=====] - 1s 12ms/step - loss: 2.2671 - accuracy: 0.3425 - val_loss: 7.1301 - val_accuracy: 0.1050 - lr: 3.0000e-04
Epoch 4/100
50/50 [=====] - 1s 11ms/step - loss: 1.9795 - accuracy: 0.4541 - val_loss: 8.6366 - val_accuracy: 0.0900 - lr: 3.0000e-04
Epoch 5/100
48/50 [=====>..] - ETA: 0s - loss: 1.6543 - accuracy: 0.5866
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
50/50 [=====] - 1s 11ms/step - loss: 1.6540 - accuracy: 0.5881 - val_loss: 10.0500 - val_accuracy: 0.0900 - lr: 3.0000e-04
Epoch 6/100
50/50 [=====] - 1s 11ms/step - loss: 1.3764 - accuracy: 0.6997 - val_loss: 10.4054 - val_accuracy: 0.0900 - lr: 1.5000e-04
Epoch 7/100
50/50 [=====] - 1s 11ms/step - loss: 1.2063 - accuracy: 0.7697 - val_loss: 9.8760 - val_accuracy: 0.1187 - lr: 1.5000e-04
Epoch 8/100
50/50 [=====] - 1s 11ms/step - loss: 1.0802 - accuracy: 0.8275 - val_loss: 10.6038 - val_accuracy: 0.0900 - lr: 1.5000e-04
Epoch 9/100
45/50 [=====>...] - ETA: 0s - loss: 0.9836 - accuracy: 0.8538
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
50/50 [=====] - 1s 10ms/step - loss: 0.9761 - accuracy: 0.8559 - val_loss: 8.8589 - val_accuracy: 0.0975 - lr: 1.5000e-04
Epoch 10/100
50/50 [=====] - 1s 10ms/step - loss: 0.8819 - accuracy: 0.8981 - val_loss: 7.3814 - val_accuracy: 0.2387 - lr: 7.5000e-05
Epoch 11/100
50/50 [=====] - 1s 10ms/step - loss: 0.8272 - accuracy: 0.9147 - val_loss: 7.0700 - val_accuracy: 0.0900 - lr: 7.5000e-05
Epoch 12/100
50/50 [=====] - 1s 11ms/step - loss: 0.7792 - accuracy: 0.9306 - val_loss: 9.2118 - val_accuracy: 0.0900 - lr: 7.5000e-05
Epoch 13/100
49/50 [=====>.] - ETA: 0s - loss: 0.7500 - accuracy: 0.9385
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
50/50 [=====] - 1s 11ms/step - loss: 0.7489 - accuracy: 0.9381 - val_loss: 4.1036 - val_accuracy: 0.1575 - lr: 7.5000e-05
Epoch 14/100
50/50 [=====] - 1s 11ms/step - loss: 0.7160 - accuracy: 0.9453 - val_loss: 2.1702 - val_accuracy: 0.5088 - lr: 3.7500e-05
Epoch 15/100
50/50 [=====] - 1s 10ms/step - loss: 0.6868 - accuracy: 0.9581 - val_loss: 3.5058 - val_accuracy: 0.3162 - lr: 3.7500e-05
Epoch 16/100
50/50 [=====] - 1s 11ms/step - loss: 0.6785 - accuracy: 0.9600 - val_loss: 1.1598 - val_accuracy: 0.7625 - lr: 3.7500e-05
Epoch 17/100
50/50 [=====] - 1s 11ms/step - loss: 0.6614 - accuracy: 0.9656 - val_loss: 1.4984 - val_accuracy: 0.5900 - lr: 3.7500e-05
Epoch 18/100
50/50 [=====] - 1s 11ms/step - loss: 0.6532 - accuracy: 0.9656 - val_loss: 1.2045 - val_accuracy: 0.6637 - lr: 3.7500e-05

Epoch 19/100
50/50 [=====] - 1s 11ms/step - loss: 0.6369 - accuracy: 0.9694 - val_loss: 2.4858 - val_accuracy: 0.4025 - lr: 3.7500e-05
Epoch 20/100
48/50 [=====>..] - ETA: 0s - loss: 0.6210 - accuracy: 0.9688
Epoch 20: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
50/50 [=====] - 1s 11ms/step - loss: 0.6212 - accuracy: 0.9681 - val_loss: 3.4506 - val_accuracy: 0.3187 - lr: 3.7500e-05
Epoch 21/100
50/50 [=====] - 1s 12ms/step - loss: 0.6035 - accuracy: 0.9762 - val_loss: 0.8468 - val_accuracy: 0.8813 - lr: 1.8750e-05
Epoch 22/100
50/50 [=====] - 1s 12ms/step - loss: 0.6079 - accuracy: 0.9750 - val_loss: 1.5512 - val_accuracy: 0.5525 - lr: 1.8750e-05
Epoch 23/100
50/50 [=====] - 1s 11ms/step - loss: 0.5938 - accuracy: 0.9781 - val_loss: 1.4103 - val_accuracy: 0.5888 - lr: 1.8750e-05
Epoch 24/100
50/50 [=====] - 1s 12ms/step - loss: 0.5910 - accuracy: 0.9762 - val_loss: 0.7467 - val_accuracy: 0.9100 - lr: 1.8750e-05
Epoch 25/100
50/50 [=====] - 1s 11ms/step - loss: 0.5824 - accuracy: 0.9784 - val_loss: 0.7228 - val_accuracy: 0.9413 - lr: 1.8750e-05
Epoch 26/100
50/50 [=====] - 0s 10ms/step - loss: 0.5804 - accuracy: 0.9772 - val_loss: 3.3638 - val_accuracy: 0.3550 - lr: 1.8750e-05
Epoch 27/100
50/50 [=====] - 1s 10ms/step - loss: 0.5736 - accuracy: 0.9825 - val_loss: 0.7381 - val_accuracy: 0.9362 - lr: 1.8750e-05
Epoch 28/100
50/50 [=====] - 0s 10ms/step - loss: 0.5692 - accuracy: 0.9822 - val_loss: 1.7926 - val_accuracy: 0.4675 - lr: 1.8750e-05
Epoch 29/100
44/50 [=====>....] - ETA: 0s - loss: 0.5679 - accuracy: 0.9805
Epoch 29: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
50/50 [=====] - 1s 10ms/step - loss: 0.5665 - accuracy: 0.9809 - val_loss: 1.6056 - val_accuracy: 0.5738 - lr: 1.8750e-05
Epoch 30/100
50/50 [=====] - 0s 10ms/step - loss: 0.5544 - accuracy: 0.9862 - val_loss: 1.0788 - val_accuracy: 0.7763 - lr: 9.3750e-06
Epoch 31/100
50/50 [=====] - 1s 10ms/step - loss: 0.5539 - accuracy: 0.9844 - val_loss: 1.4268 - val_accuracy: 0.5763 - lr: 9.3750e-06
Epoch 32/100
50/50 [=====] - 1s 11ms/step - loss: 0.5564 - accuracy: 0.9872 - val_loss: 0.6547 - val_accuracy: 0.9463 - lr: 9.3750e-06
Epoch 33/100
50/50 [=====] - 1s 11ms/step - loss: 0.5453 - accuracy: 0.9859 - val_loss: 0.8495 - val_accuracy: 0.8725 - lr: 9.3750e-06
Epoch 34/100
50/50 [=====] - 1s 11ms/step - loss: 0.5474 - accuracy: 0.9856 - val_loss: 0.6835 - val_accuracy: 0.9325 - lr: 9.3750e-06
Epoch 35/100
50/50 [=====] - 1s 11ms/step - loss: 0.5388 - accuracy: 0.9853 - val_loss: 1.6642 - val_accuracy: 0.5250 - lr: 9.3750e-06
Epoch 36/100
49/50 [=====>.] - ETA: 0s - loss: 0.5373 - accuracy: 0.9879

Epoch 36: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
50/50 [=====] - 1s 11ms/step - loss: 0.5378 - accuracy: 0.9875 - val_loss: 0.7222 - val_accuracy: 0.9300 - lr: 9.3750e-06
Epoch 37/100
50/50 [=====] - 1s 11ms/step - loss: 0.5336 - accuracy: 0.9875 - val_loss: 0.7380 - val_accuracy: 0.9187 - lr: 4.6875e-06
Epoch 38/100
50/50 [=====] - 1s 11ms/step - loss: 0.5340 - accuracy: 0.9887 - val_loss: 0.6956 - val_accuracy: 0.9287 - lr: 4.6875e-06
Epoch 39/100
50/50 [=====] - 1s 11ms/step - loss: 0.5309 - accuracy: 0.9875 - val_loss: 0.6344 - val_accuracy: 0.9563 - lr: 4.6875e-06
Epoch 40/100
50/50 [=====] - 1s 11ms/step - loss: 0.5320 - accuracy: 0.9875 - val_loss: 0.6430 - val_accuracy: 0.9563 - lr: 4.6875e-06
Epoch 41/100
50/50 [=====] - 1s 11ms/step - loss: 0.5240 - accuracy: 0.9878 - val_loss: 0.6296 - val_accuracy: 0.9488 - lr: 4.6875e-06
Epoch 42/100
50/50 [=====] - 1s 11ms/step - loss: 0.5330 - accuracy: 0.9841 - val_loss: 0.6395 - val_accuracy: 0.9463 - lr: 4.6875e-06
Epoch 43/100
50/50 [=====] - 1s 11ms/step - loss: 0.5255 - accuracy: 0.9878 - val_loss: 0.7839 - val_accuracy: 0.8900 - lr: 4.6875e-06
Epoch 44/100
50/50 [=====] - 1s 11ms/step - loss: 0.5230 - accuracy: 0.9872 - val_loss: 0.6364 - val_accuracy: 0.9538 - lr: 4.6875e-06
Epoch 45/100
50/50 [=====] - 1s 12ms/step - loss: 0.5235 - accuracy: 0.9900 - val_loss: 0.6171 - val_accuracy: 0.9550 - lr: 4.6875e-06
Epoch 46/100
50/50 [=====] - 1s 11ms/step - loss: 0.5178 - accuracy: 0.9900 - val_loss: 0.6119 - val_accuracy: 0.9563 - lr: 4.6875e-06
Epoch 47/100
50/50 [=====] - 1s 11ms/step - loss: 0.5198 - accuracy: 0.9897 - val_loss: 0.6160 - val_accuracy: 0.9538 - lr: 4.6875e-06
Epoch 48/100
50/50 [=====] - 1s 11ms/step - loss: 0.5167 - accuracy: 0.9887 - val_loss: 0.6093 - val_accuracy: 0.9575 - lr: 4.6875e-06
Epoch 49/100
50/50 [=====] - 1s 11ms/step - loss: 0.5168 - accuracy: 0.9875 - val_loss: 0.6539 - val_accuracy: 0.9425 - lr: 4.6875e-06
Epoch 50/100
50/50 [=====] - 1s 11ms/step - loss: 0.5225 - accuracy: 0.9872 - val_loss: 0.6152 - val_accuracy: 0.9588 - lr: 4.6875e-06
Epoch 51/100
50/50 [=====] - 0s 10ms/step - loss: 0.5153 - accuracy: 0.9909 - val_loss: 0.6179 - val_accuracy: 0.9500 - lr: 4.6875e-06
Epoch 52/100
46/50 [=====>...] - ETA: 0s - loss: 0.5093 - accuracy: 0.9908
Epoch 52: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
50/50 [=====] - 0s 10ms/step - loss: 0.5114 - accuracy: 0.9903 - val_loss: 0.6116 - val_accuracy: 0.9525 - lr: 4.6875e-06
Epoch 53/100
50/50 [=====] - 0s 10ms/step - loss: 0.5144 - accuracy: 0.9878 - val_loss: 0.6216 - val_accuracy: 0.9525 - lr: 2.3438e-06
Epoch 54/100
50/50 [=====] - 0s 10ms/step - loss: 0.5108 - accuracy: 0.9878 - val_loss: 0.6131 - val_accuracy: 0.9538 - lr: 2.3438e-06

Epoch 55/100
50/50 [=====] - 1s 10ms/step - loss: 0.5142 - accuracy: 0.9894 - val_loss: 0.6029 - val_accuracy: 0.9588 - lr: 2.3438e-06

Epoch 56/100
50/50 [=====] - 0s 10ms/step - loss: 0.5115 - accuracy: 0.9919 - val_loss: 0.6151 - val_accuracy: 0.9550 - lr: 2.3438e-06

Epoch 57/100
50/50 [=====] - 0s 10ms/step - loss: 0.5103 - accuracy: 0.9887 - val_loss: 0.6330 - val_accuracy: 0.9513 - lr: 2.3438e-06

Epoch 58/100
50/50 [=====] - 0s 10ms/step - loss: 0.5081 - accuracy: 0.9887 - val_loss: 0.5986 - val_accuracy: 0.9563 - lr: 2.3438e-06

Epoch 59/100
50/50 [=====] - 0s 10ms/step - loss: 0.5101 - accuracy: 0.9884 - val_loss: 0.6088 - val_accuracy: 0.9575 - lr: 2.3438e-06

Epoch 60/100
50/50 [=====] - 1s 10ms/step - loss: 0.5094 - accuracy: 0.9919 - val_loss: 0.6051 - val_accuracy: 0.9575 - lr: 2.3438e-06

Epoch 61/100
50/50 [=====] - 1s 11ms/step - loss: 0.5097 - accuracy: 0.9906 - val_loss: 0.5971 - val_accuracy: 0.9613 - lr: 2.3438e-06

Epoch 62/100
50/50 [=====] - 1s 11ms/step - loss: 0.5020 - accuracy: 0.9919 - val_loss: 0.6117 - val_accuracy: 0.9538 - lr: 2.3438e-06

Epoch 63/100
50/50 [=====] - 1s 11ms/step - loss: 0.5039 - accuracy: 0.9922 - val_loss: 0.5931 - val_accuracy: 0.9563 - lr: 2.3438e-06

Epoch 64/100
50/50 [=====] - 1s 11ms/step - loss: 0.5138 - accuracy: 0.9903 - val_loss: 0.6007 - val_accuracy: 0.9563 - lr: 2.3438e-06

Epoch 65/100
50/50 [=====] - 1s 11ms/step - loss: 0.5037 - accuracy: 0.9900 - val_loss: 0.6026 - val_accuracy: 0.9563 - lr: 2.3438e-06

Epoch 66/100
50/50 [=====] - 1s 11ms/step - loss: 0.5067 - accuracy: 0.9884 - val_loss: 0.6050 - val_accuracy: 0.9550 - lr: 2.3438e-06

Epoch 67/100
48/50 [=====>..] - ETA: 0s - loss: 0.5037 - accuracy: 0.9912

Epoch 67: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
50/50 [=====] - 1s 11ms/step - loss: 0.5025 - accuracy: 0.9916 - val_loss: 0.6087 - val_accuracy: 0.9550 - lr: 2.3438e-06

Epoch 68/100
50/50 [=====] - 1s 11ms/step - loss: 0.4962 - accuracy: 0.9916 - val_loss: 0.6033 - val_accuracy: 0.9563 - lr: 1.1719e-06

Epoch 69/100
50/50 [=====] - 1s 11ms/step - loss: 0.5016 - accuracy: 0.9891 - val_loss: 0.5876 - val_accuracy: 0.9563 - lr: 1.1719e-06

Epoch 70/100
50/50 [=====] - 1s 11ms/step - loss: 0.4977 - accuracy: 0.9887 - val_loss: 0.5877 - val_accuracy: 0.9575 - lr: 1.1719e-06

Epoch 71/100
50/50 [=====] - 1s 11ms/step - loss: 0.5026 - accuracy: 0.9900 - val_loss: 0.5967 - val_accuracy: 0.9550 - lr: 1.1719e-06

Epoch 72/100
50/50 [=====] - 1s 11ms/step - loss: 0.5044 - accuracy: 0.9887 - val_loss: 0.5902 - val_accuracy: 0.9588 - lr: 1.1719e-06

Epoch 73/100
50/50 [=====] - ETA: 0s - loss: 0.4981 - accuracy: 0.9909

Epoch 73: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.
50/50 [=====] - 1s 12ms/step - loss: 0.4981 - accuracy: 0.9909 - val_loss: 0.5887 - val_accuracy: 0.9588 - lr: 1.1719e-06
[100, 64, 0.25]
92.5%
Epoch 1/100
47/47 [=====] - 1s 18ms/step - loss: 3.1030 - accuracy: 0.1667 - val_loss: 3.3153 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 2/100
47/47 [=====] - 1s 14ms/step - loss: 2.5829 - accuracy: 0.2627 - val_loss: 4.5401 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 3/100
47/47 [=====] - 1s 14ms/step - loss: 2.3125 - accuracy: 0.3337 - val_loss: 5.5508 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 4/100
47/47 [=====] - 1s 15ms/step - loss: 1.9389 - accuracy: 0.4853 - val_loss: 7.0810 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 5/100
47/47 [=====] - ETA: 0s - loss: 1.5674 - accuracy: 0.6293
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
47/47 [=====] - 1s 13ms/step - loss: 1.5674 - accuracy: 0.6293 - val_loss: 8.1206 - val_accuracy: 0.1020 - lr: 3.0000e-04
Epoch 6/100
47/47 [=====] - 1s 13ms/step - loss: 1.2485 - accuracy: 0.7610 - val_loss: 8.8039 - val_accuracy: 0.0950 - lr: 1.5000e-04
Epoch 7/100
47/47 [=====] - 1s 14ms/step - loss: 1.0942 - accuracy: 0.8170 - val_loss: 9.3057 - val_accuracy: 0.1460 - lr: 1.5000e-04
Epoch 8/100
47/47 [=====] - 1s 12ms/step - loss: 0.9652 - accuracy: 0.8613 - val_loss: 10.5450 - val_accuracy: 0.0930 - lr: 1.5000e-04
Epoch 9/100
43/47 [=====>...] - ETA: 0s - loss: 0.8708 - accuracy: 0.9001
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
47/47 [=====] - 1s 11ms/step - loss: 0.8673 - accuracy: 0.9010 - val_loss: 10.1911 - val_accuracy: 0.1510 - lr: 1.5000e-04
Epoch 10/100
47/47 [=====] - 1s 11ms/step - loss: 0.7918 - accuracy: 0.9277 - val_loss: 10.5051 - val_accuracy: 0.1020 - lr: 7.5000e-05
Epoch 11/100
47/47 [=====] - 1s 11ms/step - loss: 0.7376 - accuracy: 0.9413 - val_loss: 6.8770 - val_accuracy: 0.2850 - lr: 7.5000e-05
Epoch 12/100
47/47 [=====] - 1s 11ms/step - loss: 0.6943 - accuracy: 0.9540 - val_loss: 5.7218 - val_accuracy: 0.3100 - lr: 7.5000e-05
Epoch 13/100
42/47 [=====>...] - ETA: 0s - loss: 0.6545 - accuracy: 0.9635
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
47/47 [=====] - 1s 11ms/step - loss: 0.6534 - accuracy: 0.9657 - val_loss: 5.7267 - val_accuracy: 0.2610 - lr: 7.5000e-05
Epoch 14/100
47/47 [=====] - 1s 13ms/step - loss: 0.6378 - accuracy: 0.9723 - val_loss: 4.1441 - val_accuracy: 0.4020 - lr: 3.7500e-05
Epoch 15/100
47/47 [=====] - 1s 13ms/step - loss: 0.6149 - accuracy: 0.9753 - val_loss: 2.5548 - val_accuracy: 0.4760 - lr: 3.7500e-05
Epoch 16/100

47/47 [=====] - 1s 14ms/step - loss: 0.6056 - accuracy: 0.9763 - val_loss: 4.6683 - val_accuracy: 0.2490 - lr: 3.7500e-05
Epoch 17/100
47/47 [=====] - 1s 14ms/step - loss: 0.5857 - accuracy: 0.9827 - val_loss: 2.5098 - val_accuracy: 0.5590 - lr: 3.7500e-05
Epoch 18/100
47/47 [=====] - 1s 13ms/step - loss: 0.5790 - accuracy: 0.9830 - val_loss: 1.4187 - val_accuracy: 0.5630 - lr: 3.7500e-05
Epoch 19/100
47/47 [=====] - 1s 13ms/step - loss: 0.5635 - accuracy: 0.9837 - val_loss: 2.2468 - val_accuracy: 0.3510 - lr: 3.7500e-05
Epoch 20/100
47/47 [=====] - 1s 14ms/step - loss: 0.5561 - accuracy: 0.9850 - val_loss: 2.0345 - val_accuracy: 0.5060 - lr: 3.7500e-05
Epoch 21/100
47/47 [=====] - 1s 13ms/step - loss: 0.5437 - accuracy: 0.9867 - val_loss: 7.2597 - val_accuracy: 0.2760 - lr: 3.7500e-05
Epoch 22/100
47/47 [=====] - 1s 14ms/step - loss: 0.5301 - accuracy: 0.9900 - val_loss: 1.4010 - val_accuracy: 0.5760 - lr: 3.7500e-05
Epoch 23/100
47/47 [=====] - 1s 14ms/step - loss: 0.5271 - accuracy: 0.9897 - val_loss: 1.1872 - val_accuracy: 0.7270 - lr: 3.7500e-05
Epoch 24/100
47/47 [=====] - 1s 13ms/step - loss: 0.5168 - accuracy: 0.9893 - val_loss: 1.4194 - val_accuracy: 0.6340 - lr: 3.7500e-05
Epoch 25/100
47/47 [=====] - 1s 14ms/step - loss: 0.5098 - accuracy: 0.9920 - val_loss: 3.1983 - val_accuracy: 0.4630 - lr: 3.7500e-05
Epoch 26/100
47/47 [=====] - 1s 13ms/step - loss: 0.5003 - accuracy: 0.9927 - val_loss: 2.5362 - val_accuracy: 0.4550 - lr: 3.7500e-05
Epoch 27/100
47/47 [=====] - 1s 15ms/step - loss: 0.4918 - accuracy: 0.9937 - val_loss: 0.8602 - val_accuracy: 0.8480 - lr: 3.7500e-05
Epoch 28/100
47/47 [=====] - 1s 12ms/step - loss: 0.4881 - accuracy: 0.9940 - val_loss: 1.3754 - val_accuracy: 0.6410 - lr: 3.7500e-05
Epoch 29/100
47/47 [=====] - 1s 14ms/step - loss: 0.4822 - accuracy: 0.9940 - val_loss: 1.9417 - val_accuracy: 0.4670 - lr: 3.7500e-05
Epoch 30/100
47/47 [=====] - 1s 14ms/step - loss: 0.4782 - accuracy: 0.9920 - val_loss: 1.5293 - val_accuracy: 0.5770 - lr: 3.7500e-05
Epoch 31/100
47/47 [=====] - ETA: 0s - loss: 0.4628 - accuracy: 0.9963
Epoch 31: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
47/47 [=====] - 1s 14ms/step - loss: 0.4628 - accuracy: 0.9963 - val_loss: 3.5360 - val_accuracy: 0.3610 - lr: 3.7500e-05
Epoch 32/100
47/47 [=====] - 1s 13ms/step - loss: 0.4606 - accuracy: 0.9957 - val_loss: 0.8495 - val_accuracy: 0.8370 - lr: 1.8750e-05
Epoch 33/100
47/47 [=====] - 1s 13ms/step - loss: 0.4571 - accuracy: 0.9960 - val_loss: 0.6416 - val_accuracy: 0.9400 - lr: 1.8750e-05
Epoch 34/100
47/47 [=====] - 1s 12ms/step - loss: 0.4505 - accuracy: 0.9963 - val_loss: 1.1902 - val_accuracy: 0.7050 - lr: 1.8750e-05
Epoch 35/100

47/47 [=====] - 1s 11ms/step - loss: 0.4499 - accuracy: 0.9967 - val_loss: 0.5422 - val_accuracy: 0.9660 - lr: 1.8750e-05
Epoch 36/100
47/47 [=====] - 1s 11ms/step - loss: 0.4487 - accuracy: 0.9950 - val_loss: 0.5365 - val_accuracy: 0.9730 - lr: 1.8750e-05
Epoch 37/100
47/47 [=====] - 1s 11ms/step - loss: 0.4410 - accuracy: 0.9963 - val_loss: 0.5115 - val_accuracy: 0.9720 - lr: 1.8750e-05
Epoch 38/100
47/47 [=====] - 0s 11ms/step - loss: 0.4398 - accuracy: 0.9967 - val_loss: 0.5647 - val_accuracy: 0.9520 - lr: 1.8750e-05
Epoch 39/100
47/47 [=====] - 1s 11ms/step - loss: 0.4373 - accuracy: 0.9980 - val_loss: 0.8956 - val_accuracy: 0.8180 - lr: 1.8750e-05
Epoch 40/100
47/47 [=====] - 1s 12ms/step - loss: 0.4341 - accuracy: 0.9970 - val_loss: 1.1920 - val_accuracy: 0.6890 - lr: 1.8750e-05
Epoch 41/100
44/47 [=====>..] - ETA: 0s - loss: 0.4322 - accuracy: 0.9972
Epoch 41: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
47/47 [=====] - 1s 13ms/step - loss: 0.4314 - accuracy: 0.9973 - val_loss: 0.7266 - val_accuracy: 0.8940 - lr: 1.8750e-05
Epoch 42/100
47/47 [=====] - 1s 12ms/step - loss: 0.4294 - accuracy: 0.9980 - val_loss: 0.6386 - val_accuracy: 0.9320 - lr: 9.3750e-06
Epoch 43/100
47/47 [=====] - 1s 13ms/step - loss: 0.4268 - accuracy: 0.9970 - val_loss: 0.7245 - val_accuracy: 0.8930 - lr: 9.3750e-06
Epoch 44/100
47/47 [=====] - 1s 13ms/step - loss: 0.4241 - accuracy: 0.9977 - val_loss: 0.6933 - val_accuracy: 0.9040 - lr: 9.3750e-06
Epoch 45/100
47/47 [=====] - 1s 13ms/step - loss: 0.4259 - accuracy: 0.9987 - val_loss: 0.4915 - val_accuracy: 0.9720 - lr: 9.3750e-06
Epoch 46/100
47/47 [=====] - 1s 12ms/step - loss: 0.4209 - accuracy: 0.9973 - val_loss: 0.6615 - val_accuracy: 0.9210 - lr: 9.3750e-06
Epoch 47/100
47/47 [=====] - 1s 14ms/step - loss: 0.4196 - accuracy: 0.9980 - val_loss: 0.7350 - val_accuracy: 0.8840 - lr: 9.3750e-06
Epoch 48/100
47/47 [=====] - 1s 13ms/step - loss: 0.4203 - accuracy: 0.9980 - val_loss: 0.4837 - val_accuracy: 0.9750 - lr: 9.3750e-06
Epoch 49/100
47/47 [=====] - 1s 12ms/step - loss: 0.4174 - accuracy: 0.9983 - val_loss: 0.4977 - val_accuracy: 0.9720 - lr: 9.3750e-06
Epoch 50/100
47/47 [=====] - 1s 14ms/step - loss: 0.4152 - accuracy: 0.9993 - val_loss: 0.4758 - val_accuracy: 0.9750 - lr: 9.3750e-06
Epoch 51/100
47/47 [=====] - 1s 13ms/step - loss: 0.4142 - accuracy: 0.9983 - val_loss: 0.5017 - val_accuracy: 0.9730 - lr: 9.3750e-06
Epoch 52/100
47/47 [=====] - 1s 13ms/step - loss: 0.4121 - accuracy: 0.9980 - val_loss: 0.5815 - val_accuracy: 0.9430 - lr: 9.3750e-06
Epoch 53/100
47/47 [=====] - 1s 13ms/step - loss: 0.4119 - accuracy: 0.9987 - val_loss: 0.5912 - val_accuracy: 0.9420 - lr: 9.3750e-06
Epoch 54/100

46/47 [=====>.] - ETA: 0s - loss: 0.4093 - accuracy: 0.9993
Epoch 54: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
47/47 [=====] - 1s 14ms/step - loss: 0.4091 - accuracy: 0.9993 - val_loss: 0.5015 - val_accuracy: 0.9650 - lr: 9.3750e-06
Epoch 55/100
47/47 [=====] - 1s 14ms/step - loss: 0.4079 - accuracy: 0.9987 - val_loss: 0.4642 - val_accuracy: 0.9790 - lr: 4.6875e-06
Epoch 56/100
47/47 [=====] - 1s 14ms/step - loss: 0.4107 - accuracy: 0.9990 - val_loss: 0.4990 - val_accuracy: 0.9710 - lr: 4.6875e-06
Epoch 57/100
47/47 [=====] - 1s 14ms/step - loss: 0.4096 - accuracy: 0.9990 - val_loss: 0.4945 - val_accuracy: 0.9750 - lr: 4.6875e-06
Epoch 58/100
47/47 [=====] - 1s 14ms/step - loss: 0.4069 - accuracy: 0.9993 - val_loss: 0.4734 - val_accuracy: 0.9720 - lr: 4.6875e-06
Epoch 59/100
42/47 [=====>....] - ETA: 0s - loss: 0.4062 - accuracy: 0.9978
Epoch 59: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
47/47 [=====] - 1s 14ms/step - loss: 0.4059 - accuracy: 0.9977 - val_loss: 0.4675 - val_accuracy: 0.9730 - lr: 4.6875e-06
Epoch 60/100
47/47 [=====] - 1s 14ms/step - loss: 0.4060 - accuracy: 0.9983 - val_loss: 0.4587 - val_accuracy: 0.9810 - lr: 2.3438e-06
Epoch 61/100
47/47 [=====] - 1s 11ms/step - loss: 0.4037 - accuracy: 0.9997 - val_loss: 0.4595 - val_accuracy: 0.9790 - lr: 2.3438e-06
Epoch 62/100
47/47 [=====] - 1s 11ms/step - loss: 0.4046 - accuracy: 0.9987 - val_loss: 0.4591 - val_accuracy: 0.9780 - lr: 2.3438e-06
Epoch 63/100
47/47 [=====] - 1s 11ms/step - loss: 0.4067 - accuracy: 0.9983 - val_loss: 0.4586 - val_accuracy: 0.9770 - lr: 2.3438e-06
Epoch 64/100
43/47 [=====>...] - ETA: 0s - loss: 0.4033 - accuracy: 0.9978
Epoch 64: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
47/47 [=====] - 0s 11ms/step - loss: 0.4031 - accuracy: 0.9980 - val_loss: 0.4601 - val_accuracy: 0.9760 - lr: 2.3438e-06
Epoch 65/100
47/47 [=====] - 1s 11ms/step - loss: 0.4014 - accuracy: 0.9987 - val_loss: 0.4580 - val_accuracy: 0.9790 - lr: 1.1719e-06
Epoch 66/100
47/47 [=====] - 1s 11ms/step - loss: 0.4053 - accuracy: 0.9990 - val_loss: 0.4572 - val_accuracy: 0.9780 - lr: 1.1719e-06
Epoch 67/100
47/47 [=====] - 1s 11ms/step - loss: 0.4037 - accuracy: 0.9987 - val_loss: 0.4594 - val_accuracy: 0.9750 - lr: 1.1719e-06
Epoch 68/100
47/47 [=====] - 1s 11ms/step - loss: 0.4019 - accuracy: 0.9987 - val_loss: 0.4655 - val_accuracy: 0.9740 - lr: 1.1719e-06
Epoch 69/100
47/47 [=====] - 1s 11ms/step - loss: 0.4043 - accuracy: 0.9987 - val_loss: 0.4640 - val_accuracy: 0.9750 - lr: 1.1719e-06
Epoch 70/100
43/47 [=====>...] - ETA: 0s - loss: 0.4037 - accuracy: 0.9993
Epoch 70: ReduceLROnPlateau reducing learning rate to 5.859375278305379e-07.

47/47 [=====] - 0s 11ms/step - loss: 0.4040 - accuracy: 0.9990 - val_loss: 0.4608 - val_accuracy: 0.9730 - lr: 1.1719e-06
Epoch 71/100
47/47 [=====] - 1s 14ms/step - loss: 0.3994 - accuracy: 0.9993 - val_loss: 0.4700 - val_accuracy: 0.9700 - lr: 5.8594e-07
Epoch 72/100
47/47 [=====] - 1s 14ms/step - loss: 0.4000 - accuracy: 0.9990 - val_loss: 0.4624 - val_accuracy: 0.9760 - lr: 5.8594e-07
[100, 64, 0.30000000000000004]
95.0%
Epoch 1/100
44/44 [=====] - 2s 22ms/step - loss: 3.0960 - accuracy: 0.1396 - val_loss: 3.0965 - val_accuracy: 0.1033 - lr: 3.0000e-04
Epoch 2/100
44/44 [=====] - 1s 16ms/step - loss: 2.6377 - accuracy: 0.2557 - val_loss: 3.5786 - val_accuracy: 0.1417 - lr: 3.0000e-04
Epoch 3/100
44/44 [=====] - 1s 12ms/step - loss: 2.3020 - accuracy: 0.3379 - val_loss: 4.3056 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 4/100
44/44 [=====] - 1s 13ms/step - loss: 1.9702 - accuracy: 0.4632 - val_loss: 5.7654 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 5/100
42/44 [=====>..] - ETA: 0s - loss: 1.6396 - accuracy: 0.6112
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
44/44 [=====] - 1s 12ms/step - loss: 1.6307 - accuracy: 0.6154 - val_loss: 6.9661 - val_accuracy: 0.1133 - lr: 3.0000e-04
Epoch 6/100
44/44 [=====] - 1s 12ms/step - loss: 1.3506 - accuracy: 0.7186 - val_loss: 8.3401 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 7/100
44/44 [=====] - 1s 11ms/step - loss: 1.2128 - accuracy: 0.7711 - val_loss: 9.3157 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 8/100
44/44 [=====] - 1s 13ms/step - loss: 1.0820 - accuracy: 0.8150 - val_loss: 9.6697 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 9/100
42/44 [=====>..] - ETA: 0s - loss: 0.9752 - accuracy: 0.8538
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
44/44 [=====] - 1s 14ms/step - loss: 0.9738 - accuracy: 0.8557 - val_loss: 10.1460 - val_accuracy: 0.1133 - lr: 1.5000e-04
Epoch 10/100
44/44 [=====] - 1s 15ms/step - loss: 0.8847 - accuracy: 0.8918 - val_loss: 8.9106 - val_accuracy: 0.1500 - lr: 7.5000e-05
Epoch 11/100
44/44 [=====] - 1s 13ms/step - loss: 0.8315 - accuracy: 0.9025 - val_loss: 7.7895 - val_accuracy: 0.2367 - lr: 7.5000e-05
Epoch 12/100
44/44 [=====] - 1s 15ms/step - loss: 0.7878 - accuracy: 0.9171 - val_loss: 8.5648 - val_accuracy: 0.1625 - lr: 7.5000e-05
Epoch 13/100
43/44 [=====>..] - ETA: 0s - loss: 0.7397 - accuracy: 0.9324
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
44/44 [=====] - 1s 14ms/step - loss: 0.7407 - accuracy: 0.9314 - val_loss: 8.6516 - val_accuracy: 0.1025 - lr: 7.5000e-05
Epoch 14/100
44/44 [=====] - 1s 13ms/step - loss: 0.7154 - accuracy:

0.9421 - val_loss: 7.3629 - val_accuracy: 0.2017 - lr: 3.7500e-05
Epoch 15/100
44/44 [=====] - 0s 11ms/step - loss: 0.6884 - accuracy:
0.9539 - val_loss: 5.0661 - val_accuracy: 0.4500 - lr: 3.7500e-05
Epoch 16/100
44/44 [=====] - 1s 11ms/step - loss: 0.6698 - accuracy:
0.9546 - val_loss: 2.6066 - val_accuracy: 0.4133 - lr: 3.7500e-05
Epoch 17/100
44/44 [=====] - 0s 11ms/step - loss: 0.6626 - accuracy:
0.9536 - val_loss: 3.7310 - val_accuracy: 0.3050 - lr: 3.7500e-05
Epoch 18/100
44/44 [=====] - 0s 11ms/step - loss: 0.6440 - accuracy:
0.9596 - val_loss: 2.2568 - val_accuracy: 0.4783 - lr: 3.7500e-05
Epoch 19/100
44/44 [=====] - 1s 12ms/step - loss: 0.6285 - accuracy:
0.9643 - val_loss: 1.8178 - val_accuracy: 0.4583 - lr: 3.7500e-05
Epoch 20/100
44/44 [=====] - 0s 11ms/step - loss: 0.6055 - accuracy:
0.9689 - val_loss: 2.9406 - val_accuracy: 0.4775 - lr: 3.7500e-05
Epoch 21/100
44/44 [=====] - 1s 14ms/step - loss: 0.5955 - accuracy:
0.9736 - val_loss: 1.0891 - val_accuracy: 0.7600 - lr: 3.7500e-05
Epoch 22/100
44/44 [=====] - 1s 14ms/step - loss: 0.5869 - accuracy:
0.9721 - val_loss: 3.8834 - val_accuracy: 0.3700 - lr: 3.7500e-05
Epoch 23/100
44/44 [=====] - 1s 14ms/step - loss: 0.5720 - accuracy:
0.9789 - val_loss: 1.2841 - val_accuracy: 0.6850 - lr: 3.7500e-05
Epoch 24/100
44/44 [=====] - 1s 13ms/step - loss: 0.5607 - accuracy:
0.9804 - val_loss: 1.2311 - val_accuracy: 0.7258 - lr: 3.7500e-05
Epoch 25/100
40/44 [=====>...] - ETA: 0s - loss: 0.5474 - accuracy: 0.984
0
Epoch 25: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
44/44 [=====] - 1s 13ms/step - loss: 0.5478 - accuracy:
0.9829 - val_loss: 2.6924 - val_accuracy: 0.3683 - lr: 3.7500e-05
Epoch 26/100
44/44 [=====] - 1s 12ms/step - loss: 0.5371 - accuracy:
0.9832 - val_loss: 0.7638 - val_accuracy: 0.8992 - lr: 1.8750e-05
Epoch 27/100
44/44 [=====] - 1s 13ms/step - loss: 0.5288 - accuracy:
0.9839 - val_loss: 0.7568 - val_accuracy: 0.9050 - lr: 1.8750e-05
Epoch 28/100
44/44 [=====] - 1s 12ms/step - loss: 0.5269 - accuracy:
0.9850 - val_loss: 0.7484 - val_accuracy: 0.8967 - lr: 1.8750e-05
Epoch 29/100
44/44 [=====] - 1s 13ms/step - loss: 0.5172 - accuracy:
0.9889 - val_loss: 0.8482 - val_accuracy: 0.8633 - lr: 1.8750e-05
Epoch 30/100
44/44 [=====] - 1s 13ms/step - loss: 0.5104 - accuracy:
0.9857 - val_loss: 0.8818 - val_accuracy: 0.8483 - lr: 1.8750e-05
Epoch 31/100
44/44 [=====] - 1s 13ms/step - loss: 0.5143 - accuracy:
0.9846 - val_loss: 1.0299 - val_accuracy: 0.7683 - lr: 1.8750e-05
Epoch 32/100
44/44 [=====] - ETA: 0s - loss: 0.4987 - accuracy: 0.989
6
Epoch 32: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
44/44 [=====] - 1s 13ms/step - loss: 0.4987 - accuracy:

0.9896 - val_loss: 1.1895 - val_accuracy: 0.7017 - lr: 1.8750e-05
Epoch 33/100
44/44 [=====] - 1s 13ms/step - loss: 0.4977 - accuracy:
0.9889 - val_loss: 0.7224 - val_accuracy: 0.9033 - lr: 9.3750e-06
Epoch 34/100
44/44 [=====] - 1s 14ms/step - loss: 0.4975 - accuracy:
0.9900 - val_loss: 0.6623 - val_accuracy: 0.9175 - lr: 9.3750e-06
Epoch 35/100
44/44 [=====] - 1s 12ms/step - loss: 0.4915 - accuracy:
0.9896 - val_loss: 0.7115 - val_accuracy: 0.9008 - lr: 9.3750e-06
Epoch 36/100
44/44 [=====] - 1s 13ms/step - loss: 0.4893 - accuracy:
0.9900 - val_loss: 0.6999 - val_accuracy: 0.9133 - lr: 9.3750e-06
Epoch 37/100
44/44 [=====] - 1s 13ms/step - loss: 0.4899 - accuracy:
0.9882 - val_loss: 0.6587 - val_accuracy: 0.9258 - lr: 9.3750e-06
Epoch 38/100
44/44 [=====] - 1s 12ms/step - loss: 0.4860 - accuracy:
0.9914 - val_loss: 0.7098 - val_accuracy: 0.9042 - lr: 9.3750e-06
Epoch 39/100
44/44 [=====] - 1s 12ms/step - loss: 0.4830 - accuracy:
0.9889 - val_loss: 0.6517 - val_accuracy: 0.9175 - lr: 9.3750e-06
Epoch 40/100
44/44 [=====] - 1s 12ms/step - loss: 0.4833 - accuracy:
0.9911 - val_loss: 0.6400 - val_accuracy: 0.9292 - lr: 9.3750e-06
Epoch 41/100
44/44 [=====] - 1s 12ms/step - loss: 0.4791 - accuracy:
0.9904 - val_loss: 0.6819 - val_accuracy: 0.9217 - lr: 9.3750e-06
Epoch 42/100
44/44 [=====] - 1s 14ms/step - loss: 0.4810 - accuracy:
0.9896 - val_loss: 0.6409 - val_accuracy: 0.9300 - lr: 9.3750e-06
Epoch 43/100
44/44 [=====] - 1s 12ms/step - loss: 0.4748 - accuracy:
0.9921 - val_loss: 0.6656 - val_accuracy: 0.9258 - lr: 9.3750e-06
Epoch 44/100
43/44 [=====>.] - ETA: 0s - loss: 0.4701 - accuracy: 0.990
2
Epoch 44: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
44/44 [=====] - 1s 12ms/step - loss: 0.4706 - accuracy:
0.9900 - val_loss: 0.6653 - val_accuracy: 0.9242 - lr: 9.3750e-06
Epoch 45/100
44/44 [=====] - 1s 12ms/step - loss: 0.4699 - accuracy:
0.9914 - val_loss: 0.6407 - val_accuracy: 0.9292 - lr: 4.6875e-06
Epoch 46/100
44/44 [=====] - 1s 11ms/step - loss: 0.4704 - accuracy:
0.9911 - val_loss: 0.6423 - val_accuracy: 0.9225 - lr: 4.6875e-06
Epoch 47/100
44/44 [=====] - 0s 11ms/step - loss: 0.4731 - accuracy:
0.9921 - val_loss: 0.6555 - val_accuracy: 0.9267 - lr: 4.6875e-06
Epoch 48/100
44/44 [=====] - 0s 11ms/step - loss: 0.4671 - accuracy:
0.9929 - val_loss: 0.6301 - val_accuracy: 0.9350 - lr: 4.6875e-06
Epoch 49/100
44/44 [=====] - 1s 13ms/step - loss: 0.4700 - accuracy:
0.9911 - val_loss: 0.6414 - val_accuracy: 0.9275 - lr: 4.6875e-06
Epoch 50/100
44/44 [=====] - 1s 13ms/step - loss: 0.4608 - accuracy:
0.9943 - val_loss: 0.6255 - val_accuracy: 0.9267 - lr: 4.6875e-06
Epoch 51/100
44/44 [=====] - 1s 12ms/step - loss: 0.4612 - accuracy:

0.9946 - val_loss: 0.6855 - val_accuracy: 0.9067 - lr: 4.6875e-06
Epoch 52/100
44/44 [=====] - 1s 12ms/step - loss: 0.4646 - accuracy:
0.9918 - val_loss: 0.6078 - val_accuracy: 0.9333 - lr: 4.6875e-06
Epoch 53/100
44/44 [=====] - 1s 12ms/step - loss: 0.4595 - accuracy:
0.9939 - val_loss: 0.6078 - val_accuracy: 0.9325 - lr: 4.6875e-06
Epoch 54/100
44/44 [=====] - 1s 12ms/step - loss: 0.4587 - accuracy:
0.9918 - val_loss: 0.6090 - val_accuracy: 0.9350 - lr: 4.6875e-06
Epoch 55/100
44/44 [=====] - 1s 12ms/step - loss: 0.4585 - accuracy:
0.9936 - val_loss: 0.6240 - val_accuracy: 0.9292 - lr: 4.6875e-06
Epoch 56/100
44/44 [=====] - 1s 13ms/step - loss: 0.4610 - accuracy:
0.9929 - val_loss: 0.6074 - val_accuracy: 0.9383 - lr: 4.6875e-06
Epoch 57/100
44/44 [=====] - 1s 13ms/step - loss: 0.4540 - accuracy:
0.9932 - val_loss: 0.6259 - val_accuracy: 0.9342 - lr: 4.6875e-06
Epoch 58/100
44/44 [=====] - 1s 14ms/step - loss: 0.4598 - accuracy:
0.9918 - val_loss: 0.6496 - val_accuracy: 0.9192 - lr: 4.6875e-06
Epoch 59/100
44/44 [=====] - 1s 12ms/step - loss: 0.4530 - accuracy:
0.9921 - val_loss: 0.6144 - val_accuracy: 0.9342 - lr: 4.6875e-06
Epoch 60/100
39/44 [=====>....] - ETA: 0s - loss: 0.4474 - accuracy: 0.994
8
Epoch 60: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
44/44 [=====] - 1s 12ms/step - loss: 0.4480 - accuracy:
0.9946 - val_loss: 0.6190 - val_accuracy: 0.9367 - lr: 4.6875e-06
Epoch 61/100
44/44 [=====] - 1s 13ms/step - loss: 0.4484 - accuracy:
0.9929 - val_loss: 0.6057 - val_accuracy: 0.9392 - lr: 2.3438e-06
Epoch 62/100
44/44 [=====] - 1s 13ms/step - loss: 0.4496 - accuracy:
0.9950 - val_loss: 0.5979 - val_accuracy: 0.9400 - lr: 2.3438e-06
Epoch 63/100
44/44 [=====] - 1s 13ms/step - loss: 0.4470 - accuracy:
0.9939 - val_loss: 0.5995 - val_accuracy: 0.9375 - lr: 2.3438e-06
Epoch 64/100
44/44 [=====] - 1s 12ms/step - loss: 0.4475 - accuracy:
0.9961 - val_loss: 0.5985 - val_accuracy: 0.9392 - lr: 2.3438e-06
Epoch 65/100
44/44 [=====] - 1s 13ms/step - loss: 0.4448 - accuracy:
0.9929 - val_loss: 0.5917 - val_accuracy: 0.9383 - lr: 2.3438e-06
Epoch 66/100
44/44 [=====] - 1s 12ms/step - loss: 0.4441 - accuracy:
0.9946 - val_loss: 0.5993 - val_accuracy: 0.9383 - lr: 2.3438e-06
Epoch 67/100
44/44 [=====] - 1s 13ms/step - loss: 0.4501 - accuracy:
0.9918 - val_loss: 0.5952 - val_accuracy: 0.9358 - lr: 2.3438e-06
Epoch 68/100
44/44 [=====] - 1s 14ms/step - loss: 0.4473 - accuracy:
0.9936 - val_loss: 0.5990 - val_accuracy: 0.9367 - lr: 2.3438e-06
Epoch 69/100
44/44 [=====] - ETA: 0s - loss: 0.4427 - accuracy: 0.991
4
Epoch 69: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
44/44 [=====] - 1s 13ms/step - loss: 0.4427 - accuracy:

0.9914 - val_loss: 0.5944 - val_accuracy: 0.9342 - lr: 2.3438e-06
Epoch 70/100
44/44 [=====] - 1s 15ms/step - loss: 0.4453 - accuracy:
0.9939 - val_loss: 0.6012 - val_accuracy: 0.9300 - lr: 1.1719e-06
Epoch 71/100
44/44 [=====] - 1s 12ms/step - loss: 0.4389 - accuracy:
0.9954 - val_loss: 0.5915 - val_accuracy: 0.9358 - lr: 1.1719e-06
Epoch 72/100
44/44 [=====] - 1s 12ms/step - loss: 0.4432 - accuracy:
0.9961 - val_loss: 0.5914 - val_accuracy: 0.9342 - lr: 1.1719e-06
Epoch 73/100
44/44 [=====] - 0s 11ms/step - loss: 0.4410 - accuracy:
0.9932 - val_loss: 0.5928 - val_accuracy: 0.9342 - lr: 1.1719e-06
Epoch 74/100
44/44 [=====] - 1s 12ms/step - loss: 0.4379 - accuracy:
0.9946 - val_loss: 0.5888 - val_accuracy: 0.9317 - lr: 1.1719e-06
[100, 64, 0.35000000000000003]
97.5%
Epoch 1/100
41/41 [=====] - 2s 20ms/step - loss: 3.3164 - accuracy:
0.1374 - val_loss: 4.1455 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 2/100
41/41 [=====] - 1s 15ms/step - loss: 2.7631 - accuracy:
0.2482 - val_loss: 5.0216 - val_accuracy: 0.1006 - lr: 3.0000e-04
Epoch 3/100
41/41 [=====] - 1s 15ms/step - loss: 2.4558 - accuracy:
0.3128 - val_loss: 4.6530 - val_accuracy: 0.0971 - lr: 3.0000e-04
Epoch 4/100
41/41 [=====] - 1s 14ms/step - loss: 2.1822 - accuracy:
0.4221 - val_loss: 4.1500 - val_accuracy: 0.0992 - lr: 3.0000e-04
Epoch 5/100
40/41 [=====>.] - ETA: 0s - loss: 1.8683 - accuracy: 0.524
6
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
41/41 [=====] - 1s 13ms/step - loss: 1.8676 - accuracy:
0.5237 - val_loss: 6.3231 - val_accuracy: 0.0914 - lr: 3.0000e-04
Epoch 6/100
41/41 [=====] - 1s 15ms/step - loss: 1.5586 - accuracy:
0.6533 - val_loss: 8.0424 - val_accuracy: 0.0914 - lr: 1.5000e-04
Epoch 7/100
41/41 [=====] - 1s 14ms/step - loss: 1.3815 - accuracy:
0.7226 - val_loss: 10.2685 - val_accuracy: 0.0914 - lr: 1.5000e-04
Epoch 8/100
41/41 [=====] - 1s 13ms/step - loss: 1.1982 - accuracy:
0.7895 - val_loss: 12.6893 - val_accuracy: 0.0914 - lr: 1.5000e-04
Epoch 9/100
39/41 [=====>..] - ETA: 0s - loss: 1.0712 - accuracy: 0.841
3
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
41/41 [=====] - 1s 15ms/step - loss: 1.0680 - accuracy:
0.8426 - val_loss: 13.7211 - val_accuracy: 0.0914 - lr: 1.5000e-04
Epoch 10/100
41/41 [=====] - 1s 15ms/step - loss: 0.9682 - accuracy:
0.8807 - val_loss: 14.0112 - val_accuracy: 0.0914 - lr: 7.5000e-05
Epoch 11/100
41/41 [=====] - 1s 15ms/step - loss: 0.9067 - accuracy:
0.9007 - val_loss: 13.6222 - val_accuracy: 0.0914 - lr: 7.5000e-05
Epoch 12/100
41/41 [=====] - 1s 14ms/step - loss: 0.8620 - accuracy:
0.9169 - val_loss: 12.4467 - val_accuracy: 0.0914 - lr: 7.5000e-05

Epoch 13/100
38/41 [=====>...] - ETA: 0s - loss: 0.8209 - accuracy: 0.9326
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
41/41 [=====] - 1s 16ms/step - loss: 0.8219 - accuracy: 0.9323 - val_loss: 12.0071 - val_accuracy: 0.1014 - lr: 7.5000e-05
Epoch 14/100
41/41 [=====] - 1s 14ms/step - loss: 0.7807 - accuracy: 0.9450 - val_loss: 8.6461 - val_accuracy: 0.0978 - lr: 3.7500e-05
Epoch 15/100
41/41 [=====] - 1s 13ms/step - loss: 0.7485 - accuracy: 0.9561 - val_loss: 9.5000 - val_accuracy: 0.1706 - lr: 3.7500e-05
Epoch 16/100
41/41 [=====] - 1s 14ms/step - loss: 0.7437 - accuracy: 0.9569 - val_loss: 3.9683 - val_accuracy: 0.2284 - lr: 3.7500e-05
Epoch 17/100
41/41 [=====] - 1s 12ms/step - loss: 0.7215 - accuracy: 0.9661 - val_loss: 3.9994 - val_accuracy: 0.2056 - lr: 3.7500e-05
Epoch 18/100
41/41 [=====] - 1s 13ms/step - loss: 0.7186 - accuracy: 0.9600 - val_loss: 1.6719 - val_accuracy: 0.5575 - lr: 3.7500e-05
Epoch 19/100
41/41 [=====] - 1s 13ms/step - loss: 0.7056 - accuracy: 0.9654 - val_loss: 3.2966 - val_accuracy: 0.2120 - lr: 3.7500e-05
Epoch 20/100
41/41 [=====] - 1s 14ms/step - loss: 0.6911 - accuracy: 0.9654 - val_loss: 2.1921 - val_accuracy: 0.6003 - lr: 3.7500e-05
Epoch 21/100
41/41 [=====] - 1s 12ms/step - loss: 0.6762 - accuracy: 0.9696 - val_loss: 3.4749 - val_accuracy: 0.1156 - lr: 3.7500e-05
Epoch 22/100
37/41 [=====>...] - ETA: 0s - loss: 0.6722 - accuracy: 0.9692
Epoch 22: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
41/41 [=====] - 1s 15ms/step - loss: 0.6720 - accuracy: 0.9692 - val_loss: 2.6774 - val_accuracy: 0.3790 - lr: 3.7500e-05
Epoch 23/100
41/41 [=====] - 1s 12ms/step - loss: 0.6508 - accuracy: 0.9746 - val_loss: 2.2754 - val_accuracy: 0.4154 - lr: 1.8750e-05
Epoch 24/100
41/41 [=====] - 0s 12ms/step - loss: 0.6476 - accuracy: 0.9754 - val_loss: 1.1926 - val_accuracy: 0.7316 - lr: 1.8750e-05
Epoch 25/100
41/41 [=====] - 0s 12ms/step - loss: 0.6430 - accuracy: 0.9785 - val_loss: 1.7535 - val_accuracy: 0.5318 - lr: 1.8750e-05
Epoch 26/100
41/41 [=====] - 0s 11ms/step - loss: 0.6400 - accuracy: 0.9777 - val_loss: 1.4442 - val_accuracy: 0.6317 - lr: 1.8750e-05
Epoch 27/100
41/41 [=====] - 0s 12ms/step - loss: 0.6374 - accuracy: 0.9785 - val_loss: 1.3381 - val_accuracy: 0.6859 - lr: 1.8750e-05
Epoch 28/100
41/41 [=====] - 0s 12ms/step - loss: 0.6269 - accuracy: 0.9765 - val_loss: 1.0862 - val_accuracy: 0.7809 - lr: 1.8750e-05
Epoch 29/100
41/41 [=====] - 0s 12ms/step - loss: 0.6221 - accuracy: 0.9781 - val_loss: 1.2835 - val_accuracy: 0.7123 - lr: 1.8750e-05
Epoch 30/100
41/41 [=====] - 1s 13ms/step - loss: 0.6135 - accuracy: 0.9819 - val_loss: 1.9229 - val_accuracy: 0.4383 - lr: 1.8750e-05

Epoch 31/100
41/41 [=====] - 1s 14ms/step - loss: 0.6088 - accuracy: 0.9846 - val_loss: 0.8459 - val_accuracy: 0.8944 - lr: 1.8750e-05
Epoch 32/100
41/41 [=====] - 1s 15ms/step - loss: 0.6011 - accuracy: 0.9854 - val_loss: 1.3948 - val_accuracy: 0.6674 - lr: 1.8750e-05
Epoch 33/100
41/41 [=====] - 1s 14ms/step - loss: 0.5962 - accuracy: 0.9869 - val_loss: 3.0558 - val_accuracy: 0.2527 - lr: 1.8750e-05
Epoch 34/100
41/41 [=====] - 1s 14ms/step - loss: 0.5962 - accuracy: 0.9835 - val_loss: 1.0484 - val_accuracy: 0.7923 - lr: 1.8750e-05
Epoch 35/100
37/41 [=====>...] - ETA: 0s - loss: 0.5896 - accuracy: 0.9865
Epoch 35: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
41/41 [=====] - 1s 14ms/step - loss: 0.5900 - accuracy: 0.9865 - val_loss: 0.8582 - val_accuracy: 0.8744 - lr: 1.8750e-05
Epoch 36/100
41/41 [=====] - 1s 13ms/step - loss: 0.5840 - accuracy: 0.9858 - val_loss: 0.8401 - val_accuracy: 0.8829 - lr: 9.3750e-06
Epoch 37/100
41/41 [=====] - 1s 14ms/step - loss: 0.5851 - accuracy: 0.9869 - val_loss: 0.8982 - val_accuracy: 0.8744 - lr: 9.3750e-06
Epoch 38/100
41/41 [=====] - 1s 15ms/step - loss: 0.5762 - accuracy: 0.9873 - val_loss: 0.9577 - val_accuracy: 0.8373 - lr: 9.3750e-06
Epoch 39/100
41/41 [=====] - 1s 14ms/step - loss: 0.5759 - accuracy: 0.9873 - val_loss: 0.7936 - val_accuracy: 0.9001 - lr: 9.3750e-06
Epoch 40/100
41/41 [=====] - 1s 15ms/step - loss: 0.5795 - accuracy: 0.9838 - val_loss: 0.7320 - val_accuracy: 0.9186 - lr: 9.3750e-06
Epoch 41/100
41/41 [=====] - 1s 14ms/step - loss: 0.5770 - accuracy: 0.9865 - val_loss: 0.7415 - val_accuracy: 0.9236 - lr: 9.3750e-06
Epoch 42/100
41/41 [=====] - 1s 14ms/step - loss: 0.5716 - accuracy: 0.9885 - val_loss: 0.9002 - val_accuracy: 0.8480 - lr: 9.3750e-06
Epoch 43/100
41/41 [=====] - 1s 12ms/step - loss: 0.5720 - accuracy: 0.9904 - val_loss: 0.8435 - val_accuracy: 0.8622 - lr: 9.3750e-06
Epoch 44/100
37/41 [=====>...] - ETA: 0s - loss: 0.5741 - accuracy: 0.9894
Epoch 44: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
41/41 [=====] - 1s 13ms/step - loss: 0.5729 - accuracy: 0.9892 - val_loss: 0.7820 - val_accuracy: 0.8944 - lr: 9.3750e-06
Epoch 45/100
41/41 [=====] - 1s 14ms/step - loss: 0.5659 - accuracy: 0.9896 - val_loss: 0.7175 - val_accuracy: 0.9265 - lr: 4.6875e-06
Epoch 46/100
41/41 [=====] - 1s 15ms/step - loss: 0.5694 - accuracy: 0.9873 - val_loss: 0.7098 - val_accuracy: 0.9272 - lr: 4.6875e-06
Epoch 47/100
41/41 [=====] - 1s 13ms/step - loss: 0.5707 - accuracy: 0.9900 - val_loss: 0.7090 - val_accuracy: 0.9308 - lr: 4.6875e-06
Epoch 48/100
41/41 [=====] - 1s 15ms/step - loss: 0.5594 - accuracy: 0.9919 - val_loss: 0.7062 - val_accuracy: 0.9322 - lr: 4.6875e-06

Epoch 49/100
41/41 [=====] - 1s 15ms/step - loss: 0.5662 - accuracy: 0.9838 - val_loss: 0.7273 - val_accuracy: 0.9265 - lr: 4.6875e-06
Epoch 50/100
41/41 [=====] - 1s 14ms/step - loss: 0.5576 - accuracy: 0.9908 - val_loss: 0.6928 - val_accuracy: 0.9408 - lr: 4.6875e-06
Epoch 51/100
41/41 [=====] - 1s 16ms/step - loss: 0.5597 - accuracy: 0.9885 - val_loss: 0.7450 - val_accuracy: 0.9179 - lr: 4.6875e-06
Epoch 52/100
41/41 [=====] - 1s 14ms/step - loss: 0.5609 - accuracy: 0.9873 - val_loss: 0.9208 - val_accuracy: 0.8365 - lr: 4.6875e-06
Epoch 53/100
41/41 [=====] - 0s 12ms/step - loss: 0.5575 - accuracy: 0.9900 - val_loss: 0.7817 - val_accuracy: 0.9058 - lr: 4.6875e-06
Epoch 54/100
41/41 [=====] - 0s 12ms/step - loss: 0.5522 - accuracy: 0.9896 - val_loss: 0.6884 - val_accuracy: 0.9293 - lr: 4.6875e-06
Epoch 55/100
41/41 [=====] - 0s 12ms/step - loss: 0.5538 - accuracy: 0.9904 - val_loss: 0.6903 - val_accuracy: 0.9393 - lr: 4.6875e-06
Epoch 56/100
41/41 [=====] - 0s 11ms/step - loss: 0.5523 - accuracy: 0.9912 - val_loss: 0.6932 - val_accuracy: 0.9308 - lr: 4.6875e-06
Epoch 57/100
41/41 [=====] - 0s 11ms/step - loss: 0.5537 - accuracy: 0.9900 - val_loss: 0.6989 - val_accuracy: 0.9293 - lr: 4.6875e-06
Epoch 58/100
41/41 [=====] - 1s 13ms/step - loss: 0.5507 - accuracy: 0.9896 - val_loss: 0.6862 - val_accuracy: 0.9393 - lr: 4.6875e-06
Epoch 59/100
41/41 [=====] - 1s 15ms/step - loss: 0.5494 - accuracy: 0.9908 - val_loss: 0.6898 - val_accuracy: 0.9350 - lr: 4.6875e-06
Epoch 60/100
41/41 [=====] - 1s 13ms/step - loss: 0.5491 - accuracy: 0.9915 - val_loss: 0.6872 - val_accuracy: 0.9350 - lr: 4.6875e-06
Epoch 61/100
41/41 [=====] - 1s 14ms/step - loss: 0.5460 - accuracy: 0.9915 - val_loss: 0.6972 - val_accuracy: 0.9308 - lr: 4.6875e-06
Epoch 62/100
37/41 [=====>...] - ETA: 0s - loss: 0.5463 - accuracy: 0.9903
Epoch 62: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
41/41 [=====] - 1s 13ms/step - loss: 0.5457 - accuracy: 0.9908 - val_loss: 0.8035 - val_accuracy: 0.8851 - lr: 4.6875e-06
[100, 64, 0.4]
100.0%
Epoch 1/100
38/38 [=====] - 1s 19ms/step - loss: 3.2345 - accuracy: 0.1471 - val_loss: 2.9766 - val_accuracy: 0.0956 - lr: 3.0000e-04
Epoch 2/100
38/38 [=====] - 1s 14ms/step - loss: 2.7438 - accuracy: 0.2433 - val_loss: 3.3185 - val_accuracy: 0.0956 - lr: 3.0000e-04
Epoch 3/100
38/38 [=====] - 0s 12ms/step - loss: 2.5057 - accuracy: 0.2904 - val_loss: 3.8069 - val_accuracy: 0.0956 - lr: 3.0000e-04
Epoch 4/100
38/38 [=====] - 1s 14ms/step - loss: 2.2952 - accuracy: 0.3487 - val_loss: 4.4648 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 5/100

34/38 [=====>....] - ETA: 0s - loss: 2.0694 - accuracy: 0.434
3
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.000150000071246177.
38/38 [=====] - 1s 14ms/step - loss: 2.0733 - accuracy:
0.4342 - val_loss: 5.3642 - val_accuracy: 0.1081 - lr: 3.0000e-04
Epoch 6/100
38/38 [=====] - 1s 15ms/step - loss: 1.8218 - accuracy:
0.5317 - val_loss: 6.7538 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 7/100
38/38 [=====] - 0s 13ms/step - loss: 1.6495 - accuracy:
0.6054 - val_loss: 7.5868 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 8/100
38/38 [=====] - 0s 12ms/step - loss: 1.5364 - accuracy:
0.6471 - val_loss: 8.5346 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 9/100
34/38 [=====>....] - ETA: 0s - loss: 1.4389 - accuracy: 0.692
6
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
38/38 [=====] - 1s 13ms/step - loss: 1.4353 - accuracy:
0.6938 - val_loss: 9.2711 - val_accuracy: 0.1081 - lr: 1.5000e-04
Epoch 10/100
38/38 [=====] - 1s 14ms/step - loss: 1.2976 - accuracy:
0.7487 - val_loss: 9.5258 - val_accuracy: 0.1125 - lr: 7.5000e-05
Epoch 11/100
38/38 [=====] - 1s 13ms/step - loss: 1.2361 - accuracy:
0.7779 - val_loss: 9.6763 - val_accuracy: 0.1312 - lr: 7.5000e-05
Epoch 12/100
38/38 [=====] - 0s 13ms/step - loss: 1.1677 - accuracy:
0.7975 - val_loss: 10.0824 - val_accuracy: 0.1375 - lr: 7.5000e-05
Epoch 13/100
37/38 [=====>.] - ETA: 0s - loss: 1.1282 - accuracy: 0.812
5
Epoch 13: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
38/38 [=====] - 1s 15ms/step - loss: 1.1263 - accuracy:
0.8138 - val_loss: 9.4755 - val_accuracy: 0.1450 - lr: 7.5000e-05
Epoch 14/100
38/38 [=====] - 1s 15ms/step - loss: 1.0539 - accuracy:
0.8512 - val_loss: 8.4687 - val_accuracy: 0.1412 - lr: 3.7500e-05
Epoch 15/100
38/38 [=====] - 1s 15ms/step - loss: 1.0400 - accuracy:
0.8467 - val_loss: 7.5579 - val_accuracy: 0.1619 - lr: 3.7500e-05
Epoch 16/100
38/38 [=====] - 1s 14ms/step - loss: 1.0160 - accuracy:
0.8562 - val_loss: 5.8184 - val_accuracy: 0.1875 - lr: 3.7500e-05
Epoch 17/100
34/38 [=====>....] - ETA: 0s - loss: 0.9813 - accuracy: 0.875
5
Epoch 17: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
38/38 [=====] - 0s 13ms/step - loss: 0.9803 - accuracy:
0.8767 - val_loss: 5.3505 - val_accuracy: 0.1800 - lr: 3.7500e-05
Epoch 18/100
38/38 [=====] - 1s 14ms/step - loss: 0.9428 - accuracy:
0.8888 - val_loss: 4.8214 - val_accuracy: 0.2075 - lr: 1.8750e-05
Epoch 19/100
38/38 [=====] - 1s 14ms/step - loss: 0.9275 - accuracy:
0.8954 - val_loss: 3.1047 - val_accuracy: 0.2944 - lr: 1.8750e-05
Epoch 20/100
38/38 [=====] - 0s 12ms/step - loss: 0.9262 - accuracy:
0.8925 - val_loss: 2.6669 - val_accuracy: 0.3681 - lr: 1.8750e-05
Epoch 21/100

38/38 [=====] - 0s 12ms/step - loss: 0.9139 - accuracy: 0.8917 - val_loss: 2.2847 - val_accuracy: 0.3825 - lr: 1.8750e-05
Epoch 22/100
38/38 [=====] - 0s 12ms/step - loss: 0.9171 - accuracy: 0.8900 - val_loss: 1.9033 - val_accuracy: 0.4725 - lr: 1.8750e-05
Epoch 23/100
38/38 [=====] - 0s 12ms/step - loss: 0.8968 - accuracy: 0.8929 - val_loss: 1.6992 - val_accuracy: 0.5519 - lr: 1.8750e-05
Epoch 24/100
38/38 [=====] - 0s 13ms/step - loss: 0.8809 - accuracy: 0.9079 - val_loss: 1.4626 - val_accuracy: 0.6675 - lr: 1.8750e-05
Epoch 25/100
38/38 [=====] - 1s 15ms/step - loss: 0.8793 - accuracy: 0.9087 - val_loss: 1.4957 - val_accuracy: 0.6206 - lr: 1.8750e-05
Epoch 26/100
38/38 [=====] - 1s 15ms/step - loss: 0.8661 - accuracy: 0.9092 - val_loss: 1.9755 - val_accuracy: 0.4406 - lr: 1.8750e-05
Epoch 27/100
38/38 [=====] - 1s 14ms/step - loss: 0.8558 - accuracy: 0.9146 - val_loss: 1.7328 - val_accuracy: 0.5175 - lr: 1.8750e-05
Epoch 28/100
38/38 [=====] - 1s 14ms/step - loss: 0.8464 - accuracy: 0.9179 - val_loss: 1.2548 - val_accuracy: 0.7212 - lr: 1.8750e-05
Epoch 29/100
38/38 [=====] - 1s 14ms/step - loss: 0.8494 - accuracy: 0.9167 - val_loss: 1.1468 - val_accuracy: 0.7756 - lr: 1.8750e-05
Epoch 30/100
38/38 [=====] - 1s 16ms/step - loss: 0.8239 - accuracy: 0.9208 - val_loss: 1.3745 - val_accuracy: 0.6475 - lr: 1.8750e-05
Epoch 31/100
38/38 [=====] - 1s 14ms/step - loss: 0.8333 - accuracy: 0.9167 - val_loss: 1.2808 - val_accuracy: 0.6938 - lr: 1.8750e-05
Epoch 32/100
38/38 [=====] - 1s 15ms/step - loss: 0.8058 - accuracy: 0.9271 - val_loss: 1.1511 - val_accuracy: 0.7788 - lr: 1.8750e-05
Epoch 33/100
38/38 [=====] - ETA: 0s - loss: 0.8109 - accuracy: 0.9250
Epoch 33: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
38/38 [=====] - 1s 14ms/step - loss: 0.8109 - accuracy: 0.9250 - val_loss: 1.2393 - val_accuracy: 0.7481 - lr: 1.8750e-05
Epoch 34/100
38/38 [=====] - 1s 14ms/step - loss: 0.7948 - accuracy: 0.9304 - val_loss: 1.2849 - val_accuracy: 0.7025 - lr: 9.3750e-06
Epoch 35/100
38/38 [=====] - 1s 14ms/step - loss: 0.7940 - accuracy: 0.9279 - val_loss: 1.2097 - val_accuracy: 0.7538 - lr: 9.3750e-06
Epoch 36/100
38/38 [=====] - 1s 14ms/step - loss: 0.7877 - accuracy: 0.9383 - val_loss: 1.0123 - val_accuracy: 0.8381 - lr: 9.3750e-06
Epoch 37/100
38/38 [=====] - 1s 15ms/step - loss: 0.7848 - accuracy: 0.9325 - val_loss: 1.0059 - val_accuracy: 0.8381 - lr: 9.3750e-06
Epoch 38/100
38/38 [=====] - 0s 13ms/step - loss: 0.7850 - accuracy: 0.9383 - val_loss: 1.0557 - val_accuracy: 0.8125 - lr: 9.3750e-06
Epoch 39/100
38/38 [=====] - 1s 14ms/step - loss: 0.7711 - accuracy: 0.9408 - val_loss: 1.0870 - val_accuracy: 0.7850 - lr: 9.3750e-06
Epoch 40/100

38/38 [=====] - 0s 13ms/step - loss: 0.7700 - accuracy: 0.9379 - val_loss: 1.0059 - val_accuracy: 0.8350 - lr: 9.3750e-06
Epoch 41/100
38/38 [=====] - 1s 14ms/step - loss: 0.7700 - accuracy: 0.9425 - val_loss: 0.9811 - val_accuracy: 0.8500 - lr: 9.3750e-06
Epoch 42/100
38/38 [=====] - 1s 14ms/step - loss: 0.7675 - accuracy: 0.9421 - val_loss: 1.0180 - val_accuracy: 0.8338 - lr: 9.3750e-06
Epoch 43/100
38/38 [=====] - 1s 15ms/step - loss: 0.7639 - accuracy: 0.9400 - val_loss: 0.9774 - val_accuracy: 0.8444 - lr: 9.3750e-06
Epoch 44/100
38/38 [=====] - 1s 15ms/step - loss: 0.7574 - accuracy: 0.9454 - val_loss: 0.9999 - val_accuracy: 0.8356 - lr: 9.3750e-06
Epoch 45/100
38/38 [=====] - 1s 14ms/step - loss: 0.7555 - accuracy: 0.9425 - val_loss: 0.9684 - val_accuracy: 0.8487 - lr: 9.3750e-06
Epoch 46/100
38/38 [=====] - 1s 13ms/step - loss: 0.7539 - accuracy: 0.9421 - val_loss: 0.9927 - val_accuracy: 0.8356 - lr: 9.3750e-06
Epoch 47/100
38/38 [=====] - 0s 13ms/step - loss: 0.7426 - accuracy: 0.9433 - val_loss: 1.1094 - val_accuracy: 0.7887 - lr: 9.3750e-06
Epoch 48/100
38/38 [=====] - 1s 13ms/step - loss: 0.7381 - accuracy: 0.9542 - val_loss: 1.0931 - val_accuracy: 0.7837 - lr: 9.3750e-06
Epoch 49/100
37/38 [=====>.] - ETA: 0s - loss: 0.7375 - accuracy: 0.9510
Epoch 49: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
38/38 [=====] - 0s 12ms/step - loss: 0.7387 - accuracy: 0.9500 - val_loss: 1.0340 - val_accuracy: 0.8125 - lr: 9.3750e-06
Epoch 50/100
38/38 [=====] - 0s 12ms/step - loss: 0.7396 - accuracy: 0.9475 - val_loss: 0.9609 - val_accuracy: 0.8619 - lr: 4.6875e-06
Epoch 51/100
38/38 [=====] - 0s 12ms/step - loss: 0.7345 - accuracy: 0.9496 - val_loss: 0.9860 - val_accuracy: 0.8419 - lr: 4.6875e-06
Epoch 52/100
38/38 [=====] - 0s 12ms/step - loss: 0.7410 - accuracy: 0.9467 - val_loss: 0.9524 - val_accuracy: 0.8581 - lr: 4.6875e-06
Epoch 53/100
38/38 [=====] - 0s 12ms/step - loss: 0.7339 - accuracy: 0.9542 - val_loss: 0.9717 - val_accuracy: 0.8500 - lr: 4.6875e-06
Epoch 54/100
38/38 [=====] - 0s 12ms/step - loss: 0.7147 - accuracy: 0.9533 - val_loss: 1.0035 - val_accuracy: 0.8250 - lr: 4.6875e-06
Epoch 55/100
38/38 [=====] - 0s 13ms/step - loss: 0.7159 - accuracy: 0.9608 - val_loss: 0.9524 - val_accuracy: 0.8619 - lr: 4.6875e-06
Epoch 56/100
36/38 [=====>..] - ETA: 0s - loss: 0.7172 - accuracy: 0.9579
Epoch 56: ReduceLROnPlateau reducing learning rate to 2.3437501113221515e-06.
38/38 [=====] - 1s 14ms/step - loss: 0.7189 - accuracy: 0.9571 - val_loss: 0.9670 - val_accuracy: 0.8512 - lr: 4.6875e-06
Epoch 57/100
38/38 [=====] - 1s 14ms/step - loss: 0.7146 - accuracy: 0.9546 - val_loss: 0.9398 - val_accuracy: 0.8619 - lr: 2.3438e-06
Epoch 58/100

```

38/38 [=====] - 1s 14ms/step - loss: 0.7229 - accuracy:
0.9475 - val_loss: 0.9286 - val_accuracy: 0.8662 - lr: 2.3438e-06
Epoch 59/100
38/38 [=====] - 0s 12ms/step - loss: 0.7140 - accuracy:
0.9596 - val_loss: 0.9295 - val_accuracy: 0.8675 - lr: 2.3438e-06
Epoch 60/100
38/38 [=====] - 1s 14ms/step - loss: 0.7185 - accuracy:
0.9546 - val_loss: 0.9246 - val_accuracy: 0.8675 - lr: 2.3438e-06
Epoch 61/100
38/38 [=====] - 1s 14ms/step - loss: 0.7148 - accuracy:
0.9542 - val_loss: 0.9252 - val_accuracy: 0.8650 - lr: 2.3438e-06
Epoch 62/100
38/38 [=====] - 1s 15ms/step - loss: 0.7145 - accuracy:
0.9600 - val_loss: 0.9275 - val_accuracy: 0.8644 - lr: 2.3438e-06
Epoch 63/100
38/38 [=====] - 1s 15ms/step - loss: 0.7158 - accuracy:
0.9533 - val_loss: 0.9308 - val_accuracy: 0.8650 - lr: 2.3438e-06
Epoch 64/100
38/38 [=====] - ETA: 0s - loss: 0.7110 - accuracy: 0.958
3
Epoch 64: ReduceLROnPlateau reducing learning rate to 1.1718750556610757e-06.
38/38 [=====] - 1s 14ms/step - loss: 0.7110 - accuracy:
0.9583 - val_loss: 0.9354 - val_accuracy: 0.8587 - lr: 2.3438e-06
Epoch 65/100
38/38 [=====] - 1s 15ms/step - loss: 0.7134 - accuracy:
0.9550 - val_loss: 0.9285 - val_accuracy: 0.8637 - lr: 1.1719e-06
Epoch 66/100
38/38 [=====] - 1s 14ms/step - loss: 0.7105 - accuracy:
0.9583 - val_loss: 0.9225 - val_accuracy: 0.8644 - lr: 1.1719e-06
Epoch 67/100
38/38 [=====] - 1s 14ms/step - loss: 0.7180 - accuracy:
0.9542 - val_loss: 0.9210 - val_accuracy: 0.8650 - lr: 1.1719e-06
Epoch 68/100
38/38 [=====] - 1s 15ms/step - loss: 0.7052 - accuracy:
0.9550 - val_loss: 0.9254 - val_accuracy: 0.8675 - lr: 1.1719e-06
Epoch 69/100
38/38 [=====] - 1s 15ms/step - loss: 0.7106 - accuracy:
0.9554 - val_loss: 0.9213 - val_accuracy: 0.8637 - lr: 1.1719e-06
Epoch 70/100
38/38 [=====] - 1s 14ms/step - loss: 0.7077 - accuracy:
0.9521 - val_loss: 0.9162 - val_accuracy: 0.8650 - lr: 1.1719e-06
Epoch 71/100
38/38 [=====] - 1s 13ms/step - loss: 0.7088 - accuracy:
0.9646 - val_loss: 0.9147 - val_accuracy: 0.8662 - lr: 1.1719e-06

```

[100, 8, 0.2] seem the best, so well try it:

```
In [81]: best_model, history_cnn = single_model_trainer(100,8,0.2)
```

Epoch 1/100
400/400 [=====] - 5s 11ms/step - loss: 2.8837 - accuracy
: 0.1787 - val_loss: 3.1952 - val_accuracy: 0.1488 - lr: 3.0000e-04

Epoch 2/100
400/400 [=====] - 4s 11ms/step - loss: 2.2900 - accuracy
: 0.3131 - val_loss: 13.5822 - val_accuracy: 0.1213 - lr: 3.0000e-04

Epoch 3/100
400/400 [=====] - 6s 14ms/step - loss: 1.6535 - accuracy
: 0.5491 - val_loss: 34.0757 - val_accuracy: 0.1075 - lr: 3.0000e-04

Epoch 4/100
400/400 [=====] - 4s 11ms/step - loss: 1.0296 - accuracy
: 0.7906 - val_loss: 7.9944 - val_accuracy: 0.2250 - lr: 3.0000e-04

Epoch 5/100
397/400 [=====>.] - ETA: 0s - loss: 0.6864 - accuracy: 0.9
203
Epoch 5: ReduceLROnPlateau reducing learning rate to 0.0001500000071246177.
400/400 [=====] - 4s 9ms/step - loss: 0.6856 - accuracy:
0.9206 - val_loss: 6.8680 - val_accuracy: 0.3125 - lr: 3.0000e-04

Epoch 6/100
400/400 [=====] - 5s 12ms/step - loss: 0.5254 - accuracy
: 0.9638 - val_loss: 4.6406 - val_accuracy: 0.3162 - lr: 1.5000e-04

Epoch 7/100
400/400 [=====] - 4s 9ms/step - loss: 0.4484 - accuracy:
0.9766 - val_loss: 4.0665 - val_accuracy: 0.4975 - lr: 1.5000e-04

Epoch 8/100
400/400 [=====] - 4s 9ms/step - loss: 0.4064 - accuracy:
0.9806 - val_loss: 7.8265 - val_accuracy: 0.1462 - lr: 1.5000e-04

Epoch 9/100
397/400 [=====>.] - ETA: 0s - loss: 0.3609 - accuracy: 0.9
852
Epoch 9: ReduceLROnPlateau reducing learning rate to 7.500000356230885e-05.
400/400 [=====] - 4s 9ms/step - loss: 0.3607 - accuracy:
0.9853 - val_loss: 5.1135 - val_accuracy: 0.2387 - lr: 1.5000e-04

Epoch 10/100
400/400 [=====] - 5s 13ms/step - loss: 0.3321 - accuracy
: 0.9884 - val_loss: 1.1996 - val_accuracy: 0.6775 - lr: 7.5000e-05

Epoch 11/100
400/400 [=====] - 4s 10ms/step - loss: 0.3067 - accuracy
: 0.9919 - val_loss: 1.3974 - val_accuracy: 0.7237 - lr: 7.5000e-05

Epoch 12/100
400/400 [=====] - 4s 11ms/step - loss: 0.2953 - accuracy
: 0.9925 - val_loss: 1.3358 - val_accuracy: 0.7462 - lr: 7.5000e-05

Epoch 13/100
400/400 [=====] - 4s 10ms/step - loss: 0.2752 - accuracy
: 0.9941 - val_loss: 0.5759 - val_accuracy: 0.8625 - lr: 7.5000e-05

Epoch 14/100
400/400 [=====] - 3s 8ms/step - loss: 0.2547 - accuracy:
0.9944 - val_loss: 0.4346 - val_accuracy: 0.9237 - lr: 7.5000e-05

Epoch 15/100
400/400 [=====] - 13s 33ms/step - loss: 0.2437 - accurac
y: 0.9944 - val_loss: 4.0893 - val_accuracy: 0.5150 - lr: 7.5000e-05

Epoch 16/100
400/400 [=====] - 4s 11ms/step - loss: 0.2252 - accuracy
: 0.9972 - val_loss: 23.2301 - val_accuracy: 0.1587 - lr: 7.5000e-05

Epoch 17/100
400/400 [=====] - 4s 9ms/step - loss: 0.2074 - accuracy:
0.9966 - val_loss: 0.7597 - val_accuracy: 0.8138 - lr: 7.5000e-05

Epoch 18/100
393/400 [=====>.] - ETA: 0s - loss: 0.2032 - accuracy: 0.9
962

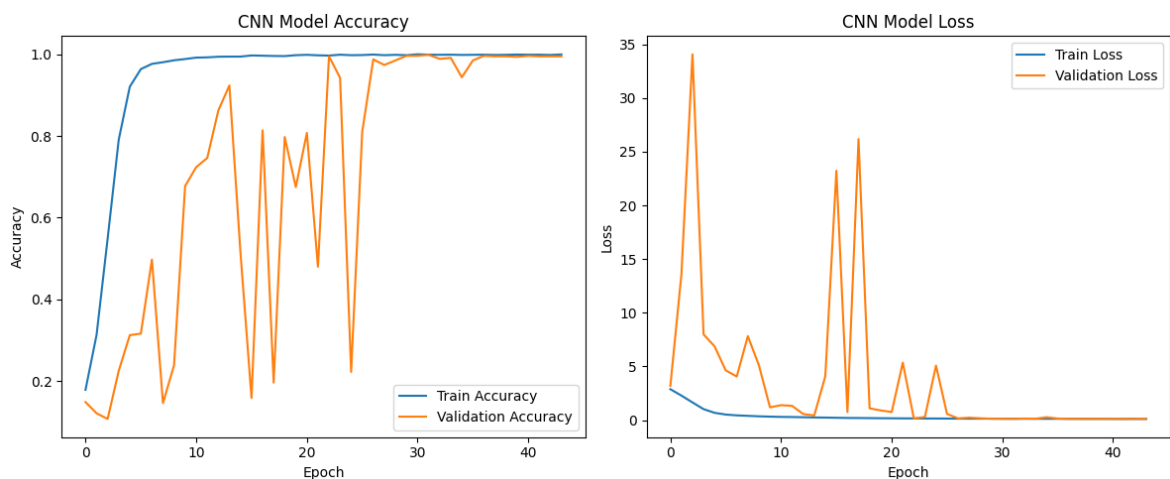
Epoch 18: ReduceLROnPlateau reducing learning rate to 3.7500001781154424e-05.
400/400 [=====] - 4s 9ms/step - loss: 0.2031 - accuracy: 0.9959 - val_loss: 26.1864 - val_accuracy: 0.1963 - lr: 7.5000e-05
Epoch 19/100
400/400 [=====] - 4s 9ms/step - loss: 0.1937 - accuracy: 0.9956 - val_loss: 1.1039 - val_accuracy: 0.7975 - lr: 3.7500e-05
Epoch 20/100
400/400 [=====] - 4s 10ms/step - loss: 0.1827 - accuracy: 0.9978 - val_loss: 0.9094 - val_accuracy: 0.6750 - lr: 3.7500e-05
Epoch 21/100
400/400 [=====] - 7s 17ms/step - loss: 0.1765 - accuracy: 0.9987 - val_loss: 0.7595 - val_accuracy: 0.8075 - lr: 3.7500e-05
Epoch 22/100
397/400 [=====>.] - ETA: 0s - loss: 0.1695 - accuracy: 0.9975
Epoch 22: ReduceLROnPlateau reducing learning rate to 1.8750000890577212e-05.
400/400 [=====] - 3s 9ms/step - loss: 0.1694 - accuracy: 0.9975 - val_loss: 5.3630 - val_accuracy: 0.4800 - lr: 3.7500e-05
Epoch 23/100
400/400 [=====] - 9s 23ms/step - loss: 0.1646 - accuracy: 0.9969 - val_loss: 0.1552 - val_accuracy: 0.9950 - lr: 1.8750e-05
Epoch 24/100
400/400 [=====] - 3s 9ms/step - loss: 0.1572 - accuracy: 0.9991 - val_loss: 0.3052 - val_accuracy: 0.9425 - lr: 1.8750e-05
Epoch 25/100
400/400 [=====] - 4s 9ms/step - loss: 0.1558 - accuracy: 0.9978 - val_loss: 5.0841 - val_accuracy: 0.2225 - lr: 1.8750e-05
Epoch 26/100
400/400 [=====] - 4s 9ms/step - loss: 0.1510 - accuracy: 0.9981 - val_loss: 0.5805 - val_accuracy: 0.8100 - lr: 1.8750e-05
Epoch 27/100
398/400 [=====>.] - ETA: 0s - loss: 0.1433 - accuracy: 0.9994
Epoch 27: ReduceLROnPlateau reducing learning rate to 9.375000445288606e-06.
400/400 [=====] - 4s 9ms/step - loss: 0.1432 - accuracy: 0.9994 - val_loss: 0.1674 - val_accuracy: 0.9875 - lr: 1.8750e-05
Epoch 28/100
400/400 [=====] - 5s 14ms/step - loss: 0.1407 - accuracy: 0.9978 - val_loss: 0.2376 - val_accuracy: 0.9737 - lr: 9.3750e-06
Epoch 29/100
400/400 [=====] - 4s 9ms/step - loss: 0.1416 - accuracy: 0.9987 - val_loss: 0.1780 - val_accuracy: 0.9850 - lr: 9.3750e-06
Epoch 30/100
400/400 [=====] - 4s 9ms/step - loss: 0.1408 - accuracy: 0.9978 - val_loss: 0.1365 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 31/100
400/400 [=====] - 3s 8ms/step - loss: 0.1365 - accuracy: 1.0000 - val_loss: 0.1311 - val_accuracy: 0.9962 - lr: 9.3750e-06
Epoch 32/100
400/400 [=====] - 4s 10ms/step - loss: 0.1324 - accuracy: 0.9991 - val_loss: 0.1279 - val_accuracy: 0.9987 - lr: 9.3750e-06
Epoch 33/100
400/400 [=====] - 4s 9ms/step - loss: 0.1324 - accuracy: 0.9987 - val_loss: 0.1541 - val_accuracy: 0.9887 - lr: 9.3750e-06
Epoch 34/100
400/400 [=====] - 4s 10ms/step - loss: 0.1308 - accuracy: 0.9991 - val_loss: 0.1376 - val_accuracy: 0.9912 - lr: 9.3750e-06
Epoch 35/100
400/400 [=====] - 4s 11ms/step - loss: 0.1305 - accuracy: 0.9984 - val_loss: 0.2780 - val_accuracy: 0.9438 - lr: 9.3750e-06

```

Epoch 36/100
398/400 [=====>.] - ETA: 0s - loss: 0.1289 - accuracy: 0.9987
Epoch 36: ReduceLROnPlateau reducing learning rate to 4.687500222644303e-06.
400/400 [=====] - 4s 9ms/step - loss: 0.1289 - accuracy: 0.9987 - val_loss: 0.1540 - val_accuracy: 0.9850 - lr: 9.3750e-06
Epoch 37/100
400/400 [=====] - 4s 11ms/step - loss: 0.1262 - accuracy: 0.9991 - val_loss: 0.1286 - val_accuracy: 0.9962 - lr: 4.6875e-06
Epoch 38/100
400/400 [=====] - 4s 10ms/step - loss: 0.1249 - accuracy: 0.9984 - val_loss: 0.1252 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 39/100
400/400 [=====] - 4s 11ms/step - loss: 0.1249 - accuracy: 0.9987 - val_loss: 0.1209 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 40/100
400/400 [=====] - 3s 7ms/step - loss: 0.1224 - accuracy: 0.9994 - val_loss: 0.1281 - val_accuracy: 0.9937 - lr: 4.6875e-06
Epoch 41/100
400/400 [=====] - 4s 11ms/step - loss: 0.1210 - accuracy: 0.9991 - val_loss: 0.1210 - val_accuracy: 0.9962 - lr: 4.6875e-06
Epoch 42/100
400/400 [=====] - 3s 8ms/step - loss: 0.1188 - accuracy: 0.9991 - val_loss: 0.1175 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 43/100
400/400 [=====] - 4s 9ms/step - loss: 0.1223 - accuracy: 0.9984 - val_loss: 0.1192 - val_accuracy: 0.9950 - lr: 4.6875e-06
Epoch 44/100
400/400 [=====] - 4s 11ms/step - loss: 0.1177 - accuracy: 0.9997 - val_loss: 0.1208 - val_accuracy: 0.9950 - lr: 4.6875e-06

```

```
In [82]: training_progress(history_cnn)
```



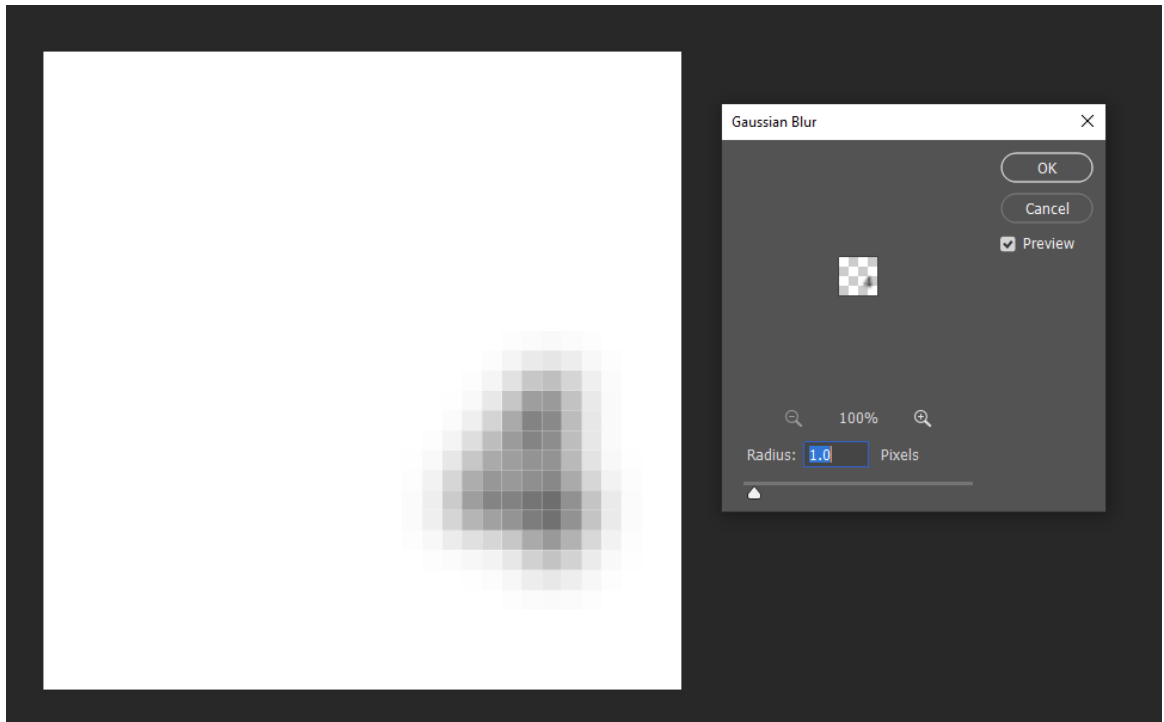
It "jumps" more because of the 8 size batch size, less data for every epoch can make big differences.

Although it "jumps" a lot, we can "catch" or "save" the weights of the best epoch to get the best.

Lets (for fun) test it:

```
In [91]: Image(filename='making of image_test.png')
```

Out[91]:



```
In [88]: image_path = 'image_test.png'

img = tf.keras.utils.load_img(image_path, target_size=(32, 32))

img_array = tf.keras.utils.img_to_array(img)

img_array = np.expand_dims(img_array, axis=0)

predictions = best_model.predict(img_array)

predicted_class = np.argmax(predictions[0])
confidence = predictions[0][predicted_class] * 100

print(f"Model think the number is: {predicted_class}")
print(f"Confidence: {confidence:.2f}%")
```

Model think the number is: 4
Confidence: 97.22%

:)